

Book XI of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Introduction

This volume documents the formal reconstruction of Euclid’s Book XI through Proposition XI.25. Its subject is not the Lean source code as such, but the mathematical structure exposed by formalization: the spatial configurations used by Euclid, the hidden incidence and order obligations, the dependence on earlier propositions, and the synthetic mechanisms required to make the classical arguments precise.

Each proposition is studied on three related but distinct levels:

- Euclid’s historical proof;
- the mathematical proposition reconstructed in a modern synthetic setting;
- the production proof formalized in Lean.

These levels need not coincide line by line. A classical diagram may suppress carrier planes, coplanarity, betweenness, plane intersections, proper-part relations, or comparison data that must become explicit in a formal proof.

The reconstruction is based primarily on Hilbert geometry. Planar arguments are carried out inside explicit plane slices, while genuinely spatial incidence remains ambient. Where it is mathematically useful, a second route uses the carrier and flat viewpoint associated with Wyler. The purpose of that route is not to duplicate the Hilbert development, but to identify which parts of a proof belong to spatial incidence and which belong to a common metric or magnitude core.

This distinction becomes especially visible in the later propositions. Proposition XI.24 admits both a direct Hilbert reconstruction and a Hilbert–Wyler reconstruction with a different spatial incidence front end. Proposition XI.25 then shows that this change of front end can survive through a substantially larger argument: the construction of repeated slabs, geometric comparison of bases and solids, and finally the Eudoxian proportion of Book V.

Accordingly, XI.25 is treated as a theorem about geometric magnitudes, not as a calculation with a previously defined numerical volume. The formal proof retains Euclid’s comparison-of-multiples architecture and makes explicit the proper-part and order arguments that are compressed in the classical text.

The proposition chapters are therefore intended as mathematical documentation of the formal proofs. They explain why the formal argument has the shape it does, which earlier results it uses, what is genuinely three-dimensional, and where the formal reconstruction departs from or clarifies Euclid’s presentation. Lean remains authoritative for exact theorem statements, hypotheses, names, and kernel dependencies.

The detailed implementation architecture of the library is intentionally not repeated here. It appears only where it is needed to understand a particular proof. The aim of the present volume

is to keep the mathematical argument in the foreground.

Chapter 1

Proposition XI.1

1.1 Introduction

Proposition XI.1 opens Euclid’s solid geometry with a statement whose mathematical content is fundamentally incidence-theoretic. No angle, congruence, order, parallelism, or continuity is involved. The intended content is that once two distinct points of a straight line lie in a plane, the whole straight line lies in that plane.

This proposition is therefore an especially useful comparison point between Euclid’s organization of Book XI and Hilbert’s axiomatic organization of space. In the present formal reconstruction the proposition is not proved by reproducing Euclid’s historical circle argument. Its precise incidence content is already one of the basic spatial incidence principles: Hilbert I.6.

The production file is

```
CGJteamLab/Proposition11_1.lean
```

and it imports only

```
CGJteamLab.Hilbert3DInterface
```

from the project-level spatial infrastructure.

1.2 Euclid’s Statement

Euclid states Proposition XI.1 as follows:

A part of a straight line cannot be in the plane of reference and a part in plane more elevated.

The natural precise incidence reading is:

If two distinct points of a straight line lie in a plane, then the whole straight line lies in that plane.

This is the interpretation used by the formal reconstruction.

1.3 Classical Configuration

Let l be a straight line and let π be the reference plane. Choose two distinct points A, B on l such that

$$A \in \pi, \quad B \in \pi.$$

The formal target is

$$l \subset \pi.$$

Euclid describes the same situation diagrammatically by regarding a segment AB as lying in the reference plane and supposing, for contradiction, that a continuation of the same straight line leaves that plane.

There is no essential metric configuration. The proposition concerns only the relation between a line, two of its points, and a plane.

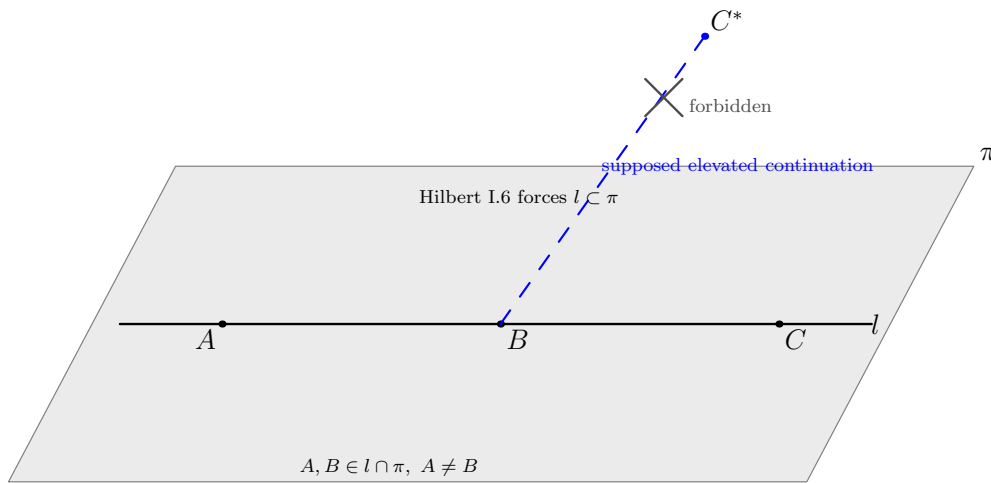


Figure 1.1: The incidence content of Proposition XI.1. The black carrier l contains the distinct points A, B of the reference plane π , hence Hilbert I.6 forces the whole line to remain in π . The blue dashed segment toward C^* does not represent a realizable straight continuation; it records only Euclid's contradictory supposition that the same straight line could leave the plane.

1.4 Euclid's Classical Proof

The received proof proceeds by contradiction. Euclid supposes that a part AB of a straight line ABC lies in the reference plane while the continuation BC is elevated out of that plane.

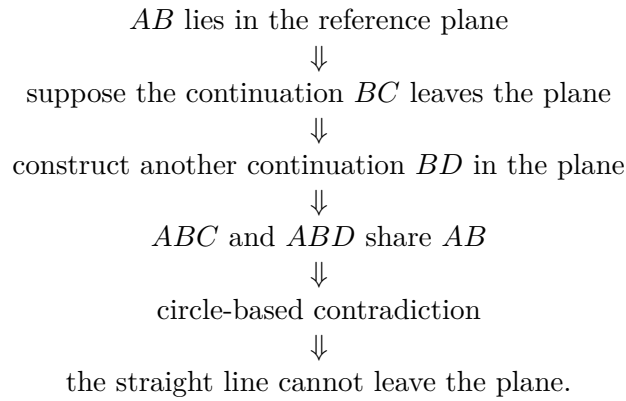
He then chooses in the reference plane another straight continuation BD of AB . The lines ABC and ABD would therefore share the segment AB . Euclid declares this impossible and appeals to a circle with center B and radius AB .

The logical difficulty is that this circle argument is not an adequate spatial incidence foundation. In three-dimensional space the center and radius do not determine a unique circle unless its plane is also specified. More importantly, the desired conclusion is itself an elementary closure property of line–plane incidence and should not require metric circle geometry.

For the formal reconstruction we therefore retain the mathematical incidence content of XI.1 but not this historical argument.

1.5 Classical Dependency Pattern

At the level of the printed proof, the structure is roughly



This is not the dependency graph used by the formal proof.

1.6 What Is Genuinely Three-Dimensional Here?

XI.1 is spatial only because the primitive object π is an ambient plane and the conclusion concerns containment of an ambient line in that plane. There is no change of plane and no interaction between two distinct planes.

The genuine three-dimensional datum is simply

$$A, B \in l \cap \pi, \quad A \neq B.$$

From these data the conclusion is the closure statement

$$l \subset \pi.$$

Thus XI.1 belongs entirely to the spatial incidence layer. It does not require an induced planar geometry $\text{PlaneGeo}(\pi)$, because there is no planar theorem to prove inside π .

1.7 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_1
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (pi : S.Plane)
  (l : Geo.Line)
  (A B : Geo.Point)
  (hAB : Ne A B)
  (hA1 : H.OnLine A l)
  (hB1 : H.OnLine B l)
  (hApi : S.OnPlane A pi)
  (hBpi : S.OnPlane B pi) :
  HilbertLineInPlane Geo l pi := by
exact
  HilbertSpaceIncidence.line_in_plane
    (Geo := Geo)
    A B hAB
    l hA1 hB1
    pi hApi hBpi

```

The conclusion

```
HilbertLineInPlane Geo l pi
```

means that every point incident with l is incident with π .

1.8 Hilbert Incidence Architecture

The decisive declaration is the spatial incidence field

```

HilbertSpaceIncidence.line_in_plane :
forall A B : Geo.Point,
  Ne A B ->
forall l : Geo.Line,
  H.OnLine A l ->
  H.OnLine B l ->
forall pi : S.Plane,
  S.OnPlane A pi ->
  S.OnPlane B pi ->
  HilbertLineInPlane Geo l pi

```

This compatibility field has exactly the geometric content of Hilbert incidence axiom I.6:

If two distinct points of a line lie in a plane, then every point of the line lies in that plane.

After the Hilbert foundation refactor, the source-faithful axiom itself is

```
HilbertGrundlagenGroupI.I6_line_in_plane
```

in `HilbertAxioms.lean`. The field `HilbertSpaceIncidence.line_in_plane` is the legacy-compatible Book XI interface retained in `HilbertSpaceIncidence.lean`. Consequently the formal Euclid theorem remains intentionally a thin wrapper, but the compatibility field should no longer be mistaken for a separate source-level 3D axiom system.

This is structurally analogous to several Book I reconstruction points where a Euclid proposition becomes short after its reusable mathematical content has already been isolated in a general Hilbert theorem or axiom interface.

1.9 Ambient Space and PlaneGeo

No `PlaneGeo` slice is created in XI.1.

The objects remain ambient throughout:

```
A B : Geo.Point
l   : Geo.Line
pi  : S.Plane
```

and the incidence relations are

```
H.OnLine A l
H.OnLine B l
S.OnPlane A pi
S.OnPlane B pi
```

The proof does not require

```
HilbertPlaneIncidence Geo
HilbertSpaceOrder Geo
HilbertSpaceCongruence Geo
HilbertSpaceEuclidean Geo
```

as theorem parameters.

This is the minimal spatial layer appropriate to the proposition.

1.10 Lean Proof Architecture

The Lean proof has exactly one mathematical step:

$$\begin{array}{c}
 A \neq B \\
 A \in l, \quad B \in l \\
 A \in \pi, \quad B \in \pi \\
 \Downarrow \\
 \text{HilbertSpaceIncidence.line_in_plane} \\
 \Downarrow \\
 l \subset \pi.
 \end{array}$$

There are no proposition-local helper lemmas and no case distinctions.

The formal proof does not construct a second line, a circle, an auxiliary plane, or any metric object. The historical contradiction is replaced by the exact spatial incidence principle that expresses the intended content.

1.11 Wyler-Style Flat Reconstruction

The Wyler-inspired reading of XI.1 replaces the informal idea that a line cannot “leave” a plane by a closure statement about generated flats. This is a structural reinterpretation of the proof, not a historical claim about Wyler’s own treatment of Euclid.

Let A, B be the two distinct points of l already known to lie in π . The relevant rank-one chain is

$$\text{carrier}(l) \subseteq \text{Span}\{A, B\} \subseteq \text{carrier}(\pi).$$

The first inclusion expresses that the line carrier is generated by any two of its distinct points; in the flat API the corresponding span is in fact the same rank-one carrier. The second inclusion is the least-flat property of the span: since the plane carrier is a flat containing both generators, it must contain their generated flat.

Thus the public conclusion $l \subseteq \pi$ is recovered by composition:

```

A,B in l,  A != B
  |
  v
carrier(l) subset Span{A,B}
  |
A,B in pi and carrier(pi) is flat
  |
  v
Span{A,B} subset carrier(pi)
  |
  v
l subset pi

```

XI.1 is therefore the smallest nontrivial test of the flat vocabulary. A classical incidence statement becomes a rank-one closure statement, while the public theorem remains exactly Euclid XI.1 rather than being replaced by a span identity.

1.12 Hidden Formal Data

Although the theorem is short, two pieces of data that are implicit in Euclid’s prose are explicit in Lean.

1.12.1 The two plane points must be distinct

The hypothesis

```
hAB : Ne A B
```

is essential. A single point shared by a line and a plane does not force the whole line to lie in that plane.

1.12.2 The line carrier is explicit

Euclid speaks of the straight line ABC . Lean receives an explicit carrier

```
l : Geo.Line
```

together with the incidence witnesses

```
hA1 : H.OnLine A l
hB1 : H.OnLine B l
```

The conclusion is not merely that a third named point lies in the plane; it is the universal carrier statement `HilbertLineInPlane Geo l pi`.

1.13 Relationship with Earlier and Later Book XI Propositions

XI.1 has no earlier Book XI proposition dependency.

Its significance is forward-looking. The closure principle expressed by XI.1 is repeatedly needed whenever a later spatial argument knows two points of a line to lie in a plane and must promote that pointwise information to whole-line containment.

In the present Hilbert architecture later propositions normally use the general declaration

```
HilbertSpaceIncidence.line_in_plane
```

directly rather than importing `euclid_proposition_11_1`. Thus the mathematical content of XI.1 is foundational, while the Euclid-numbered theorem records the correspondence between the *Elements* and the general 3D incidence layer.

XI.2 is the first immediate classical consumer of XI.1: Euclid uses the principle that a line determined inside a plane cannot subsequently leave that plane. In the formal development XI.2 can instead call the stronger general spatial theorem for two intersecting lines.

1.14 Relationship with Hilbert's Group I

XI.1 is a particularly direct meeting point between Euclid and Hilbert.

In the current project, Hilbert's spatial incidence layer contains:

- existence of points on lines;
- existence and uniqueness of planes through noncollinear points;
- line closure under two plane points;

- the common-point principle for two planes;
- existence of noncoplanar points.

XI.1 corresponds exactly to the line-closure clause, Hilbert I.6.

No order axiom from Group II is used. No congruence axiom from Group III is used. No Euclidean parallel axiom from Group IV is used. No continuity assumption is used.

1.15 Classical DAG versus Formal DAG

The central structural difference can be summarized as follows.

Euclid.

supposed line leaving the reference plane
 $\Downarrow$
 second continuation in the reference plane
 $\Downarrow$
 shared segment
 $\Downarrow$
 circle-based contradiction
 $\Downarrow$
 XI.1.

Lean / Hilbert.

$A \neq B, \quad A, B \in l, \quad A, B \in \pi$
 $\Downarrow$
 Hilbert I.6
 HilbertGrundlagenGroupI.I6_line_in_plane
 $\Downarrow$
 compatibility facade
 HilbertSpaceIncidence.line_in_plane
 $\Downarrow$
 euclid_proposition_11_1.

The mathematical DAG remains short because the required incidence principle is source-level Hilbert I.6. The extra formal node records only the current compatibility boundary between the source-faithful foundation and the unchanged Book XI theorem signature.

1.16 Structural Lesson

XI.1 exposes a foundational issue that is easy to miss when reading Book XI diagrammatically. To begin solid geometry one needs rules governing how lines and planes are incident. A planar circle construction cannot replace those rules.

The Hilbert reconstruction makes this boundary explicit. The proposition contains no metric content to reconstruct: its correct mathematical home is the spatial incidence system itself.

This also explains why the production theorem is deliberately thin. The formalization is not avoiding Euclid’s mathematical content; it is locating that content at the level where it functions as a foundational rule for all later spatial arguments.

1.17 Conclusion

Proposition XI.1 is the incidence threshold of Book XI. Its intended mathematical statement is simple:

$$A \neq B, \quad A, B \in l, \quad A, B \in \pi \quad \implies \quad l \subset \pi.$$

In the current formal architecture this is exactly Hilbert I.6. The Lean theorem therefore contains no auxiliary construction and no proposition-local proof machinery: it exposes the already explicit spatial incidence rule under Euclid’s proposition number.

The contrast with the historical proof is instructive. Euclid attempts to exclude a line leaving the plane by a circle-based contradiction. The Hilbert reconstruction instead identifies the missing foundational content directly as line–plane incidence closure. That is both logically cleaner and exactly the form required by the subsequent development of solid geometry.

Chapter 2

Proposition XI.2

2.1 Introduction

Proposition XI.2 contains two closely related incidence statements:

1. two distinct straight lines which intersect lie in one plane;
2. every nondegenerate triangle lies in one plane.

The proposition is still entirely incidence-theoretic. No order, congruence, perpendicularity, parallelism, or continuity is needed.

Its importance is structural. Euclid wants to begin reasoning with configurations of intersecting lines and triangles in space, but such reasoning first requires an existence principle for planes. The present Hilbert reconstruction makes that existence explicit.

The production file is

```
CGJteamLab/Proposition11_2.lean
```

and the decisive reusable theorem is

```
hilbert_plane_through_two_intersecting_lines
```

from

```
CGJteamLab/Hilbert3DInterface.lean
```

2.2 Euclid's Statement

Euclid states:

If two straight lines cut one another, then they lie in one plane; and every triangle lies in one plane.

Thus XI.2 has two conclusions, not merely one. In the formal development they are exposed as two public theorems:

```
euclid_proposition_11_2
euclid_proposition_11_2_triangle
```

The first handles intersecting lines. The second isolates the triangle clause.

2.3 Classical Configuration

Let the two straight lines AB and CD intersect at E .

Euclid then selects points on the rays EC and EB and considers the triangle ECB . The intended plane containing this triangle becomes the plane in which the two intersecting carriers AB and CD must lie.

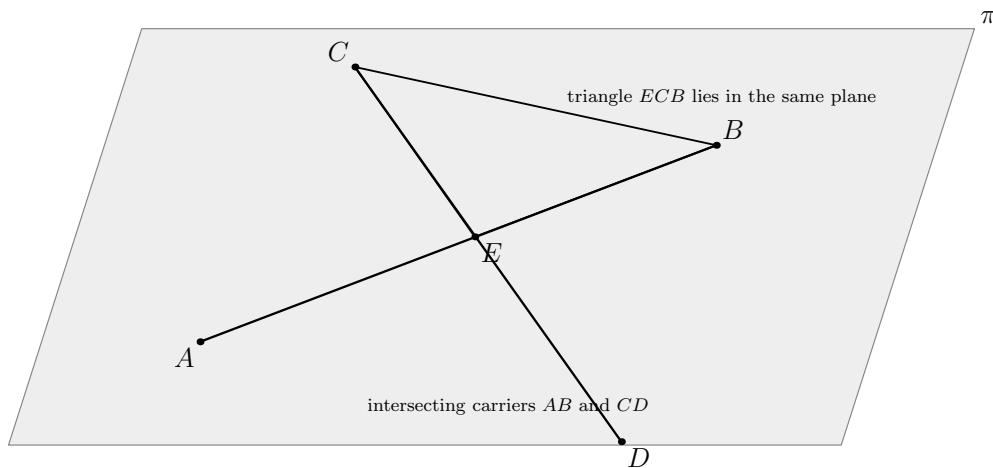


Figure 2.1: The incidence configuration of Proposition XI.2. The two distinct carriers AB and CD meet at E . The triangle ECB determines the plane π ; once EC and EB lie in that plane, the full carriers CD and AB lie there as well.

The essential data are therefore

$$E \in AB, \quad E \in CD, \quad AB \neq CD.$$

For the triangle clause, the corresponding data are simply three noncollinear points.

2.4 Euclid's Classical Proof

Euclid first considers the triangle ECB . He argues that no part of this triangle can lie in one plane while another part lies in another, because that would force part of one of its straight sides to lie in one plane and part in another, contradicting XI.1.

Once the triangle ECB is taken to lie in one plane, its two sides EC and EB lie in that plane. Since EC is part of the carrier CD and EB is part of the carrier AB , XI.1 is used again to

promote the side segments to the whole straight lines. Hence both intersecting lines lie in one plane.

The second conclusion, that every triangle lies in one plane, is treated by the same idea.

The classical local dependency is therefore

$$\begin{array}{c}
 AB \cap CD = \{E\} \\
 \Downarrow \\
 \text{consider triangle } ECB \\
 \Downarrow \\
 \text{XI.1} \\
 \Downarrow \\
 ECB \text{ lies in one plane} \\
 \Downarrow \\
 EC, EB \text{ lie in that plane} \\
 \Downarrow \\
 \text{XI.1} \\
 \Downarrow \\
 AB, CD \text{ lie in that plane.}
 \end{array}$$

2.5 The Missing Plane-Existence Step

The classical argument contains a foundational difficulty. It speaks of “the plane of reference” and reasons about parts of the triangle lying in that plane, but no plane containing the relevant spatial configuration has yet been constructed.

This is exactly the sort of obligation that a modern incidence axiomatization must make explicit.

A natural repair is the Hilbert incidence principle that three noncollinear points determine a plane. Once such a plane exists, XI.1-like line closure can be applied rigorously.

The formal reconstruction therefore does not imitate the circular “reference plane” language. It explicitly produces the required plane from incidence data.

2.6 What Is Genuinely Three-Dimensional Here?

XI.2 is the first Book XI proposition whose conclusion is an existential statement about a plane determined by spatial data.

For the intersecting-line clause the input consists of two ambient lines

$$l, m$$

and a common point

$$P \in l \cap m, \quad l \neq m.$$

The conclusion is the existence of an ambient plane

$$\pi$$

such that

$$l \subset \pi, \quad m \subset \pi.$$

For the triangle clause the input is a noncollinear triple

$$A, B, C,$$

and the conclusion is a plane containing all three points.

No induced planar theorem is needed after the plane has been obtained. Thus XI.2 is genuinely spatial, but purely at the incidence level.

2.7 Formal Statement: Intersecting Lines

The first production theorem is:

```

theorem euclid_proposition_11_2
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (l m : Geo.Line)
  (P : Geo.Point)
  (hlm : Ne l m)
  (hPl : H.OnLine P l)
  (hPm : H.OnLine P m) :
  exists pi : S.Plane,
    HilbertLineInPlane Geo l pi /\
    HilbertLineInPlane Geo m pi := by

  have h :=
    hilbert_plane_through_two_intersecting_lines
      (Geo := Geo)
      l m hlm P hPl hPm

  cases h with
  | intro pi hData =>
    exact
      Exists.intro pi
        (And.intro
          hData.1
          hData.2.1)

```

The theorem intentionally exposes only the existence conclusion required by Euclid.

The reusable Hilbert theorem is stronger: it also proves uniqueness of the plane.

2.8 The Stronger Hilbert Theorem

The interface theorem has the form

```

theorem hilbert_plane_through_two_intersecting_lines
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  (l m : Geo.Line)
  (hlm : Ne l m)
  (P : Geo.Point)
  (hPl : H.OnLine P l)
  (hPm : H.OnLine P m) :
  exists pi : S.Plane,
    HilbertLineInPlane Geo l pi /\
    HilbertLineInPlane Geo m pi /\
  forall rho : S.Plane,
    HilbertLineInPlane Geo l rho ->
    HilbertLineInPlane Geo m rho ->
    rho = pi

```

Thus the Hilbert layer proves not only

$$\exists \pi, \quad l, m \subset \pi,$$

but also

$$\forall \rho, \quad l, m \subset \rho \implies \rho = \pi.$$

The Euclid theorem discards this uniqueness component because it is not part of XI.2's stated conclusion.

2.9 How the Stronger Theorem Is Proved

The reusable theorem is a genuine derived incidence result.

Starting from the common point P , it chooses a second point A on l and a second point B on m , with

$$A \neq P, \quad B \neq P.$$

Because $l \neq m$, the three points A, P, B are noncollinear. Hilbert spatial incidence then supplies a plane through them.

The two line-containment conclusions follow from the XI.1-type incidence principle:

$$A, P \in l \cap \pi \implies l \subset \pi,$$

$$B, P \in m \cap \pi \implies m \subset \pi.$$

Finally, if another plane ρ contained both l and m , then it would contain the same noncollinear triple A, P, B . Plane uniqueness therefore gives

$$\rho = \pi.$$

Schematically,

$$\begin{array}{c}
 l \neq m, \quad P \in l \cap m \\
 \Downarrow \\
 A \in l, A \neq P, \quad B \in m, B \neq P \\
 \Downarrow \\
 A, P, B \text{ noncollinear} \\
 \Downarrow \\
 \text{Hilbert I.4} \\
 \Downarrow \\
 \exists \pi : A, P, B \in \pi \\
 \Downarrow \\
 \text{Hilbert I.6} \\
 \Downarrow \\
 l, m \subset \pi \\
 \Downarrow \\
 \text{Hilbert I.5} \\
 \Downarrow \\
 \pi \text{ is unique.}
 \end{array}$$

This is the real formal incidence content behind the short public wrapper.

2.10 Formal Statement: Triangle Clause

The second public theorem is:

```

theorem euclid_proposition_11_2_triangle
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (A B C : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C)) :
  exists pi : S.Plane,
    S.OnPlane A pi /\
    S.OnPlane B pi /\
    S.OnPlane C pi := by

exact
  HilbertSpaceIncidence.plane_through
    (Geo := Geo)
    A B C hABC

```

This is precisely the existence part needed to say that a nondegenerate triangle lies in a plane.

The theorem returns the three vertex-incidence facts. If explicit side carriers are later introduced, Hilbert I.6 promotes each pair of vertex incidences to whole-line containment.

2.11 Why the Triangle Theorem Is Minimal

It would be possible to formulate a larger theorem returning explicit side lines AB , BC , and CA together with proofs that all three lie in the same plane.

The current theorem deliberately avoids that extra representation layer. At the incidence level a nondegenerate triangle is already determined by three noncollinear vertices. The existence of their plane is the exact content of the second clause of XI.2.

Whenever side carriers are required, they can be constructed separately by the ordinary two-point line theorem and placed in the same plane using `line_in_plane`.

2.12 Spatial Architecture

The intersecting-line theorem uses only:

```
HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

The triangle theorem uses an even smaller signature:

```
HilbertIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

There is no use of

```
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
PlaneGeo
```

Thus XI.2 belongs completely to Hilbert's spatial Group I incidence layer.

2.13 Wyler-Style Flat Reconstruction

XI.2 exhibits the first genuine rank jump. The two clauses of the proposition have two parallel normal forms in the flat calculus.

For two distinct intersecting lines l, m , the common point guarantees that they generate a plane

and the exact carrier identity is

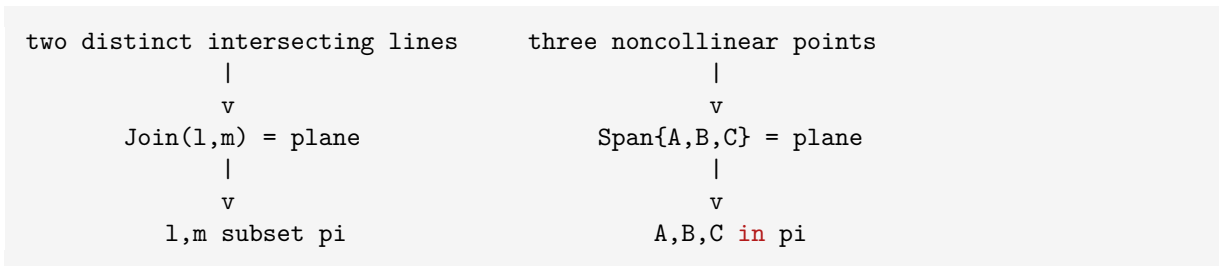
$$\text{Join}(\text{carrier}(l), \text{carrier}(m)) = \text{carrier}(\pi).$$

The public incidence conclusion is then recovered because both generators are contained in their join.

For the triangle clause, three noncollinear points form a rank-two generating set:

$$\text{Span}\{A, B, C\} = \text{carrier}(\pi).$$

Hence the two parts of XI.2 may be read as the two standard ways to generate a plane in three-space:



This is the first place where the point–line–plane hierarchy becomes a rank ladder rather than a list of unrelated incidence constructions. No stronger flat statement replaces the theorem: the exact Euclidean clauses are still the public outputs.

2.14 Hidden Formal Data

Several facts suppressed by the classical diagram must be explicit in Lean.

2.14.1 The two carriers must be distinct

The hypothesis

```
hlm : Ne l m
```

is essential. If $l = m$, infinitely many planes may contain that one line, so the unique-plane theorem is false.

2.14.2 The common point must be explicit

The intersection is represented by

```
P : Geo.Point
hPl : H.OnLine P l
hPm : H.OnLine P m
```

rather than by a primitive relation saying merely that the lines intersect.

2.14.3 Noncollinearity must be proved

The stronger helper does not assume three ready-made noncollinear points. It constructs one auxiliary point on each line and derives noncollinearity from the distinctness of the two carriers.

2.14.4 The witness plane survives

The conclusion is existential:

```
exists pi : S.Plane, ...
```

so the plane is a genuine proof object, not an implicit background surface supplied by the diagram.

2.15 Relationship with Proposition XI.1

Classically XI.1 is the immediate dependency of XI.2.

Euclid repeatedly reasons that if part of a side lies in a plane, then the whole straight line must lie there. That is precisely the content of XI.1.

The formal DAG is organized differently. The public theorem `euclid_proposition_11_2` does not call `euclid_proposition_11_1`. Instead, the stronger general theorem `hilbert_plane_through_two_inter` uses the underlying Hilbert I.6 field directly.

Hence:

classical dependency: $\text{XI.1} \rightarrow \text{XI.2}$,

whereas

formal dependency: $\text{Hilbert I.2--I.6} \rightarrow \text{hilbert_plane_through_two_intersecting_lines} \rightarrow \text{XI.2}$.

This is not a loss of historical structure. It records the fact that XI.1's mathematical content has already been moved into the reusable spatial incidence interface.

2.16 Relationship with Later Book XI Propositions

XI.2 is a basic coplanarity theorem and becomes part of the classical infrastructure for later Book XI arguments.

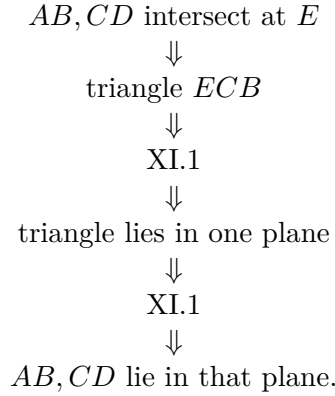
In particular, later propositions repeatedly need to know that two intersecting carriers determine a common plane before a planar theorem can be applied to them.

In the formal development the stronger interface theorem often replaces direct calls to XI.2. This is analogous to XI.1: the Euclid-numbered theorem records the correspondence with the *Elements*, while downstream Lean proofs are free to consume the stronger reusable result.

2.17 Classical DAG versus Formal DAG

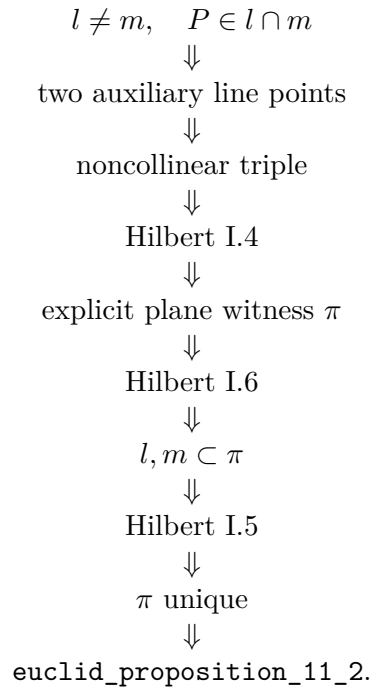
The distinction is especially clear for XI.2.

Euclid.



The weak point is that the plane itself is never produced from an explicit plane-existence principle.

Lean / Hilbert.



For the triangle clause the formal DAG is even shorter:

$$A, B, C \text{ noncollinear} \implies \text{Hilbert I.4} \implies \exists \pi, A, B, C \in \pi.$$

2.18 Structural Lesson

XI.2 makes explicit why solid geometry cannot begin merely by reusing planar reasoning in an unspecified ambient setting.

Before one can say that two intersecting spatial lines are a planar configuration, one must prove the existence of the plane containing them. Before one can treat a triangle as planar, one must prove the existence of the plane through its vertices.

The Hilbert incidence system supplies exactly these missing existence and uniqueness principles. Consequently XI.2 becomes a short Euclid interface over a stronger and reusable spatial theorem.

This proposition is therefore an early example of a recurring pattern in Book XI:

$$\text{diagrammatic coplanarity} \longrightarrow \text{explicit existential plane witness.}$$

That distinction becomes increasingly important in later propositions, where several different planes coexist and their identities or intersections can no longer be left implicit.

2.19 Conclusion

Proposition XI.2 is the first explicit plane-existence theorem of the Euclidean solid-geometry sequence.

Its two conclusions are

$$l \neq m, \quad P \in l \cap m \implies \exists \pi, \quad l, m \subset \pi,$$

and

$$A, B, C \text{ noncollinear} \implies \exists \pi, \quad A, B, C \in \pi.$$

The historical proof tries to obtain these conclusions through XI.1 while speaking of a reference plane that has not itself been produced. The Hilbert reconstruction resolves the issue at the incidence level: noncollinear points determine a plane, two plane points force their whole line into the plane, and three noncollinear common plane points determine that plane uniquely.

The public Euclid theorem is therefore short, but unlike XI.1 its first clause sits on top of a genuine derived spatial theorem. That stronger theorem is the correct reusable form for the remainder of the formal development.

Chapter 3

Proposition XI.3

3.1 Introduction

Proposition XI.3 states that if two distinct planes meet, then their common section is a straight line. In Euclid’s sequence this is the natural continuation of XI.2: once planes containing spatial configurations have been introduced, one must also know how two such planes intersect.

At first sight the proposition seems elementary. In fact it is one of the first explicitly dimension-sensitive steps of Book XI. In genuine three-dimensional incidence geometry, two distinct planes with a common point intersect in a line. In higher-dimensional ambient settings this need not be true: two distinct 2-planes may meet in a single point only. Thus XI.3 is not merely a convenient diagrammatic fact. It is a structural property of the three-dimensional spatial layer.

The production file is

```
CGJteamLab/Proposition11_3.lean
```

and the decisive reusable theorem is

```
hilbert_plane_intersection_line
```

from

```
CGJteamLab/Hilbert3DInterface.lean
```

3.2 Euclid’s Statement

Euclid states:

If two planes cut one another, then their common section is a straight line.

The key phrase is “common section”. In modern incidence language this means the full intersection of the two planes, not merely the existence of one line lying in both.

The present formal reconstruction therefore states XI.3 extensionally: it returns a line l contained in both planes and proves that a point belongs to both planes if and only if it lies on l .

3.3 Classical Configuration

Let π and ρ be two distinct planes having a common point A .

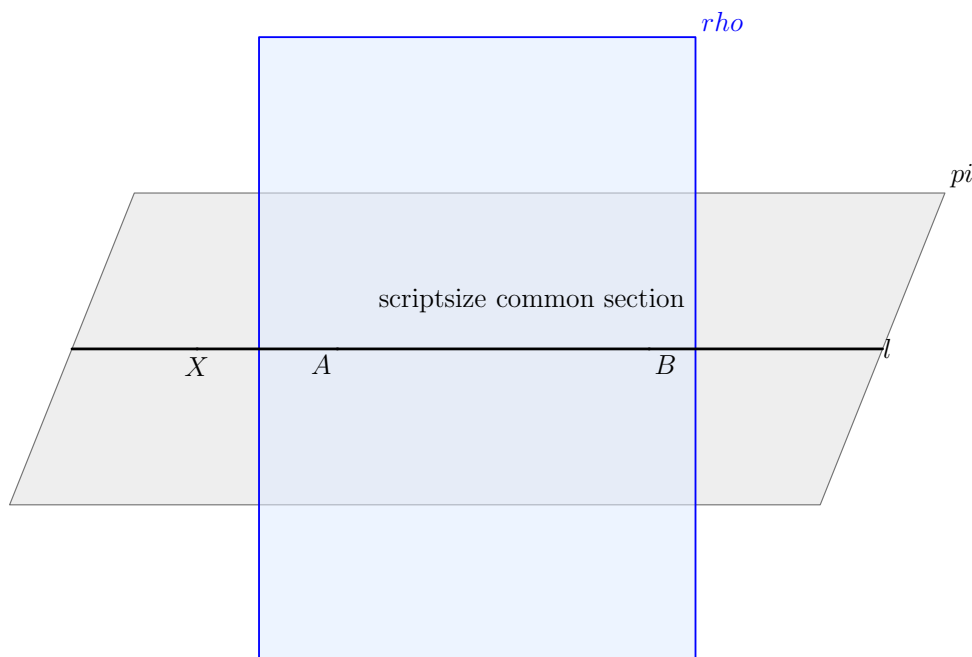
The intended conclusion is that there exists a line l such that

$$A \in l, \quad l \subset \pi, \quad l \subset \rho,$$

and indeed that the whole common section satisfies

$$\pi \cap \rho = l$$

in the extensional incidence sense.



every common point lies on l

Figure 3.1: The incidence configuration of Proposition XI.3. The two distinct planes π and ρ have the common point A . Their common section is the black line l . Any point belonging simultaneously to π and ρ lies on this same line.

The important point is that the line is not background decoration. It is the entire common section to be proved.

3.4 Euclid's Classical Proof

Euclid argues by choosing another common point of the two planes in addition to the given one, and then taking the straight line through the two common points. He then reasons that any

point common to both planes must lie on this same straight line, so the common section is a straight line.

At the historical level the proof is short and intuitive. But once one asks what is required for a formal derivation, two hidden obligations appear:

1. from one common point A , one must obtain a second common point B of the two distinct planes;
2. one must prove not only that some line lies in both planes, but that every common point lies on that line.

These are exactly the obligations discharged by the Hilbert 3D interface.

3.5 Why XI.3 Is Dimension-Sensitive

XI.3 is the first place in Book XI where the three-dimensional nature of the ambient setting becomes mathematically decisive.

In ordinary 3-space, two distinct planes sharing one point must share a whole line. But in a higher-dimensional ambient geometry two distinct 2-planes can intersect in a single point only. Thus the implication

$$A \in \pi \cap \rho, \quad \pi \neq \rho \implies \exists l, \pi \cap \rho = l$$

is not dimension-free.

In the present project this fact is not left to geometric intuition. It is encoded in the spatial incidence architecture through the principle that two distinct planes with one common point admit a second common point, from which the intersection line is then constructed.

This is the first Book XI proposition whose formal meaning strongly depends on our decision to work in synthetic 3-space rather than in an arbitrary ambient dimension.

3.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_3
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (pi rho : S.Plane)
  (hneq : Ne pi rho)
  (A : Geo.Point)
  (hApi : S.OnPlane A pi)
  (hArho : S.OnPlane A rho) :
  exists l : Geo.Line,
    H.OnLine A l /\
    HilbertLineInPlane Geo l pi /\

```

```

HilbertLineInPlane Geo l rho /\
forall X : Geo.Point,
  (S.OnPlane X pi /\ S.OnPlane X rho) <=>
    H.OnLine X l := by

exact
hilbert_plane_intersection_line
  (Geo := Geo)
  pi rho hneq
  A hApi hArho

```

This is a deliberately faithful formalization of the phrase “their common section is a straight line”. The theorem does not stop at

$$\exists l, l \subset \pi, l \subset \rho.$$

It also includes the extensional characterization

$$X \in \pi \cap \rho \iff X \in l.$$

3.7 The Stronger Intersection Characterization

The theorem used in the proof is not proposition-local infrastructure. It is the general spatial theorem

```

theorem hilbert_plane_intersection_line
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  (pi rho : S.Plane)
  (hneq : Ne pi rho)
  (A : Geo.Point)
  (hApi : S.OnPlane A pi)
  (hArho : S.OnPlane A rho) :
  exists l : Geo.Line,
    H.OnLine A l /\
    HilbertLineInPlane Geo l pi /\
    HilbertLineInPlane Geo l rho /\
  forall X : Geo.Point,
    (S.OnPlane X pi /\ S.OnPlane X rho) <=>
      H.OnLine X l

```

Thus XI.3 is a thin Euclid wrapper over a reusable theorem whose conclusion already has exactly the right extensional strength.

3.8 How the Hilbert Theorem Works

The proof of `hilbert_plane_intersection_line` reveals the real incidence structure hidden behind Euclid's short argument.

Starting from the common point A , the theorem first uses the spatial incidence clause

```
plane_second_common_point
```

to obtain a second common point B such that

$$B \neq A, \quad B \in \pi, \quad B \in \rho.$$

Then the ordinary two-point line theorem produces the line l through A and B . Since both points lie in π , Hilbert I.6 implies

$$l \subset \pi.$$

The same argument gives

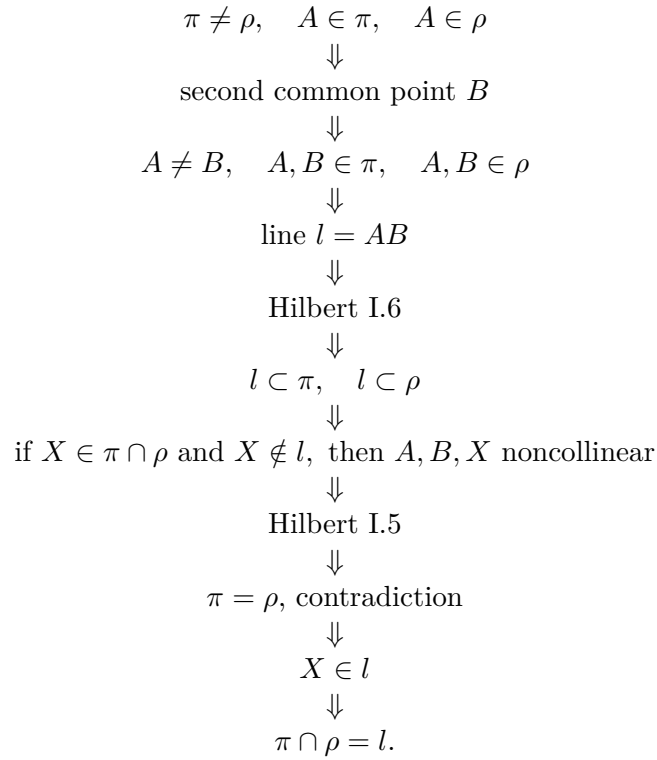
$$l \subset \rho.$$

The remaining task is extensionality. Suppose X lies in both planes. If X were not on l , then A, B, X would be noncollinear, so by plane uniqueness the plane through A, B, X would have to be both π and ρ . This would contradict $\pi \neq \rho$. Therefore every common point X lies on l .

Conversely, if $X \in l$, then the two line-in-plane containments immediately imply $X \in \pi$ and $X \in \rho$.

3.9 Formal Dependency Chain

The real formal dependency pattern is therefore



This is the real geometric content behind the short public theorem.

3.10 Spatial Architecture

The theorem uses only the incidence-side spatial structure:

```

HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo

```

It does not require

```

HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
PlaneGeo

```

as theorem parameters.

Thus XI.3 still lies fully inside the Group I incidence layer. What makes it subtle is not order or congruence, but the need to express a genuinely three-dimensional intersection theorem at the incidence level.

3.11 Ambient Space and the Absence of Plane_{Geo}

No induced planar geometry is created in the proof.

All objects remain ambient:

```
pi rho : S.Plane
A X    : Geo.Point
l      : Geo.Line
```

and all relations are ambient incidence relations. The theorem does not enter a slice $\text{PlaneGeo}(\pi)$ or $\text{PlaneGeo}(\rho)$ because the result is not a planar theorem inside one chosen plane. It is a statement about the intersection pattern of two ambient planes.

3.12 Wyler-Style Flat Reconstruction

XI.3 is naturally a meet theorem. Once $\pi \neq \rho$ and a common point are known, the flat layer produces a line l whose carrier is exactly the set-theoretic intersection of the two plane carriers:

$$\text{carrier}(\pi) \cap \text{carrier}(\rho) = \text{carrier}(l).$$

This identity is stronger than merely returning one line lying in both planes: it says that every common point is on the same line and that every point of the line is common to both planes.

The proposition is therefore recovered by a single extensional passage:

```
pi != rho, A in pi cap rho
|
v
Meet(pi,rho) = carrier(l)
|
v
X in pi and X in rho <-> X in l
```

In the Wyler-style vocabulary the meet identity is the normal form of XI.3. It also records why the proposition is specifically three-dimensional: in higher rank, the meet of two codimension-one flats need not have the same rank-one form without additional rank information.

3.13 Hidden Formal Data

Several facts that are visually obvious in a traditional diagram must be present as proof data in Lean.

3.13.1 The planes must be distinct

The hypothesis

```
hneq : Ne pi rho
```

is essential. If $\pi = \rho$, then the common section is the whole plane, not a line.

3.13.2 The common point must be explicit

The theorem takes a witness

```
A : Geo.Point
hApi : S.OnPlane A pi
hArho : S.OnPlane A rho
```

rather than a primitive relation saying merely that the planes intersect.

3.13.3 The second common point is nontrivial

Euclid's picture encourages one to imagine the common line immediately. Lean must first prove that a second common point exists. This is the hidden three-dimensional step.

3.13.4 The line must capture all common points

It is not enough to produce one line lying in both planes. The formal conclusion contains the equivalence

```
(S.OnPlane X pi /\ S.OnPlane X rho) <=> H.OnLine X l
```

which states that the line is exactly the whole common section.

3.14 Relationship with XI.1 and XI.2

Classically XI.3 depends on the coplanarity infrastructure established in XI.1–XI.2.

In the formal development the immediate public theorem does not call `euclid_proposition_11_1` or `euclid_proposition_11_2`. Instead, it uses the stronger general theorem `hilbert_plane_intersection_line` whose proof internally consumes the corresponding Hilbert incidence principles.

The conceptual relation is nevertheless clear:

- XI.1 supplies the line-in-plane closure pattern;
- XI.2 supplies the idea that noncollinear or intersecting data determine a plane;
- XI.3 turns from plane determination to plane intersection.

Thus the classical sequence XI.1–XI.3 is reflected formally in the spatial incidence layer, even though later public theorems may call stronger reusable interface results directly.

3.15 Relationship with Later Book XI Propositions

XI.3 is a major infrastructure proposition for the remainder of Book XI.

Whenever a later argument introduces an auxiliary plane and then needs its section with another plane, XI.3 is the conceptual bridge. This becomes especially important in propositions such as XI.13, where the proof passes to the auxiliary plane determined by two lines and then studies its intersection with a reference plane.

In the Lean development, later propositions typically consume the reusable theorem `hilbert_plane_intersection` rather than the Euclid wrapper `euclid_proposition_11_3`. The mathematical content, however, is exactly XI.3.

3.16 Classical DAG versus Formal DAG

The difference between the historical and formal organizations is again instructive.

Euclid.

$$\begin{array}{c}
 \pi, \rho \text{ meet} \\
 \Downarrow \\
 \text{take another common point} \\
 \Downarrow \\
 \text{join the two common points} \\
 \Downarrow \\
 \text{any further common point lies on the same join} \\
 \Downarrow \\
 \text{XI.3.}
 \end{array}$$

Lean / Hilbert.

$$\begin{array}{c}
 \pi \neq \rho, \quad A \in \pi \cap \rho \\
 \Downarrow \\
 \text{plane_second_common_point} \\
 \Downarrow \\
 B \neq A, \quad B \in \pi \cap \rho \\
 \Downarrow \\
 \text{line_through } A, B \\
 \Downarrow \\
 \text{line_in_plane twice} \\
 \Downarrow \\
 l \subset \pi, \quad l \subset \rho \\
 \Downarrow \\
 \text{plane_unique} \\
 \Downarrow \\
 \forall X, X \in \pi \cap \rho \leftrightarrow X \in l \\
 \Downarrow \\
 \text{euclid_proposition_11_3.}
 \end{array}$$

The formal DAG is longer because Euclid’s phrase “common section” has been made fully extensional.

3.17 Structural Lesson

XI.3 illustrates a basic principle of spatial formalization: once several planes are present, their intersections cannot remain visual background. They must be represented by explicit proof objects and characterized extensionally.

This proposition also shows that “working synthetically” does not mean leaving dimensional constraints implicit. On the contrary, the synthetic reconstruction isolates exactly the point where 3-dimensionality matters.

The theorem is still incidence-only, but it is no longer merely elementary bookkeeping. It identifies one of the structural laws that make Book XI possible as a theory of solid geometry rather than a collection of planar pictures drawn in space.

3.18 Conclusion

Proposition XI.3 states that two distinct planes with a common point have a common section which is a straight line.

In the present reconstruction this is expressed in its strongest natural incidence form:

$$\pi \neq \rho, \quad A \in \pi \cap \rho \implies \exists l, \quad A \in l, \quad l \subset \pi, \quad l \subset \rho,$$

together with the extensional characterization

$$X \in \pi \cap \rho \iff X \in l.$$

The proof is short at the public level, but the underlying helper theorem reveals the real 3D structure: from one common point one obtains a second, from two common points one constructs the section line, and plane uniqueness forces every further common point onto that same line.

Thus XI.3 is not just a visual fact about two drawn sheets. It is one of the first genuinely structural theorems of synthetic 3-space in the Book XI development.

Chapter 4

Proposition XI.4

4.1 Introduction

Euclid XI.4 gives the basic synthetic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The proposition is the first major test of the spatial Hilbert architecture. The classical picture suggests a local statement about two right angles, but the conclusion is universal: the candidate line must be perpendicular to *every* line of the plane through the common point. The formal proof therefore has to reconstruct Euclid's arbitrary-transversal argument, keep planar reasoning inside explicit plane slices, and compare triangles that may lie in different planes without introducing coordinates or an ambient vector metric.

The production reconstruction remains synthetic. Its main structural features are the balanced cross, the arbitrary transversal selected by a Pasch argument, spatial SSS/SAS, and the final conversion of the resulting right angle into the universal line-plane perpendicularity predicate.

4.2 Classical Configuration

Let m, n be two distinct straight lines of a plane π , meeting at O , and let l be a line through O perpendicular to both.

The desired conclusion is that l is perpendicular to every line $q \subset \pi$ through O .

4.3 Statement

Theorem 1 (Euclid XI.4). *Let m and n be two distinct lines of a plane π , meeting at O . If a line l through O is perpendicular to both m and n , then l is perpendicular to the plane π .*

Equivalently, the conclusion says that for every line $q \subset \pi$ through O , the line l is perpendicular to q at O . This is exactly the universal content of Euclid XI.Def.3.

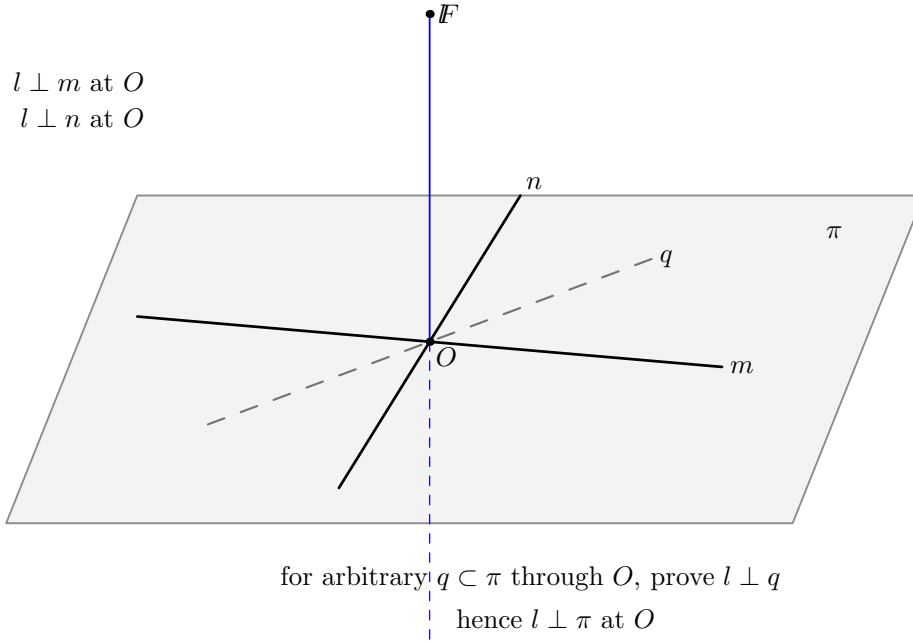


Figure 4.1: Classical configuration for Euclid XI.4. The ambient line l passes through O and is perpendicular at O to two distinct lines m, n lying in the plane π . The dashed line q represents an arbitrary line of π through O , which is the universal target of the conclusion.

4.4 Classical Proof

Euclid makes the argument constructive.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where E is the common point of the two given lines. Draw an arbitrary straight line GEH through E in the plane, choose an arbitrary point F on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

1. Vertical angles at E are equal (I.15). Together with the four equal radial segments, I.4 gives

$$AD = CB$$

and a corresponding angle equality.

2. Applying I.26 to the planar transversal gives

$$GE = EH, \quad AG = BH.$$

3. Since FE is perpendicular to both original plane lines, I.4 applied to two right triangles gives

$$FA = FB, \quad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.

5. A further I.4 gives

$$FG = FH.$$

6. Finally, from

$$GE = EH, \quad FE = FE, \quad FG = FH,$$

I.8 yields

$$\angle GEF = \angle HEF.$$

Since $G - E - H$, the adjacent equal angles are right angles.

7. The transversal GEH was arbitrary. Hence FE makes a right angle with every line through E in the plane. By XI.Def.3, $FE \perp \pi$.

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

4.5 What is genuinely three-dimensional here?

The balanced-cross and transversal constructions live in the reference plane π , but the decisive point F lies on the candidate line l outside that plane. Euclid then compares triangles such as those formed from F and points of the balanced cross. These triangles are not assumed to share one carrier plane.

Accordingly, the reconstruction alternates between two kinds of reasoning: ordinary Hilbert plane geometry inside explicit `PlaneGeo` slices, and spatial congruence arguments comparing triangles carried by different planes. This alternation is the genuinely solid-geometric content of XI.4.

4.6 Formal Target: Euclid XI.Def.3

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt
  [H : HilbertIncidence Geo]
  (l m : Geo.Line)
  (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  H.OnLine O m /\
  exists A B : Geo.Point,
    Ne A O /\
    Ne B O /\
    H.OnLine A l /\
    H.OnLine B m /\
    Not (PrimCollinear Geo A O B) /\
    HilbertRightAngle Geo A O B
```

The explicit noncollinearity condition is important. It prevents a degenerate “self-perpendicular” witness and yields the reusable theorem

```
hilbert_linesPerpendicularAt_ne
```

showing that perpendicular lines are distinct.

For a line and a plane:

```
def HilbertLinePerpendicularPlaneAt
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  S.OnPlane O pi /\
  forall m : Geo.Line,
    HilbertLineInPlane Geo m pi ->
    H.OnLine O m ->
    HilbertLinesPerpendicularAt Geo l m O
```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

4.7 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_4
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (m n : PlaneLine Geo pi)
  (l : Geo.Line)
  (O : PlanePoint Geo pi)
  (hmn : Ne m n)
  (hPerpM :
    HilbertLinesPerpendicularAt Geo l m.1 O.1)
  (hPerpN :
    HilbertLinesPerpendicularAt Geo l n.1 O.1) :
  HilbertLinePerpendicularPlaneAt Geo l pi O.1
```

The theorem does not assume separately that $l \neq m$ or $l \neq n$. Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

4.8 Spatial / Planar Architecture Used

4.8.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\text{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```
PlanePoint Geo pi
PlaneLine  Geo pi
```

and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

4.8.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid's unqualified use of plane triangle congruence

inside solid geometry.

4.9 The Balanced Cross

The first construction is the formal analogue of Euclid's

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

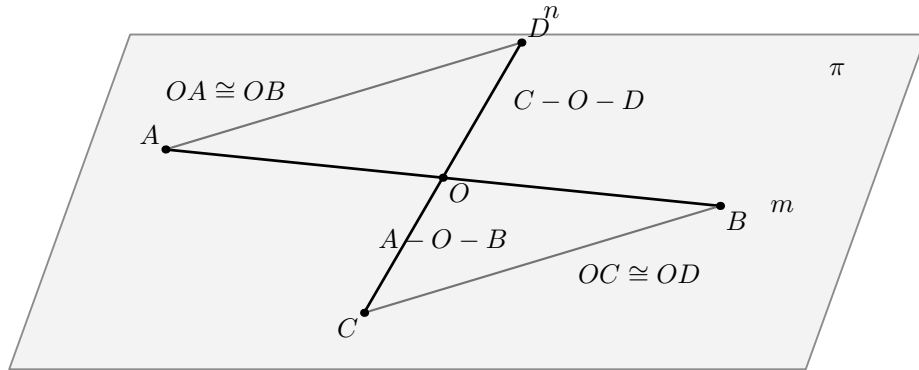
```
hilbert_XI4_balanced_cross_exists
```

and points A, B, C, D satisfying

$$A - O - B, \quad C - O - D$$

and

$$OA \cong OB, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$



all objects in this figure lie in π

Figure 4.2: Balanced cross in π . Points A, B lie on m , points C, D lie on n , with $A - O - B$, $C - O - D$, $OA \cong OB$, and $OC \cong OD$. The connector segments AD and BC are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \quad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

4.10 Fixed-Transversal Prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \quad B - J - C, \quad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \quad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point $F \in l$, together with the two original perpendicularities, then gives

$$AF \cong BF, \quad CF \cong DF.$$

The proof packages this in

```
hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant
```

The spatial SSS/SAS chain is then:

```
hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal
```

Its mathematical content is

$$\begin{aligned}
& AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\
& \quad \Downarrow \\
& \angle DAF \cong \angle CBF \\
& \quad \Downarrow \\
& A - G - D, \quad B - J - C \\
& \quad \Downarrow \\
& \angle GAF \cong \angle JBF \\
& \quad \Downarrow \\
& AG \cong BJ, \quad AF \cong BF \\
& \quad \Downarrow \\
& GF \cong JF \\
& \quad \Downarrow \\
& OG \cong OJ, \quad OF \cong OF, \quad GF \cong JF \\
& \quad \Downarrow \\
& \angle GOF \cong \angle JOF \\
& \quad \Downarrow \\
& G - O - J \\
& \quad \Downarrow \\
& \text{RightAngle}(G, O, F).
\end{aligned}$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

4.11 The Arbitrary Transversal: the Real Formal Difficulty

The public conclusion quantifies over *every* line of π through O . A conventional drawing suggests that such a transversal meets the two connector segments AD and BC , but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \quad \text{or} \quad BD/AC.$$

For a transversal $q \neq m, n$, apply Pasch to the triangle ABD . The line q enters through AB at O , hence it exits either through AD or through BD .

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point J opposite G on q with

$$G - O - J, \quad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```

hilbert_XI4_first_exit_opposite Lands_on_BC
hilbert_XI4_second_exit_opposite Lands_on_AC

```

so that the opposite point lands on the correct paired connector. The combined package is

```

hilbert_XI4_arbitrary_transversal_connector_package

```

and returns

$$\begin{aligned}
 &G, J \in q, \quad G - O - J, \quad OG \cong OJ, \\
 &\quad \text{and either} \\
 &\quad A - G - D, \quad B - J - C, \\
 &\quad \text{or} \\
 &\quad B - G - D, \quad A - J - C.
 \end{aligned}$$

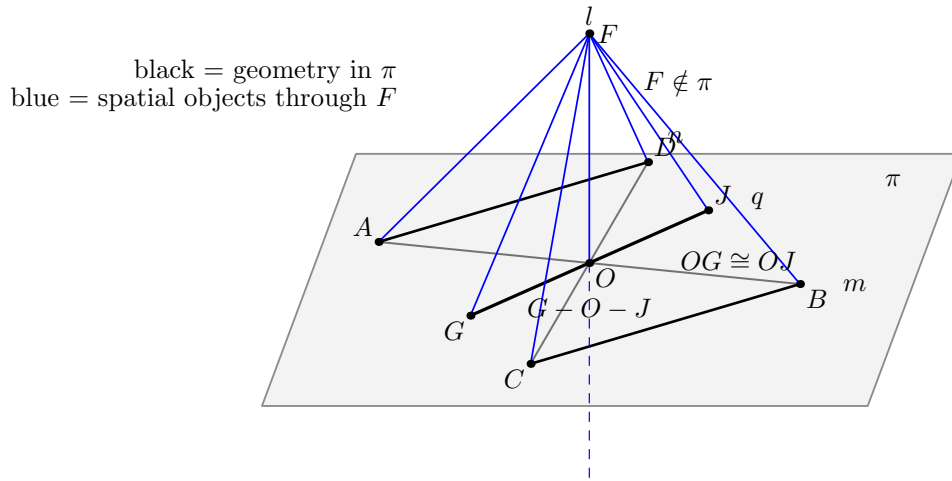


Figure 4.3: Arbitrary-transversal stage. For $q \subset \pi$ through O , Pasch selects one of the two connector families. The opposite point J is constructed with $G - O - J$ and $OG \cong OJ$. The spatial SSS/SAS chain then proves that $\angle GOF$ is right.

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

4.12 Right Angle on Every Transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```

hilbert_XI4_arbitrary_transversal_right_angle

```

states, in substance:

$$\begin{aligned}
 &q \subset \pi, \quad O \in q, \quad q \neq m, n, \\
 &F \notin \pi, \quad AF \cong BF, \quad CF \cong DF
 \end{aligned}$$

imply the existence of $G \in q$, $G \neq O$, such that

$$\text{RightAngle}(G, O, F).$$

The two exceptional cases $q = m$ and $q = n$ are discharged directly from the hypotheses of XI.4.

4.13 Why the Point F Is Outside the Plane

The core proof chooses an arbitrary nonvertex point F on l . To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume $F \in \pi$. Then the line l , determined by O and F , is itself a line of π . Transfer the two ambient perpendicularities to **PlaneGeo** π . Choose nonvertex points $A \in m$ and $C \in n$. Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that A, O, C are collinear. Line uniqueness then forces $m = n$, contradicting the hypothesis.

No parallel postulate is used.

4.14 Closing the Universal Line–Plane Statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose $F \neq O$ on l ;
2. construct the balanced cross A, B, C, D in π ;
3. prove $F \notin \pi$;
4. prove $AF \cong BF$ and $CF \cong DF$;
5. let q be an arbitrary line of π through O ;
6. if $q = m$ or $q = n$, use the hypotheses;

7. otherwise invoke the arbitrary-transversal theorem and obtain a point $G \in q$ such that $\angle GOF$ is right;
8. construct the witness required by `HilbertLinesPerpendicularAt Geo l q O`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since l and q intersect at O , they determine an explicit plane ρ . Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

4.15 Wyler-Style Flat Reconstruction

XI.4 is the first proposition in which the flat language supplies only the incidence skeleton and the metric layer performs the decisive step. The two distinct plane lines m, n through O are rank-two generators of the reference plane. Conceptually,

$$\text{Join}(\text{carrier}(m), \text{carrier}(n)) = \text{carrier}(\pi).$$

The theorem then says that perpendicularity to two independent generating directions propagates to the whole generated plane.

The proof architecture can therefore be compressed to

```

m,n distinct generators of pi
l perp m at O
l perp n at O
  |
  +-- normalize l != m, l != n
  +-- normalize O in l,m,n
  |
  v
XI.4 metric core
  |
  v
l perp pi at O

```

This is not a theorem of flats alone: `join` tells us which plane is generated, but no closure axiom turns line–line perpendicularity into line–plane perpendicularity. The proposition therefore marks the first clean interface between the incidence calculus and the spatial congruence layer.

4.16 Formalization Notes

4.16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

4.16.2 The point F is not a normal-vector surrogate

The point F is merely a nonvertex point chosen on the candidate line l . It is used to form ordinary synthetic triangles such as AOF , BOF , GOF , and JOF . The proof never assigns coordinates to F and never interprets OF as a vector.

4.16.3 Why PlaneGeo is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \longleftrightarrow \text{PlaneGeo}(\pi) \quad \text{or} \quad \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

4.16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$ and $C - O - D$ are strict orders;

- the auxiliary point F is outside π ;
- a general transversal need not meet the fixed pair AD, BC ;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through l and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

4.16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through O meets both segments AD and BC . The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle ABD . This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \quad \text{or} \quad (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

4.17 Reusable Infrastructure Produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

4.18 Classical Dependencies versus Lean Dependencies

Role	Euclid	Lean reconstruction
Four equal radial segments	cut off AE, EB, CE, ED equal	opposite-point construction on each of the two lines inside PlaneGeo pi
Vertical angles	I.15	developed vertical-angle and angle-congruence infrastructure
First triangle congruence	I.4	balanced-cross SAS helper
Transversal equalities	I.26	Pasch, transversal ASA, and the two landing branches
Distances from F	I.4 on right triangles	spatial perpendicular-bisector/equidistance helper
Cross-plane triangle comparison	I.8 and I.4	general spatial SSS/SAS API through explicit carrier planes
Arbitrary line in the plane	“drawn at random”	universal q ; cases $q = m$, $q = n$, or the Pasch branch
Line perpendicular to plane	XI.Def.3	the universal line–plane perpendicularity predicate

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

4.19 Complete Formal DAG

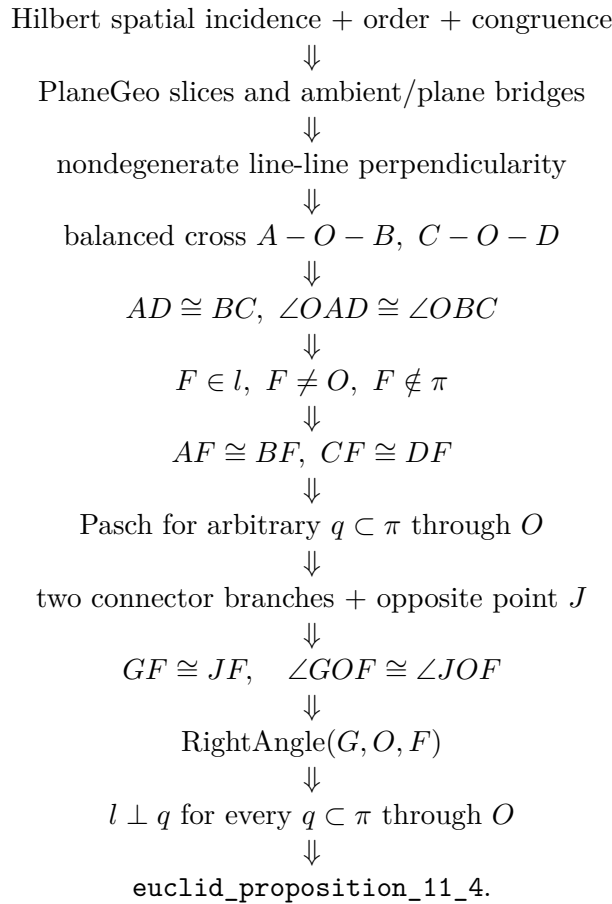
The proof architecture can be summarized as follows.

```

two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD = BC and angle OAD = angle OBC
-> choose F != O on l
-> F notin pi
-> AF = BF, CF = DF
-> arbitrary q in pi through O
-> if q=m or q=n: hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG = OJ
-> landing on BC or AC
-> ASA: AG = BJ (or BG = AJ)
-> spatial SSS/SAS: GF = JF
-> spatial SSS: angle GOF = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.

```

4.20 Dependency Summary



4.21 Relationship with the Foundations

The public theorem uses spatial incidence, order, and congruence together with induced Hilbert geometry on explicit plane slices. It does not assume Hilbert Group IV: no Euclidean parallel axiom is used in the proof of XI.4. No continuity principle is needed either.

This is mathematically significant. The arbitrary-transversal argument is closed by Pasch and congruence, not by parallel-line theory. The theorem is therefore neutral in the sense relevant to the Hilbert hierarchy used by the project.

4.22 Conclusion

Euclid XI.4 is the first major test of the project's spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid's proof: equal-segment constructions, vertical angles, triangle congruence, an arbitrary transversal, and the definition of a perpendicular to a plane.

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate

logical content.

Chapter 5

Proposition XI.5

5.1 Introduction

Proposition XI.5 is the first Book XI proposition whose conclusion is a genuinely spatial coplanarity statement forced by perpendicularity.

A line l passes through a point O and is perpendicular there to three lines m, n, p . Two of these lines, m and n , are assumed distinct. The conclusion is that m, n, p lie in one plane.

The formal reconstruction is not a line-by-line transcription of Euclid's proof. The current spatial interface makes it possible to use Proposition XI.4 twice and thereby obtain a short contradiction argument.

5.2 Euclid's Statement

Euclid's Proposition XI.5 states:

If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

In Euclid's notation the common perpendicular is AB , and the three lines through the common point B are BC , BD , and BE .

5.3 Classical Configuration

Euclid assumes, for contradiction, that two of the three lines, say BD and BE , lie in a reference plane while BC lies in a more elevated plane.

He then considers the plane through AB and BC . Its intersection with the reference plane is a line BF . Thus AB, BC, BF lie in one plane.

At the same time, since AB is perpendicular to BD and BE , Proposition XI.4 implies that AB is perpendicular to the reference plane. Therefore AB is also perpendicular to BF .

The contradiction is that, in the plane through AB and BC , the lines BC and BF would both form a right angle with AB at the same point.

5.4 Classical Proof

The local classical dependency pattern is:

$$\text{XI.3} \longrightarrow \text{intersection line } BF,$$

$$\text{XI.4} \longrightarrow AB \perp (\text{reference plane}),$$

followed by Euclid’s Definition XI.3, which turns line–plane perpendicularity into perpendicularity to every line of the plane meeting the perpendicular.

Thus the classical proof is a contradiction proof based on the intersection of two planes and on the impossibility of two distinct perpendiculars to the same line at the same point in one plane.

5.5 What is genuinely three-dimensional here?

The essential spatial content is not the elementary fact about right angles. It is the management of planes.

The proof must establish:

- a plane through two intersecting lines;
- a second plane through another pair of intersecting lines;
- that these two planes are distinct;
- a line of intersection of the two planes;
- containment of relevant lines in the appropriate planes.

The current Lean proof does not open a new planar `PlaneGeo` argument of its own. Instead, its planar geometric content is inherited through the already proved Proposition XI.4.

5.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_5
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
```



```

[_HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
(1 m n p : Geo.Line)
(0 : Geo.Point)
(hmn : Ne m n)
(hPerpM :
  HilbertLinesPerpendicularAt Geo 1 m 0)
(hPerpN :
  HilbertLinesPerpendicularAt Geo 1 n 0)
(hPerpP :
  HilbertLinesPerpendicularAt Geo 1 p 0) :
exists pi : S.Plane,
  HilbertLineInPlane Geo m pi /\
  HilbertLineInPlane Geo n pi /\
  HilbertLineInPlane Geo p pi

```

The conclusion is constructive at the level of incidence: Lean returns an explicit plane π containing all three lines.

The hypothesis `hmn : Ne m n` makes explicit a point that is usually left implicit in the classical diagram. The two intersecting lines m, n must be distinct in order to determine the reference plane.

5.7 Spatial / Planar Architecture Used

The proposition uses the ambient three-dimensional layer:

- `HilbertSpacePrimitive` for planes and point–plane incidence;
- `HilbertSpaceIncidence` for spatial plane constructions;
- `HilbertLineInPlane` for line–plane containment;
- `HilbertLinesPerpendicularAt` for spatial line–line perpendicularity;
- `HilbertLinePerpendicularPlaneAt` for line–plane perpendicularity.

The ambient geometry does not install the planar order and congruence classes as global instances on the spatial object. The architectural distinction is:

```

-- not global ambient instances:
HilbertOrder Geo
HilbertCongruence Geo

-- spatial layers:
HilbertSpaceOrder
HilbertSpaceCongruence

```

Planar order and congruence enter only through the established `PlaneGeo` machinery used inside Proposition XI.4.

5.8 Proof Architecture of the Reconstruction

Let m and n be the first two lines perpendicular to l at O .

5.8.1 Step 1: construct the reference plane

Since $m \neq n$ and both pass through O , the theorem

```
hilbert_plane_through_two_intersecting_lines
```

constructs a plane π containing m and n .

5.8.2 Step 2: apply Proposition XI.4

The lines m and n are packaged as `PlaneLines` of π , and O as a `PlanePoint` of π .

Proposition XI.4 then yields

$$l \perp \pi \quad \text{at } O.$$

In Lean this is the call to

```
euclid_proposition_11_4
```

with the two given perpendicularities $l \perp m$ and $l \perp n$.

5.8.3 Step 3: split on whether p already lies in π

If

$$p \subset \pi,$$

the theorem is finished immediately.

Assume therefore

$$p \not\subset \pi.$$

Since $l \perp p$, the lines l and p are distinct. They meet at O , so they determine another plane σ .

5.8.4 Step 4: intersect the two planes

The planes π and σ must be distinct. If $\pi = \sigma$, then $p \subset \sigma$ would imply $p \subset \pi$, contrary to the case assumption.

The theorem

```
hilbert_plane_intersection_line
```

therefore gives a line q through O contained in both planes:

$$q \subset \pi, \quad q \subset \sigma.$$

Since $l \perp \pi$ and q is a line of π through O ,

$$l \perp q.$$

This is obtained directly from

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

5.8.5 Step 5: the second use of XI.4

Now p and q both lie in σ , and

$$l \perp p, \quad l \perp q.$$

Suppose $p \neq q$. Applying Proposition XI.4 inside σ gives

$$l \perp \sigma.$$

But $l \subset \sigma$ by construction. This is impossible.

Hence

$$p = q.$$

Since $q \subset \pi$, it follows that $p \subset \pi$, and therefore

$$m, n, p \subset \pi.$$

5.9 Wyler-Style Flat Reconstruction

XI.5 is the first proposition whose complete synthetic structure is naturally expressed by both join and meet. The three lines m, n, p are all perpendicular to the same line l at O , and $m \neq n$.

First m, n generate the reference plane:

$$\text{Join}(m, n) = \pi.$$

XI.4 gives $l \perp \pi$. If $p \subseteq \pi$, the proof is finished. In the nontrivial branch, l and p generate a second plane σ , and π and σ meet in a line q through O . Since $q \subseteq \pi$, one has $l \perp q$. If $p \neq q$,

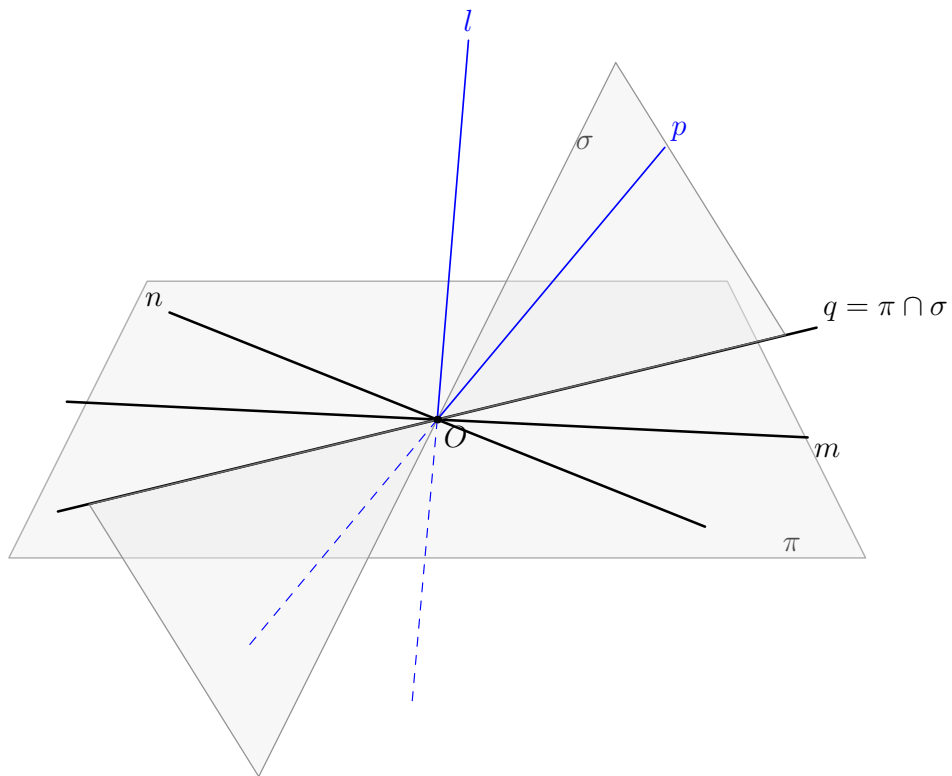


Figure 5.1: The formal contradiction argument for Proposition XI.5. The black lines m, n, q lie in the reference plane π . The spatial lines l, p are blue. Under the temporary assumption $p \not\subset \pi$, the lines l, p determine the second plane σ , and $q = \pi \cap \sigma$. If $p \neq q$, a second application of XI.4 would make l perpendicular to σ , although $l \subset \sigma$.

then the two directions p, q generate σ , and XI.4 would force $l \perp \sigma$; this contradicts $l \subseteq \sigma$. Hence $p = q$, so $p \subseteq \pi$.

The complete structural graph is

```

m,n --join--> pi
|
|           +--- XI.4 --> l perp pi
|
p not subset pi
|
l,p --join--> sigma
|
pi,sigma --meet--> q
|
l perp q
|
assume p != q
|
XI.4 in sigma --> l perp sigma
|
l subset sigma
|
contradiction
|
p = q subset pi

```

The point of the reformulation is not merely shorter code. The auxiliary planes and the section line become canonical generated objects rather than anonymous witnesses inside a long incidence argument.

5.10 Lean Formalization

The small proposition-local helper is:

```

theorem hilbert_linePerpendicularPlaneAt_not_in_plane
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi O) :
  Not (HilbertLineInPlane Geo l pi) := by

intro hlp

have hSelf :
  HilbertLinesPerpendicularAt Geo l l O :=
  HilbertLinePerpendicularPlaneAt.perpendicular_to_line
    (Geo := Geo)
    hPerp
    hlp
    hPerp.1

```

```
exact
(hilbert_linesPerpendicularAt_ne
  Geo 1 1 0 hSelf) rfl
```

This helper is purely definitional in geometric content. If a line perpendicular to a plane were itself contained in the plane, then the universal line–plane perpendicularity condition would make the line perpendicular to itself. The predicate `HilbertLinesPerpendicularAt` is nondegenerate, so this is impossible.

The theorem body then consists of two plane constructions, one plane-intersection construction, and two calls to XI.4.

5.11 Hidden Formal Data

Several facts visible in a classical spatial diagram must be supplied explicitly in Lean.

- O lies in the plane determined by m, n .
- The ambient lines m, n must be packaged as `PlaneLine Geo pi`.
- Their distinctness must be transported to the subtype level.
- If $p \not\subset \pi$, the plane through l, p must be constructed explicitly.
- The two planes must be proved distinct before their intersection line can be used.
- The intersection line q must carry proofs that it lies in both π and σ .
- For the second XI.4 call, p, q and O are repackaged as `PlaneLine` and `PlanePoint` objects of σ .

These are mathematical incidence obligations, not merely syntactic casts. They encode information that Euclid’s diagram leaves implicit.

5.12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and XI.4.

The Lean proof imports Proposition XI.4 and uses the current spatial intersection theorem

```
hilbert_plane_intersection_line
```

instead of importing a separate formal Proposition XI.3.

Thus the two dependency graphs are not identical.

The actual Lean Book XI dependency is:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

Proposition XI.5 is then the coplanarity tool required by the classical route to Proposition XI.6.

5.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by the XI.5 theorem body.

The inherited planar Hilbert and Book Zero infrastructure enters transitively through Proposition XI.4. In particular, XI.4 internally uses induced plane geometry, while XI.5 itself is primarily a spatial incidence argument.

The new helper `hilbert_linePerpendicularPlaneAt_not_in_plane` does not add a geometric axiom; it follows from the existing definitions and nondegeneracy of line–line perpendicularity.

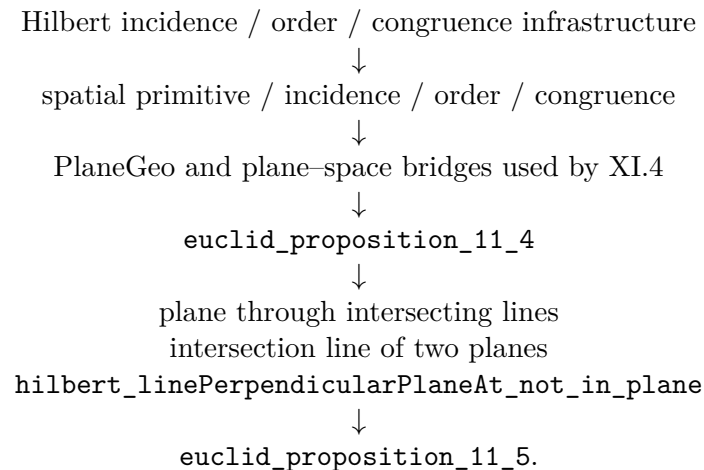
5.14 Relationship with the Foundations

XI.5 is neutral in the current reconstruction. Its proof uses spatial incidence, order, congruence, Proposition XI.4, and explicit plane intersection arguments, but no Hilbert Group IV assumption and no continuity principle.

The important foundational point is therefore architectural rather than axiomatic: coplanarity is never inferred from the picture. The proof must construct or identify the relevant plane and carry the corresponding line-in-plane witnesses explicitly.

5.15 Dependency Summary

The actual formal dependency chain is:



At the current checkpoint the production theorem compiles locally and its axiom audit has been run successfully. A full repository `lake build` is a separate checkpoint and should be recorded only after it is run.

5.16 Conclusion

Proposition XI.5 identifies the plane perpendicular to a given line at a given point as the unique carrier of all lines through that point perpendicular to the line, in the precise three-line form

stated by Euclid.

The formal proof is shorter than the classical local proof graph because the present 3D interface already contains an intersection-line theorem for two planes and because XI.4 can be applied a second time to exclude the hypothetical second plane.

No coordinates, vectors, scalar products, continuity principle, or new spatial axiom enter the proof.

Chapter 6

Proposition XI.6

6.1 Introduction

Proposition XI.6 is the first proposition in the present Book XI development whose conclusion is a genuinely spatial parallelism statement. Two ambient lines l and m are perpendicular to the same plane π , at distinct feet B and D . Euclid concludes that the two lines are parallel.

In three-dimensional geometry this conclusion contains two logically different assertions. The lines must first be shown to be *coplanar*; only then can planar geometry prove that they do not meet. This distinction is essential in the formal reconstruction, because two disjoint ambient lines may be skew.

The current Lean proof follows the metric core of Euclid's construction closely:

$$\text{auxiliary perpendicular and equal segment} \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5.}$$

After XI.5 has supplied the missing common plane, the proof changes deliberately to an induced planar geometry and closes the disjointness argument by the neutral alternate-angle criterion.

6.2 Euclid's Statement

Euclid's Proposition XI.6 states:

If two straight lines be at right angles to the same plane, the straight lines will be parallel.

In the classical configuration the two perpendiculars are AB and CD , meeting the reference plane at B and D . The formal theorem makes the nondegenerate case explicit by assuming

$$B \neq D.$$

6.3 Classical Configuration

Join the feet B and D . In the reference plane construct through D a line DE perpendicular to BD , and choose E so that

$$DE \cong AB.$$

Join BE , AE , and AD .

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through B , in particular to BD and BE . Hence

$$\angle ABD, \quad \angle ABE$$

are right angles.

Likewise CD is perpendicular to the reference plane at D , so it is perpendicular to both DB and DE .

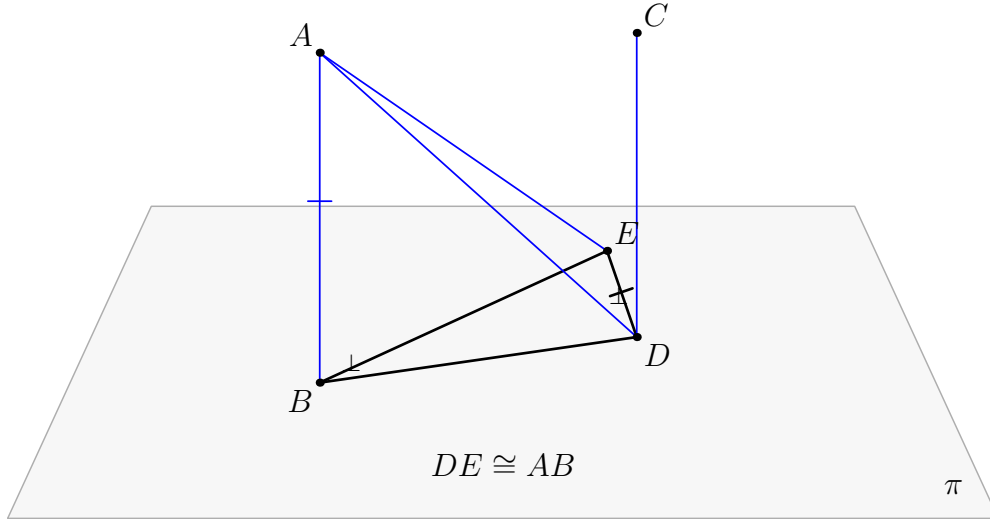


Figure 6.1: The classical configuration of Proposition XI.6. The black geometry lies in the reference plane π : B, D, E , the transversal BD , and the auxiliary perpendicular DE . The blue lines and connectors leave π . The construction $DE \cong AB$ prepares the SAS comparison of BAD and DEB ; the resulting equality $AD \cong EB$ prepares the SSS comparison of DEA and BAE .

6.4 Classical Proof

The classical metric part consists of two triangle-congruence steps.

First,

$$AB \cong DE, \quad BD \cong DB,$$

and the included angles

$$\angle ABD, \quad \angle EDB$$

are both right. Proposition I.4 therefore gives

$$AD \cong EB.$$

Second, the triangles DEA and BAE have

$$DE \cong BA, \quad EA \cong AE, \quad DA \cong BE.$$

Proposition I.8 gives

$$\angle EDA \cong \angle ABE.$$

Since $\angle ABE$ is right, $\angle EDA$ is right; hence

$$ED \perp DA.$$

Now at the point D the line ED is perpendicular to the three lines

$$DB, \quad DA, \quad DC.$$

Proposition XI.5 therefore places those three lines in one plane. Proposition XI.2 identifies the triangle plane determined by the intersecting lines, so AB is brought into the same plane as BD and DC . Finally

$$\angle ABD = \angle BDC = 90^\circ,$$

and Proposition I.28 yields

$$AB \parallel CD.$$

A compact classical dependency graph is therefore

$$\begin{array}{c}
 \text{construct } DE \perp BD, \quad DE \cong AB \\
 \Downarrow \\
 \text{I.4} \\
 \Downarrow \\
 AD \cong EB \\
 \Downarrow \\
 \text{I.8} \\
 \Downarrow \\
 ED \perp DA \\
 \Downarrow \\
 \text{XI.5} \\
 \Downarrow \\
 DB, DA, DC \text{ coplanar} \\
 \Downarrow \\
 \text{XI.2} \\
 \Downarrow \\
 AB, BD, DC \text{ coplanar} \\
 \Downarrow \\
 \text{I.28} \\
 \Downarrow \\
 AB \parallel CD.
 \end{array}$$

The construction of the auxiliary perpendicular and equal segment is part of the earlier Book I construction machinery. The substantive local congruence steps are I.4 and I.8.

6.5 What is genuinely three-dimensional here?

The difficult point is not the planar theorem that two perpendiculars to one transversal are parallel. That statement cannot even be applied until one knows that the two candidate parallel lines lie in one plane.

In ambient three-space the information

$$l \perp BD, \quad m \perp BD$$

does not by itself constitute a parallelism proof: without a coplanarity theorem the lines could still be skew.

The formal proof therefore separates XI.6 into

$$\boxed{\text{coplanarity}} \quad + \quad \boxed{\text{planar disjointness}}.$$

The first box is the genuinely spatial content and is exactly where Proposition XI.5 is used.

6.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_6
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (l m : Geo.Line)
  (B D : Geo.Point)
  (hBD : Ne B D)
  (hLperp :
    HilbertLinePerpendicularPlaneAt
      Geo l pi B)
  (hMperp :
    HilbertLinePerpendicularPlaneAt
      Geo m pi D) :
  HilbertSpaceLinesParallel Geo l m

```

The target relation is part of the reusable spatial interface `Hilbert3DInterface`:

```

def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma /\
  HilbertLinesDisjoint Geo l m

```

Thus “parallel in space” means exactly:

1. there is a plane containing both lines;
2. the two lines are disjoint.

The first clause is not redundant. If spatial parallelism were defined only by ambient disjointness, skew lines would incorrectly satisfy the definition.

6.7 Spatial / Planar Architecture Used

The proof uses three layers which must remain distinct.

6.7.1 Ambient three-space

The ambient objects and relations include

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

In particular, segment congruence between configurations carried by different planes is handled through `HilbertSpaceCongruence`.

6.7.2 Induced plane geometry

Whenever a planar theorem is needed, the relevant ambient plane σ or π is converted into

```
PlanePoint
PlaneLine
PlaneGeo
```

and ordinary planar incidence, order, right-angle, side-of-line, and triangle theorems are used there.

There is deliberately no global ambient instance

```
HilbertOrder Geo
HilbertCongruence Geo
```

for the three-dimensional space.

6.7.3 Plane-space bridges

The proof repeatedly uses bridges such as

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_sameRay_iff_ambient
```

These are not cosmetic casts. They state which geometric fact is being used in which plane slice.

6.8 Proof Architecture of the Reconstruction

6.8.1 Step 1: Euclid's auxiliary construction in the reference plane

Choose a point $A \neq B$ on the first normal l . Inside the reference plane π , construct the line

$$d = BD.$$

Through D construct a line e perpendicular to d , and lay off on e a point E such that

$$DE \cong AB.$$

This is packaged by

```
hilbert_XI6_auxiliary_perpendicular_equal_segment
```

The line d and the auxiliary line e are constructed in `PlaneGeo pi`. The segment copied onto DE , however, may have the spatial endpoint A ; that copy is therefore performed by `HilbertSpaceCongruence.segment_construction`.

6.8.2 Step 2: the first spatial SAS

Since $l \perp \pi$ at B and $d \subset \pi$ passes through B ,

$$l \perp d.$$

The auxiliary construction gives

$$e \perp d$$

at D .

The helper

```
hilbert_XI6_space_linesPerpendicularAt_right_angle_of_points
```

normalizes these line–line perpendicularities to the concrete right angles

$$\angle ABD, \quad \angle EDB.$$

The spatial right-angle comparison then supplies

$$\angle ABD \cong \angle EDB.$$

With

$$BA \cong DE, \quad BD \cong DB,$$

the spatial SAS theorem gives

$$AD \cong EB.$$

This whole block is packaged by

```
hilbert_XI6_first_SAS
```

and uses

```
hilbert_space_sas_third_side_and_angle
```

for the actual triangle comparison.

6.8.3 Step 3: the second spatial SSS and the new right angle

The next comparison is between the triangles DEA and BAE . The three side relations are

$$DE \cong BA, \quad EA \cong AE, \quad DA \cong BE.$$

The spatial SSS theorem gives

$$\angle EDA \cong \angle ABE.$$

Because $AB \perp \pi$ and $BE \subset \pi$, the angle ABE is right. The rightness is transported to EDA , yielding

$$ED \perp DA.$$

This block is packaged by

```
hilbert_XI6_second_SSS_right_angle
```

The proof must create explicit carrier planes for the relevant triangles. The right-angle transport is handled by

```
hilbert_XI6_space_right_angle_transport_in_plane
```

rather than by installing a forbidden ambient `HilbertCongruence` `Geo` instance.

6.8.4 Step 4: Proposition XI.5 supplies coplanarity

At D , the line $e = DE$ is now perpendicular to

$$d = DB, \quad a = DA, \quad m.$$

The last perpendicularity follows because $m \perp \pi$ and $e \subset \pi$.

Proposition XI.5 is applied exactly here:

```
euclid_proposition_11_5
```

It produces a plane σ containing

$$d, \quad a, \quad m.$$

Since $A \in a$ and $B \in d$, both A and B lie in σ . The line $l = AB$ therefore also lies in σ .

The result is isolated as

```
hilbert_XI6_normals_to_same_plane_coplanar
```

with conclusion

```
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma
```

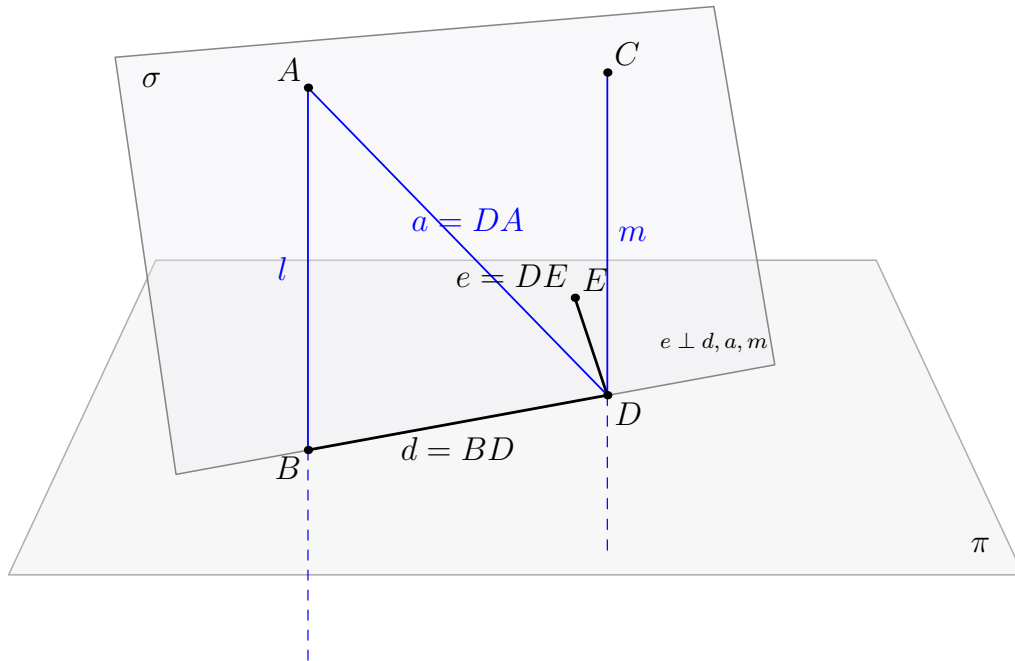


Figure 6.2: The genuinely spatial step in the formal XI.6 proof. The black lines $d = BD$ and $e = DE$ lie in the reference plane π . The blue lines $a = DA$, $m = DC$, and $l = AB$ leave π . At D , the line e is perpendicular to d, a, m , so XI.5 produces the second plane σ containing d, a, m . Because σ contains A and B , it also contains l . The lines l and m are therefore coplanar.

6.8.5 Step 5: the remaining problem is planar

After Step 4, both l and m lie in σ . The common transversal $d = BD$ lies in σ as well, and

$$l \perp d \text{ at } B, \quad m \perp d \text{ at } D.$$

The rest of the proof is carried out entirely inside

PlaneGeo Geo sigma.

The helper

```
hilbert_XI6_coplanar_normals_disjoint
```


chooses nonfoot points on l and m , establishes the required side-of-line orientation relative to d , compares the two right angles, and applies

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

to obtain planar parallelism.

The final conversion proves that the actual ambient carriers l and m are disjoint.

6.8.6 Step 6: assemble spatial parallelism

The public theorem now has exactly the two pieces required by `HilbertSpaceLinesParallel`:

$$l, m \subset \sigma$$

and

$$\text{HilbertLinesDisjoint}(l, m).$$

Therefore

$$\text{HilbertSpaceLinesParallel}(l, m).$$

6.9 Wyler-Style Flat Reconstruction

The target notion of spatial parallelism already decomposes into two geometric components: common-plane incidence and disjointness. XI.6 is therefore read in the flat calculus as a two-stage theorem about two normals to the same plane at distinct feet.

The first stage produces a plane σ containing both normals:

$$l \subseteq \sigma, \quad m \subseteq \sigma.$$

Equivalently, once $l \neq m$ is known, their join is the carrier of σ . The second stage proves

$$l \cap m = \emptyset.$$

The public parallelism object is exactly the package of these two facts.

In the Wyler-style second pass the theorem is therefore read as

```
l perp pi at B
m perp pi at D
B != D
  |
  +--> common carrier plane sigma
  |
  +--> l,m disjoint
  |
  v
HilbertSpaceLinesParallel l m
```

The deeper metric work that establishes coplanarity and disjointness remains synthetic and is not replaced by lattice theory. The flat language identifies the correct output object: a rank-two carrier containing the two rank-one carriers, together with the separate order/Euclidean fact that they do not meet.

6.10 Lean Formalization

The final theorem body is short because the geometric work has been isolated into two major helpers. First,

```
hilbert_XI6_normals_to_same_plane_coplanar
```

returns a plane `sigma` containing both ambient normals. Then

```
hilbert_XI6_coplanar_normals_disjoint
```

proves that their ambient carriers are disjoint. The public theorem packages exactly these three fields of `HilbertSpaceLinesParallel`: containment of l , containment of m , and disjointness.

The production module imports

```
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition27
```

The first import supplies the decisive spatial coplanarity theorem. The second supplies the neutral alternate-angle machinery used in the final induced plane.

6.11 Hidden Formal Data

The classical diagram suppresses several obligations which the Lean proof must make explicit.

- The feet B, D are distinct.
- The point A selected on l is outside the reference plane π .
- The auxiliary point E is nondegenerate and lies both on e and in π .
- Every triangle used in SAS or SSS carries an explicit noncollinearity proof.
- Spatial triangle comparisons require an explicit target carrier plane.
- Right-angle transport between ambient configurations is performed through an explicit `PlaneGeo`, not through an ambient planar congruence instance.
- Before XI.5 can be applied, the three lines d, a, m must be proved distinct in the required places and must all pass through D .
- The plane σ returned by XI.5 must be shown to contain both A and B , so that the whole carrier $l = AB$ lies in σ .
- In the final planar proof, the chosen points on l and m must be oriented on opposite sides of the transversal d . The proof performs an explicit `SameSide/OppositeSide` case split.
- Point-pair parallelism inside `PlaneGeo sigma` must finally be converted into disjointness of the original ambient carriers.

These obligations are geometric. They are the incidence, order, and plane-membership data that a conventional three-dimensional drawing usually leaves implicit.

6.12 Relationship with Earlier Book XI Propositions

Proposition XI.5 is a direct formal dependency:

$$\boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

XI.5 itself depends on XI.4, and XI.6 also reuses several XI.4 normalization helpers for perpendicular lines and right angles. Consequently the effective Book XI chain is

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

The classical proof also invokes XI.2 when it moves from the three coplanar lines through D to the plane containing the triangle ABD . The Lean proof does not need a separate formal call to Proposition XI.2: the spatial incidence API directly constructs and tracks the relevant carrier plane.

6.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The metric part of the formal proof reproduces the roles of Euclid I.4 and I.8 through the spatial SAS and SSS infrastructure:

```
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

Right angles and their transport are handled synthetically; no angle measure is introduced.

The classical final step is I.28. The production Lean file instead imports `Proposition27` and uses the established neutral alternate-angle theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

inside the explicitly constructed plane σ . Thus the formal module dependency and the classical Euclid proposition-number dependency are not identical, even though they close the same planar configuration.

Book Zero enters only transitively through the established Hilbert geometric infrastructure. No new Book Zero axiom or theorem is added by XI.6.

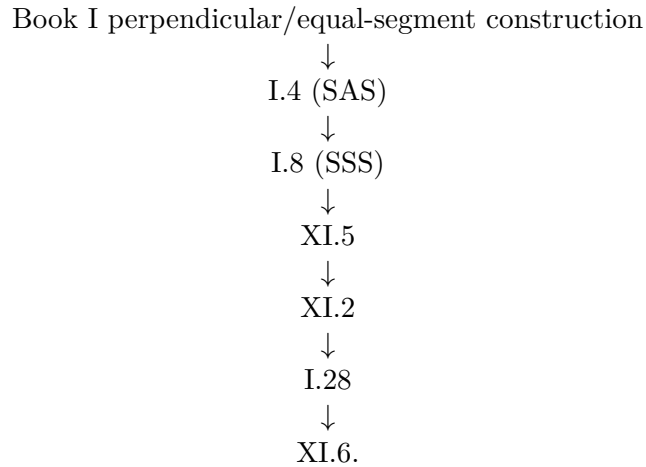
6.14 Relationship with the Foundations

XI.6 is also neutral in the current reconstruction. The proof uses spatial incidence, order, congruence, the induced geometry of explicit plane slices, and the earlier perpendicularity results, but it does not invoke Hilbert Group IV or continuity.

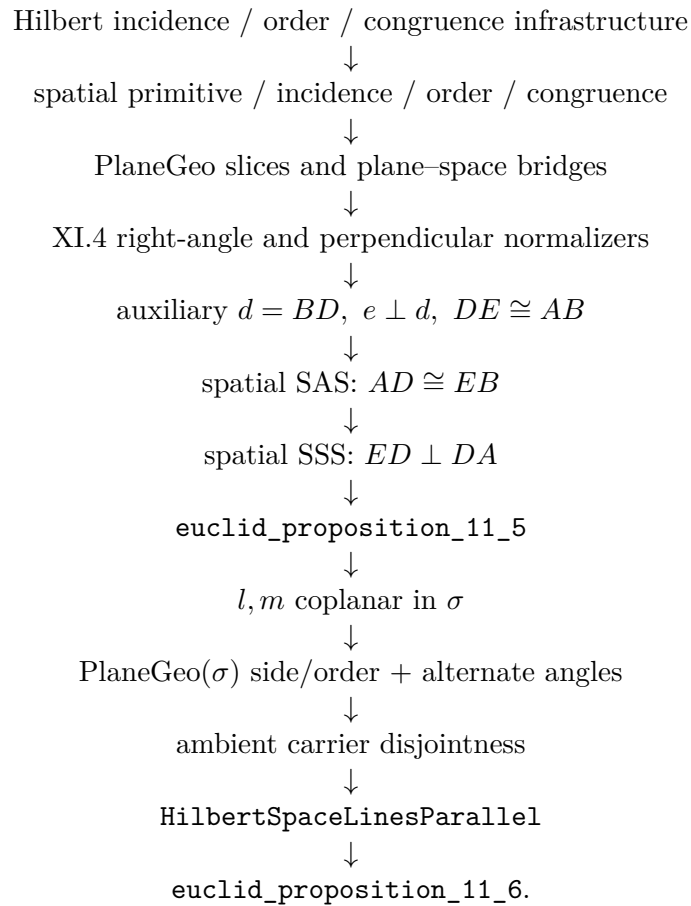
A reusable spatial parallelism predicate is introduced at this stage for later propositions. Its role is representational: it packages coplanarity and disjointness of ambient lines without turning the ambient three-space into a globally planar geometry.

6.15 Dependency Summary

The classical local chain is



The actual formal chain is



6.16 Conclusion

Proposition XI.6 is a useful boundary theorem for the three-dimensional architecture. Its metric core is recognizably Euclidean: construct an auxiliary perpendicular and an equal segment, use SAS, then SSS, and obtain a third perpendicular at D .

The genuinely spatial step is XI.5. It converts three perpendicularities through one point into the common plane in which the final parallelism argument is allowed to take place. Lean makes this logical boundary explicit rather than allowing the diagram to hide it.

The final definition of spatial parallelism records the same distinction: coplanarity is proved first and disjointness second. No coordinates, vectors, scalar products, numerical angle measure, continuity principle, new spatial axiom, or Euclidean parallel axiom is introduced.

Chapter 7

Proposition XI.7

7.1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert’s spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid’s *reductio*. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

7.2 Euclid’s Statement

Euclid’s Proposition XI.7 states:

If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD . The claim is that the joining straight line EF lies in the plane of AB and CD .

7.3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD . The points E and F are arbitrary points on the two lines.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

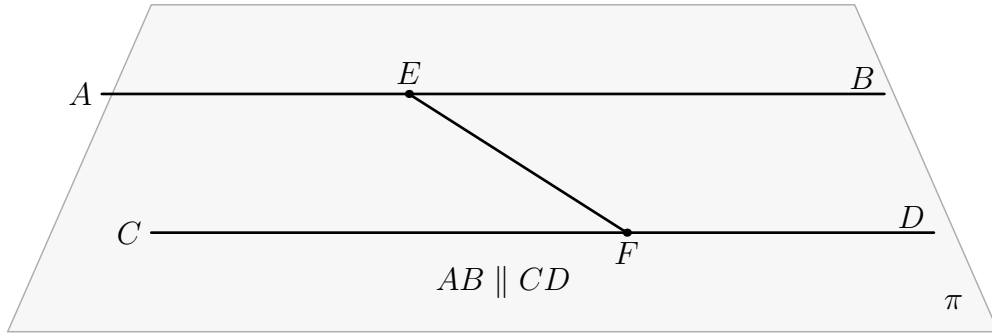


Figure 7.1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π , with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π . All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

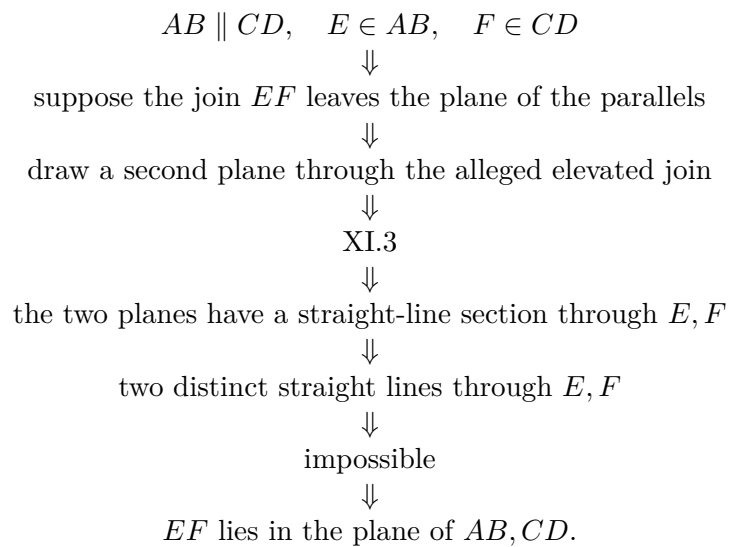
7.4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F . Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore



The only explicit Book XI proposition cited in the proof is XI.3.

7.5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

coplanarity and disjointness.

XI.7 uses the coplanarity component directly. No metric geometry is needed.

7.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_7
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (l m : Geo.Line)
  (E F : Geo.Point)
  (hParallel : HilbertSpaceLinesParallel Geo l m)
  (hEl : H.OnLine E l)
  (hFm : H.OnLine F m) :
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
  exists n : Geo.Line,
    H.OnLine E n /\
    H.OnLine F n /\
    HilbertLineInPlane Geo n sigma

```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through `E` and `F`, together with the proof that `n` lies in `sigma`.

7.7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```

HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel

```


No induced **PlaneGeo** is constructed. There is no planar betweenness, SameRay, SameSide/OppositeSide, angle congruence, right angle, SAS, SSS, or plane–space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

7.8 Proof Architecture of the Reconstruction

7.8.1 Step 1: unpack spatial parallelism

From

$$\text{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \quad m \subset \sigma,$$

together with

$$\text{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general **Hilbert3DInterface** rather than a proposition-local declaration.

7.8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

7.8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid’s diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m , contradicting

$$\text{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

7.8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with

$$E \in n, \quad F \in n.$$

7.8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n , and both lie in σ . The decisive step is:

```
HilbertSpaceIncidence.line_in_plane
```

which gives

$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert's incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

7.9 Wyler-Style Flat Reconstruction

XI.7 isolates a particularly reusable generator statement. If $E \in l$ and $F \in m$, then independently of parallelism one has

$$\text{Span}\{E, F\} \subseteq \text{Join}(l, m).$$

This is pure monotonicity of generated flats: the two point generators already belong to the join of the two line carriers.

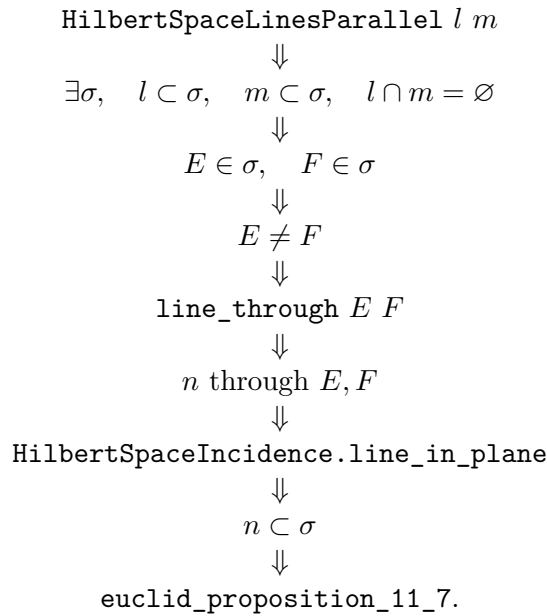
Parallelism enters only afterwards. It supplies a common plane σ for l, m , identifies their join with the carrier of σ , and guarantees that $E \neq F$. The span of the two points is therefore the carrier of the connector line $n = EF$, and the generic inclusion becomes $n \subseteq \sigma$.

```
E in l, F in m
  |
  v
Span{E,F} subset Join(l,m)
  |
l || m gives Join(l,m)=carrier(sigma)
  |
E != F gives Span{E,F}=carrier(EF)
  |
  v
EF subset sigma
```

This is a useful example of abstraction at the correct level: the central flat lemma is more general than XI.7, but the public proposition still records the classical parallel-line configuration.

7.10 Lean Formalization

The proof DAG is extremely short:



There are no proposition-local helper theorems.

7.11 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase “the straight line joining E and F ” is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

7.12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. The formal spatial incidence interface instead provides the line-in-plane closure principle directly.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

7.13 Relationship with Book I / Spatial Incidence Infrastructure

Euclid’s definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive formal ingredient is the spatial line-in-plane closure principle used by `HilbertSpaceIncidence.line_in_plane`. Thus the proof lands entirely in the spatial incidence layer and nowhere higher in the hierarchy. The source-level identification of this clause inside the classical incidence systems is collected in the Book XI source appendices.

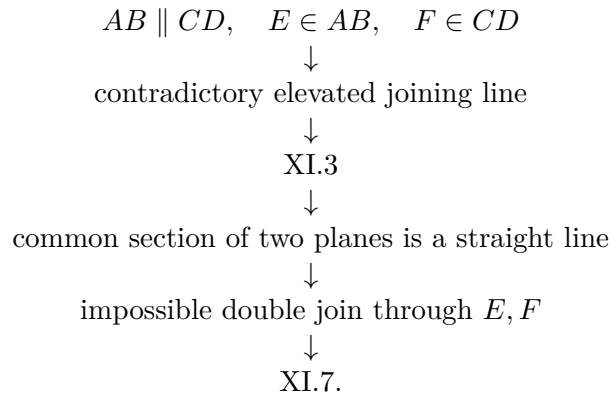
7.14 Relationship with the Foundations

XI.7 is incidence-only at the geometric level. The proof does not use spatial order, spatial congruence, induced `PlaneGeo`, Hilbert Group IV, or continuity. Once the two parallel lines are unpacked into a common carrier plane, the joining line is obtained by ordinary spatial incidence and the line-closure principle.

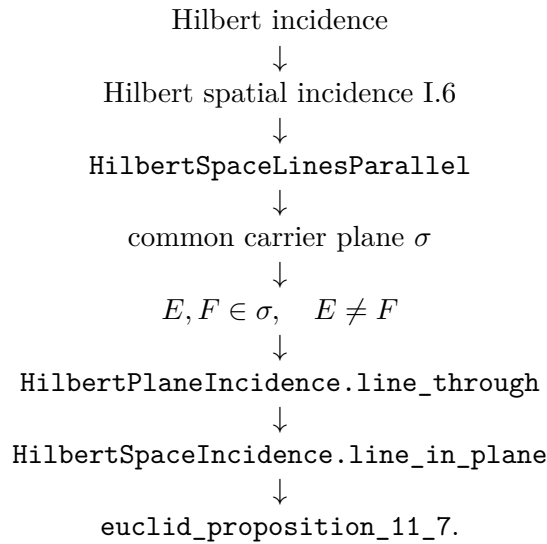
This makes XI.7 a useful contrast with XI.6 and XI.8: its content is almost entirely about retaining the correct carrier plane through the construction.

7.15 Dependency Summary

The classical local chain is



The actual formal chain is



7.16 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The formal reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and `HilbertSpaceIncidence.line_in_plane` closes the line under two distinct points of that plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.

Chapter 8

Proposition XI.8

8.1 Introduction

Proposition XI.8 is the Euclidean converse companion to XI.6. Proposition XI.6 proved that two straight lines perpendicular to the same plane are parallel. Proposition XI.8 starts with two parallel spatial lines and assumes that one of them is perpendicular to a reference plane; the conclusion is that the second parallel is perpendicular to the same plane.

The proposition is also a structural boundary in the reconstruction. The formalizations XI.4–XI.7 did not require Hilbert Group IV. XI.8 does. The reason is already visible in Euclid’s classical proof: the decisive transversal step uses Proposition I.29, the converse direction of the neutral parallel-angle criterion.

The Lean proof preserves the synthetic geometry but makes the Euclidean dependency more explicit. Instead of calling a planar I.29 wrapper, it separates the argument into

neutral I.27 + uniqueness of the parallel through a point.

The second component is precisely the plane-local Group IV assumption.

A second feature is genuinely three-dimensional. Euclid begins by saying that the two parallel lines meet the reference plane at points B and D . The first intersection is given by the hypothesis that the first line is perpendicular to the plane. The second intersection is not automatic in an abstract incidence space. The formal proof derives it.

8.2 Euclid’s Statement

Euclid’s Proposition XI.8 states:

If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

Let AB and CD be parallel straight lines, and suppose that AB is perpendicular to the reference plane π at B . The claim is that CD meets π at a point D and is perpendicular to π there.

8.3 Classical Configuration

Euclid lets the parallel lines AB and CD meet the reference plane at B and D . He joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

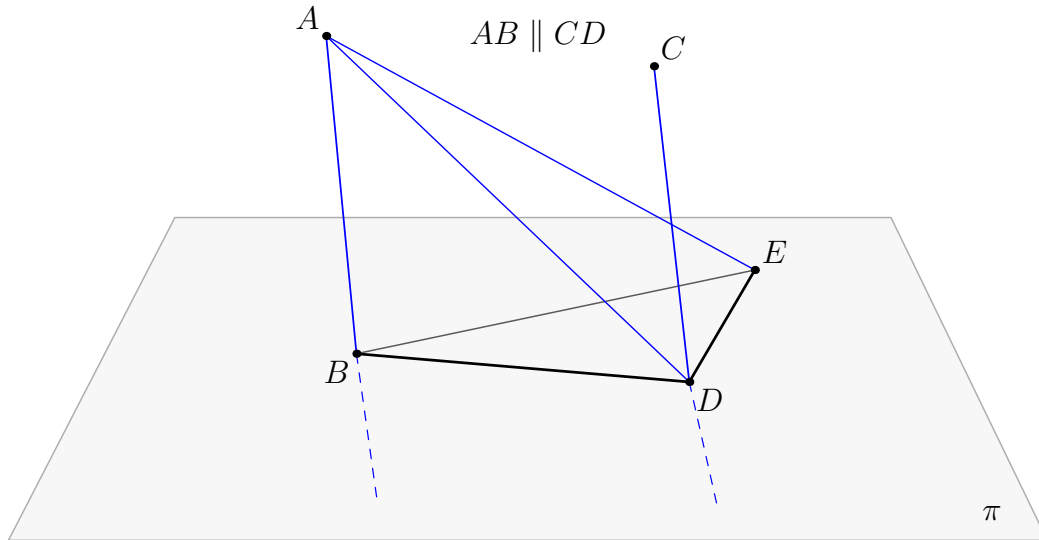


Figure 8.1: The classical configuration of Proposition XI.8. Black lines lie in the reference plane π . Blue lines or segments leave π . The hidden continuations of the two spatial parallel carriers are shown blue and dashed. The auxiliary construction reproduces the metric core already used in XI.6: $DE \perp BD$ and $DE = AB$.

The line BD plays two roles. It is a line of the reference plane and also lies in the common plane of the spatial parallels. This is the transversal on which the classical I.29 step acts.

8.4 Classical Proof

The local classical proof has five main stages.

First, since $AB \parallel CD$, Proposition XI.7 places AB , CD , and the joining line BD in one plane. The opening phrase that the parallel lines meet the reference plane at B, D is diagrammatic in the classical text.

Second, Euclid constructs DE in the reference plane with

$$DE \perp BD, \quad DE = AB.$$

Since $AB \perp \pi$, the angles

$$\angle ABD, \quad \angle ABE$$

are right.

Third, BD falls on the parallel lines AB and CD . Proposition I.29 says that the appropriate interior angles sum to two right angles. Since $\angle ABD$ is right, $\angle CDB$ is right. Hence

$$CD \perp BD.$$

Fourth, Euclid repeats the metric triangle argument familiar from XI.6. From

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles, Proposition I.4 gives

$$AD = BE.$$

Then

$$AB = DE, \quad BE = AD, \quad AE = EA,$$

so Proposition I.8 gives

$$\angle ABE = \angle EDA.$$

The former is right, hence the latter is right:

$$ED \perp AD.$$

Now ED is perpendicular to the two intersecting lines DB and DA . Proposition XI.4 therefore gives

$$ED \perp \text{plane}(BDA).$$

The line CD lies in this plane, so

$$ED \perp CD.$$

Finally, in the reference plane π the two lines DB and DE meet at D , and CD is perpendicular to both:

$$CD \perp DB, \quad CD \perp DE.$$

A second application of XI.4 yields

$$CD \perp \pi.$$

The classical local dependency chain is therefore

$$\begin{array}{c}
 AB \parallel CD, \quad AB \perp \pi \\
 \Downarrow \\
 \text{XI.7} \quad \text{and the joining line } BD \\
 \Downarrow \\
 DE \perp BD, \quad DE = AB \quad (\text{I.11, I.3}) \\
 \Downarrow \\
 AB \perp \pi \Rightarrow \angle ABD, \angle ABE \text{ right} \\
 \Downarrow \\
 \text{I.29} \\
 \Downarrow \\
 CD \perp BD \\
 \Downarrow \\
 \text{I.4, I.8} \\
 \Downarrow \\
 ED \perp AD \\
 \Downarrow \\
 \text{XI.4} \\
 \Downarrow \\
 ED \perp CD \\
 \Downarrow \\
 \text{XI.4} \\
 \Downarrow \\
 CD \perp \pi.
 \end{array}$$

The incidence assertion that CD belongs to the plane through BD, DA also uses the fact that the triangle ABD is planar and that AB, CD, BD share the plane supplied by XI.7. In a formal proof this plane identification cannot be left to the printed diagram.

8.5 What is genuinely three-dimensional here?

There are two distinguished planes.

The first is the reference plane π , to which AB is perpendicular. The second is the common plane σ containing the parallel spatial lines AB and CD . Since AB is not contained in π , the planes are distinct.

The points B and D lie in both planes. Consequently their joining line

$$d = BD$$

lies in both π and σ ; in the formal proof it is the relevant line of intersection of the two plane slices.

The auxiliary line

$$e = DE$$

lies in π , while the line

$$a = DA$$

lies in σ . The proof therefore alternates between two induced plane geometries.

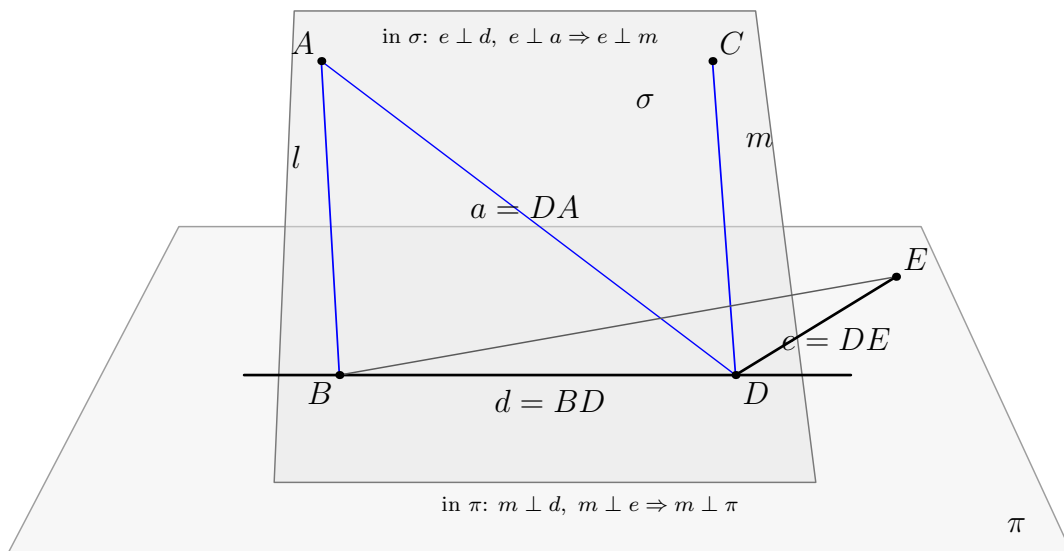


Figure 8.2: The two plane slices used by the reconstruction. The common line $d = BD$ lies in both π and σ . The first XI.4 application uses the two lines $d, a \subset \sigma$ to prove $e \perp \sigma$, hence $e \perp m$. The final XI.4 application uses $d, e \subset \pi$ to prove $m \perp \pi$.

This change of plane is not bookkeeping. It is the spatial geometry of the proposition.

8.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_8
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
```

```

[HSI : HilbertSpaceIncidence Geo]
[_HSO : HilbertSpaceOrder
  (Geo := Geo) (H := H) (S := S)]
[HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(l m : Geo.Line)
(pi : S.Plane)
(B : Geo.Point)
(hParallel : HilbertSpaceLinesParallel Geo l m)
(hPerp : HilbertLinePerpendicularPlaneAt Geo l pi B) :
exists D : Geo.Point,
  HilbertLinePerpendicularPlaneAt Geo m pi D

```

The point D is deliberately not an input. Its existence is part of the theorem. This is stronger as a formal interface than assuming in advance that the second parallel already meets the plane.

8.7 Spatial / Planar Architecture Used

The ambient spatial layer consists of

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
HilbertLinesPerpendicularAt

```

The proof then uses induced planar geometry in two different fixed planes:

$$\text{PlaneGeo}(Geo, \sigma) \quad \text{and} \quad \text{PlaneGeo}(Geo, \pi).$$

Inside σ , the proof uses neutral planar order, right angles, and the I.27 alternate-angle criterion. Inside π , the final configuration supplies two distinct intersecting lines d, e for XI.4.

The plane-space bridges used in the argument include

```

planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_linesDisjoint_iff_ambient

```

The last bridge was added with the Group IV layer. It identifies disjointness of two lines in a fixed **PlaneGeo** with ambient disjointness of their underlying spatial carriers.

8.8 The Euclidean Layer: Group IV

XI.8 is the first proposition in the present Book XI sequence whose formal proof requires the Euclidean parallel axiom.

The source-faithful Group IV axiom now belongs to `HilbertAxioms.lean`, as part of the single Hilbert foundation. The current Book XI production signature still consumes the plane-local compatibility record `HilbertSpaceEuclidean`, retained in `HilbertSpaceIncidence.lean`. It is not installed as a global `HilbertEuclideanPlaneGeo` instance, because such an instance would incorrectly treat the ambient three-space as a plane.

The compatibility record exposes Group IV in the form required by the existing spatial proofs:

```
class HilbertSpaceEuclidean
  (Geo : Geometry.Geo)
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertSpaceOrder Geo]
  [HilbertSpaceCongruence Geo] : Prop where

parallel_unique_in_plane :
  forall pi : S.Plane,
    forall l : Geo.Line,
      HilbertLineInPlane Geo l pi ->
        forall A : Geo.Point,
          S.OnPlane A pi ->
            Not (H.OnLine A l) ->
              forall b c : Geo.Line,
                HilbertLineInPlane Geo b pi ->
                  HilbertLineInPlane Geo c pi ->
                    H.OnLine A b ->
                      HilbertLinesDisjoint Geo b l ->
                        H.OnLine A c ->
                          HilbertLinesDisjoint Geo c l ->
                            b = c
```

For every fixed ambient plane, the bridge instance

```
instance planeGeoHilbertEuclidean
  ...
  (pi : S.Plane) :
    HilbertEuclideanPlane (PlaneGeo Geo pi)
```

recovers the existing planar Group IV API. The architectural rule is therefore preserved:

Euclidean parallelism is planar, even inside three-space.

8.9 Proof Architecture of the Reconstruction

8.9.1 Step 1: derive the hidden intersection point D

The helper

```
hilbert_XI8_parallel_meets_perpendicular_plane
```

proves that the second parallel m meets the reference plane.

Unpacking

$\text{HilbertSpaceLinesParallel}(l, m)$

gives a common plane σ containing l, m , with the two lines disjoint. The foot B of $l \perp \pi$ lies in both π and σ .

The planes are distinct. If $\sigma = \pi$, then $l \subset \pi$, contradicting the theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

The general plane-intersection theorem then supplies a line

$$q = \pi \cap \sigma$$

through B .

Suppose m did not meet q . Then both l and q would be lines through B in σ disjoint from m . Group IV uniqueness would force $l = q$, but $q \subset \pi$, again contradicting $l \not\subset \pi$.

Hence m meets q at some point D , and therefore

$$D \in m \cap \pi.$$

This is the formal proof of a fact Euclid leaves implicit at the opening of XI.8.

8.9.2 Step 2: reuse the XI.6 metric core

With D fixed, the proof reuses

```
hilbert_XI6_second_SSS_right_angle
```

from Proposition XI.6.

The helper constructs points and lines corresponding to Euclid's auxiliary configuration and returns, in particular,

$$d = BD, \quad e = DE,$$

with

$$d, e \subset \pi, \quad e \perp d,$$

and the right angle

$$\angle EDA.$$

Thus the SAS/SSS part of XI.8 is not duplicated. The earlier XI.6 development is used as reusable spatial congruence infrastructure.

8.9.3 Step 3: first application of XI.4 in σ

Let a be the line DA . The proof shows

$$d, a \subset \sigma, \quad d \neq a.$$

The inequality is geometrically meaningful: A lies on the spatial line l but outside the reference plane π , while $d \subset \pi$.

The XI.6 metric result gives

$$e \perp d, \quad e \perp a.$$

Therefore XI.4, applied with the induced geometry of σ , gives

$$e \perp \sigma.$$

Since $m \subset \sigma$ and $D \in m$,

$$e \perp m.$$

This is the formal counterpart of Euclid's first XI.4 step.

8.9.4 Step 4: replace I.29 by neutral I.27 plus Group IV

The remaining task is to prove

$$m \perp d.$$

The proof does not call a public I.29 theorem. Instead, inside $\text{PlaneGeo}(\text{Geo}, \sigma)$, it constructs a line p through D perpendicular to d .

The neutral helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```

shows that p and l are disjoint. Its proof is an I.27 argument: two right angles are normalized to equal alternate angles and the neutral alternate-angle theorem produces nonintersection.

But m is also disjoint from l , because $l \parallel m$. Thus m and p are two lines of the same plane σ , both passing through D , both disjoint from l . Group IV uniqueness forces

$$m = p.$$

Since $p \perp d$,

$$m \perp d.$$

This factorization isolates the exact Euclidean content:

$$\underbrace{\text{right angles} \Rightarrow \text{nonintersection}}_{\text{neutral I.27}} + \underbrace{\text{unique parallel through } D}_{\text{Group IV}} \implies m \perp d.$$

It is the formal counterpart of Euclid's I.29 step.

8.9.5 Step 5: final application of XI.4 in π

At this point

$$d, e \subset \pi, \quad m \perp d, \quad m \perp e.$$

The lines d and e are distinct because $e \perp d$.

The final call to

```
euclid_proposition_11_4
```

uses d, e as the two distinct intersecting lines of the reference plane. It returns

$$m \perp \pi$$

at D , which is the required conclusion.

8.10 Wyler-Style Flat Reconstruction

XI.8 shows the exact boundary of the flat calculus. The common-plane and join structure organizes the auxiliary configuration, but incidence alone cannot transfer perpendicularity from one parallel line to the other. The Euclidean parallel axiom is essential.

The second-pass proof separates three layers:

```
flat / incidence layer
  construct the auxiliary planes and lines at m
    |
    v
Euclidean layer (Group IV)
  transfer the fixed-transversal perpendicularity
    |
    v
metric layer
  m perp d, m perp e
    |
    v
  XI.4 metric core
    |
    v
  m perp pi
```

Thus XI.8 is not absorbed into join and meet. Rather, join and carrier language make the incidence skeleton explicit, Group IV provides the genuinely Euclidean step, and XI.4 converts two independent perpendicular directions into perpendicularity to the reference plane. This separation is one of the most important architectural checks in Book XI.

8.11 Lean Formalization

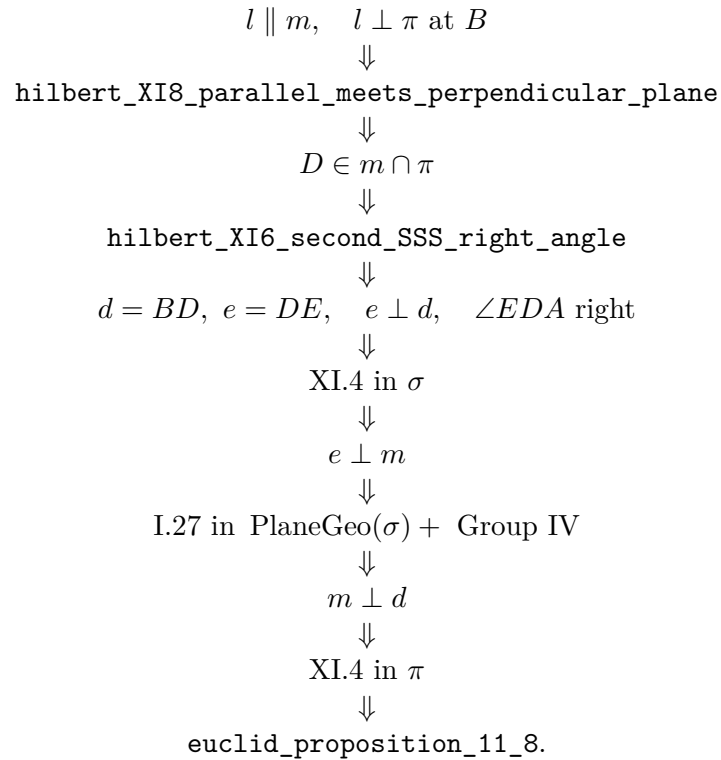
The proposition-local proof is organized around four named helpers:

```

hilbert_XI8_parallel_meets_perpendicular_plane
hilbert_XI8_auxiliary_line_perpendicular_to_parallel
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
hilbert_XI8_parallel_line_perpendicular_to_given_BD

```

The actual Lean DAG is



The production file imports XI.4, XI.5, and XI.6. It does not import XI.7. The common plane that Euclid obtains through XI.7 is already contained in the formal hypothesis `HilbertSpaceLinesParallel`.

8.12 Hidden Formal Data

XI.8 contains substantially more hidden incidence information than its classical diagram suggests.

- The second parallel must be proved to meet the reference plane. The point D is an existential output, not an assumed input.
- The common plane σ of the parallels must be proved distinct from π .
- The intersection line $q = \pi \cap \sigma$ is constructed explicitly.
- The points B, D must be proved distinct from disjointness of the parallel carriers.
- The line $d = BD$ must be shown to lie both in π and in σ .

- The point A selected on l must be proved to lie outside π ; otherwise the line l would lie in π , contrary to $l \perp \pi$.
- The line $a = DA$ must be constructed and shown distinct from d .
- Right-angle and perpendicularity statements must be transported between ambient geometry and `PlaneGeo`.
- The neutral I.27 argument requires explicit `SameSide` / `OppositeSide` and `SameRay` normalization that Euclid suppresses in the diagram.
- The two disjointness orientations required by Group IV are supplied explicitly.
- Before the final XI.4 application, $d \neq e$ is derived from $e \perp d$.

Subtype projections and record fields occur in the Lean code, but they are implementation details. The mathematically significant hidden data are the plane memberships, distinctness statements, side orientations, and the derived intersection point D .

8.13 Relationship with Earlier Book XI Propositions

The classical and formal dependency graphs differ in a controlled way.

Classically, XI.7 is an explicit opening dependency: it places AB, CD, BD in one plane. XI.4 is then used twice. The classical plane-identification step also relies on the ordinary fact from XI.2 that a triangle lies in one plane.

The production Lean file imports

```
import CGJteamLab.Proposition11_4
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition11_6
```

but does not import Proposition XI.7.

XI.4 is used twice exactly as in the classical architecture: first in the common plane σ , then in the reference plane π .

XI.5 contributes established line–plane perpendicularity infrastructure, including the theorem that a line perpendicular to a plane is not contained in that plane.

XI.6 contributes the reusable metric construction and the spatial SAS/SSS right-angle machinery. This reuse is substantial: the central auxiliary configuration of XI.8 is almost the same as the configuration built for XI.6.

XI.7 is absent from the Lean import graph because `HilbertSpaceLinesParallel` already stores the common carrier plane as part of the definition of spatial parallelism. Thus the formal proof consumes the conclusion-like data that XI.7 exposes, without invoking the proposition theorem itself.

8.14 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical proof uses I.11 and I.3 to construct the auxiliary perpendicular and equal segment. In the Lean reconstruction these operations are already packaged in the XI.6 helper chain over the Hilbert/Book Zero construction interface.

The metric part corresponds to I.4 and I.8 and is likewise inherited from XI.6.

The parallel step is the important logical boundary. The direction represented by I.27,

$$\text{equal alternate angles} \implies \text{nonintersection},$$

belongs to the neutral congruence/order layer. The reverse use of parallelism represented by I.29 requires the Euclidean parallel axiom. The formal proof displays this by using the neutral I.27 theorem first and applying Group IV only to identify the unique line through D disjoint from l .

No coordinate system, vector space, scalar product, distance formula, trigonometric function, or numerical angle measure is used.

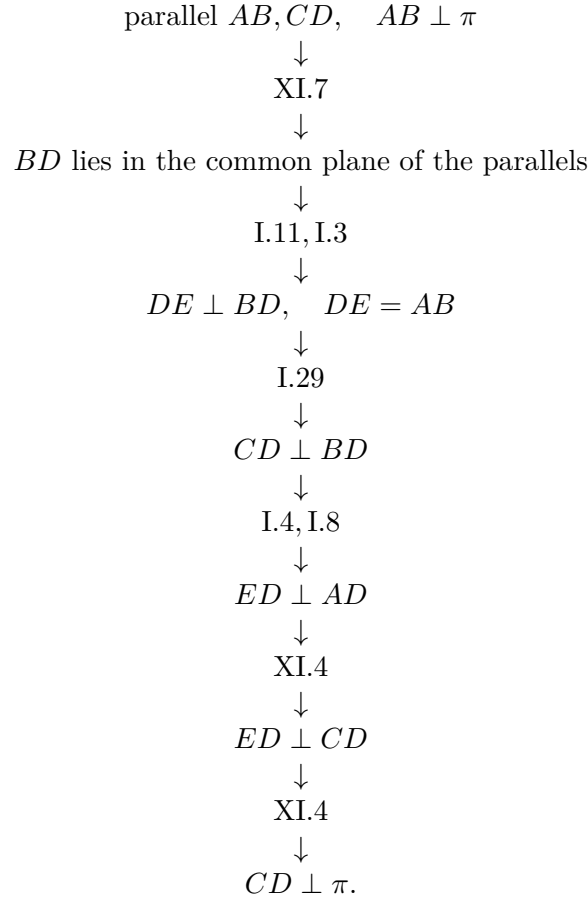
8.15 Relationship with the Foundations

XI.8 is the first proposition in this local sequence whose current Euclid-faithful reconstruction genuinely uses Hilbert Group IV. The public theorem assumes `HilbertSpaceEuclidean Geo`, and the proof consumes that structure through the induced Euclidean geometry of an explicit plane slice.

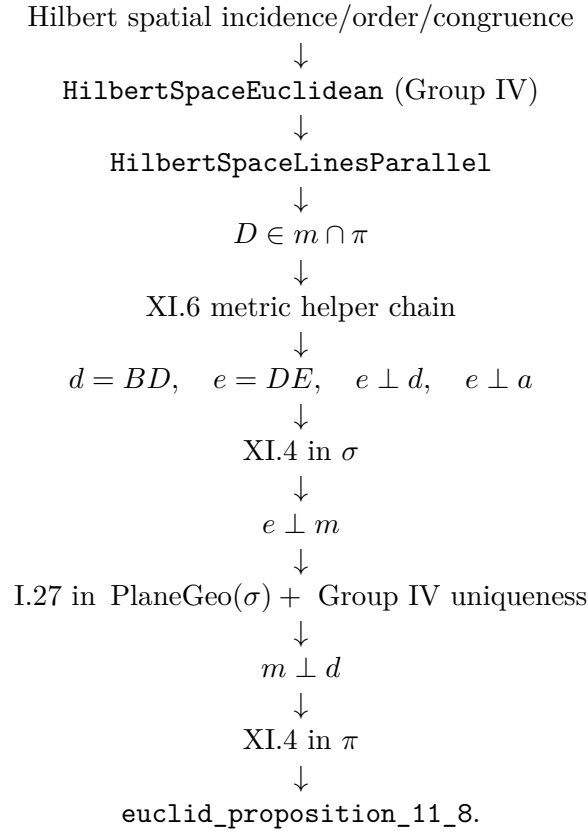
This use of Group IV is already analyzed in the section “The Euclidean Layer: Group IV” above. The essential point is mathematical: the proof needs the uniqueness/existence behavior of parallels in the relevant plane; no global ambient planar instance is installed. No continuity principle is used.

8.16 Dependency Summary

The classical local chain is



The actual formal chain is



The dependency classification is therefore:

New proposition-specific global axiom	no
New proposition-specific spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via <code>HilbertSpaceEuclidean</code>
Earlier Book XI production imports	XI.4, XI.5, XI.6
Classical explicit Book XI dependency	XI.7, XI.4
New PlaneGeo Euclidean bridge	yes
Coordinates / vectors / scalar product	no
sorry/admit	no

8.17 Conclusion

Proposition XI.8 is the first point in the present Book XI reconstruction where Euclidean parallel uniqueness is mathematically necessary. The formal proof makes that boundary sharper than the classical text: neutral I.27 proves nonintersection, while Group IV supplies the uniqueness needed to recover the I.29 direction.

At the same time, the proof remains fully synthetic. It reuses the metric construction of XI.6, applies XI.4 in two different plane slices, and derives the hidden intersection point of the second parallel with the reference plane instead of assuming it from the diagram.

The result therefore tests three parts of the spatial architecture at once: plane-local Euclidean parallelism, transport between ambient three-space and induced plane geometry, and the universal line–plane perpendicularity predicate. No continuity or analytic representation is introduced.

Chapter 9

Proposition XI.9

9.1 Introduction

Proposition XI.9 is the spatial counterpart of Euclid I.30. It asserts that if two lines are each parallel to a third line, then they are parallel to one another, provided that the three lines are not all contained in one plane.

This extra non-coplanarity clause is the essential new feature. In a plane, Euclid I.30 already proves transitivity of parallelism. In space, however, the theorem must distinguish two cases:

- the planar case, where the three lines lie in one plane and I.30 is the relevant model;
- the genuinely spatial case, where the three lines are distributed across different planes and the conclusion must be rebuilt from 3D incidence.

The current formal proof treats XI.9 as a theorem of synthetic spatial incidence plus one plane-local use of the Euclidean parallel axiom. Contrary to the historical proof, it does not proceed through a chain of perpendicular constructions.

The production file is

```
CGJteamLab/Proposition11_9.lean
```

and XI.9 introduces two proposition-local helpers:

```
hilbert_XI9_point_on_second_line_off_first  
hilbert_XI9_planes_eq_of_common_line_and_external_point
```

9.2 Euclid's Statement

Euclid states:

If two straight lines are parallel to the same straight line, then they are parallel to one another, if they do not lie in the same plane with it.

In the present library, spatial parallelism is represented by

```
HilbertSpaceLinesParallel Geo l m
```

which packages:

- a plane containing both lines, and
- ambient disjointness of the two lines.

Thus the formal conclusion of XI.9 is not merely “coplanar with the same direction” but the full structured statement that l and m are parallel lines in the project’s spatial sense.

The phrase “if they do not lie in the same plane with it” is formalized as

```
Not (exists omega : S.Plane,
    HilbertLineInPlane Geo l omega /\
    HilbertLineInPlane Geo m omega /\
    HilbertLineInPlane Geo n omega)
```

That is: there is no single plane containing all three lines l, m, n . This is the exact ambient reading of Euclid’s proviso.

9.3 Classical Configuration

We are given three lines l, m, n such that

$$l \parallel n, \quad m \parallel n,$$

and the three are not all coplanar.

Let σ be a plane containing l and n , and let τ be a plane containing m and n . Because all three lines are not contained in one plane, these two carrier planes are distinct:

$$\sigma \neq \tau.$$

The formal proof then chooses a point B on m not lying on l , forms the plane ω through l and B , and studies the intersection

$$r = \omega \cap \tau.$$

The line r passes through B . Inside τ , both r and m pass through B and are parallel to n . Euclidean uniqueness in the plane τ forces

$$r = m.$$

Hence both l and m lie in ω . The final step proves they are disjoint, so

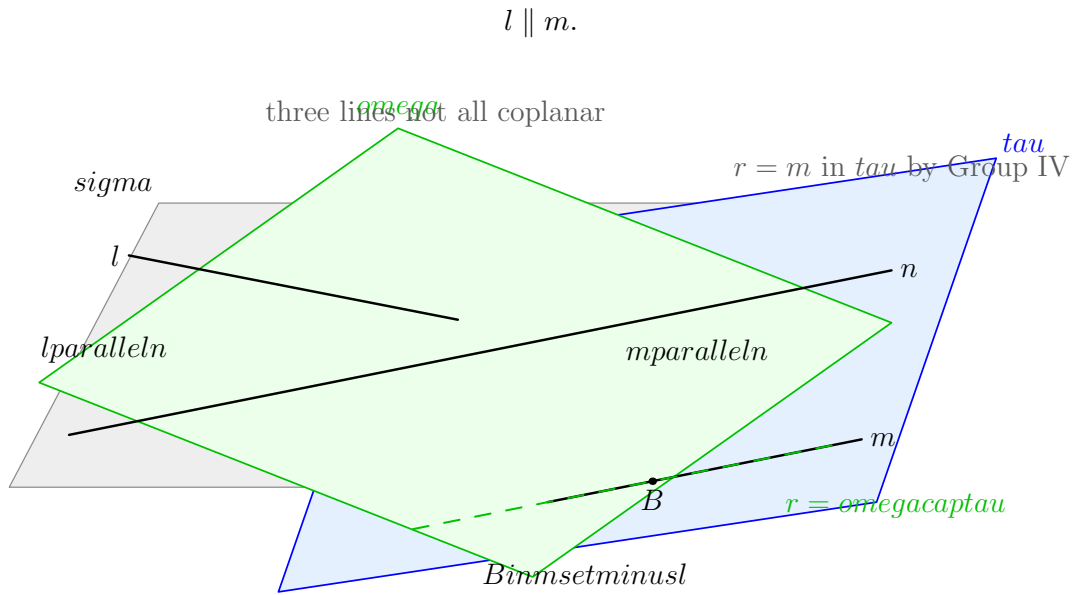


Figure 9.1: A schematic picture for Proposition XI.9. The line n lies in both carrier planes σ and τ . The line l lies in σ , the line m lies in τ , and the three lines are not all coplanar. Choosing $B \in m \setminus l$, the plane ω through l and B intersects τ in a line r . Since both r and m pass through B and are parallel to n inside τ , Group IV gives $r = m$.

9.4 Euclid's Classical Proof Pattern

Euclid's printed proof is not organized in this affine way. Its characteristic strategy is to convert the parallel problem into a perpendicular problem. Schematically, the classical route is:

$$\text{I.11 in two planes} \longrightarrow \text{common perpendiculars} \longrightarrow \text{XI.4} \longrightarrow \text{XI.8} \longrightarrow \text{XI.6} \longrightarrow \text{XI.9}.$$

In more explicit local terms:

1. in suitable planes, erect lines perpendicular to the common line;
2. use XI.4 to pass from right-angle data in one plane to perpendicularity to the plane;
3. use XI.8 to transport plane-perpendicularity across parallels;
4. use XI.6 to conclude that the resulting perpendiculars are parallel;
5. conclude that the original lines are parallel.

This is entirely reasonable in Euclid's sequence, since Book XI is gradually building a theory of perpendicularity and parallelism for lines and planes.

Nevertheless, from the standpoint of the current formal library, this is not the most efficient reconstruction. The actual theorem can be proved more directly from spatial incidence and planar parallel uniqueness.

9.5 Why the Formal Reconstruction Takes a Different Route

The project already contains a precise synthetic notion of ambient line parallelism, a theorem on the intersection of two planes, and a plane-local Euclidean parallel uniqueness axiom.

Once these are available, XI.9 can be reconstructed with a cleaner structure:

- first create the target coplanarity of l and m ;
- then prove their disjointness;
- package both facts as `HilbertSpaceLinesParallel Geo l m`.

This proof is still fully synthetic. It uses no coordinates, vectors, or scalar products. The difference is conceptual, not analytic: the formal proof exposes XI.9 as a theorem about how distinct planes and parallel lines fit together in 3-space.

9.5.1 First local incidence lemma

The first local helper is

```
theorem hilbert_XI9_point_on_second_line_off_first
...
(l m : Geo.Line)
(hlm : Ne l m) :
exists P : Geo.Point,
  H.OnLine P m /\
  Not (H.OnLine P l)
```

Its content is elementary but useful: if two lines are distinct, then the second line has a point not lying on the first.

The proof uses two points on m . If both were also on l , line uniqueness would force $m = l$, contradiction.

This helper is a pure incidence lemma and abstracts away a routine but repeatedly needed argument.

9.5.2 Plane uniqueness from a line and an external point

The second helper is

```
theorem hilbert_XI9_planes_eq_of_common_line_and_external_point
...
(l : Geo.Line)
(P : Geo.Point)
(hPl : Not (H.OnLine P l))
(pi rho : S.Plane)
(hlpi : HilbertLineInPlane Geo l pi)
(hlrho : HilbertLineInPlane Geo l rho)
(hPpi : S.OnPlane P pi)
(hPrho : S.OnPlane P rho) :
```

```
pi = rho
```

This is the uniqueness clause of the theorem “a line and an external point determine a unique plane,” extracted into a convenient reusable form.

Its role in XI.9 is structural. Whenever two planes both contain the same line and the same point outside that line, the planes coincide. This is the key contradiction engine used twice in the proof:

- first to show that the auxiliary intersection line r is disjoint from n ;
- then to show that the final target lines l and m are disjoint.

9.6 Formal Statement

The public theorem is:

```
theorem euclid_proposition_11_9
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (l m n : Geo.Line)
  (hParallelLN : HilbertSpaceLinesParallel Geo l n)
  (hParallelMN : HilbertSpaceLinesParallel Geo m n)
  (hNoCommonPlane :
    Not (exists omega : S.Plane,
      HilbertLineInPlane Geo l omega /\
      HilbertLineInPlane Geo m omega /\
      HilbertLineInPlane Geo n omega)) :
  HilbertSpaceLinesParallel Geo l m
```

The assumptions separate clearly into three layers:

- spatial incidence and plane structure;
- spatial order/congruence, needed through the Group IV interface;
- the Euclidean parallel uniqueness axiom `HilbertSpaceEuclidean Geo`.

9.7 Proof Architecture of the Reconstruction

The proof has six main steps.

Step 1: unpack the two given parallels

From

$$l \parallel n, \quad m \parallel n,$$

we obtain planes σ and τ such that

$$l, n \subset \sigma, \quad m, n \subset \tau,$$

with the corresponding disjointness facts

$$l \cap n = \emptyset, \quad m \cap n = \emptyset.$$

Step 2: prove $\sigma \neq \tau$ and $l \neq m$

If $\sigma = \tau$, then the three lines would lie in one plane, contrary to the explicit hypothesis `hNoCommonPlane`. Likewise, if $l = m$, then again all three would lie in one plane.

Step 3: choose $B \in m \setminus l$ and form ω

Using `hilbert_XI9_point_on_second_line_off_first`, choose a point B on m which does not lie on l . Then the theorem

```
hilbert_plane_through_line_and_external_point
```

produces a plane ω containing l and B .

Step 4: intersect ω and τ

The planes ω and τ are distinct, and both contain B . By XI.3 in its strengthened Hilbert form, they intersect in a line r through B :

$$r = \omega \cap \tau.$$

Step 5: identify r with m

The crucial point is that r is disjoint from n . For if a point P belonged to both r and n , then P would lie in both ω and σ , while being outside l . Since both planes also contain l , helper 2 would force

$$\sigma = \omega,$$

contradiction.

Thus in the plane τ , both lines r and m :

- pass through the point B , and

- are disjoint from the line n .

By plane-local Euclidean uniqueness of a parallel through a point,

$$r = m.$$

Hence $m \subset \omega$, and the target coplanarity of l and m is established.

Step 6: prove l and m are disjoint

Suppose a point P lay on both l and m . Since l is disjoint from n , this point is outside n . But P lies in σ (because $P \in l \subset \sigma$) and in τ (because $P \in m \subset \tau$). Since both planes also contain n , helper 2 forces

$$\sigma = \tau,$$

again contradicting the non-coplanarity assumption. Therefore

$$l \cap m = \emptyset.$$

Since l and m lie in the common plane ω and are disjoint, the packaged conclusion follows:

$$l \parallel m.$$

9.8 Formal DAG

The actual dependency chain of the current reconstruction is:

```

hParallelLN, hParallelMN, hNoCommonPlane
  |
  +-- hilbert_XI9_point_on_second_line_off_first
  |
  +-- hilbert_plane_through_line_and_external_point
  |
  +-- hilbert_plane_intersection_line    (XI.3 infrastructure)
  |
  +-- HilbertSpaceEuclidean.parallel_unique_in_plane
  |
  +-- hilbert_XI9_planes_eq_of_common_line_and_external_point
  |
  v
euclid_proposition_11_9

```

This should be contrasted with the historical local chain:

I.11

- > XI.4
- > XI.8
- > XI.6
- > XI.9

The formal DAG is not less synthetic. It simply exposes the spatial incidence content more directly.

9.9 Wyler-Style Flat Reconstruction

The production reconstruction above is affine and synthetic. The Wyler-style second pass changes the way the auxiliary planes are identified: instead of a proposition-local uniqueness lemma, two planes are shown equal because they are the same generated join of a line and an external point.

The resulting structural graph is

```

sigma contains l,n
tau  contains m,n
sigma != tau
    |
    v
B in m, B notin l
omega = Join(l,{B})
    |
    v
r = Meet(omega,tau)
    |
    v
r disjoint n
    |
    v
Group IV in tau: r = m
    |
    v
m subset omega
l,m disjoint
    |
    v
l || m

```

The key normalization is the generated plane

$$\omega = \text{Join}(l, \{B\}).$$

Whenever another plane contains the same line l and the same point $B \notin l$, equality is detected by equality with this join. Likewise the section line r is treated as the exact meet of ω and τ . Only the identification $r = m$ requires the plane-local Euclidean uniqueness of a parallel through B .

This graph is proposition-specific and therefore belongs here rather than in the general Wyler appendix.

9.10 Spatial / Planar Architecture Used

The theorem depends on the following ambient structure:

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
```

It also reuses the previously developed spatial theorems:

```
hilbert_plane_through_line_and_external_point
hilbert_plane_intersection_line
```

No coordinates, vectors, affine spaces, or scalar products are introduced. No induced `PlaneGeo` slice is required in the public proof, even though the decisive Euclidean axiom is conceptually plane-local.

9.11 Role of Group IV

XI.9 uses the Euclidean parallel axiom exactly once, in the form

```
HilbertSpaceEuclidean.parallel_unique_in_plane
```

applied in the plane τ .

This is the precise place where Euclidean strength enters. Everything before that point is incidence-only: choose a point off a line, construct a plane, intersect two planes, and compare two candidate planes by uniqueness.

The proof therefore gives a clean conceptual reading of XI.9:

3D incidence geometry + one planar uniqueness step $\implies$ spatial transitivity of parallelism under Euclid’s prov

9.12 Hidden Formal Data

Several facts that are implicit in the classical picture must be made explicit in Lean.

The carrier planes are proof objects

The hypotheses `hParallelLN` and `hParallelMN` do not merely say “parallel” in a bare relational sense. They supply witness planes σ and τ .

The non-coplanarity clause is genuinely global

The condition `hNoCommonPlane` is not a statement about one chosen pair of lines; it quantifies over all candidate planes ω .

The point B is not diagrammatic decoration

The point off l but on m is the key to constructing the new plane ω that will eventually contain both target lines.

Disjointness must be proved twice

The proof must explicitly certify both

$$r \cap n = \emptyset \quad \text{and} \quad l \cap m = \emptyset.$$

In a traditional diagram these facts often look visually obvious, but in a synthetic proof they are independent obligations.

9.13 Mathematical Significance of the Reconstruction

XI.9 is a good example of the difference between the order of exposition in the *Elements* and the most transparent dependency structure of the formal theory.

Euclid's proof moves through perpendicularity because Book XI is organized as a gradual synthetic development of the geometry of lines and planes. The modern formal system, however, already has explicit infrastructure for planes through lines, intersections of planes, and plane-local uniqueness of parallels.

Once those pieces are isolated, the theorem turns out to be structurally about *coplanarity creation* and *disjointness control*. The core problem is not to manufacture right angles, but to produce the unique plane in which the two target lines must lie.

In this sense XI.9 is one of the first propositions of Book XI where the formal reconstruction becomes conceptually more informative than the classical proof: it reveals the theorem as a hybrid of spatial incidence and planar Euclidean uniqueness.

9.14 Conclusion

Proposition XI.9 states that if two lines are each parallel to a third line, and the three are not all contained in one plane, then the first two are parallel to one another.

The present Lean proof establishes this in a direct synthetic way:

$$l \parallel n, \quad m \parallel n, \quad \text{no common plane for } l, m, n \implies l \parallel m.$$

The proof proceeds by choosing a point on m outside l , constructing a plane through l and that point, intersecting it with the plane containing m and n , and invoking Euclidean uniqueness

of a parallel through a point. A second plane-uniqueness argument then supplies the required disjointness of l and m .

Thus XI.9 is formalized not as a proposition about perpendicular gadgets, but as a structural theorem about three lines distributed across several planes. It remains fully synthetic, uses Group IV at exactly one point, and achieves a very clean axiom profile.

Chapter 10

Proposition XI.10

10.1 Introduction

Proposition XI.10 transfers an angle from one plane to another by means of two pairs of corresponding parallel lines. In Euclid’s configuration, AB and BC meet at B , while DE and EF meet at E . The first pair is parallel to the second pair, and the two angle configurations are not all contained in one plane. The conclusion is

$$\angle ABC \cong \angle DEF.$$

The proposition is easy to underestimate. Ordinary, unoriented parallelism is not sufficient to determine the correct angle. Reversing exactly one of the four rays changes the desired angle to its supplement. The formal reconstruction therefore has to retain directional information which Euclid leaves to the diagram and to the phrase “parallel to” in a particular direction.

The production proof remains fully synthetic. It uses no coordinates, vectors, scalar product, normal vectors, trigonometric functions, or numerical angle measure. Its central ingredients are:

- Euclid I.33 in explicitly chosen plane slices;
- Euclid XI.9 for spatial parallelism;
- a plane-separation / section-line bridge;
- spatial SSS;
- a final SameRay transport used by the public wrapper.

The proof also illustrates a distinction that is especially important in Book XI. The note intentionally contains several schematic figures: a main spatial overview, the I.33–XI.9 connector state, the directed-parallel slice, the section-line / plane-side bridge, the public-wrapper normalization, and the final SameRay/SSS states. Each figure is generated from an auditable Asymptote source.

unoriented carrier parallelism $\neq$ directed parallel segments.

The former is enough for XI.9. The latter is needed to recover exactly the angle intended by XI.10.

10.2 Euclid's Statement

Euclid's Proposition XI.10 states:

If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

Let AB and BC meet at B , and let DE and EF meet at E . Assume that the corresponding straight lines are parallel and that the two intersecting pairs are not contained in one common plane. Euclid's claim is

$$\angle ABC = \angle DEF.$$

The phrase “parallel to” here carries directional content. If one ray is reversed while the other three are kept fixed, the conclusion is generally not equality of the two displayed angles. This directional issue is part of the mathematical statement, not merely a formalization detail.

10.3 Classical Configuration

Euclid first cuts four equal segments:

$$BA = BC = ED = EF.$$

He then joins

$$AD, \quad CF, \quad BE, \quad AC, \quad DF.$$

The resulting configuration contains three successive parallelogram-type steps.

First,

$$BA \parallel ED, \quad BA = ED$$

implies by I.33 that

$$AD \parallel BE, \quad AD = BE.$$

Second,

$$BC \parallel EF, \quad BC = EF$$

implies, again by I.33,

$$CF \parallel BE, \quad CF = BE.$$

Third, XI.9 gives

$$AD \parallel CF,$$

and equality follows through the common segment BE :

$$AD = CF.$$

A final use of I.33 yields

$$AC \parallel DF, \quad AC = DF.$$

The triangles

$$\triangle ABC, \quad \triangle DEF$$

therefore have three corresponding equal sides, and I.8 gives

$$\angle ABC = \angle DEF.$$

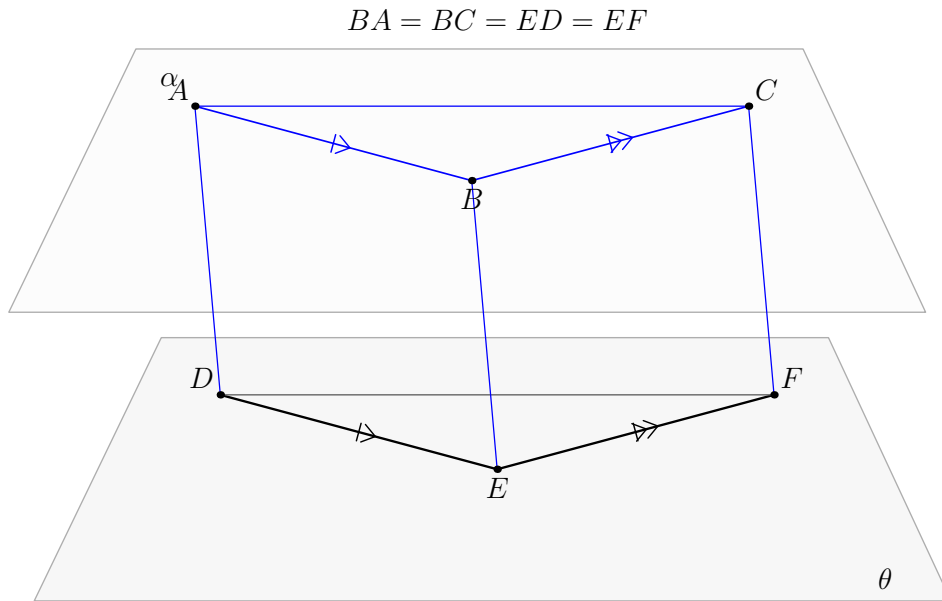


Figure 10.1: Classical spatial configuration for Proposition XI.10. The two lines meeting at B are compared with the two lines meeting at E ; the two angle configurations are not contained in one common plane. This figure is the global reference picture for the proposition.

10.4 Classical Proof

The classical local chain is

$$\begin{aligned}
 &BA = BC = ED = EF \\
 &\quad \Downarrow \\
 &BA \parallel ED, \quad BC \parallel EF \\
 &\quad \Downarrow \\
 &\text{I.33 twice} \\
 &\quad \Downarrow \\
 &AD \parallel BE, \quad AD = BE, \\
 &CF \parallel BE, \quad CF = BE \\
 &\quad \Downarrow \\
 &\text{XI.9} \\
 &\quad \Downarrow \\
 &AD \parallel CF, \quad AD = CF \\
 &\quad \Downarrow \\
 &\text{I.33} \\
 &\quad \Downarrow \\
 &AC \parallel DF, \quad AC = DF \\
 &\quad \Downarrow \\
 &\text{I.8} \\
 &\quad \Downarrow \\
 &\angle ABC = \angle DEF.
 \end{aligned}$$

Thus the explicit classical proposition dependencies are

$$\text{I.3}, \quad \text{I.33}, \quad \text{XI.9}, \quad \text{I.8}.$$

The first step, I.3, is a normalization: equal pieces are laid off on the four rays. The geometric core begins after those equal segments have been chosen.

10.5 Why Ordinary Spatial Parallelism Is Not Enough

The project definition

```
HilbertSpaceLinesParallel Geo l m
```

records that two carrier lines are coplanar and disjoint. This is exactly the correct spatial analogue of ordinary line parallelism: skew disjoint lines are excluded.

XI.10 needs more. I.33 is directional: if two equal parallel segments are taken in corresponding directions, then the joining segments are equal and parallel in the corresponding direction. The same carriers with one endpoint order reversed describe the supplementary configuration.

For this reason XI.10 introduces

```
def HilbertSpaceDirectedParallelSegments
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (A B C D : Geo.Point) : Prop :=
  exists pi : S.Plane,
    exists l m t : Geo.Line,
      H.OnLine A l /\
      H.OnLine B l /\
      H.OnLine C m /\
      H.OnLine D m /\
      HilbertLineInPlane Geo l pi /\
      HilbertLineInPlane Geo m pi /\
      HilbertLinesDisjoint Geo l m /\
      H.OnLine B t /\
      H.OnLine D t /\
      HilbertLineInPlane Geo t pi /\
      HilbertSameSideInPlane Geo A C t pi
```

The ordered segments are $A \rightarrow B$ and $C \rightarrow D$. The connector t passes through the terminal points B, D . The initial points A, C must lie on the same side of that connector in the common plane of the two parallel carriers.

This is a synthetic orientation condition. It contains no coordinates and no selected global orientation of space.

10.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_10
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
```

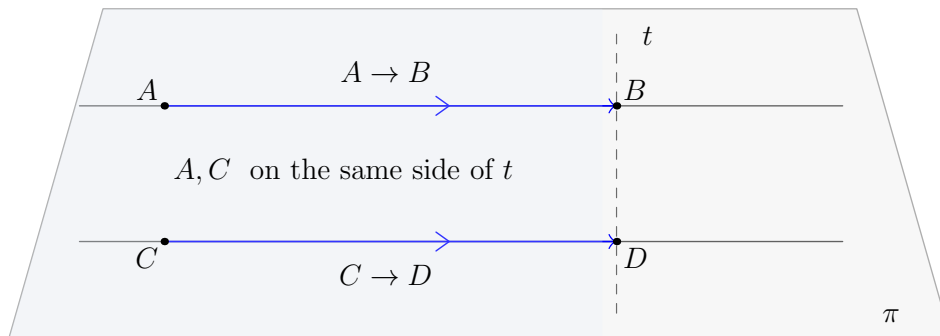


Figure 10.2: Directed parallel segments in a common plane slice. The carriers are parallel, the connector joins the terminal endpoints, and the initial endpoints lie on the same side of the connector. This is the synthetic orientation datum encoded by `HilbertSpaceDirectedParallelSegments`.

```
(Geo := Geo) (H := H) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(A B C D E F : Geo.Point)
(hDirAB_DE :
  HilbertSpaceDirectedParallelSegments Geo A B D E)
(hDirCB_FE :
  HilbertSpaceDirectedParallelSegments Geo C B F E)
(hABC : Not (PrimCollinear Geo A B C))
(hDEF : Not (PrimCollinear Geo D E F))
(hNoCommonPlane :
  Not (exists omega : S.Plane,
    S.OnPlane A omega /\
    S.OnPlane B omega /\
    S.OnPlane C omega /\
    S.OnPlane D omega /\
    S.OnPlane E omega /\
    S.OnPlane F omega)) :
Geo.AngleCongruent A B C D E F
```

The hypotheses separate four kinds of information.

1. `hDirAB_DE` and `hDirCB_FE` encode the two corresponding directed parallel pairs.
2. `hABC` and `hDEF` ensure that both displayed angles are nondegenerate.
3. `hNoCommonPlane` states the genuinely spatial condition that the six selected points do not all belong to one plane.
4. The Hilbert typeclass hypotheses provide ambient incidence, spatial order, congruence, and plane-local Euclidean parallel uniqueness.

The conclusion is directly:

```
Geo.AngleCongruent A B C D E F
```

10.7 Spatial / Planar Architecture Used

The ambient layer is:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

The proof never assumes

```
HilbertOrder Geo
HilbertCongruence Geo
```

globally on ambient space.

Whenever an ordinary planar theorem is needed, an explicit plane slice is used:

```
PlanePoint Geo pi
PlaneLine Geo pi
PlaneGeo Geo pi
```

The principal bridges used by XI.10 include:

```
planeGeo_sameSide_iff_space
planeGeo_sameRay_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeoAngleToAmbient_angle
hilbert_onPlane_of_primCollinear_with_two_on_plane
hilbert_plane_intersection_line
```

The formalization therefore alternates between ambient three-space and ordinary planar Hilbert geometry, but it never conflates the two.

10.8 Proof Architecture of the Reconstruction

10.8.1 The normalized XI.10 core

The proof was first closed in a normalized form, in which the four selected arms already have the equal lengths used by Euclid:

$$BA \cong ED, \quad BC \cong EF.$$

At this stage the first two I.33 applications are represented by

```
hilbert_XI10_I33_directed_output
```

and packaged by

```
hilbert_XI10_first_two_I33_directed_package
```

The package supplies explicit carriers AD , CF , and BE , together with

$$AD \parallel BE, \quad AD \cong BE,$$

$$CF \parallel BE, \quad CF \cong BE,$$

and, crucially, the directed versions

$$BE \longrightarrow AD, \quad BE \longrightarrow CF.$$

The ordinary carrier information is enough for XI.9. The directed information is retained for the later orientation step.

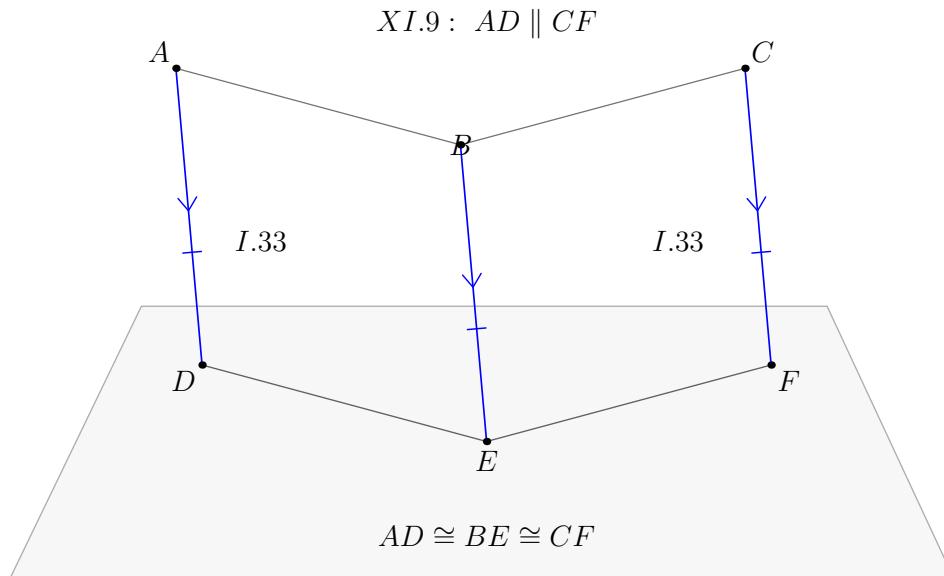


Figure 10.3: The connector state after the first two I.33 applications. The three joining segments AD , BE , and CF are congruent, while XI.9 supplies the carrier relation $AD \parallel CF$. This is the ordinary-parallel part of the normalized core.

10.8.2 The XI.9 step

The two first I.33 applications give two lines parallel to the same line BE . The hypothesis that the six selected points do not belong to one common plane supplies the noncoplanarity clause needed by XI.9.

Thus

```
euclid_proposition_11_9
```

gives

$$AD \parallel CF.$$

Segment congruence transitivity through BE gives

$$AD \cong CF.$$

At this point the unoriented part of the classical proof has been recovered. What remains is to prove that AD and CF have the correct *relative direction* for the third I.33 application.

10.8.3 Plane separation and the section-line bridge

This is the main formal point which is almost invisible in Euclid's printed proof.

Let θ be the plane of the second angle, containing D, E, F . The first two directed I.33 steps imply, in two auxiliary planes, that A and B , and respectively B and C , lie on the appropriate sides of the section lines ED and EF .

To combine these data in ambient space the proof introduces:

```
def HilbertSegmentMeetsPlane ...
def HilbertSameSideOfPlane ...
```

The essential bridge is

```
hilbert_XI10_borsuk_szmielew_52
```

Its content is the following. Suppose two planes π, θ meet along a line t , and $P, Q \in \pi$. Then the segment PQ crosses θ exactly when it crosses the section line t inside π . Equivalently, under the usual off-boundary conditions, same-side in the plane π with respect to t is the same as same-half-space in ambient space with respect to θ .

The source-level identification of this bridge is collected in the Book XI source appendices. Here only its formal geometric role in XI.10 is needed.

No topological separation axiom is added. The bridge is derived from the existing spatial incidence and order structure.

10.8.4 Transitivity of same side of a plane

The next helper is

```
hilbert_XI10_sameSideOfPlane_trans_noncollinear
```

It proves the transitivity needed for three noncollinear points:

$$A \sim_{\theta} B, \quad B \sim_{\theta} C \quad \implies \quad A \sim_{\theta} C.$$

The proof has two geometric cases.

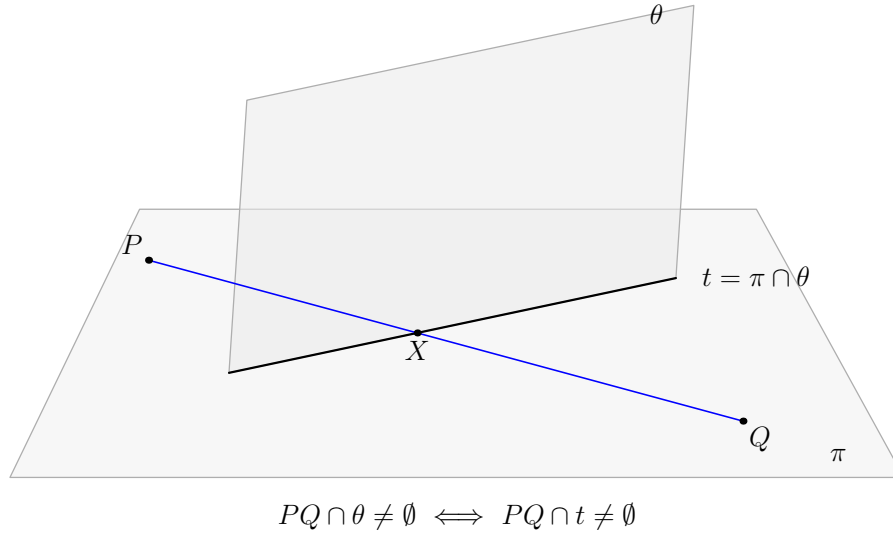


Figure 10.4: A segment PQ in a plane slice crosses the ambient plane θ exactly when it crosses the section line $t = \pi \cap \theta$ inside the slice plane π . This is the geometric bridge between plane-side and half-space information.

If the plane $\alpha = \text{plane}(A, B, C)$ meets θ , their intersection is a line. The section-line / plane-side bridge converts both half-space relations into ordinary same-side relations in α ; the planar same-side paths compose there, and the result is converted back.

If α and θ have no common point, every point of the open segment AC remains in α , so that segment cannot meet θ . Hence A and C are automatically on the same side of θ .

This disjoint-plane branch is essential. One cannot assume that the two angle planes intersect.

10.8.5 Recovering the direction $AD \rightarrow CF$

The central orientation theorem is

```
hilbert_XI10_directed_AD_CF_from_common_BE
```

It combines

$$BE \longrightarrow AD, \quad BE \longrightarrow CF,$$

the ordinary conclusion

$$AD \parallel CF,$$

and the noncoplanarity of the two angle configurations.

The proof lifts the two local same-side statements to same-side-of-plane relations with respect to $\theta = \text{plane}(D, E, F)$, composes them there, and then cuts the result back down in the plane containing AD and CF . The section line is DF .

The conclusion is the exact directed statement

$$AD \longrightarrow CF.$$

This is the missing orientation datum which ordinary XI.9 cannot provide.

10.8.6 The third I.33 and spatial SSS

Now both ingredients required by the third I.33 are available:

$$AD \longrightarrow CF, \quad AD \cong CF.$$

The helper

```
hilbert_XI10_third_I33_AC_DF
```

applies the directed I.33 package and obtains

$$AC \cong DF.$$

The three side congruences are therefore

$$BA \cong ED, \quad BC \cong EF, \quad AC \cong DF.$$

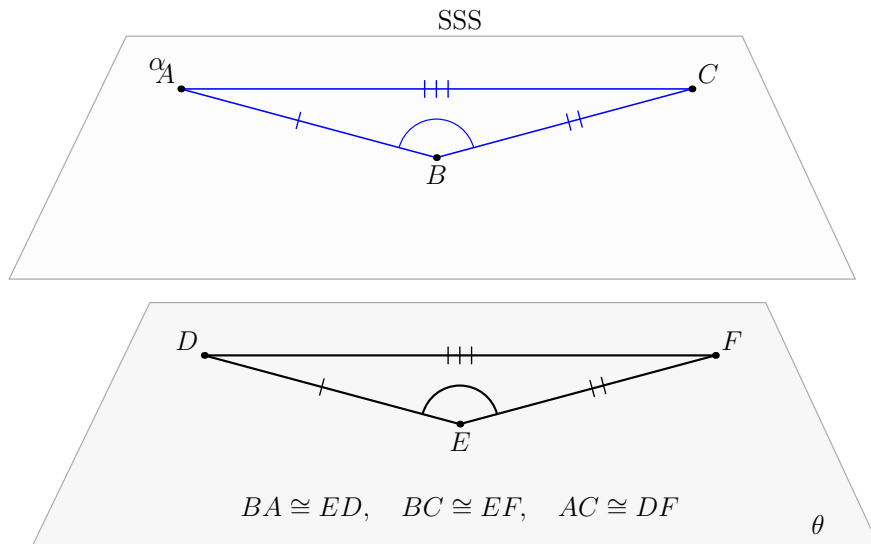


Figure 10.5: The final normalized SSS state. The three pairs of corresponding sides are marked separately: $BA \cong ED$, $BC \cong EF$, and $AC \cong DF$. The two triangles lie in distinct parallel plane slices.

The final normalized step is spatial SSS:

```
hilbert_space_sss_angleA_in_plane
```

Applied to the triangles BAC and EDF , it yields exactly

$$\angle ABC \cong \angle DEF.$$

The target triangle is represented inside the explicit slice $\text{PlaneGeo}(\text{plane}(D, E, F))$, while the first triangle remains ambient. The SSS bridge is designed precisely for this cross-plane comparison.

10.8.7 Fixed-connector normalization

For the public theorem a slightly different endpoint orientation is more convenient:

$$AB \longrightarrow DE, \quad CB \longrightarrow FE.$$

This is the theorem

```
euclid_proposition_11_10_normalized_fixed_connector
```

The advantage is that the connector in both directed-parallel relations is the same fixed line BE . When the outer endpoints are moved along their rays during segment construction, BE does not change.

Two small transport results support this form:

```
hilbert_XI10_directedParallelSegments_swap_pairs
hilbert_XI10_directedParallelSegments_transport_first_endpoints
```

This is a representation choice only. It does not change the geometry of Euclid's proposition.

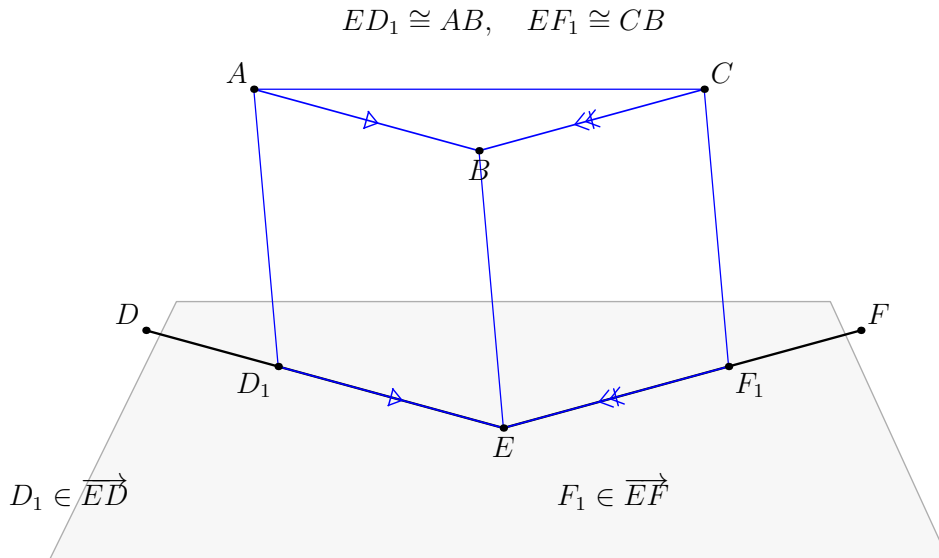


Figure 10.6: Wrapper-level segment construction. Only the points D_1 and F_1 are laid off on the rays ED and EF , with $ED_1 \cong AB$ and $EF_1 \cong CB$. This yields the normalized configuration used by the internal core theorem.

10.8.8 The public wrapper: Euclid's I.3 normalization

The public theorem does not assume equal selected arms.

Instead, spatial segment construction lays off on the rays ED and EF new points D_1, F_1 such that

$$ED_1 \cong AB, \quad EF_1 \cong CB.$$

Only the second angle is normalized; the points A, B, C are left unchanged. This is formally more economical than reproducing all four layoffs of the classical prose.

Because

$$\text{SameRay}(E, D, D_1), \quad \text{SameRay}(E, F, F_1),$$

the directed parallel hypotheses transport to

$$AB \longrightarrow D_1E, \quad CB \longrightarrow F_1E.$$

The fixed-connector normalized theorem gives

$$\angle ABC \cong \angle D_1EF_1.$$

Finally, the SameRay angle bridge is applied inside the genuine plane $\text{plane}(D, E, F)$. This is an important architectural point: the proof does *not* install a global

[HilbertOrder Geo]

on ambient three-space. The ray-angle transport occurs legally in the induced planar geometry and is then mapped back to ambient angle data.

Thus

$$\angle D_1EF_1 = \angle DEF,$$

and the public conclusion follows.

$$\angle D_1EF_1 = \angle DEF$$

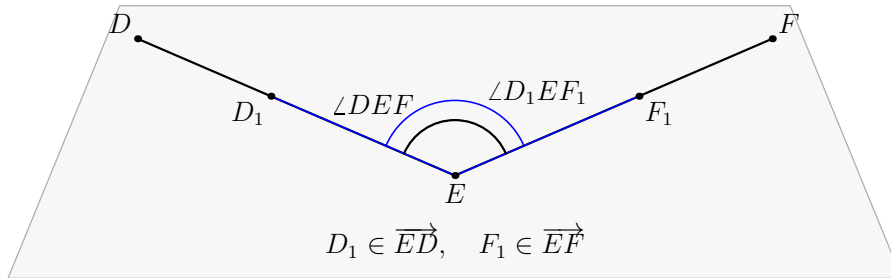


Figure 10.7: Final SameRay angle transport in the plane of DEF . After the normalized proof gives $\angle ABC \cong \angle D_1EF_1$, the facts that D_1 lies on the ray ED and F_1 lies on the ray EF identify the copied angle with the original target angle $\angle DEF$.

10.8.9 Normalized proof spine

The following diagram is intentionally kept in ASCII form. It is the actual proof spine of the normalized Lean reconstruction and should not be replaced by a cosmetically simplified diagram.

```

BA ~= ED, BC ~= EF
directed parallels
|
I.33 twice
|
AD ~= BE, CF ~= BE
|
XI.9
|

```

```

AD || CF
  |
Borsuk-Szmielew 52
  |
AD directed-parallel CF
  |
I.33
  |
AC ~= DF
  |
spatial SSS
  |
angle ABC ~= angle DEF

```

This DAG is not merely a restatement of Euclid's local theorem references. The additional section-line / plane-side stage is the formal replacement for directional plane-separation information that is carried silently by the classical spatial diagram.

10.9 Wyler-Style Flat Reconstruction

XI.10 requires particular care because the flat statement is only a spatial certificate inside a theorem whose actual conclusion is angle congruence. The public target remains

$$\angle ABC \cong \angle DEF.$$

The noncoplanarity assumptions first produce two rank-two carriers:

$$\text{Span}\{A, B, C\} = \text{carrier}(\alpha), \quad \text{Span}\{D, E, F\} = \text{carrier}(\theta).$$

If $\alpha = \theta$, all six points would be coplanar, contrary to the public hypothesis. Hence

$$\alpha \neq \theta, \quad \text{Join}(\alpha, \theta) = E^3.$$

This is the rank-three certificate of the configuration.

```

ABC noncollinear      DEF noncollinear
  |                    |
  v                    v
Span ABC = alpha      Span DEF = theta
  \                    /
   alpha != theta
     |
     v
Join(alpha, theta)=E^3
  |
  v
full directed/metric XI.10 proof
  |
  v
AngleCongruent ABC DEF

```

The final two lines are essential. Rank three explains why the configuration is genuinely spatial, but it does not prove the directed-parallel transport or the angle congruence. XI.10 is therefore the main warning against confusing a useful flat invariant with the theorem itself.

10.10 Hidden Formal Data

The classical diagram suppresses several obligations which are explicit in Lean.

10.10.1 Direction on each parallel pair

Carrier parallelism does not determine whether the selected ordered segments point in corresponding directions. XI.10 must store this information explicitly.

10.10.2 The connector line

Each directed parallel pair contains an explicit connector through the terminal endpoints. For the fixed-connector wrapper this line is BE .

10.10.3 The planes of the parallel pairs

The common plane of each pair of spatial parallels is an explicit witness. Disjointness alone would also admit skew lines and is therefore insufficient.

10.10.4 The plane of the second angle

The points D, E, F determine a genuine plane because

$$\neg \text{PrimCollinear}(D, E, F).$$

This plane is needed both for the half-space argument and for the final SameRay angle transport.

10.10.5 The intersection line in the section bridge

When two auxiliary planes meet, their common line must be constructed by

```
hilbert_plane_intersection_line
```

and characterized pointwise. It is never read off from a perspective drawing.

10.10.6 Disjoint angle planes are possible

The proof of same-side-of-plane transitivity cannot assume that the two angle planes meet. Write

$$\alpha = \text{plane}(A, B, C), \quad \theta = \text{plane}(D, E, F).$$

The branch $\alpha \cap \theta = \emptyset$ is a genuine part of the formal proof.

10.10.7 Noncoplanarity after segment layoff

The public wrapper replaces D, F by D_1, F_1 on the same rays. It must prove that a common plane containing A, B, C, D_1, E, F_1 would also contain the original D, F , contradicting the input hypothesis.

10.10.8 No ambient planar order instance

Several old SameRay utilities require `HilbertOrder Geo`. XI.10 does not introduce such an instance. The final ray-angle transport is performed inside the plane DEF , where the induced `PlaneGeo` legitimately has planar Hilbert order.

10.11 Relationship with Earlier Propositions

The direct Book XI dependency is XI.9:

$$AD \parallel BE, \quad CF \parallel BE \implies AD \parallel CF.$$

The proof also uses I.33 three times in mathematical substance:

1. BA, ED produce AD, BE ;
2. BC, EF produce CF, BE ;
3. AD, CF produce AC, DF .

The final triangle argument is I.8 / spatial SSS.

Euclid explicitly starts with I.3. In the current public Lean wrapper this role is played directly by

```
HilbertSpaceCongruence.segment_construction
```

rather than by importing a numbered I.3 theorem.

Thus there are two useful dependency descriptions:

Euclid's local DAG	current Lean route
I.3	<code>segment_construction</code>
I.33	directed I.33 helper
XI.9	<code>euclid_proposition_11_9</code>
I.33	directed I.33 helper
I.8	<code>hilbert_space_sss_angleA_in_plane</code>

The additional plane-separation stage has no separate Euclid proposition number. It exposes the section-line content needed to make the directional I.33 chain formally valid.

10.12 Relationship with the Foundations

The production theorem uses spatial incidence, order, congruence, and the Euclidean layer. Hilbert Group IV enters through the directed I.33 machinery and Proposition XI.9, both of which require plane-local uniqueness of parallels. No continuity principle is used.

The additional section-line / plane-side bridge does not introduce a new separation axiom. It is derived from the existing spatial incidence and order structure and serves only to make explicit the directional information which the classical three-dimensional diagram carries visually.

No coordinates, vectors, scalar products, global orientation of space, or numerical angle measures are used.

10.13 Dependency Summary

The classical proof spine is:

```
I.3
|
I.33 twice
|
XI.9
|
I.33
|
I.8
|
angle ABC = angle DEF
```

The actual normalized Lean spine is:

```
BA ~= ED, BC ~= EF
directed parallels
|
I.33 twice
|
AD ~= BE, CF ~= BE
|
XI.9
|
AD || CF
|
Borsuk-Szmielew 52
|
AD directed-parallel CF
|
I.33
|
AC ~= DF
|
spatial SSS
|
angle ABC ~= angle DEF
```

The public-wrapper layer is:

```
original rays ED, EF
|
segment construction
|
ED1 ~= AB
EF1 ~= CB
```

```

      |
SameRay transport
      |
AB -> D1E
CB -> F1E
      |
normalized XI.10
      |
angle ABC ~= angle D1EF1
      |
SameRay angle transport
      |
angle ABC ~= angle DEF

```

These three diagrams should be read together. The first is Euclid’s local theorem DAG. The second is the mathematical core actually realized by the Lean reconstruction. The third records the normalization wrapper needed to expose a public theorem with no prior equal-length hypotheses.

10.14 Conclusion

Proposition XI.10 is the first proposition in this part of Book XI where orientation of spatial parallels becomes unavoidable. Ordinary coplanar-disjoint parallelism suffices for XI.9, but XI.10 must remember which rays correspond. The dedicated `HilbertSpaceDirectedParallelSegments` relation supplies exactly that missing information without adding coordinates or a global orientation of space.

The formal proof preserves Euclid’s structure: two I.33 steps create a common parallel BE , XI.9 compares the opposite joining segments, a third I.33 creates the third side equality, and SSS closes the angle comparison. The additional plane-separation bridge explains rigorously how the directional information passes between the relevant plane slices.

The production theorem compiles, the full project build is clean, and the axiom audit is

```

[proptext, Classical.choice,
 Geometry.bookZero_nullSegment1, Quot.sound]

```

with no new spatial or continuity axiom.

Chapter 11

Proposition XI.11

11.1 Introduction

Proposition XI.11 is the first Book XI construction in the present reconstruction whose target is the existence of a perpendicular from an arbitrary point outside a plane. Given a plane π and an elevated point $A \notin \pi$, the theorem constructs a line through A that is perpendicular to π .

The proposition is a natural test of the architecture developed in XI.4–XI.8. Its final statement is spatial, but the construction repeatedly moves into ordinary planar Hilbert geometry on explicitly chosen plane slices. In the nontrivial branch three planes are relevant:

$$\pi, \quad \sigma = \text{plane through } A \text{ and } BC, \quad \tau = \text{plane through } AD \text{ and } DE.$$

The proof also exposes an important distinction between the theorem proved and the proof route used. The current production proof follows Euclid’s construction closely and therefore invokes XI.8. Since XI.8 uses the plane-local Euclidean parallel axiom, the public Lean theorem assumes `HilbertSpaceEuclidean`. This records the strength of the current proof, not a claim that the existence statement itself is axiomatically minimal.

11.2 Euclid’s Statement

Euclid’s Proposition XI.11 states:

From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the elevated point and let π be the plane of reference. The required construction is a line AF through A such that

$$AF \perp \pi.$$

11.3 Classical Configuration

Choose a straight line BC at random in the plane of reference. Draw AD from A perpendicular to BC . If AD is already perpendicular to the reference plane, the construction is complete.

Otherwise draw DE in the reference plane through D , with

$$DE \perp BC.$$

Then draw AF from A perpendicular to DE , and through F draw GH parallel to BC .

Thus the reference plane contains

$$BC, \quad DE, \quad GH,$$

while the two lines

$$AD, \quad AF$$

leave the reference plane and meet it at D and F , respectively.

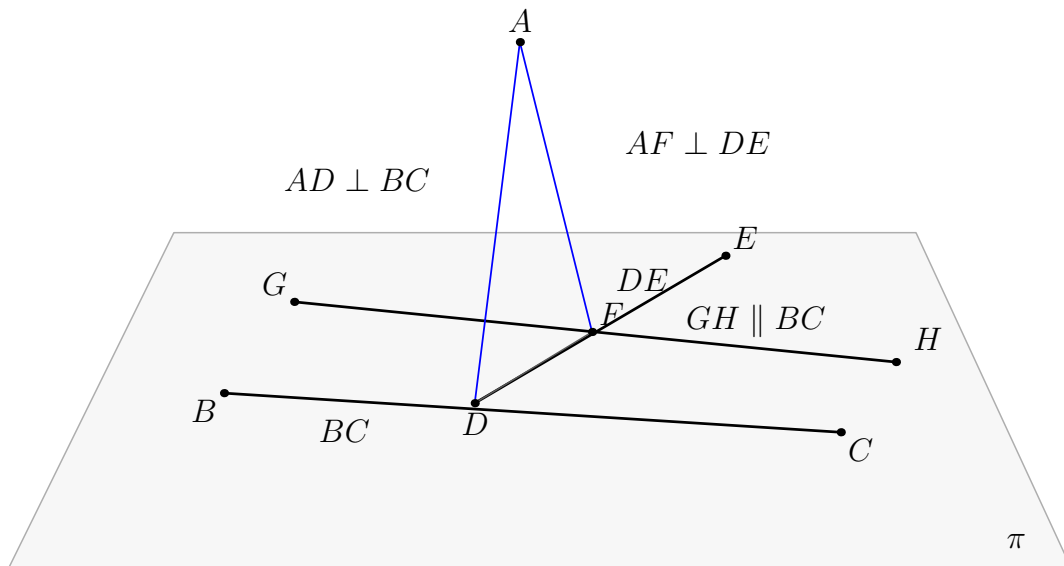


Figure 11.1: The classical configuration of Proposition XI.11. Black lines lie in the reference plane π ; blue segments leave π . The construction starts with $AD \perp BC$, then uses $DE \perp BC$, $AF \perp DE$, and $GH \parallel BC$. The desired conclusion is $AF \perp \pi$.

11.4 Classical Proof

The classical proof is short but structurally dense.

First, choose $BC \subset \pi$ and draw

$$AD \perp BC$$

from the elevated point A by Proposition I.12. Euclid immediately splits into two cases.

If $AD \perp \pi$, the required perpendicular has already been drawn.

Assume therefore that AD is not perpendicular to π . In the reference plane draw

$$DE \perp BC$$

through D by Proposition I.11. Next draw

$$AF \perp DE$$

from A by a second application of I.12. Finally draw through F

$$GH \parallel BC$$

by I.31.

Now BC is perpendicular to both DA and DE . These two lines intersect at D , so Proposition XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Since $GH \parallel BC$, Proposition XI.8 gives

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, a line perpendicular to a plane is perpendicular to every line of that plane through the foot. The line AF belongs to $\text{plane}(ED, DA)$ and meets GH at F . Hence

$$GH \perp AF,$$

and therefore also

$$AF \perp GH.$$

But $AF \perp DE$ by construction. Thus AF is perpendicular at F to the two intersecting lines DE and GH of the reference plane. A second application of XI.4 yields

$$AF \perp \text{plane}(DE, GH).$$

The plane through DE and GH is the plane of reference, hence

$$AF \perp \pi.$$

The local classical dependency chain is therefore

$$\begin{array}{c}
A \notin \pi, \quad BC \subset \pi \\
\Downarrow \\
\text{I.12 : } AD \perp BC \\
\Downarrow \\
AD \perp \pi \quad \text{or} \quad AD \not\perp \pi \\
\Downarrow \\
\text{I.11 : } DE \perp BC \\
\Downarrow \\
\text{I.12 : } AF \perp DE \\
\Downarrow \\
\text{I.31 : } GH \parallel BC \\
\Downarrow \\
\text{XI.4 : } BC \perp \text{plane}(ED, DA) \\
\Downarrow \\
\text{XI.8 : } GH \perp \text{plane}(ED, DA) \\
\Downarrow \\
\text{XI.Def.3 : } GH \perp AF \\
\Downarrow \\
AF \perp DE, \quad AF \perp GH \\
\Downarrow \\
\text{XI.4 : } AF \perp \pi.
\end{array}$$

No use of XI.9 or XI.10 occurs in this classical chain. The Book XI dependencies needed locally are XI.4 and XI.8.

11.5 What is genuinely three-dimensional here?

The proof is not simply a planar perpendicular construction repeated in space. The essential issue is selecting the correct plane for each planar construction and then transporting its conclusion back to ambient space.

The first auxiliary plane is

$$\sigma = \text{plane}(A, BC).$$

It exists because $A \notin BC$. Inside σ , the point A and the line BC are ordinary planar objects, so I.12 can be applied in the induced geometry $\text{PlaneGeo}(\sigma)$.

After the foot D is obtained, the line DE is constructed inside the reference plane π .

The second auxiliary plane is

$$\tau = \text{plane}(AD, DE).$$

The lines AD and DE intersect at D and are distinct. Inside $\text{PlaneGeo}(\tau)$, the second I.12 construction produces the line $AF \perp DE$.

The three plane slices have a particularly transparent intersection pattern:

$$\pi \cap \sigma = BC, \quad \sigma \cap \tau = AD, \quad \pi \cap \tau = DE,$$

The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi F
```

is the spatial target relation established for XI.4. It expresses XI.Def.3 synthetically: F lies on the line and in the plane, and the line is perpendicular to every line of the plane through F .

11.7 Spatial / Planar Architecture Used

11.7.1 Ambient spatial layer

The ambient construction uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

There is no global ambient instance

```
HilbertOrder Geo
```

and no global ambient instance

```
HilbertCongruence Geo
```

installed merely to make planar theorems available in space.

11.7.2 Induced planar geometry

Planar constructions are performed in explicitly named plane slices using

```
PlanePoint
PlaneLine
PlaneGeo
```

The proof uses three instances of induced planar geometry:

$$\text{PlaneGeo}(\sigma), \quad \text{PlaneGeo}(\pi), \quad \text{PlaneGeo}(\tau).$$

The first I.12 construction is performed in $\text{PlaneGeo}(\sigma)$. The construction of DE and the I.31

parallel construction are performed in $\text{PlaneGeo}(\pi)$. The second I.12 construction is performed in $\text{PlaneGeo}(\tau)$.

11.7.3 Plane-space bridges

The principal bridges used by the proof include

```
planeGeo_rightAngle_iff_ambient
planeGeo_linesDisjoint_iff_ambient
HilbertSpaceIncidence.line_in_plane
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

The proof also uses the ambient symmetry normalizer

```
hilbert_space_linesPerpendicularAt_symm
```

when XI.4 requires the perpendicularity relation in the opposite carrier orientation.

11.8 Proof Architecture of the Reconstruction

The formal proof follows the classical route but expands the incidence and plane-selection steps that the printed diagram leaves implicit.

11.8.1 1. Choosing the reference line

The proof first obtains three noncollinear points of the reference plane:

```
hilbert_three_noncollinear_on_plane
```

Two of them, B, C , determine the line bc . The theorem

```
HilbertSpaceIncidence.line_in_plane
```

then proves

$$bc \subset \pi.$$

Since $A \notin \pi$, it follows immediately that

$$A \notin bc.$$

11.8.2 2. Constructing the first auxiliary plane

The helper

```
hilbert_plane_through_line_and_external_point
```

constructs the plane σ containing bc and A .

This step supplies exactly the missing incidence premise needed before a planar I.12 theorem can be applied. The classical phrase “draw AD from A perpendicular to BC ” presupposes that A and BC have first been put in a common plane.

11.8.3 3. First I.12 construction in σ

Inside

PlaneGeo(σ)

the theorem

```
hilbert_perpendicular_from_point_exists
```

constructs the foot D on bc and gives a right-angle witness. The ambient carrier ad through D, A is then constructed explicitly and proved to lie in σ .

The right-angle data are packaged as

```
HilbertLinesPerpendicularAt Geo bc ad D
```

using the bridge

```
planeGeo_rightAngle_iff_ambient
```

and explicit noncollinearity.

11.8.4 4. Euclid’s case split

The formal proof uses the explicit split

```
by_cases hADpi :  
  HilbertLinePerpendicularPlaneAt Geo ad pi D
```

If this proposition holds, the proof returns ad, D immediately.

This is not a Lean-specific mathematical invention. It is exactly Euclid’s sentence: if AD is also perpendicular to the plane of reference, the construction is already complete.

11.8.5 5. Constructing DE in the reference plane

In the nontrivial branch the point D is known to lie in π . The formal proof constructs opposite points on bc in $\text{PlaneGeo}(\pi)$ and then uses

```
hilbert_right_angle_exists_nondegenerate
```

to obtain a nondegenerate right-angle direction through D . The resulting carrier de satisfies

$$de \subset \pi, \quad bc \perp de \text{ at } D.$$

Thus the formal proof realizes the role of Euclid I.11 without importing a separate I.11 proposition wrapper in the production file.

11.8.6 6. Constructing the second auxiliary plane

The proof establishes

$$ad \neq de.$$

Indeed, if $ad = de$, then A would lie on $de \subset \pi$, contrary to $A \notin \pi$.

The theorem

```
hilbert_plane_through_two_intersecting_lines
```

then constructs the plane τ containing ad and de .

11.8.7 7. Second I.12 construction in τ

Inside

$$\text{PlaneGeo}(\tau)$$

the proof applies

```
hilbert_perpendicular_from_point_exists
```

a second time, now from A to de . This yields the foot F and the carrier af , with

$$af \subset \tau, \quad de \perp af \text{ at } F.$$

Because $de \subset \pi$, the point F also lies in the reference plane.

11.8.8 8. The hidden fact $F \notin BC$

Before I.31 can be applied through F , the formal proof must establish that F does not lie on bc .

Assume $F \in bc$. Since D, F lie on both bc and de , line uniqueness forces $F = D$. The two lines ad and af then contain the two distinct points D, A , so line uniqueness forces

$$ad = af.$$

The proof already knows

$$ad \perp bc, \quad ad \perp de$$

at D . Since $bc, de \subset \pi$, XI.4 would imply

$$ad \perp \pi,$$

contradicting the nontrivial branch assumption.

Therefore

$$F \notin bc.$$

This is one of the main pieces of hidden formal data in XI.11.

11.8.9 9. I.31 in the reference plane

Now B, C, F are noncollinear in $\text{PlaneGeo}(\pi)$. The proof applies

```
euclid_proposition_31
```

to obtain a point G and a line gh through F, G parallel to bc in the induced plane.

The result of I.31 is initially planar point-pair parallelism. The proof converts it to disjointness of the actual carriers using

```
hilbert_mem_pointLine_iff_onLine
```

and then transports planar disjointness back to ambient space by

```
planeGeo_linesDisjoint_iff_ambient
```

Finally it packages

```
HilbertSpaceLinesParallel Geo bc gh
```

with the reference plane π as the explicit common carrier plane.

11.8.10 10. First XI.4: $BC \perp \tau$

In the plane τ , the lines ad and de are distinct and meet at D . The proof has

$$bc \perp ad, \quad bc \perp de.$$

Therefore

```
euclid_proposition_11_4
```

gives

$$bc \perp \tau.$$

This is the first use of XI.4 in the construction.

11.8.11 11. XI.8: transfer to the parallel GH

The proof now has

$$bc \parallel gh, \quad bc \perp \tau.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point K such that

$$gh \perp \tau \text{ at } K.$$

This is exactly the classical XI.8 step, but the formal theorem returns the foot existentially. Euclid's diagram already identifies the relevant intersection as F ; Lean requires that identification to be proved.

11.8.12 12. Normalizing the XI.8 foot: $K = F$

The proof knows from the XI.8 conclusion that

$$K \in gh, \quad K \in \tau,$$

and from the construction that

$$F \in gh, \quad F \in \tau.$$

Suppose $K \neq F$. Since the two distinct points K, F lie on gh and in τ , spatial incidence gives

$$gh \subset \tau.$$

But $gh \perp \tau$ at K . Applying the universal line–plane-perpendicularity condition to the line gh itself would give

$$gh \perp gh.$$

The theorem

```
hilbert_linesPerpendicularAt_ne
```

then yields the absurd conclusion $gh \neq gh$.

Hence

$$K = F.$$

This argument is purely synthetic and uses the nondegeneracy already built into `HilbertLinesPerpendicularAt`.

11.8.13 13. XI.Def.3 gives $AF \perp GH$

After replacing K by F , the proof has

$$gh \perp \tau \text{ at } F.$$

Since

$$af \subset \tau, \quad F \in af,$$

the universal line condition is unpacked by

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

to obtain

$$gh \perp af.$$

The symmetry helper

```
hilbert_space_linesPerpendicularAt_symm
```

normalizes this to

$$af \perp gh.$$

The same helper normalizes the previously constructed $de \perp af$ to

$$af \perp de.$$

11.8.14 14. Proving $DE \neq GH$

The final XI.4 application requires two distinct lines of the reference plane. This distinctness is not left to the diagram.

Suppose

$$de = gh.$$

Since $D \in de$, it follows that $D \in gh$. But $D \in bc$, while the spatial parallelism package $bc \parallel gh$ contains disjointness of these two ambient carriers. Contradiction.

Thus

$$de \neq gh.$$

11.8.15 15. Final XI.4 in the reference plane

Both de and gh lie in π , meet at F , and are distinct. The proof has

$$af \perp de, \quad af \perp gh.$$

A final call to

```
euclid_proposition_11_4
```

therefore gives

$$af \perp \pi \text{ at } F.$$

The returned witnesses are af and F .

11.9 Wyler-Style Flat Reconstruction

XI.11 begins with the canonical rank jump from a plane and an external point. For $A \notin \pi$, the flat layer records

$$\text{Join}(\text{carrier}(\pi), \{A\}) = E^3.$$

The converse characterization is also natural: if the point were already in the plane, adjoining it would not increase the generated flat. Thus externality is precisely the condition that the plane plus the point generate the whole three-space.

The perpendicular construction itself remains genuinely metric. The Wyler certificate organizes the ambient position but does not replace Euclid’s construction:

```

A outside pi
  |
  v
Join(pi,{A}) = E^3
  |
  choose a line bc in pi
  |
  construct the required auxiliary plane slices
  |
  I.12, I.31, XI.4, XI.8
  |
  v
AF perp pi

```

The value of the flat statement is therefore diagnostic: it proves at the outset that the construction uses the full ambient rank. The remaining steps explain how the perpendicular is actually obtained synthetically.

11.10 Lean Formalization

The production module imports

```

import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition12
import CGJteamLab.Proposition31

```

The direct imports therefore expose XI.8, the Book I perpendicular-from- external-point construction used as I.12, and I.31.

XI.4 is invoked twice by the proof and is available transitively through the established Book XI import chain. The production theorem does not import or invoke XI.9 or XI.10.

The principal reusable declarations consumed by XI.11 are:

```

hilbert_three_noncollinear_on_plane
hilbert_plane_through_line_and_external_point
hilbert_perpendicular_from_point_exists
planeGeo_rightAngle_iff_ambient
planeGeo_opposite_points_on_line_congruent_to
hilbert_right_angle_exists_nondegenerate
hilbert_plane_through_two_intersecting_lines
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_31
hilbert_mem_pointLine_iff_onLine
planeGeo_linesDisjoint_iff_ambient
euclid_proposition_11_4
euclid_proposition_11_8
HilbertLinePerpendicularPlaneAt.incidence
HilbertLinePerpendicularPlaneAt.perpendicular_to_line

```

```
hilbert_linesPerpendicularAt_ne
```

No XI.11-specific helper theorem is exported. The full construction is contained in the proof of `euclid_proposition_11_11`.

11.11 Hidden Formal Data

The classical diagram suppresses several obligations that are explicit in the Lean proof.

11.11.1 The first I.12 needs a plane

The statement $A \notin \pi$ does not by itself put A and the chosen line $BC \subset \pi$ in a common planar geometry. The plane σ through A and BC must be constructed before planar I.12 can be invoked.

11.11.2 Every constructed line needs a carrier proof

After constructing ambient lines ad , de , af , and gh , the proof records explicitly which plane contains each line. These are the data that permit later calls to XI.4, XI.8, and induced-plane theorems.

11.11.3 The two feet are not automatically the same

XI.8 returns a foot K of gh on τ . The earlier construction already supplied the point $F \in gh \cap \tau$. The equality $K = F$ is a real incidence theorem, not a definitional simplification.

11.11.4 Distinctness for the final XI.4

The statement $de \neq gh$ follows from the disjointness part of $bc \parallel gh$, but it must be proved explicitly before the two lines can be packaged as distinct `PlaneLines` of π .

11.11.5 Orientation of perpendicularity

The project relation `HilbertLinesPerpendicularAt` carries ordered line arguments. Several classical sentences silently use symmetry of perpendicularity. The proof normalizes these orientations explicitly with `hilbert_space_linesPerpendicularAt_symm`.

11.12 Relationship with Earlier Book XI Propositions

The only earlier Book XI propositions invoked by the mathematical proof are XI.4 and XI.8.

XI.4 is used twice:

$$bc \perp ad, \quad bc \perp de \implies bc \perp \tau,$$

and later

$$af \perp de, \quad af \perp gh \implies af \perp \pi.$$

XI.8 is used once:

$$bc \parallel gh, \quad bc \perp \tau \implies gh \perp \tau.$$

XI.9 and XI.10 are not local dependencies of XI.11. The production Lean source neither imports nor invokes them, so they are absent from the actual Lean DAG for XI.11 as well as from the classical local dependency chain.

XI.7 is also not a local dependency of XI.11. The spatial parallelism used here is created directly by I.31 inside the reference plane and then packaged as `HilbertSpaceLinesParallel`.

11.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are:

$$\text{I.12, I.11, I.12, I.31.}$$

The formal proof calls the established I.12 theorem

```
hilbert_perpendicular_from_point_exists
```

twice, once in σ and once in τ .

The role of I.11 is reconstructed from the lower-level right-angle construction interface in `PlaneGeo( $\pi$ )`. The proof does not import a separate `proposition-I.11` wrapper.

The I.31 step is represented directly by

```
euclid_proposition_31
```

inside `PlaneGeo( $\pi$ )`.

Book Zero enters transitively through the established planar Hilbert construction and right-angle infrastructure. XI.11 introduces no new Book Zero declaration.

The ambient three-space remains separated from planar order and congruence. All ordinary planar constructions are carried out in a named `PlaneGeo`; the spatial classes provide only the incidence, congruence transport, and plane-local Euclidean data required to connect those slices.

11.14 Relationship with the Foundations

The current Euclid-faithful proof of XI.11 uses Hilbert Group IV because it invokes Proposition XI.8. Thus the public theorem carries `HilbertSpaceEuclidean Geo` as a genuine geometric

hypothesis of this proof route. No continuity principle is used.

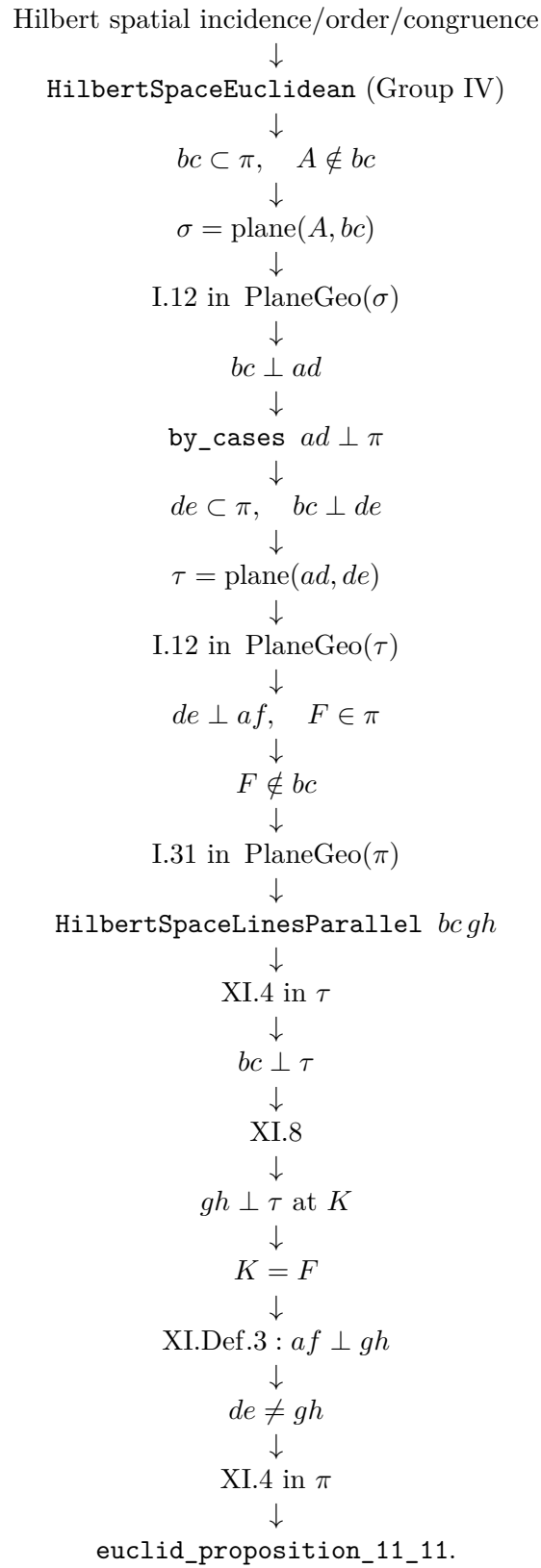
This records the strength of the present reconstruction, not a minimality claim about the abstract target statement. The rest of the argument keeps planar order and congruence inside explicit plane slices rather than installing global planar instances on ambient three-space.

11.15 Dependency Summary

The classical local chain is

$$\begin{array}{c}
 A \notin \pi, \quad BC \subset \pi \\
 \downarrow \\
 \text{I.12} \\
 \downarrow \\
 AD \perp BC \\
 \downarrow \\
 \text{case } AD \perp \pi \text{ or not} \\
 \downarrow \\
 \text{I.11, I.12, I.31} \\
 \downarrow \\
 DE \perp BC, \quad AF \perp DE, \quad GH \parallel BC \\
 \downarrow \\
 \text{XI.4} \\
 \downarrow \\
 BC \perp \text{plane}(ED, DA) \\
 \downarrow \\
 \text{XI.8} \\
 \downarrow \\
 GH \perp \text{plane}(ED, DA) \\
 \downarrow \\
 \text{XI.Def.3} \\
 \downarrow \\
 AF \perp GH \\
 \downarrow \\
 \text{XI.4} \\
 \downarrow \\
 AF \perp \pi.
 \end{array}$$

The actual formal chain is



The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.8
Direct production imports	XI.8, I.12 module, I.31
Classical explicit Book XI dependencies	XI.4, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
sorry/admit	no

11.16 Conclusion

Proposition XI.11 is a compact classical construction whose formal reconstruction exercises almost the entire plane-space interface developed so far. The key mathematical idea remains Euclid's: construct two successive planar perpendiculars, introduce a parallel in the reference plane, transfer a line-plane perpendicularity by XI.8, and close with XI.4.

Lean makes the spatial bookkeeping visible. The first I.12 needs the plane σ ; the second needs the plane τ ; the I.31 construction must be converted from planar parallelism to ambient spatial parallelism; the foot returned by XI.8 must be proved equal to F ; and the final two reference-plane carriers must be proved distinct.

No analytic geometry enters. The production theorem is fully synthetic, compiles in the current project, and has the inherited axiom audit

```
[propext, Classical.choice, Geometry.bookZero_nullSegment1, Quot.sound].
```

Its present Group IV hypothesis is inherited from the Euclid-faithful use of XI.8. Whether that hypothesis can be reduced is a separate strengthening question; the source-side comparison is kept in the appendices rather than repeated in this proposition chapter.

Chapter 12

Proposition XI.12

12.1 Introduction

Proposition XI.12 is the companion construction to XI.11. Proposition XI.11 constructs a perpendicular to a plane from a point outside the plane; XI.12 starts with a point A already lying in the reference plane π and constructs a line through A perpendicular to π .

The classical proof is exceptionally short. Euclid chooses an elevated point, uses XI.11 to obtain one perpendicular to the reference plane, draws through A a parallel to that perpendicular by I.31, and then uses XI.8 to transfer the line–plane perpendicularity to the new line.

The Lean reconstruction follows this classical route, but two incidence obligations that are invisible in the printed proof become explicit. First, the perpendicular obtained from XI.11 may already meet the reference plane at A . In that case it is itself the required line, and there is no new parallel to construct through A . Second, XI.8 returns the foot of the new perpendicular existentially; Lean must prove that this foot is the already specified point A .

These two obligations are handled by the proposition-local theorem `hilbert_XI12_perpendicular_foot_unique`. The other proposition-local helper, `hilbert_XI12_parallel_through_point_in_plane`, isolates the planar I.31 construction and packages its result as ambient spatial parallelism.

12.2 Euclid’s Statement

Euclid’s Proposition XI.12 states:

To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point of the reference plane π . The required construction is a straight line through A satisfying

$$AD \perp \pi.$$

12.3 Classical Configuration

Let B be an arbitrary elevated point. By XI.11 draw BC perpendicular to the reference plane, with $C \in \pi$. Through the given point $A \in \pi$, draw AD parallel to BC .

The generic classical configuration is shown in Figure 12.1. The two blue lines leave the reference plane. Their hidden continuations below the plane are dashed, following the standing Book XI figure convention.

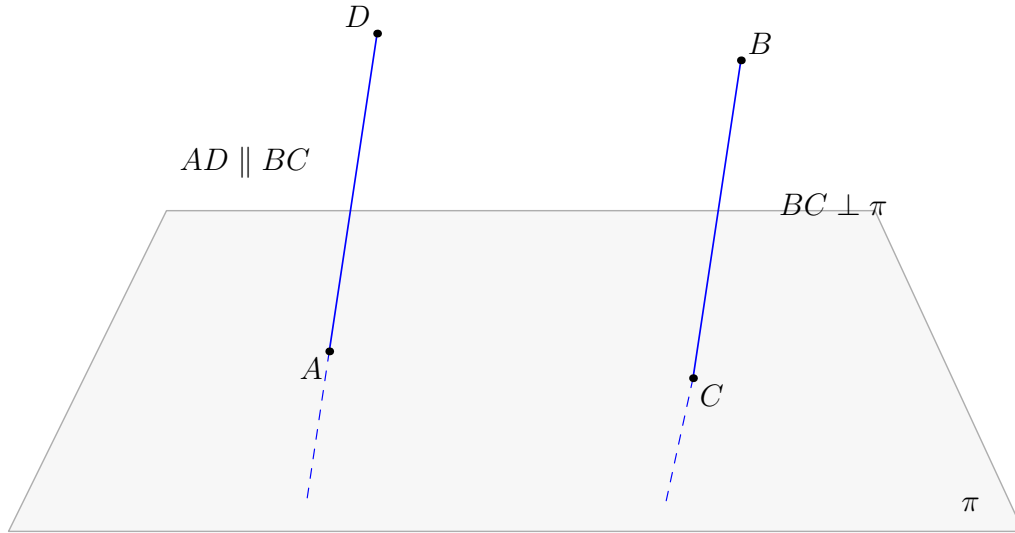


Figure 12.1: The classical configuration of Proposition XI.12. The points A, C lie in the reference plane π . XI.11 gives $BC \perp \pi$, and I.31 draws $AD \parallel BC$ through A . XI.8 then gives $AD \perp \pi$. The figure shows the generic case $A \neq C$; if $A = C$, the line BC already solves the construction.

12.4 Classical Proof

Let A be the given point in the reference plane. Choose any elevated point B . By Proposition XI.11 draw

$$BC \perp \pi.$$

Through A , draw

$$AD \parallel BC$$

by Proposition I.31. Since AD and BC are parallel and BC is perpendicular to the reference plane, Proposition XI.8 gives

$$AD \perp \pi.$$

Thus AD is the required straight line.

The local classical dependency chain is therefore

$$\begin{array}{c}
A \in \pi \\
\Downarrow \\
\text{choose an elevated point } B \\
\Downarrow \\
\text{XI.11} \\
\Downarrow \\
BC \perp \pi \\
\Downarrow \\
\text{I.31} \\
\Downarrow \\
AD \parallel BC, \quad A \in AD \\
\Downarrow \\
\text{XI.8} \\
\Downarrow \\
AD \perp \pi.
\end{array}$$

Neither XI.9 nor XI.10 occurs in this local proof. The only earlier Book XI propositions used explicitly are XI.11 and XI.8.

12.5 What is genuinely three-dimensional here?

The I.31 step is planar, but the input line BC is not a line of the reference plane: it is perpendicular to that plane and meets it only at its foot C . To apply a planar parallel theorem through A , the formal reconstruction first has to place the spatial line and the point A in a common auxiliary plane.

In the nontrivial branch this plane is

$$\sigma = \text{plane}(BC, A).$$

Inside $\text{PlaneGeo}(\sigma)$, I.31 becomes an ordinary planar construction. Its output must then be transported back to ambient three-space and packaged as

$$\text{HilbertSpaceLinesParallel}(BC, AD).$$

Only after this packaging can XI.8 be applied.

The other genuinely spatial ingredient is the meaning of a line perpendicular to a plane. The predicate

```
HilbertLinePerpendicularPlaneAt Geo l pi A
```

contains the incidence of the foot with both line and plane and the universal assertion that the line is perpendicular to every line of the plane through the foot. This universal structure is what makes the foot-uniqueness argument possible.

12.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_12
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi : S.Plane)
  (A : Geo.Point)
  (hApi : S.OnPlane A pi) :
  exists l : Geo.Line,
    HilbertLinePerpendicularPlaneAt Geo l pi A

```

The theorem therefore returns an ambient line l that is perpendicular to the given plane exactly at the prescribed point A .

12.7 Spatial / Planar Architecture Used

12.7.1 Ambient spatial layer

The ambient proof uses

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt

```

As throughout the current Book XI architecture, there is no global ambient

```

HilbertOrder Geo
HilbertCongruence Geo

```

installed to make plane geometry available in space.

12.7.2 Induced planar geometry

The only new plane slice constructed specifically by XI.12 is σ , the plane through the XI.11 perpendicular and the point A , in the branch where A does not already lie on that line.

The I.31 construction is then carried out in

$$\text{PlaneGeo}(\sigma).$$

The helper `hilbert_XI12_parallel_through_point_in_plane` selects two points of the ambient line, views them together with A as `PlanePoints` of σ , applies the established planar parallel-existence theorem, and converts the result back to an ambient line.

12.7.3 Plane-space bridges

The principal bridge operations used locally are

```
planeGeo_primCollinear_to_ambient
hilbert_on_line_of_primCollinear_with_two_on_line
hilbert_mem_pointLine_iff_onLine
HilbertSpaceIncidence.line_in_plane
```

The planar I.31 output is not identified with ambient spatial parallelism by definition. The proof explicitly establishes ambient disjointness and supplies σ as the common carrier plane.

12.8 Proof Architecture of the Reconstruction

12.8.1 1. Choosing an elevated point and invoking XI.11

The proof first calls

```
hilbert_point_off_plane
```

to choose a point $B \notin \pi$. It then invokes

```
euclid_proposition_11_11
```

and obtains an ambient line l and a point F such that

$$B \in l, \quad l \perp \pi \text{ at } F.$$

At this point the formal proof has exactly the spatial object that Euclid denotes by BC .

12.8.2 2. The exceptional incidence branch

The classical text immediately asks for a parallel through A . Formally, however, one must first decide whether A already lies on the XI.11 line l :

```
by_cases hA1 : H.OnLine A l
```

If $A \in l$, then both A and the perpendicular foot F lie in the reference plane and on l . The helper

```
hilbert_XI12_perpendicular_foot_unique
```

proves

$$A = F.$$

Hence $l \perp \pi$ at A , and the theorem is complete in this branch. No I.31 construction is needed.

12.8.3 3. Why the perpendicular foot is unique

The helper has the normalized signature

```
theorem hilbert_XI12_perpendicular_foot_unique
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (pi : S.Plane)
  (l : Geo.Line)
  (C A : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi C)
  (hA1 : H.OnLine A l)
  (hApi : S.OnPlane A pi) :
  A = C
```

Suppose $A \neq C$. Since the two distinct points A, C lie both on l and in π , the spatial incidence axiom `line_in_plane` forces

$$l \subset \pi.$$

But $l \perp \pi$ at C . The universal defining clause of line–plane perpendicularity can therefore be applied to l itself, giving

$$l \perp l.$$

The established theorem `hilbert_linesPerpendicularAt_ne` then yields $l \neq l$, a contradiction. Thus the common point of a perpendicular line and its plane is unique.

12.8.4 4. Constructing the auxiliary carrier plane

In the second branch $A \notin l$. The theorem

```
hilbert_plane_through_line_and_external_point
```


constructs a plane σ satisfying

$$l \subset \sigma, \quad A \in \sigma.$$

This is the precise formal replacement for the plane that the classical diagram leaves implicit when it draws a line through A parallel to the spatial line BC .

12.8.5 5. I.31 inside the induced plane

The helper

```
hilbert_XI12_parallel_through_point_in_plane
```

has the normalized signature

```
theorem hilbert_XI12_parallel_through_point_in_plane
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (sigma : S.Plane)
  (l : Geo.Line)
  (A : Geo.Point)
  (hlsigma : HilbertLineInPlane Geo l sigma)
  (hAsigma : S.OnPlane A sigma)
  (hA1 : Not (H.OnLine A l)) :
  exists m : Geo.Line,
    H.OnLine A m /\
    HilbertSpaceLinesParallel Geo l m
```

The proof chooses two distinct points B, C on l , proves that B, C, A are noncollinear in $\text{PlaneGeo}(\sigma)$, and then calls

```
hilbert_parallel_through_point_exists
```

inside the induced plane.

This is the formal I.31 step. The production module imports **Proposition31**, but the local helper calls the established lower-level parallel-existence theorem directly.

12.8.6 6. Packaging planar parallelism as spatial parallelism

The planar theorem initially gives disjoint **PointLine** carriers in $\text{PlaneGeo}(\sigma)$. The proof constructs the actual ambient carrier m , proves that it lies in σ , and shows that an ambient intersection point of l and m would contradict the planar disjointness.

It can therefore package

$$l \parallel m$$

in the spatial sense

```
HilbertSpaceLinesParallel Geo 1 m
```

with σ as the explicit common carrier plane. This step excludes skew lines by construction.

12.8.7 7. XI.8 transfers perpendicularity

The proof now has

$$l \parallel m, \quad l \perp \pi.$$

It applies

```
euclid_proposition_11_8
```

and obtains a point D such that

$$m \perp \pi \text{ at } D.$$

This is the exact classical XI.8 step. The only extra issue is that the formal theorem returns D existentially, whereas the target of XI.12 requires the foot to be the prescribed point A .

12.8.8 8. The second foot normalization

The constructed line m passes through A , and the original hypothesis gives $A \in \pi$. Applying `hilbert_XI12_perpendicular_foot_unique` to the XI.8 conclusion therefore gives

$$A = D.$$

After this substitution the conclusion is exactly

$$m \perp \pi \text{ at } A.$$

This closes the nontrivial branch and completes XI.12.

12.9 Wyler-Style Flat Reconstruction

The nontrivial branch of XI.12 has a particularly clean generated-flat normal form. XI.11 first supplies one normal l to π through an external point. If the prescribed point $A \in \pi$ does not

already lie on l , the line and point generate an auxiliary plane

$$\text{Join}(l, \{A\}) = \text{carrier}(\sigma).$$

Inside σ , a line m through A is constructed parallel to l . The same carrier can then be written

$$\text{Join}(l, m) = \text{carrier}(\sigma).$$

Thus the middle of the proposition is normalized by the exact identity

$$\text{Join}(l, \{A\}) = \text{Join}(l, m) = \text{carrier}(\sigma).$$

```

XI.11: l perp pi at F
      |
A in l? +-- yes --> foot uniqueness
      |
      no
      v
sigma = Join(l, {A})
      |
construct m through A, m || l
      |
Join(l, m) = sigma
      |
XI.8 transfers perpendicularity
      |
foot uniqueness
      |
      v
m perp pi at A

```

Here the flat calculus does more than shorten an incidence proof: it names the precise auxiliary slice that is preserved across the parallel construction.

12.10 Lean Formalization

The production module imports

```

import CGJteamLab.Proposition11_11
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition31

```

The two proposition-local declarations introduced by XI.12 are

```

hilbert_XI12_perpendicular_foot_unique
hilbert_XI12_parallel_through_point_in_plane

```

and the public theorem is

```

euclid_proposition_11_12

```

The first helper is a general uniqueness fact about a perpendicular line and its plane. The second is a general plane-slice packaging fact for parallel construction. Both are currently located in the proposition module; their signatures are sufficiently general that later promotion to `Hilbert3DInterface` may be considered if subsequent Book XI proofs reuse them.

12.11 Hidden Formal Data

12.11.1 The classical parallel construction has an exceptional case

Euclid chooses an arbitrary elevated point B , drops the XI.11 perpendicular, and then draws through A a parallel to it. The printed proof does not discuss the possibility that the foot of the XI.11 perpendicular is already A .

The formal proof cannot ignore this case. If $A \in l$, foot uniqueness shows that A is the foot and l is already the required perpendicular. Only when $A \notin l$ is the I.31 construction invoked.

12.11.2 I.31 needs an explicit plane slice

A planar parallel theorem cannot be applied directly to an arbitrary ambient spatial line and a point. The carrier plane σ through l and A is constructed first, and the entire I.31 step is performed inside `PlaneGeo( $\sigma$ )`.

12.11.3 Planar and spatial parallelism are different data

The I.31 result is initially a planar point-pair parallel relation. XI.8 expects `HilbertSpaceLinesParallel`, which includes an explicit common plane and ambient disjointness. The helper proves and packages these data rather than treating the two relations as definitionally identical.

12.11.4 XI.8 returns a new foot

XI.8 proves that the parallel line is perpendicular to π , but its formal conclusion contains an existentially returned foot D . The equality $D = A$ is a genuine incidence theorem and is discharged by the same foot-uniqueness helper used in the first branch.

12.12 Relationship with Earlier Book XI Propositions

The classical local Book XI dependencies are exactly XI.11 and XI.8.

XI.11 supplies a line perpendicular to the reference plane from an elevated point:

$$B \notin \pi \implies \exists l, F, \quad B \in l, \quad l \perp \pi \text{ at } F.$$

XI.8 supplies the transfer along spatial parallel lines:

$$l \parallel m, \quad l \perp \pi \implies m \perp \pi.$$

The production file imports both propositions directly. XI.9 and XI.10 are neither classical local dependencies nor Lean dependencies of XI.12.

XI.4 is not called directly by XI.12, but it remains in the transitive proof history through XI.11 and XI.8.

12.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The only explicit classical Book I proposition in XI.12 is I.31. In the formal proof its role is implemented by

```
hilbert_parallel_through_point_exists
```

inside `PlaneGeo( $\sigma$ )`.

The XI.12-specific parallel helper does not itself require `HilbertSpaceEuclidean`; its theorem signature stops at spatial order and congruence. The Euclidean Group IV assumption of the public XI.12 theorem is inherited from the calls to XI.11 and XI.8.

Book Zero contributes no new XI.12 declaration. The final public axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`, whereas both new proposition-local helpers audit without that dependency.

The ambient three-space continues to obey the Book XI instance discipline: ordinary order and congruence are used only inside explicit `PlaneGeo` slices, while the ambient layer uses the spatial classes.

12.14 Relationship with the Foundations

The current production proof of XI.12 uses Hilbert Group IV through the established XI.11–XI.8 chain. The public theorem therefore carries `HilbertSpaceEuclidean Geo`. No continuity principle is required.

Two proposition-local helpers isolate perpendicular-foot uniqueness and the construction of a parallel through a point in a fixed plane. Their role is mathematical and architectural: they make explicit two steps which the classical proof treats as immediate from the diagram and earlier Euclidean plane theory.

12.15 Dependency Summary

The classical local chain is

$$\begin{array}{c}
A \in \pi \\
\downarrow \\
B \text{ elevated} \\
\downarrow \\
\text{XI.11} \\
\downarrow \\
BC \perp \pi \\
\downarrow \\
\text{I.31} \\
\downarrow \\
AD \parallel BC \\
\downarrow \\
\text{XI.8} \\
\downarrow \\
AD \perp \pi.
\end{array}$$

The actual formal chain is

$$\begin{array}{c}
A \in \pi \\
\downarrow \\
\text{hilbert_point_off_plane} \\
\downarrow \\
B \notin \pi \\
\downarrow \\
\text{XI.11 : } l \perp \pi \text{ at } F \\
\downarrow \\
\text{by_cases } A \in l \\
\downarrow \\
\begin{array}{c|c}
\begin{array}{c}
A \in l \\
\downarrow \\
A = F \\
\downarrow \\
l \perp \pi \text{ at } A
\end{array}
&
\begin{array}{c}
A \notin l \\
\downarrow \\
\sigma = \text{plane}(l, A) \\
\downarrow \\
\text{I.31 in PlaneGeo}(\sigma) \\
\downarrow \\
\text{HilbertSpaceLinesParallel } l m \\
\downarrow \\
\text{XI.8 : } m \perp \pi \text{ at } D \\
\downarrow \\
D = A
\end{array}
\end{array} \\
\downarrow \\
\text{euclid_proposition_11_12.}
\end{array}$$

The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	yes, via XI.11 and XI.8
Direct production imports	XI.11, XI.8, I.31 module
Classical explicit Book XI dependencies	XI.11, XI.8
XI.9 / XI.10	not used
New proposition-local helper theorem	yes, two
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
sorry/admit	no

12.16 Conclusion

Proposition XI.12 has a minimal classical proof but a revealing formal shape. Euclid's three-step route remains visible:

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8}.$$

Lean adds no alternative geometry. It makes explicit the carrier plane needed for the planar parallel construction, the exceptional case in which the XI.11 perpendicular already passes through the prescribed point, the conversion from planar to spatial parallelism, and the identification of the foot returned by XI.8.

The two new helpers are axiomatically clean relative to the inherited Book Zero dependency. The public theorem has the same global axiom profile as XI.11 and XI.8:

```
[propxt, Classical.choice, Geometry.bookZero_nullSegment1, Quot.sound].
```

The current result remains fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure. Its Group IV hypothesis belongs to the chosen Euclid-faithful route through XI.11 and XI.8; it is not a claim that the abstract perpendicular-existence statement intrinsically requires Euclidean parallelism.

Chapter 13

Proposition XI.13

13.1 Introduction

Proposition XI.13 is the uniqueness theorem that follows the two perpendicular-construction propositions XI.11 and XI.12. Euclid now assumes that two straight lines have been erected from the same point at right angles to the same plane and proves that, on the same side of the plane, the configuration is impossible.

The classical proof is short but genuinely spatial. The two supposed perpendiculars determine an auxiliary plane. That plane intersects the reference plane in a straight line through the common foot. Each of the two perpendiculars is therefore perpendicular to the same intersection line. Euclid then reaches a planar impossibility inside the auxiliary plane.

The formal reconstruction keeps the same spatial core but packages the last planar uniqueness argument differently. Since `Geo.Line` is an unoriented extensional carrier, the phrase “on the same side” has no independent content at the level of line equality: opposite rays through the common point belong to the same ambient line. The formal target is therefore equality of the two line carriers.

Rather than reconstructing a separate proposition-local theorem for uniqueness of a perpendicular ray in the auxiliary plane, the Lean proof reuses XI.4. If the intersection line of the two planes were perpendicular to two distinct lines of the auxiliary plane, XI.4 would make that intersection line perpendicular to the whole auxiliary plane. But the intersection line itself lies in that plane, which is impossible.

13.2 Euclid’s Statement

Euclid’s Proposition XI.13 states:

From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume, for contradiction, that AB and AC are two such straight lines erected from the common point A of the reference plane π .

13.3 Classical Configuration

Draw the plane through AB and AC . Denote it by σ . By XI.3 its section with the reference plane is a straight line through A ; Euclid writes this line as DAE .

Thus the classical configuration contains

$$AB, AC, DAE \subset \sigma, \quad DAE \subset \pi, \quad A \in \pi \cap \sigma.$$

The configuration is shown in Figure 13.1. The reference plane π is the light gray plane. The auxiliary plane σ contains the two blue spatial lines and intersects π along the black line DAE .

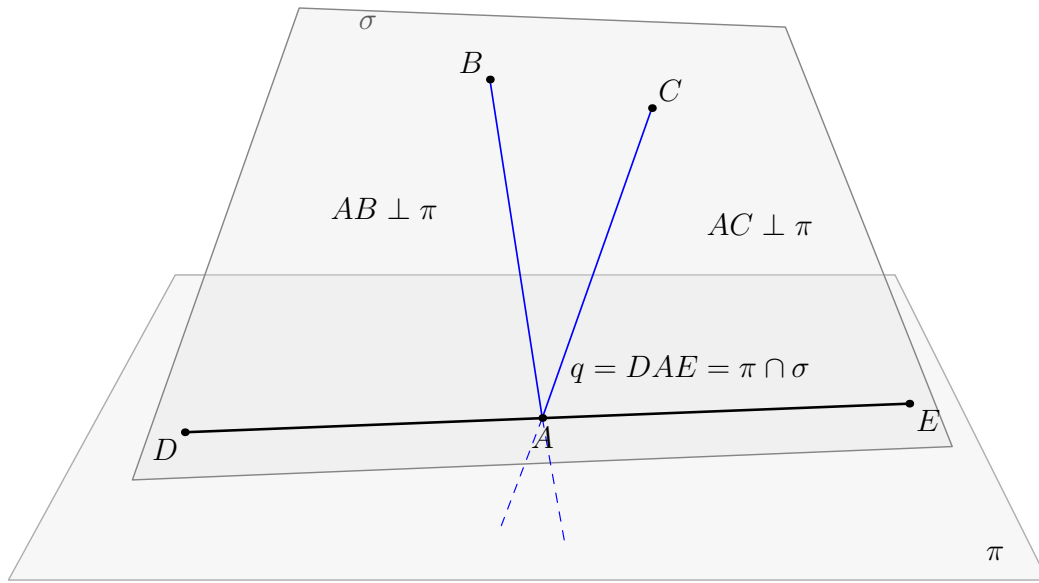


Figure 13.1: The classical configuration of Proposition XI.13. The two supposed perpendiculars AB and AC meet the reference plane π at the same point A and lie in the auxiliary plane σ . By XI.3 the planes π and σ intersect in the straight line DAE through A . Since both spatial lines are perpendicular to π , each is perpendicular to DAE .

13.4 Classical Proof

Suppose that AB and AC are distinct straight lines erected from A at right angles to the same plane π , on the same side.

Draw the plane σ through AB and AC . By XI.3 the intersection of σ with π is a straight line through A ; write it as DAE .

Since $AC \perp \pi$, Definition XI.3 says that AC is perpendicular to every straight line in π meeting it at the foot. Hence

$$AC \perp DAE.$$

For the same reason,

$$AB \perp DAE.$$

Therefore both $\angle CAE$ and $\angle BAE$ are right angles and hence equal. The two angles lie in the same plane σ , share the same side AE , and the remaining sides AC and AB are assumed to lie on the same side. Euclid concludes that this is impossible.

The local classical dependency chain is therefore

$$\begin{array}{c}
 AB \perp \pi, \quad AC \perp \pi, \quad A \in \pi \\
 \Downarrow \\
 \sigma = \text{plane}(AB, AC) \\
 \Downarrow \\
 \text{XI.3} \\
 \Downarrow \\
 DAE = \pi \cap \sigma \\
 \Downarrow \\
 \text{XI.Def.3} \\
 \Downarrow \\
 AB \perp DAE, \quad AC \perp DAE \\
 \Downarrow \\
 \angle BAE, \angle CAE \text{ are equal right angles in } \sigma \\
 \Downarrow \\
 \text{planar uniqueness / contradiction} \\
 \Downarrow \\
 AB = AC \text{ as a line carrier.}
 \end{array}$$

Neither XI.9 nor XI.10 occurs in this proof. XI.11 and XI.12 are also not dependencies of XI.13: they establish existence, whereas XI.13 is a uniqueness statement.

13.5 What is genuinely three-dimensional here?

The proof has one essential spatial move and one spatial intersection move.

First, the two supposed distinct perpendicular lines determine the auxiliary plane

$$\sigma = \text{plane}(l, m).$$

Second, this plane must be distinguished from the reference plane π , after which their common section is a line

$$q = \pi \cap \sigma$$

through the common point A .

Once q has been obtained, the classical argument is planar inside σ . The formal proof does not need to enter $\text{PlaneGeo}(\sigma)$ explicitly, because the relevant planar conclusion has already been packaged synthetically by XI.4.

The point is therefore not that XI.13 is “really planar”. The existence of the correct carrier plane, the distinctness of the two planes, and the common intersection line are genuine three-dimensional proof obligations.

13.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_13
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (l m : Geo.Line)
  (A : Geo.Point)
  (hLperp :
    HilbertLinePerpendicularPlaneAt Geo l pi A)
  (hMperp :
    HilbertLinePerpendicularPlaneAt Geo m pi A) :
  l = m

```

There is no `HilbertSpaceEuclidean` parameter. Thus the current formal XI.13 does not use Hilbert Group IV.

The difference between Euclid’s phrase “on the same side” and the formal conclusion $l = m$ is representational. A `Geo.Line` is an unoriented carrier. Reversing the ray through A does not create a second line object, so the side qualification disappears when the target is equality of line carriers.

13.7 Spatial / Planar Architecture Used

13.7.1 Ambient spatial layer

The proof uses the ambient notions

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt

```

together with the general incidence constructors

```

hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line

```

and the universal elimination theorem

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

As in the preceding Book XI development, no global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

are installed.

13.7.2 Induced planar geometry

The final production proof makes no direct theorem call in $\text{PlaneGeo}(\sigma)$. This is a genuine simplification of the formal DAG relative to the classical exposition.

The planar content has not disappeared. It is already contained in the earlier theorem

```
euclid_proposition_11_4
```

which states synthetically that a line perpendicular to two distinct intersecting lines of a plane is perpendicular to the entire plane.

13.7.3 Plane–space bridges

XI.13 introduces no new plane–space bridge theorem. The proof stays in ambient predicates and uses the already established spatial perpendicularity symmetry

```
hilbert_space_linesPerpendicularAt_symm
```

before applying XI.4.

13.8 Proof Architecture of the Reconstruction

13.8.1 1. Contradiction hypothesis

Lean begins with

```
by_contra hlm
```

so the working assumption is

$$l \neq m.$$

The incidence components of the two line–plane perpendicularities give

$$A \in l, \quad A \in m, \quad A \in \pi.$$

13.8.2 2. Constructing the auxiliary plane

Since l and m are distinct and meet at A , the theorem

```
hilbert_plane_through_two_intersecting_lines
```

produces a plane σ such that

$$l \subset \sigma, \quad m \subset \sigma, \quad A \in \sigma.$$

This is the formal version of Euclid's instruction to draw the plane through the two supposed perpendiculars.

13.8.3 3. Proving that the planes are distinct

The intersection theorem for two planes requires an explicit proof that

$$\sigma \neq \pi.$$

Suppose instead that $\sigma = \pi$. Since $l \subset \sigma$, this would imply $l \subset \pi$. But the already established theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

says that a line perpendicular to a plane cannot itself lie in that plane. Hence $\sigma \neq \pi$.

This is one of the principal hidden spatial obligations of XI.13.

13.8.4 4. Constructing the intersection line

The theorem

```
hilbert_plane_intersection_line
```

is applied to π, σ at the common point A . It returns an ambient line q with

$$A \in q, \quad q \subset \pi, \quad q \subset \sigma.$$

This is the formal structural counterpart of Euclid's XI.3 step producing the line DAE .

13.8.5 5. Both candidate lines are perpendicular to the intersection

Since $q \subset \pi$ and $A \in q$, the universal clause of

```
HilbertLinePerpendicularPlaneAt
```

gives

$$l \perp q \text{ at } A, \quad m \perp q \text{ at } A.$$

The proof then applies

```
hilbert_space_linesPerpendicularAt_symm
```

to reverse the two relations:

$$q \perp l \text{ at } A, \quad q \perp m \text{ at } A.$$

13.8.6 6. XI.4 replaces the terminal planar uniqueness step

The lines l, m are now packaged as two distinct `PlaneLines` of σ , and A as a `PlanePoint` of σ . Proposition XI.4 applies:

$$q \perp l, \quad q \perp m, \quad l \neq m, \quad l, m \subset \sigma \implies q \perp \sigma.$$

This is the central difference between the classical and Lean DAGs. Euclid compares two right angles in the auxiliary plane and invokes their impossibility as distinct same-side angles. Lean reuses the already proved universal line–plane theorem XI.4.

The method remains synthetic: no metric formula, coordinate argument, or vector normal is introduced.

13.8.7 7. Final contradiction

The intersection construction preserved

$$q \subset \sigma.$$

But the XI.4 conclusion says

$$q \perp \sigma.$$

Applying

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

to q and σ gives the contradiction. Hence the assumption $l \neq m$ is false, and therefore

$$l = m.$$

13.9 Wyler-Style Flat Reconstruction

XI.13 gives the cleanest complete interaction of join, meet, and metric structure in the first fourteen propositions. Under the contradiction hypothesis $l \neq m$, the two candidate normals generate a plane σ . The reference plane π and σ then have an exact meet line q .

The entire proof can be read as the following graph:

```

assume l != m
  |
  v
sigma = Join(l,m)
  |
  v
q = Meet(pi,sigma)
  |
  v
l perp q, m perp q
  |
  v
q perp l, q perp m
  |
  v
XI.4 core: q perp sigma
  |
  v
q subset sigma
  |
  v
contradiction

```

The join identifies the canonical plane generated by the two supposed normals. The meet identifies the canonical section line of that plane with π . Only after these carriers have been fixed does the metric argument enter: because both normals are perpendicular to π , they are perpendicular to q ; symmetry and XI.4 then force $q \perp \sigma$, which is impossible because the meet construction preserved $q \subseteq \sigma$.

This graph belongs at the theorem level. It is the most compact description of why the formal proof remains synthetic while exposing more structure than the diagram alone.

13.10 Lean Formalization

The production module has the single direct import

```
import CGJteamLab.Proposition11_12
```

and introduces no proposition-local helper theorem before the public result. The public declaration is

```
euclid_proposition_11_13
```

The proof consumes general infrastructure and earlier Book XI theorems available transitively through the current production chain. In particular, the substantive reused declarations are

```
hilbert_plane_through_two_intersecting_lines
hilbert_plane_intersection_line
hilbert_linePerpendicularPlaneAt_not_in_plane
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
hilbert_space_linesPerpendicularAt_symm
euclid_proposition_11_4
```

XI.12 itself is not called in the proof. Its module is the immediate production import because XI.13 follows XI.12 in the current Book XI module sequence.

13.11 Hidden Formal Data

13.11.1 The auxiliary plane must be explicit

The classical diagram makes it visually immediate that the two supposed perpendiculars lie in a common plane. Lean constructs this plane from the two distinct intersecting line carriers and keeps both containment proofs.

13.11.2 The two planes must be proved distinct

Euclid immediately speaks of their section. The formal `hilbert_plane_intersection_line` theorem requires $\sigma \neq \pi$. This is proved by showing that equality of the planes would put a perpendicular line inside its reference plane.

13.11.3 The intersection carrier must survive

The proof must preserve all three facts returned with q :

$$A \in q, \quad q \subset \pi, \quad q \subset \sigma.$$

The first two are needed to extract the two line–line perpendicularities. The last is needed only at the final contradiction.

13.11.4 The “same side” phrase is not a missing formal hypothesis

The formal theorem does not introduce a relation saying that two lines are on the same side of a plane. This is not an omitted diagrammatic condition. Euclid’s condition distinguishes erected

rays. The formal objects $l, m : \text{Geo.Line}$ are unoriented carriers, so opposite rays through the foot determine the same line.

13.12 Relationship with Earlier Book XI Propositions

The classical local dependency is XI.3 together with Definition XI.3. Propositions XI.9–XI.12 are absent from the classical proof.

The Lean proof reorganizes the dependency graph:

- the content of the XI.3 plane-section step is supplied by the general theorem `hilbert_plane_intersection_line`;
- XI.4 is used directly to package the final planar impossibility;
- the helper `hilbert_linePerpendicularPlaneAt_not_in_plane`, introduced in the XI.5 production module, proves both the distinctness of the planes and the final contradiction;
- the spatial symmetry theorem `hilbert_space_linesPerpendicularAt_symm`, introduced in the XI.6 production module, normalizes the two perpendicular relations before XI.4.

Thus XI.13 is formally dependent on earlier XI.4–XI.6 infrastructure, but not on the existence propositions XI.11 and XI.12 as theorem calls.

13.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by XI.13. The planar right-angle machinery needed by XI.4 is inherited through the already proved Book XI infrastructure.

The current theorem uses spatial incidence, order, and congruence, but its signature has no

```
HilbertSpaceEuclidean Geo
```

parameter. Hence Group IV is absent from XI.13.

Book Zero contributes no new XI.13 declaration. The final axiom profile contains the already accepted inherited dependency `bookZero_nullSegment1`; XI.13 introduces no new global axiom constant.

13.14 Relationship with the Foundations

XI.13 is neutral in the current reconstruction. Its theorem signature does not contain `HilbertSpaceEuclidean Geo`; the proof uses spatial incidence, order, congruence, the plane-intersection infrastructure, and Proposition XI.4, but no Hilbert Group IV assumption and no continuity principle.

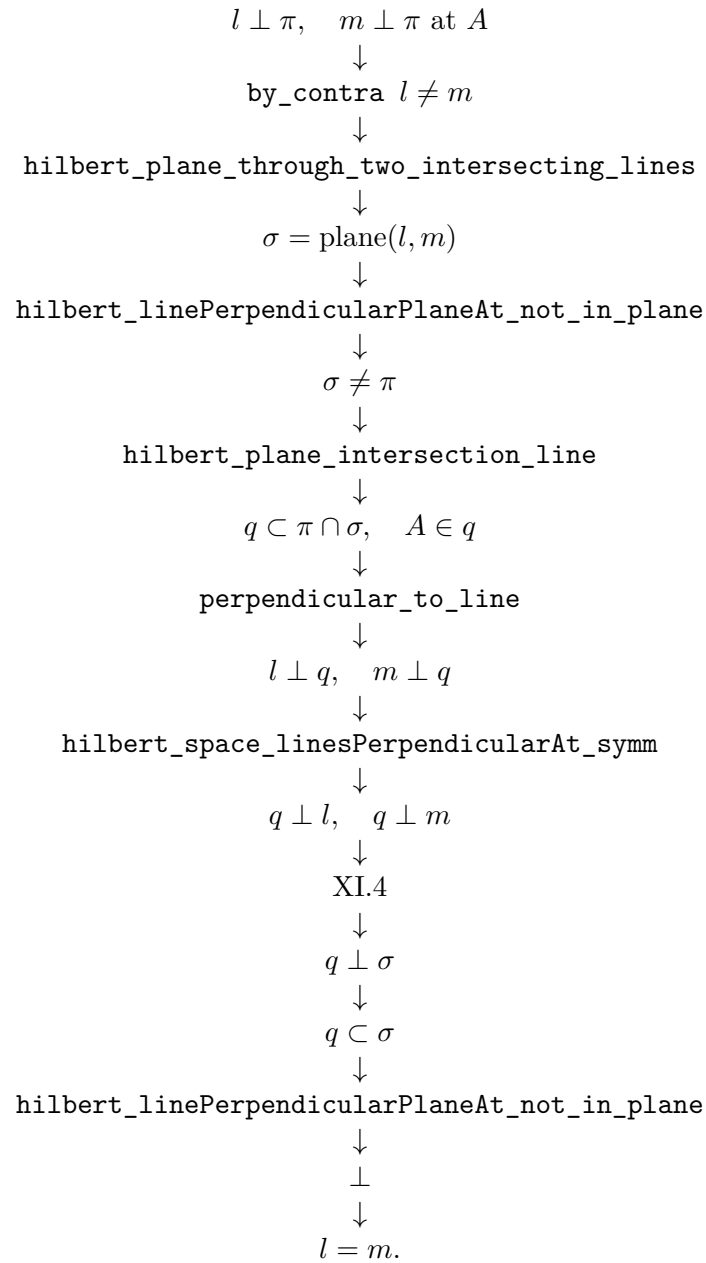
This is worth separating from the neighboring XI.11–XI.12 constructions: although all three concern perpendicularity in space, XI.13 closes by a neutral uniqueness argument inside an explicitly constructed auxiliary plane.

13.15 Dependency Summary

The classical local chain is

$$\begin{array}{c}
 l \perp \pi, \quad m \perp \pi, \quad A \in l \cap m \cap \pi \\
 \downarrow \\
 \sigma = \text{plane}(l, m) \\
 \downarrow \\
 \text{XI.3} \\
 \downarrow \\
 q = \pi \cap \sigma \\
 \downarrow \\
 \text{XI.Def.3} \\
 \downarrow \\
 l \perp q, \quad m \perp q \\
 \downarrow \\
 \text{equal right angles in one plane} \\
 \downarrow \\
 \text{contradiction.}
 \end{array}$$

The actual formal chain is



The dependency classification is:

New proposition-specific global axiom	no
New spatial axiom class	no
Continuity	no
Group IV / Euclidean parallelism	no
Direct production import	XI.12 module
Classical explicit Book XI dependency	XI.3
Classical definition dependency	XI.Def.3
XI.9 / XI.10	not used
XI.11 / XI.12 theorem calls	not used
Earlier formal Book XI theorem reused	XI.4
New proposition-local helper theorem	no
New PlaneGeo bridge	no
Coordinates / vectors / scalar product	no
sorry/admit	no

13.16 Conclusion

Proposition XI.13 is a compact uniqueness theorem with a particularly clean formal architecture. The classical proof uses an auxiliary plane, its intersection with the reference plane, and the impossibility of two distinct same-side perpendicular rays in one plane.

The Lean proof preserves the first two spatial steps exactly but reuses XI.4 for the terminal planar contradiction. The key hidden obligation is the proof that the auxiliary plane differs from the reference plane. Once the intersection line is retained with both carrier-plane proofs, the final contradiction is immediate from the fact that a line perpendicular to a plane cannot itself lie in that plane.

The theorem introduces no new helper, bridge, spatial axiom, Group IV assumption, or continuity principle. Its audited global axiom profile is

`[propext, Classical.choice, Geometry.bookZero_nullSegment1, Quot.sound]`.

The result is fully synthetic and uses no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure.

Chapter 14

Proposition XI.14

14.1 Introduction

Proposition XI.14 states that planes to which the same straight line is at right angles are parallel. The classical argument is short: if the two planes met, a point of their intersection together with the common perpendicular would produce a triangle with two right angles, contradicting Proposition I.17.

The formal reconstruction preserves this geometric core, but it exposes two points that the classical presentation leaves implicit.

First, Euclid's contradiction argument requires the two feet of the common perpendicular to be distinct. If the line meets both planes at the same point, the triangle used in the proof degenerates. The production proof therefore separates an equal-foot uniqueness theorem from the normalized case $A \neq B$.

Second, Euclid reaches a common point by first speaking about the line of intersection of the two planes. In the formal library, parallelism of two planes is defined directly as absence of a common point. Negating the target therefore gives the point K immediately; the proof does not need to construct the whole intersection line before beginning the contradiction.

These two differences make XI.14 a useful example of the project's general method. The classical geometry is retained, while diagrammatic assumptions and representational shortcuts are isolated as explicit proof obligations.

14.2 Classical Configuration

Let l be a straight line perpendicular to plane π at A and to plane ρ at B :

$$l \perp \pi \text{ at } A, \quad l \perp \rho \text{ at } B.$$

The intended nontrivial configuration has

$$\pi \neq \rho, \quad A \neq B.$$

Euclid's conclusion is that the two planes do not meet.

The formal public theorem also carries the hypothesis $\pi \neq \rho$. This matches the nontrivial reading of Euclid XI.Def.8 used in the project: parallel planes are distinct planes having no common point.

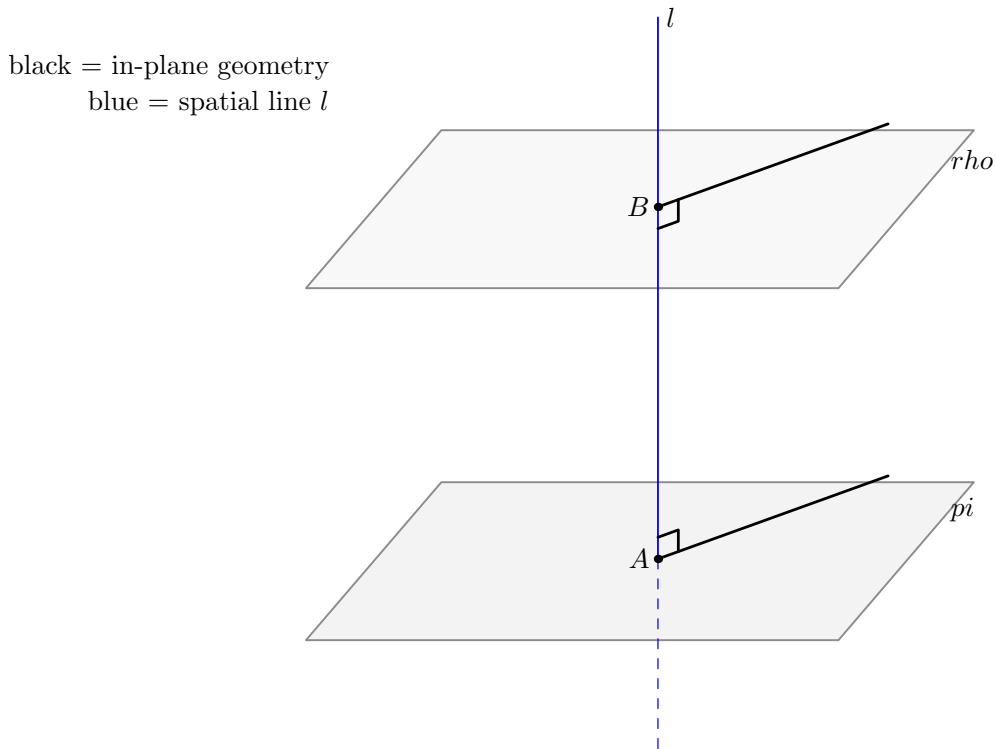


Figure 14.1: The basic spatial configuration of Proposition XI.14. The common perpendicular l meets the two reference planes at A and B . Following the Book XI drawing convention, geometry internal to a displayed plane is black, while genuinely spatial objects are blue.

Diagrammatic audit. Three stable geometric states are sufficient for the documentation of XI.14. Figure 14.1 records the input configuration. Figure 14.3 records the hypothetical common-point state and the reduction to a planar triangle. Figure 14.2 records the exceptional equal-foot uniqueness argument. No additional figure is needed for the internal SameRay/opposite-ray case split, because that split is a formal orientation normalization rather than a new stable Euclidean construction.

14.3 Statement

Theorem 2 (Euclid XI.14). *Let π and ρ be distinct planes, and let l be a straight line. If l is perpendicular to π at A and perpendicular to ρ at B , then π and ρ are parallel.*

The current production declaration is:

```
theorem euclid_proposition_11_14
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [_HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi rho : S.Plane)
  (l : Geo.Line)
```

```

(A B : Geo.Point)
(hPlanesNe : Ne pi rho)
(hPerpPi :
  HilbertLinePerpendicularPlaneAt Geo l pi A)
(hPerpRho :
  HilbertLinePerpendicularPlaneAt Geo l rho B) :
HilbertSpacePlanesParallel Geo pi rho

```

The target predicate is proposition-local:

```

def HilbertSpacePlanesParallel
  [S : HilbertSpacePrimitive Geo]
  (pi rho : S.Plane) : Prop :=
  Not
    (exists X : Geo.Point,
      S.OnPlane X pi /\
      S.OnPlane X rho)

```

Thus the conclusion is represented directly as nonintersection of the two planes.

14.4 Classical Proof

Assume, for contradiction, that the two planes meet. By XI.3 their common section is a straight line; choose a point K on this section.

Join A to K and B to K . Since $K \in \pi$, the line AK lies in π . Because $l \perp \pi$ at A , Definition XI.3 gives

$$l \perp AK.$$

Similarly, since $K \in \rho$,

$$l \perp BK.$$

The lines l and the point K determine a plane σ . The points A and B lie on l , so the triangle ABK lies in σ . Inside this plane,

$$\angle BAK \text{ is right,} \quad \angle ABK \text{ is right.}$$

But I.17 says that two angles of a triangle are together less than two right angles. A nondegenerate triangle therefore cannot have two right angles. This contradiction proves that π and ρ do not meet.

The local classical dependency pattern is

$$\begin{array}{c}
 l \perp \pi, \quad l \perp \rho \\
 \downarrow \\
 \text{assume } \pi \cap \rho \neq \emptyset \\
 \downarrow \\
 \text{XI.3} \\
 \downarrow \\
 K \in \pi \cap \rho \\
 \downarrow \\
 \text{XI.Def.3} \\
 \downarrow \\
 \angle BAK, \angle ABK \text{ right} \\
 \downarrow \\
 \text{I.17} \\
 \downarrow \\
 \perp.
 \end{array}$$

The proof is neutral. No Euclidean parallel postulate is involved.

14.5 The Equal-Foot Issue

The classical triangle argument silently presupposes $A \neq B$. If $A = B$, the side AB collapses and the I.17 contradiction cannot even be formed.

The production proof handles this branch by proving a separate uniqueness theorem: for a fixed line l and a fixed point A , there is at most one plane perpendicular to l at A .

The construction is synthetic. Choose two distinct lines m, n through A in π . Since $l \perp \pi$, both are perpendicular to l . Now let p be any line through A in ρ . Since $l \perp \rho$, the line p is also perpendicular to l . Proposition XI.5 then places m, n, p in one plane. Because the two distinct intersecting lines m, n already determine π , the line p must lie in π .

Applying the same argument to two distinct lines through A in ρ shows that two distinct intersecting lines of ρ also lie in π . The uniqueness of the plane through two intersecting lines yields

$$\rho = \pi.$$

This is not an auxiliary side remark. It is exactly what allows the public theorem to derive $A \neq B$ from the assumed plane distinctness $\pi \neq \rho$.

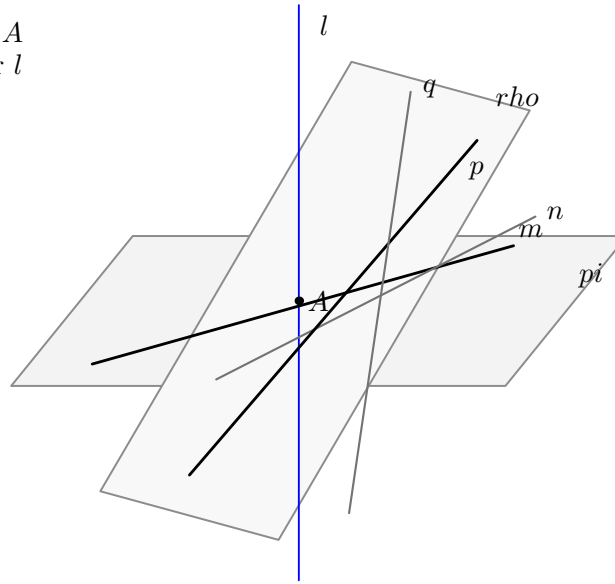
14.6 What is genuinely three-dimensional here?

XI.14 has a particularly clear spatial-to-planar transition.

The genuinely three-dimensional data are:

- two ambient planes π, ρ ;
- a single ambient line l perpendicular to each;

black = in-plane lines through A
 blue = common perpendicular l



equal-foot uniqueness case

Figure 14.2: The exceptional case in which the same line l is perpendicular to both planes at the same point A . Two distinct in-plane lines through A , together with XI.5, force every line through A in the second plane into the first plane; hence the two planes coincide.

- a hypothetical point K common to the two planes;
- the auxiliary plane σ through l and the off-line point K ;
- the two connector lines AK and BK , each simultaneously controlled by one reference plane and by σ .

Once σ has been constructed, the terminal contradiction is planar. The proof passes to

PlaneGeo(σ)

and works with the induced Hilbert incidence, order, and congruence structure. The two spatial perpendicularities are converted into the exact right-angle statements for the triangle ABK , after which I.17 is purely two-dimensional.

Thus XI.14 is not merely a planar theorem drawn in space. The proof must first manufacture the correct plane slice and transport the ambient line–plane perpendicularity data into it.

14.7 Spatial / Planar Architecture Used

The ambient layer uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

The key spatial incidence constructors are

```
hilbert_plane_through_line_and_external_point
hilbert_plane_through_two_intersecting_lines
```

For the normalized branch, the proof constructs σ from the common normal l and the hypothetical common point K . The points and lines are then re-packaged as

```
PlanePoint Geo sigma
PlaneLine Geo sigma
PlaneGeo Geo sigma
```

This passage is essential because Proposition I.17 is a planar theorem. The bridge layer transfers incidence and perpendicularity from the ambient spatial carriers to the induced plane geometry.

black = planar geometry in σ
blue = common perpendicular l

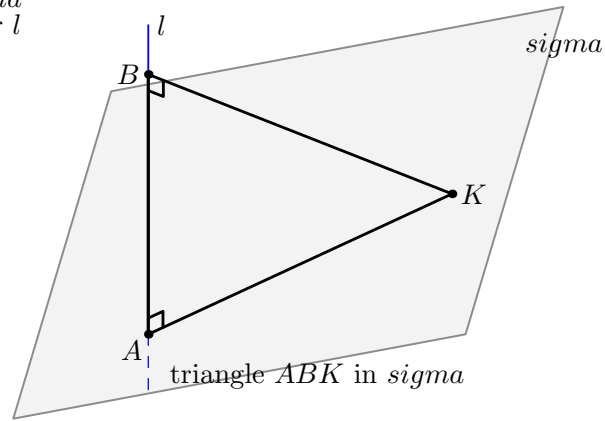


Figure 14.3: The hypothetical common point K and the common perpendicular l determine the auxiliary plane σ . In σ , the triangle ABK has right angles at both A and B . The spatial construction is therefore reduced to the planar contradiction supplied by I.17.

14.8 Proof Architecture of the Reconstruction

14.8.1 1. Excluding a common point on the normal

Assume $A \neq B$ and suppose K lies on both planes. If K also lay on l , the perpendicular-foot uniqueness theorem from the XI.12 infrastructure would give

$$K = A \quad \text{and} \quad K = B,$$

contradicting $A \neq B$.

This is formalized by `hilbert_XI14_common_point_off_normal`.

14.8.2 2. Constructing the auxiliary plane

Because $K \notin l$, the general spatial incidence theorem constructs a unique plane σ containing l and K . Since $A, B \in l$, all three vertices A, B, K lie in σ .

This stage is the formal replacement for what the classical diagram makes immediate.

14.8.3 3. Constructing the connector lines

The lines AK and BK are explicitly constructed. The first lies in both π and σ ; the second lies in both ρ and σ . Definition XI.3, encoded by `HilbertLinePerpendicularPlaneAt.perpendicular_to_line`, gives

$$l \perp AK \text{ at } A, \quad l \perp BK \text{ at } B.$$

14.8.4 4. Passing to PlaneGeo

The ambient configuration is transferred to the induced plane geometry of σ . The noncollinearity of A, B, K follows because $A, B \in l$ while $K \notin l$.

At this point the spatial part of the normalized contradiction is complete.

14.8.5 5. Recovering the exact triangle angles

The formal definition of line–line perpendicularity contains witnesses on the two carrier lines. Those witnesses need not initially be on the exact rays AB, AK and BA, BK used in the triangle.

The production proof therefore performs an orientation normalization. A collinear witness is either on the same ray from the vertex as the desired triangle point or on the opposite ray. SameRay transport and the two opposite-ray right-angle transport lemmas show that rightness is invariant under all four orientation combinations.

The result is the stable geometric statement

$$\angle BAK \text{ right}, \quad \angle ABK \text{ right}.$$

This is packaged by `hilbert_XI14_triangle_has_two_right_angles`.

14.8.6 6. Proposition I.17 closes the planar contradiction

Apply the I.17 variant `euclid_proposition_17_BAC_ABC` to the triangle ABK . It extends the side BK through B to a point E and yields

$$\angle BAK < \angle ABE.$$

Since $K - B - E$ and $\angle ABK$ is right, the angle $\angle ABE$ is also right. All right angles are congruent, so

$$\angle BAK \cong \angle ABE.$$

A strict angle inequality cannot hold between congruent angles. This gives the contradiction.

The planar closure is packaged by `hilbert_XI14_two_right_angles_impossible`.

14.8.7 7. The normalized theorem

The theorem `euclid_proposition_11_14_normalized` now has a very short public shape: assume a common point K , construct the two-right-angle triangle, and invoke the planar impossibility theorem.

14.8.8 8. Removing the normalization

The public theorem must not assume $A \neq B$. It derives this from $\pi \neq \rho$.

If $A = B$, the equal-foot uniqueness theorem `hilbert_XI14_plane_perpendicular_to_line_at_unique` gives $\pi = \rho$, contradicting the hypothesis. Therefore $A \neq B$, and the normalized theorem applies.

14.9 Wyler-Style Flat Reconstruction

For distinct planes the flat layer first records the global rank fact

$$\text{Join}(\pi, \rho) = E^3.$$

This is useful, but deliberately insufficient: two distinct planes may generate all of three-space whether or not they are parallel. XI.14 therefore requires a separate metric separation argument.

The complete structure is

```

pi != rho
|
+--> Join(pi,rho) = E^3           [flat certificate]
|
A = B ?
|
+-- yes --> uniqueness of plane normal --> pi = rho --> contradiction
|
no
v
assume K in pi cap rho
|
sigma = plane(1,K)
|
triangle ABK inside PlaneGeo(sigma)
|
right angle at A and right angle at B
|
I.17
|
contradiction
|
v
pi cap rho = empty

```

Thus the meet of the two plane carriers is forced to be empty, which is the formal spatial parallelism conclusion. The proposition is a useful boundary case for the Wyler vocabulary: join detects global generation, but empty meet is proved by the metric geometry of a common normal and the planar I.17 contradiction.

14.10 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_13
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition17
import CGJteamLab.HilbertRightAngle
```

The direct mathematical engines used by XI.14 are distributed across these imports:

- XI.5 supplies the equal-foot coplanarity argument;
- the XI.12 infrastructure supplies perpendicular-foot uniqueness;
- I.17 supplies the terminal planar angle inequality;
- the Hilbert right-angle library supplies transport and congruence of right angles.

The public proof itself is intentionally small. Almost all of the work is performed in reusable proposition-local lemmas that separate the spatial construction, the planar reduction, the angle-orientation normalization, and the equal-foot uniqueness theorem.

This organization mirrors the mathematical architecture more faithfully than a single monolithic tactic proof would.

14.11 Hidden Formal Data

14.11.1 Plane distinctness is part of the public statement

The project defines parallel planes by nonintersection, but XI.14 is intended for two distinct planes. The hypothesis `hPlanesNe` therefore appears explicitly.

It is also what rules out the equal-foot branch.

14.11.2 The common point must be off the normal

The classical picture places K away from AB automatically. Lean must prove this. The argument uses perpendicular-foot uniqueness in each reference plane.

14.11.3 The auxiliary plane must be carried explicitly

The sentence “draw the plane through AB and K ” hides several facts: $K \notin l$, the existence of the plane, containment of l , and containment of K . All are retained because later bridge lemmas depend on them.

14.11.4 Perpendicularity witnesses have orientation

A statement such as $l \perp AK$ is not yet syntactically the statement that $\angle BAK$ is right. The witness points appearing in the definition of perpendicular lines may occupy either ray of the carrier. The production proof therefore has a genuine four-case SameRay/opposite-ray normalization.

This is formal data hidden completely by the Euclidean diagram.

14.11.5 I.17 returns an angle inequality, not the final contradiction

The formal I.17 theorem yields a strict angle comparison after extending one side of the triangle. XI.14 must still prove that the resulting exterior angle is right and then use congruence of all right angles to contradict the strict inequality.

14.12 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and Definition XI.3.

The formal dependency graph is reorganized.

The content corresponding to XI.3 is supplied more economically by the definition of plane parallelism and by general spatial incidence constructors. Since the negation of `HilbertSpacePlanesParallel` already gives a common point K , the proof does not first need an explicit line representing the complete intersection $\pi \cap \rho$.

Definition XI.3 is represented directly by `HilbertLinePerpendicularPlaneAt`: every line of the reference plane through the foot is perpendicular to the given line.

XI.5 enters only in the equal-foot uniqueness branch. This is a genuine new formal dependency caused by making explicit the case $A = B$.

XI.12 contributes the previously proved uniqueness of the perpendicular foot of a fixed line in a fixed plane. XI.13 itself is not called by the proof; its module is an immediate production import in the Book XI sequence.

14.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The decisive Book I dependency is Proposition I.17. In the current library the relevant theorem is `euclid_proposition_17_BAC_ABC`. It supplies the strict comparison needed to rule out the two-right-angle triangle.

The right-angle library contributes two separate ingredients:

- transport of a right angle when one of its arms is replaced by the same or opposite ray;
- congruence of arbitrary right angles.

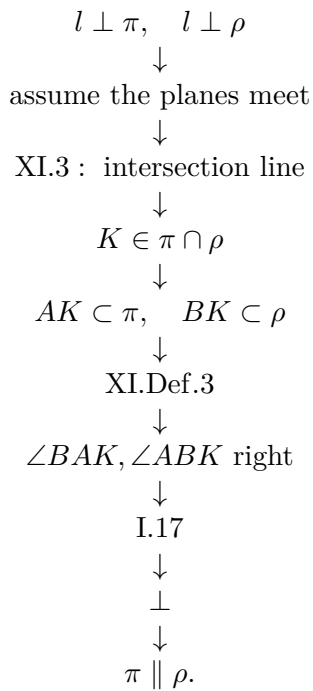
The spatial proof uses Hilbert incidence, order, and congruence only. There is no `HilbertSpaceEuclideanGeo` parameter in the theorem signature. Thus the reconstruction is neutral with respect to Hilbert Group IV.

No coordinate system, vector space, scalar product, normal vector, numerical angle measure, or trigonometric function occurs anywhere in the proof.

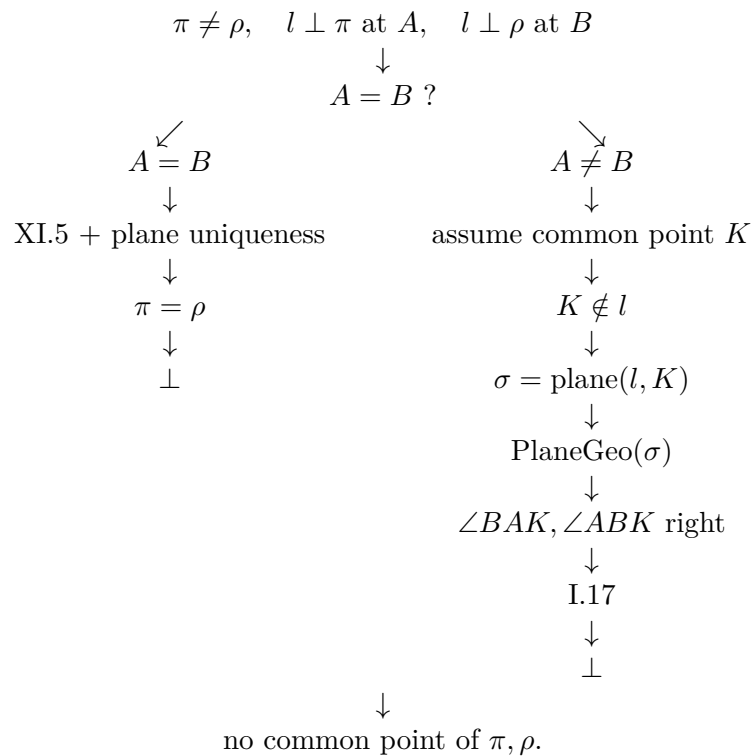
Book Zero contributes only inherited general infrastructure through the existing project layers; XI.14 introduces no new proposition-specific foundational principle.

14.14 Dependency Summary

The classical architecture is



The formal architecture has two branches:



The main conceptual difference is therefore clear. Euclid has one visible contradiction branch and suppresses the degenerate-foot possibility. The formal proof makes that possibility explicit and discharges it by a uniqueness theorem before entering the classical branch.

14.15 Conclusion

Proposition XI.14 has a simple classical idea but a rich formal anatomy.

The main Euclidean argument survives unchanged. A hypothetical common point of the two planes determines, together with the common perpendicular, an auxiliary plane. In that plane one obtains a nondegenerate triangle with two right angles, and Proposition I.17 gives the contradiction.

The formal reconstruction adds exactly what the classical diagram suppresses. It proves that the hypothetical common point is off the normal, constructs and retains the auxiliary plane explicitly, normalizes the orientation of the perpendicularity witnesses, and separates the degenerate equal-foot case. Proposition XI.5 then supplies a synthetic uniqueness theorem showing that two planes perpendicular to the same line at the same point must coincide.

The result is therefore both faithful to Euclid's spatial proof and sharper about its hidden assumptions. It also illustrates a recurring Book XI pattern: the genuinely spatial work consists in constructing and controlling the correct carrier plane; once that plane is available, the terminal argument returns to the planar Hilbert geometry already developed for Book I.

Chapter 15

Proposition XI.15

15.1 Introduction

Proposition XI.15 is a structural culmination of the parallelism and perpendicularity block developed in XI.8–XI.14. Euclid starts with two intersecting lines in one plane and two corresponding intersecting lines in another plane. If the corresponding pairs are parallel, then the two planes themselves are parallel.

The classical proof is remarkably economical. From the intersection point of the first pair, Euclid drops a perpendicular to the second plane by XI.11. Through its foot he draws two lines parallel to the two reference lines in that plane. Definition XI.3 makes the new normal perpendicular to both of these lines; XI.9 and I.29 transfer the two right angles back to the original pair. XI.4 then says that the same line is perpendicular to the first plane, and XI.14 finishes the proof.

The formal reconstruction preserves this outer architecture, but it exposes three hidden issues.

First, XI.11 requires the point B to lie outside the second plane ρ . Euclid’s diagram makes this look automatic, but Lean proves it from the two spatial parallelisms and the distinctness of the two planes.

Second, Euclid invokes I.31 twice through the foot G . The formal parallel-through-a-point construction requires the point to be outside the given line. XI.15 does not guarantee that G lies outside either reference line. The production proof therefore treats the cases $G \in m_i$ and $G \notin m_i$ uniformly by constructing a line q_i which is either literally m_i or parallel to m_i .

Third, the classical I.29 step is not used as one opaque angle theorem. The formal proof factors the transport of perpendicularity through the mechanism already developed in XI.8: neutral I.27 gives disjointness of two perpendiculars to the same transversal, and Hilbert Group IV identifies the unique parallel through the required point.

The result is a proof whose top-level DAG is very close to Euclid’s, while its local geometry is more explicit about plane carriers, exceptional incidence, and the exact place where Euclidean parallel uniqueness enters.

15.2 Classical Configuration

Let AB and BC be two straight lines meeting at B , and let DE and EF be two straight lines meeting at E . Suppose

$$AB \parallel DE, \quad BC \parallel EF,$$

and suppose that the four lines are not all contained in one plane.

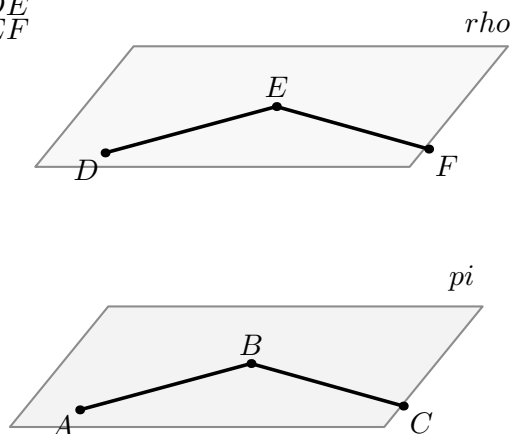
Write

$$\pi = \text{plane}(AB, BC), \quad \rho = \text{plane}(DE, EF).$$

The conclusion is

$$\pi \parallel \rho.$$

$$\begin{array}{l} AB \text{ parallel to } DE \\ BC \text{ parallel to } EF \end{array}$$



two intersecting pairs in distinct plane slices

Figure 15.1: The classical spatial configuration of Proposition XI.15. The lines AB, BC meet at B in plane π , while DE, EF meet at E in plane ρ . Corresponding carriers are parallel: $AB \parallel DE$ and $BC \parallel EF$. Geometry internal to a displayed plane is black; the separation between the two plane slices is spatial rather than an additional geometric connector.

Diagrammatic audit. Three stable geometric states are sufficient for XI.15. Figure 15.1 records the classical input. Figure 15.2 records the XI.11 normal and the two formal replacements for Euclid’s I.31 constructions, including the hidden exceptional branch $G \in m_i$. Figure 15.3 records the plane-slice mechanism which transports perpendicularity across a parallel pair.

No fourth figure is introduced for the final state

$$n \perp \pi, \quad n \perp \rho,$$

because that state is exactly the input configuration of Proposition XI.14 and would duplicate the already documented XI.14 geometry.

15.3 Statement

Euclid’s statement is:

If two straight lines meeting one another be parallel to two straight lines meeting one another, not being in the same plane, the planes through them are parallel.

The source-style mathematical form is therefore:

Theorem 3 (Euclid XI.15). *Let l_1, l_2 be distinct lines meeting in a plane π , and let m_1, m_2 be distinct lines meeting in a plane ρ . Suppose*

$$l_1 \parallel m_1, \quad l_2 \parallel m_2,$$

and suppose the four lines do not lie in one common plane. Then

$$\pi \parallel \rho.$$

The production file exposes a slightly stronger public theorem. It requires the first pair l_1, l_2 to meet at B , but it does not require an explicit second intersection point E or the inequality $m_1 \neq m_2$. Those extra data are not needed by the proof once the second pair is known to lie in ρ and the two corresponding spatial parallelisms are given.

The current public declaration is:

```

theorem euclid_proposition_11_15
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l1 l2 : PlaneLine Geo pi)
  (m1 m2 : PlaneLine Geo rho)
  (B : PlanePoint Geo pi)
  (hB11 : H.OnLine B.1 l1.1)
  (hB12 : H.OnLine B.1 l2.1)
  (hl12 : Ne l1 l2)
  (hParallel1 :
    HilbertSpaceLinesParallel Geo l1.1 m1.1)
  (hParallel2 :
    HilbertSpaceLinesParallel Geo l2.1 m2.1)
  (hNoCommonPlane :
    Not (exists omega : S.Plane,
      HilbertLineInPlane Geo l1.1 omega /\
      HilbertLineInPlane Geo l2.1 omega /\
      HilbertLineInPlane Geo m1.1 omega /\
      HilbertLineInPlane Geo m2.1 omega)) :
    HilbertSpacePlanesParallel Geo pi rho

```

For source comparison the file also provides

```
euclid_proposition_11_15_intersecting_pairs
```

which adds an explicit point $E \in m_1 \cap m_2$ and the hypothesis $m_1 \neq m_2$. Its proof is a direct projection to the stronger public theorem.

15.4 Classical Proof

Starting from B , Euclid applies XI.11 to draw

$$BG \perp \rho,$$

where $G \in \rho$.

Through G , Proposition I.31 gives two lines in ρ :

$$GH \parallel ED, \quad GK \parallel EF.$$

Since $BG \perp \rho$, Definition XI.3 gives

$$BG \perp GH, \quad BG \perp GK.$$

Now

$$BA \parallel ED \quad \text{and} \quad GH \parallel ED.$$

By XI.9,

$$BA \parallel GH.$$

Similarly,

$$BC \parallel GK.$$

Proposition I.29 transfers the two right angles:

$$BG \perp BA, \quad BG \perp BC.$$

The two lines BA and BC meet at B , so XI.4 yields

$$BG \perp \pi.$$

But by construction

$$BG \perp \rho.$$

Therefore XI.14 applies to the same straight line BG and gives

$$\pi \parallel \rho.$$

The classical local dependency pattern is

$$\begin{array}{c}
 AB \parallel DE, \quad BC \parallel EF \\
 \downarrow \\
 \text{XI.11 : } BG \perp \rho \\
 \downarrow \\
 \text{I.31 : } GH \parallel ED, \quad GK \parallel EF \\
 \downarrow \\
 \text{XI.Def.3 : } BG \perp GH, \quad BG \perp GK \\
 \downarrow \\
 \text{XI.9 + I.29} \\
 \downarrow \\
 BG \perp BA, \quad BG \perp BC \\
 \downarrow \\
 \text{XI.4} \\
 \downarrow \\
 BG \perp \pi \\
 \downarrow \\
 \text{XI.14} \\
 \downarrow \\
 \pi \parallel \rho.
 \end{array}$$

Thus XI.15 is classically a short synthesis theorem: XI.11 constructs the normal, XI.4 upgrades two line-line perpendicularities to a line-plane perpendicularity, and XI.14 converts a common normal into plane parallelism.

15.5 The Hidden I.31 Exceptional Case

The most important local correction in the formal reconstruction occurs before the classical I.31 step.

Euclid says simply: through G , draw a line parallel to ED and a line parallel to EF . In a formal Hilbert construction, however, a parallel through a point is constructed only when the point lies outside the given line. XI.15 does not imply

$$G \notin DE \quad \text{or} \quad G \notin EF.$$

The production theorem therefore uses the proposition-local helper

```
hilbert_XI15_parallel_or_equal_through_point_in_plane
```

with the following mathematical content. Given a line $m \subset \rho$ and a point $G \in \rho$, it returns a line $q \subset \rho$ through G such that

$$q = m \quad \text{or} \quad m \parallel q.$$

There are two branches:

$$\begin{aligned} G \in m &\implies q := m, \\ G \notin m &\implies \text{use the XI.12 parallel-through-point construction.} \end{aligned}$$

In the second branch, the XI.12 helper returns spatial parallelism together with a carrier plane. XI.15 then proves that the constructed line actually lies in the intended plane ρ , using uniqueness of the plane through a line and an external point.

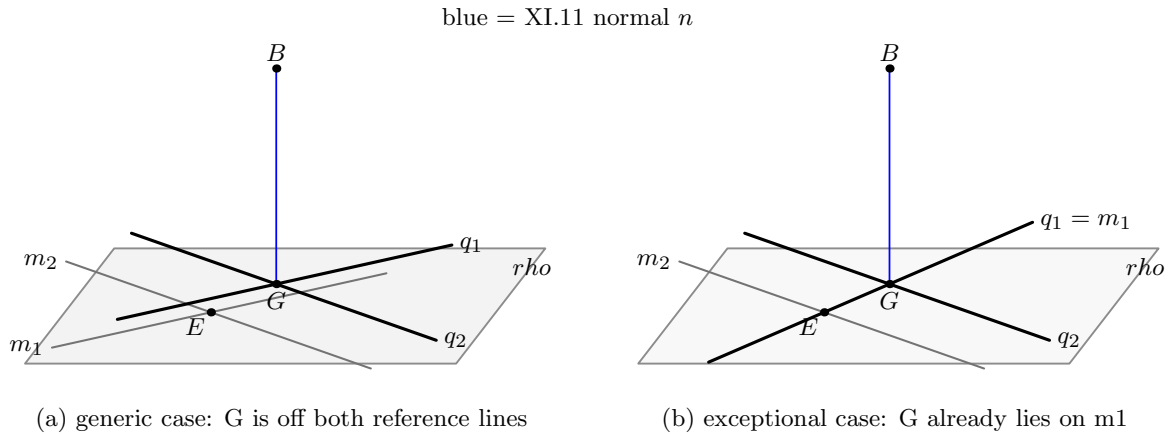


Figure 15.2: The formal replacement for Euclid's two I.31 instructions. In the generic branch, lines $q_1, q_2 \subset \rho$ through G are parallel to the reference lines m_1, m_2 . In the exceptional branch $G \in m_1$, the correct choice is simply $q_1 = m_1$; only the second reference line requires a new parallel. The blue line is the genuinely spatial normal n , while all black and gray lines lie in ρ .

This case split is not a Lean artifact. It is the precise formal content of the geometric fact that the instruction “draw a parallel through G ” degenerates to “use the given line itself” when G already lies on the carrier.

15.6 What is genuinely three-dimensional here?

XI.15 alternates repeatedly between ambient space and induced plane geometry.

The genuinely spatial data are:

- the two reference planes π, ρ ;
- the two pairs of corresponding spatially parallel lines;
- the condition that all four lines do not lie in one ambient plane;
- the point B , which must be proved to lie outside ρ ;
- the XI.11 normal n from B to ρ ;
- the final line-plane perpendicularities $n \perp \pi$ and $n \perp \rho$;
- the spatial target $\text{HilbertSpacePlanesParallel}(\pi, \rho)$.

There are also auxiliary carrier planes which are genuinely spatial. Every assertion of

$$\text{HilbertSpaceLinesParallel}(l, q)$$

contains a plane witnessing that l and q are coplanar. The perpendicularity-transport lemma uses precisely this carrier plane, denoted σ , to move from ambient line data to an induced $\text{PlaneGeo}(\sigma)$.

The planar work occurs in two distinct kinds of slices:

1. the reference plane ρ , where the lines through the foot G are constructed;
2. a carrier plane σ for each parallel pair $l \parallel q$, where the I.27/Group IV perpendicularity transport is performed.

Thus XI.15 is not one planar proof placed in space. The proof changes planes for mathematically different reasons: first to construct reference directions inside ρ , then to compare a transversal with a particular parallel pair inside its own carrier plane.

15.7 Formal Statement and Normalized Core

The substantial theorem is first isolated under explicit plane distinctness:

```
theorem euclid_proposition_11_15_normalized
...
(pi rho : S.Plane)
(l1 l2 : PlaneLine Geo pi)
(m1 m2 : PlaneLine Geo rho)
```

```

(B : PlanePoint Geo pi)
...
(hl12 : Ne l1 l2)
(hParallel1 :
  HilbertSpaceLinesParallel Geo l1.1 m1.1)
(hParallel2 :
  HilbertSpaceLinesParallel Geo l2.1 m2.1)
(hPlanesNe : Ne pi rho) :
HilbertSpacePlanesParallel Geo pi rho

```

The public theorem replaces the explicit hypothesis

$$\pi \neq \rho$$

by Euclid's source condition that the four lines are not contained in one common plane. If $\pi = \rho$, then all four displayed lines lie in that same plane, contradicting the public hypothesis. The public wrapper is therefore mathematically thin.

This separation is useful. The normalized theorem expresses the actual geometric engine. The public theorem expresses the source-facing configuration.

15.8 Spatial / Planar Architecture Used

The ambient layer is:

```

HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean

```

The principal spatial relations are:

```

HilbertLineInPlane
HilbertSpaceLinesParallel
HilbertLinesDisjoint
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
HilbertSpacePlanesParallel

```

The induced plane layer uses:

```

PlanePoint
PlaneLine
PlaneOnLine
PlaneGeo

```

and the standard ambient/plane bridges, in particular the bridge for line-line perpendicularity.

A permanent Book XI instance rule is respected throughout: the proof does not install global ambient instances

```
HilbertOrder Geo
HilbertCongruence Geo
```

for three-space. Order and planar congruence are supplied through explicit PlaneGeo slices; spatial congruence remains in HilbertSpaceCongruence.

15.9 Proof Architecture of the Reconstruction

15.9.1 Proving $B \notin \rho$

The normalized theorem cannot invoke XI.11 until it has proved that the intersection point B of l_1, l_2 lies outside the second plane.

The helper

```
hilbert_XI15_intersection_point_off_other_plane
```

is purely incidence-theoretic.

Assume $B \in \rho$. The parallelism $l_1 \parallel m_1$ provides a carrier plane σ_1 containing both lines. Since the two parallel lines are disjoint, $B \notin m_1$. Both σ_1 and ρ therefore contain the same line m_1 and the same external point B . Plane uniqueness gives

$$\sigma_1 = \rho,$$

hence

$$l_1 \subset \rho.$$

The same argument gives

$$l_2 \subset \rho.$$

But the distinct intersecting lines l_1, l_2 determine a unique plane. They already lie in π , and now they also lie in ρ . Therefore

$$\pi = \rho,$$

contradicting the normalized hypothesis.

The file also contains the symmetric package

```
hilbert_XI15_intersection_points_off_opposite_planes
```

which proves both $B \notin \rho$ and the analogous source-style statement $E \notin \pi$. The main normalized proof needs only the first half.

15.9.2 XI.11 constructs the normal

With $B \notin \rho$ established, the production proof invokes


```
euclid_proposition_11_11
```

and obtains a line n and a foot G satisfying

$$B \in n, \quad n \perp \rho \text{ at } G.$$

This is packaged by

```
hilbert_XI15_normal_from_intersection_to_other_plane
```

and then combined with the two reference-line constructions in

```
hilbert_XI15_normal_and_reference_lines
```

to produce

$$q_1, q_2 \subset \rho, \quad G \in q_1 \cap q_2,$$

with

$$q_i = m_i \quad \text{or} \quad m_i \parallel q_i.$$

15.9.3 Recovering $l_i \parallel q_i$

The original hypotheses give

$$l_i \parallel m_i.$$

If $q_i = m_i$, there is nothing to prove.

In the genuine parallel branch

$$m_i \parallel q_i,$$

the proof needs strict transitivity of spatial parallelism. This is slightly subtle because project parallelism is irreflexive: a line is not parallel to itself.

The helper

```
hilbert_XI15_spaceParallel_transitive_distinct
```

therefore assumes explicitly that the outer lines are distinct.

Its proof again separates two spatial configurations:

- if l, m, q lie in one plane, Hilbert Group IV gives uniqueness of the parallel to m through a point;
- if the three lines do not lie in one common plane, XI.9 supplies the required spatial transitivity.

In XI.15 the required distinctness $l_i \neq q_i$ follows from $B \notin \rho$. Indeed, if $l_i = q_i$, then $B \in q_i \subset \rho$.

Thus the proof obtains

$$l_1 \parallel q_1, \quad l_2 \parallel q_2.$$

15.9.4 Definition XI.3 gives the two local perpendicularities

Since

$$n \perp \rho \text{ at } G$$

and

$$q_i \subset \rho, \quad G \in q_i,$$

Definition XI.3 is represented directly by `HilbertLinePerpendicularPlaneAt.perpendicular_to_line`. It gives

$$n \perp q_1 \text{ at } G, \quad n \perp q_2 \text{ at } G.$$

The remaining problem is to transport these perpendicularities from q_i to the parallel lines l_i .

15.9.5 The formal I.29 replacement

The central transport theorem is

```
hilbert_XI15_perpendicular_transfer_across_parallel
```

and its line-plane wrapper is

```
hilbert_XI15_plane_normal_perpendicular_to_parallel_line
```

Suppose

$$l \parallel q,$$

the transversal n meets q at G and l at B , and

$$q \perp n \text{ at } G.$$

The spatial parallelism $l \parallel q$ supplies a common carrier plane σ . Since $B, G \in n$ and $B \neq G$, the whole transversal n also lies in σ .

The proof now passes to

$$\text{PlaneGeo}(\sigma).$$

Through B , it constructs a line p perpendicular to n . Hence

$$p \perp n \text{ at } B, \quad q \perp n \text{ at } G.$$

The feet are distinct. The neutral XI.8 helper representing the I.27 mechanism therefore gives

$$p \cap q = \emptyset.$$

But l is also disjoint from q , and both l and p pass through B . Hilbert Group IV gives uniqueness of the parallel through B :

$$p = l.$$

Therefore

$$l \perp n \text{ at } B,$$

and by symmetry

$$n \perp l \text{ at } B.$$

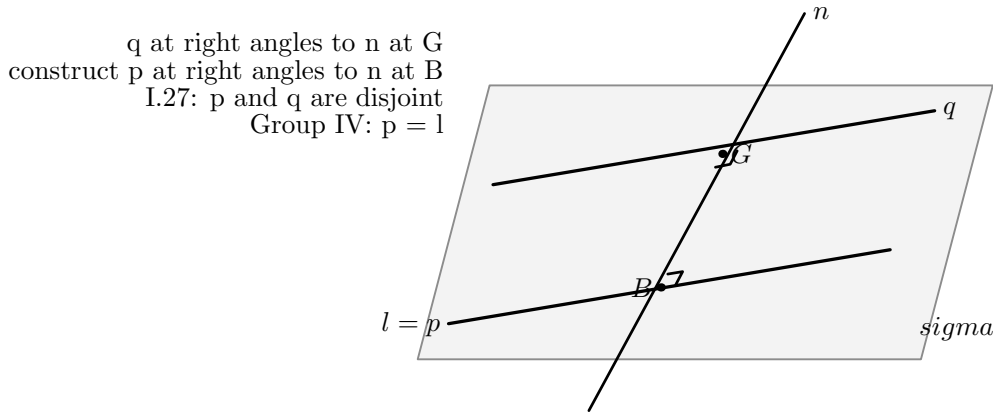


Figure 15.3: The formal replacement for Euclid’s I.29 step, carried out in the common plane σ of $l \parallel q$. The line q is perpendicular to the transversal n at G . A perpendicular p to n is erected at B . Neutral I.27 gives $p \cap q = \emptyset$; Group IV then identifies p with the already given parallel l . The final lower carrier is therefore labelled $l = p$.

This is exactly the logical role of Euclid’s I.29, but factored according to the current Hilbert architecture:

$$\text{I.27} + \text{Group IV uniqueness} \implies \text{right-angle transport across parallels.}$$

15.9.6 XI.4 makes the normal perpendicular to π

Applying the preceding transport twice gives

$$n \perp l_1 \text{ at } B, \quad n \perp l_2 \text{ at } B.$$

The lines l_1, l_2 are distinct and meet at B , so the production theorem

```
euclid_proposition_11_4
```

gives

$$n \perp \pi \text{ at } B.$$

This is one of the cleanest stages of the formal proof: the substantial work has already been normalized into exactly the XI.4 interface.

15.9.7 XI.14 closes the proof

From XI.11 we still have

$$n \perp \rho \text{ at } G.$$

From XI.4 we have

$$n \perp \pi \text{ at } B.$$

The normalized theorem assumes $\pi \neq \rho$, so XI.14 applies directly:

$$\pi \parallel \rho.$$

The final line of the normalized proof is therefore an actual invocation of

```
euclid_proposition_11_14
```

rather than a new plane-separation argument.

15.9.8 The source-facing wrapper

The public theorem assumes that the four displayed lines do not all lie in one plane. It first derives

$$\pi \neq \rho.$$

If the two planes were equal, then l_1, l_2, m_1, m_2 would all lie in π , contradicting the source hypothesis.

The public theorem then calls the normalized core.

The separate theorem

```
euclid_proposition_11_15_intersecting_pairs
```

adds the explicit second intersection point and line distinctness appearing in Euclid's wording. Since those data are not needed internally, this source-style theorem is intentionally only a wrapper.

15.10 Wyler-Style Flat Reconstruction

The Wyler reconstruction of XI.15 keeps the public Euclidean statement but changes the internal proof discipline. The main spatial states are recorded by exact generated-flat identities before the metric argument is used.

The source hypothesis that the four displayed lines are not all contained in one plane first gives

$$\pi \neq \rho.$$

The flat layer then records the global rank-three certificate

$$\text{Join}(\pi, \rho) = E^3.$$

As in XI.14, this certificate is informative but not by itself sufficient: two distinct planes generate the whole ambient three-space whether or not they are parallel. The conclusion still requires the perpendicularity and plane-separation part of the argument.

The first substantial use of the Wyler vocabulary occurs in the proof that the intersection point B of l_1, l_2 lies outside ρ . Suppose instead that $B \in \rho$. From

$$l_1 \parallel m_1$$

let τ_1 be a carrier plane of the parallel pair. Since $B \in l_1$ and l_1, m_1 are disjoint, $B \notin m_1$. Therefore the same generators m_1 and B determine both τ_1 and ρ :

$$\text{Join}(m_1, \{B\}) = \text{carrier}(\tau_1),$$

$$\text{Join}(m_1, \{B\}) = \text{carrier}(\rho).$$

Hence the two plane carriers coincide and $l_1 \subset \rho$. Repeating the same argument for l_2, m_2 gives $l_2 \subset \rho$. Since the distinct intersecting lines l_1, l_2 generate π , this forces

$$\pi = \rho,$$

a contradiction.

Thus the Wyler proof does not treat “the plane through a line and an external point is unique” as a proposition-local geometric black box. It compares the exact joins generated by the same data.

After XI.11 constructs

$$B \in n, \quad n \perp \rho \text{ at } G,$$

the same principle is used a second time. For each reference line m_i , there are two cases.

If $G \in m_i$, take

$$q_i = m_i.$$

Otherwise XI.12 constructs through G a line q_i parallel to m_i inside a generated slice σ_i , together with exact certificates

$$\text{Join}(m_i, \{G\}) = \text{carrier}(\sigma_i),$$

$$\text{Join}(m_i, q_i) = \text{carrier}(\sigma_i).$$

But $m_i \subset \rho$ and $G \in \rho$, so the same first join is also

$$\text{Join}(m_i, \{G\}) = \text{carrier}(\rho).$$

Consequently $q_i \subset \rho$. The containment of the constructed line in the intended plane is therefore recovered from equality of generated flats, not from a separate ad hoc plane-identification lemma.

The next step is the strict spatial parallelism

$$l_i \parallel q_i.$$

Again the proof separates the two possible flat configurations. If l_i, m_i, q_i lie in one plane, Hilbert Group IV gives the required uniqueness of the parallel through a point. If no common plane contains the three lines, Proposition XI.9 gives the spatial transitivity directly. The outer-line distinctness required by strict parallelism follows from $B \notin \rho$, because $B \in l_i$ whereas $q_i \subset \rho$.

Only after these incidence and flat identifications are complete does the proof enter the metric layer. Since

$$n \perp \rho \text{ at } G$$

and $q_i \subset \rho$ passes through G , Definition XI.3 gives

$$n \perp q_i \text{ at } G.$$

For each parallel pair $l_i \parallel q_i$, their common carrier plane contains both feet B, G of the transversal n , hence contains the whole line n . Inside the induced plane geometry one erects through B a line perpendicular to n . Neutral I.27 proves that this auxiliary line is disjoint from q_i , and Group IV identifies it with l_i . Therefore

$$n \perp l_i \text{ at } B.$$

The complete Wyler-style structure is therefore

```

four displayed lines not coplanar
|
v
pi != rho
|
+--> Join(pi,rho) = E^3          [rank-three certificate]
|
v
assume B in rho
|
Join(m1,{B}) = carrier(tau1) = carrier(rho)
Join(m2,{B}) = carrier(tau2) = carrier(rho)
|
l1,l2 subset rho
|
pi = rho --> contradiction
|
v
B notin rho
|
XI.11
v
n perp rho at G
|
generated slices through G
|
q1,q2 subset rho
m1 = q1 or m1 || q1
m2 = q2 or m2 || q2
|
Group IV / XI.9
|
l1 || q1, l2 || q2
|
local PlaneGeo metric transport
|
n perp l1, n perp l2
|
XI.4
v
n perp pi
|
XI.14
v
pi || rho

```

This decomposition illustrates the characteristic proof discipline of the current Wyler route. Generated-flat equalities identify the spatial configuration first; metric and Euclidean principles are invoked only after the relevant carriers have been fixed. The final theorem remains Euclid XI.15, but the proof exposes a separate incidence skeleton on which the perpendicularity argument acts.

15.11 Lean Formalization

The production file imports:

```
import CGJteamLab.Proposition11_14
import CGJteamLab.Proposition11_12
import CGJteamLab.Proposition11_9
import CGJteamLab.Proposition11_8
import CGJteamLab.Proposition11_4
```

Mathematically, the proof also calls XI.11 through the imported Book XI dependency chain.

The proposition-local helpers fall into four conceptual groups.

Spatial incidence normalization.

```
hilbert_XI15_spaceLinesParallel_symm
hilbert_XI15_intersection_point_off_other_plane
hilbert_XI15_intersection_points_off_opposite_planes
```

The XI.11/I.31 construction package.

```
hilbert_XI15_parallel_or_equal_through_point_in_plane
hilbert_XI15_normal_from_intersection_to_other_plane
hilbert_XI15_normal_and_reference_lines
```

Parallel and perpendicular transport.

```
hilbert_XI15_perpendicular_transfer_across_parallel
hilbert_XI15_plane_normal_perpendicular_to_parallel_line
hilbert_XI15_spaceParallel_transitive_distinct
```

Final theorem layer.

```
euclid_proposition_11_15_normalized
euclid_proposition_11_15
euclid_proposition_11_15_intersecting_pairs
```

This grouping reflects stable mathematical responsibilities, not the chronology of the test files used during development.

15.12 Hidden Formal Data

15.12.1 The XI.11 starting point must be outside ρ

Euclid begins XI.11 from B without a separate argument that $B \notin \rho$. The production proof derives this by a nontrivial spatial incidence argument. This is the first hidden obligation in XI.15.

15.12.2 The I.31 point may already lie on the reference line

The foot G may lie on m_1 , on m_2 , or on neither. The formal construction cannot assume the generic diagrammatic case. The equal-or-parallel helper packages all possibilities without changing the outer proof.

15.12.3 Spatial parallelism carries a plane witness

The predicate `HilbertSpaceLinesParallel` is not merely a disjointness relation. It also stores a common carrier plane. XI.15 uses that carrier plane to create the `PlaneGeo` slice in which perpendicularity transport is proved.

15.12.4 Spatial parallelism is strict

Because parallel lines are disjoint, the project relation is irreflexive. A transitivity lemma therefore needs the extra condition that the two outer lines are distinct. In XI.15 that condition is not assumed separately; it is derived from $B \notin \rho$.

15.12.5 The feet B and G must be distinct

The perpendicularity-transport theorem requires two distinct feet on the transversal n . If $B = G$, then B would lie on both parallel lines l and q , contradicting their disjointness.

15.12.6 The second source intersection is redundant

Euclid describes both pairs as intersecting. The public production theorem needs only the first explicit intersection $B \in l_1 \cap l_2$. The second intersection is mathematically compatible with the theorem but is not used by the proof.

This is a genuine strengthening, not an omission from the formal configuration. The source-style corollary records the literal Euclidean data separately.

15.13 Relationship with Earlier Book XI Propositions

XI.15 is unusually transparent as a consumer of the preceding Book XI infrastructure.

XI.4. This proposition is used exactly at the classical place. Once the formal proof has obtained

$$n \perp l_1, \quad n \perp l_2$$

at their common point B , XI.4 upgrades the pair to

$$n \perp \pi.$$

XI.8. The final theorem XI.8 is not the target of the proof, but XI.15 reuses its neutral helper for two coplanar lines perpendicular to the same transversal at distinct points. This is the formal I.27 engine inside the replacement for I.29.

XI.9. XI.9 appears in two roles. It is part of the classical proof of $BA \parallel GH$ and $BC \parallel GK$. Formally, it is used inside the strict transitivity helper when three spatial parallel lines do not have one common carrier plane.

XI.11. XI.11 supplies the normal from B to the second plane. Before it can be called, XI.15 proves the hidden hypothesis $B \notin \rho$.

XI.12. The plane-local parallel-through-point helper developed in XI.12 is reused in the generic branch $G \notin m_i$. XI.15 is therefore the first later proposition where that helper becomes genuine reusable infrastructure rather than merely proposition-local convenience.

XI.14. XI.14 is the terminal theorem. Once the same line n is known to be perpendicular to both π and ρ , the conclusion of XI.15 is immediate.

XI.10 is conceptually related because it starts from the same kind of configuration—two pairs of corresponding parallel lines in distinct spatial plane slices. But XI.15 does not need XI.10's directed segment parallelism. Orientation of selected rays matters for equality of angles in XI.10; it is irrelevant to plane parallelism in XI.15.

15.14 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical Book I dependencies are I.31 and I.29.

The formal reconstruction preserves the role of I.31 but makes its precondition explicit. The generic branch uses the existing parallel-through-point construction in a plane slice; the exceptional branch uses the original line itself.

The classical I.29 step is represented more structurally. The current proof uses:

- a neutral I.27-type theorem to show that two lines perpendicular to the same transversal at distinct points are disjoint;
- Hilbert Group IV to identify the unique parallel through the required point.

Thus XI.15 genuinely uses Euclidean parallel geometry. This is visible already in the production signature:

```
[HSE : HilbertSpaceEuclidean Geo]
```

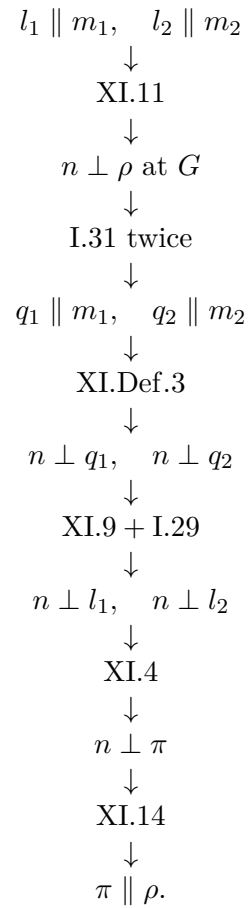
The use of Group IV is mathematically substantive, not merely inherited module noise.

No continuity principle is required. No coordinate system, vector, scalar product, numerical angle measure, or trigonometric function occurs in the reconstruction.

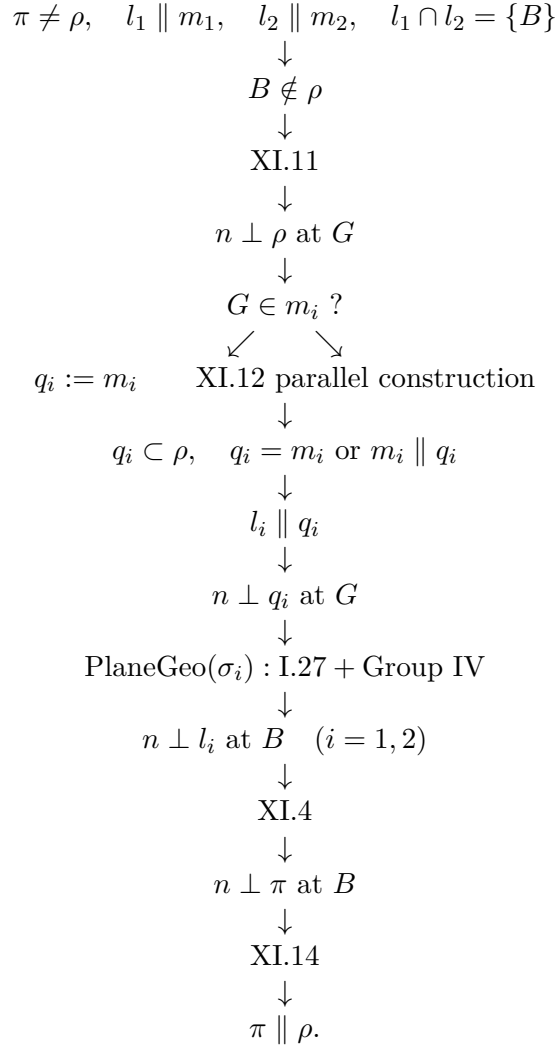
Book Zero contributes only the already established synthetic infrastructure used transitively by the right-angle and planar construction layers. XI.15 introduces no new proposition-specific geometric axiom.

15.15 Dependency Summary

The classical DAG is:



The formal DAG is:



The formal proof is longer around the two classical line constructions and the I.29 step, because the diagram suppresses exceptional incidence and carrier-plane information. It is shorter at the public boundary: the source-style second intersection point E is not required by the geometric core.

15.16 Conclusion

Proposition XI.15 is a particularly successful example of the Book XI architecture reaching maturity.

Euclid's top-level idea survives almost unchanged. Construct a normal to the second plane, transfer its right angles to the two lines of the first plane, use XI.4 to make it a normal to that plane as well, and conclude by XI.14 that the planes are parallel.

What formalization adds is not a different geometry but a precise account of the information carried by the classical diagram. It proves that the first intersection point is genuinely outside the second plane; it repairs the hidden exceptional case in the two I.31 constructions; it records the carrier planes implicit in spatial parallelism; and it factors the I.29 transport into neutral perpendicular geometry plus Group IV uniqueness.

The resulting proof also clarifies the relation between source statement and formal theorem.

Euclid presents two intersecting pairs, but the second explicit intersection is not needed by the geometric core. The production theorem therefore proves a slightly stronger statement, while a thin source-style corollary retains the literal Euclidean configuration.

At the level of the Book XI DAG, XI.15 is an important closure theorem: XI.11 supplies the common normal, XI.12 supplies the reusable parallel-through-point construction, XI.8 and XI.9 supply the parallel transport machinery, XI.4 upgrades two right angles to a line-plane perpendicularity, and XI.14 converts the common normal into the final parallelism of planes. No coordinates or metric algebra are needed; the proof remains entirely synthetic.

Chapter 16

Proposition XI.16

16.1 Introduction

Proposition XI.16 is the basic section theorem for parallel planes. If two planes π and ρ are parallel and a third plane σ meets both of them, then the two section lines cut out in σ are parallel.

Classically the argument is very short. The sections are straight lines by Book XI, and if those lines met in a point P , then P would belong to both π and ρ , contradicting the assumption that the two planes are parallel.

The formal reconstruction preserves exactly this idea, but in a more explicit incidence form. The theorem does not start with already named section lines. Instead, it begins with two witness points

$$A \in \pi \cap \sigma, \quad B \in \rho \cap \sigma,$$

constructs the corresponding intersection lines, and then proves that those lines are parallel.

This makes XI.16 an important architectural proposition in Book XI. It is genuinely spatial, because three ambient planes are involved, but it is also conceptually clean: the entire production proof lives in the incidence layer. No order, no congruence, and no Euclidean parallel axiom are used in the geometric core.

16.2 Classical Configuration

Let π and ρ be parallel planes. Let a third plane σ cut them. Write

$$l = \pi \cap \sigma, \quad m = \rho \cap \sigma.$$

Euclid's claim is that the two section lines l and m are parallel.

In the formal theorem the two intersections are represented through explicit points:

$$A \in \pi \cap \sigma, \quad B \in \rho \cap \sigma.$$

These witnesses guarantee that σ actually meets each of the two reference planes.

Diagrammatic audit. A single figure is sufficient here. The proposition has only one stable geometric state: two parallel ambient planes cut by a third plane. The proof contains no separate degenerate branch and no later planar reduction requiring an additional diagram.

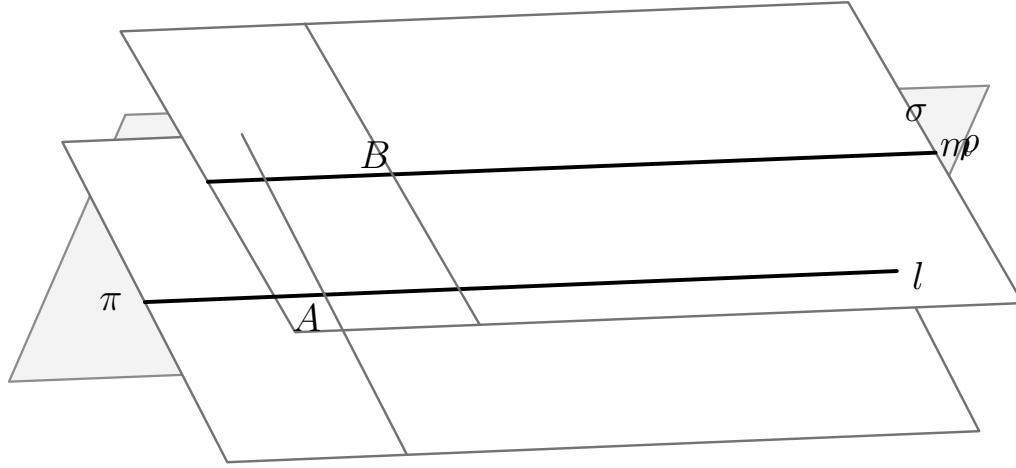


Figure 16.1: A spatial rendering of the configuration of Proposition XI.16. The displayed reference plane is σ . The two parallel section lines $l = \pi \cap \sigma$ and $m = \rho \cap \sigma$ lie in σ and are drawn in black. The ambient planes π and ρ are shown as light auxiliary planes cutting the reference plane along those lines.

16.3 Statement

Theorem 4 (Euclid XI.16). *If two parallel planes are cut by a plane, then their common sections are parallel straight lines.*

The current production declaration is:

```

theorem euclid_proposition_11_16
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (pi rho sigma : S.Plane)
  (A B : Geo.Point)
  (hPlanesParallel :
    HilbertSpacePlanesParallelIncidence Geo pi rho)
  (hApi : S.OnPlane A pi)
  (hAsigma : S.OnPlane A sigma)
  (hBrho : S.OnPlane B rho)
  (hBsigma : S.OnPlane B sigma) :
  exists l m : Geo.Line,
    H.OnLine A l /\
    H.OnLine B m /\
    (forall X : Geo.Point,
      (S.OnPlane X pi /\ S.OnPlane X sigma) <=>
      H.OnLine X l) /\
    (forall X : Geo.Point,
      (S.OnPlane X rho /\ S.OnPlane X sigma) <=>
      H.OnLine X m) /\
    HilbertSpaceLinesParallel Geo l m

```

Two features should be emphasized.

First, the public theorem is not stated with pre-existing section lines. Instead it *constructs* the

two lines l and m from the data of the plane intersections.

Second, the theorem records the intersections pointwise. For every point X , lying on the section line is equivalent to lying in both relevant planes. This is the precise formal replacement for the informal notation

$$l = \pi \cap \sigma, \quad m = \rho \cap \sigma.$$

16.4 Classical Proof

Let the common sections of π and ρ with σ be the straight lines l and m .

Assume, for contradiction, that l and m meet at a point P . Since $P \in l$ and l is the section of π with σ , the point P lies in π . Since $P \in m$ and m is the section of ρ with σ , the point P also lies in ρ .

Thus the two planes π and ρ have a common point. But parallel planes do not meet. Therefore the lines l and m do not meet. Because both lie in the same plane σ , they are parallel.

The classical dependency pattern is therefore

$$\begin{array}{c} \pi \parallel \rho, \quad l = \pi \cap \sigma, \quad m = \rho \cap \sigma \\ \downarrow \\ \text{assume } l \cap m \neq \emptyset \\ \downarrow \\ P \in \pi \cap \rho \\ \downarrow \\ \perp \\ \downarrow \\ l \parallel m. \end{array}$$

The proof is neutral: no Euclidean parallel postulate and no metric argument enter.

16.5 What is genuinely three-dimensional here?

The conclusion concerns two lines in the single plane σ , but the proposition is still genuinely spatial.

The genuinely three-dimensional data are:

- the two ambient planes π and ρ ;
- the cutting plane σ ;
- the two ambient intersections $\pi \cap \sigma$ and $\rho \cap \sigma$;
- the contradiction step that turns a hypothetical meeting point of the section lines into a common point of the two ambient planes.

What is striking in XI.16 is what *does not* happen. There is no reduction to $\text{PlaneGeo}(\sigma)$, no use of planar order, and no appeal to planar congruence or angle theory. The proposition stays entirely in the ambient incidence layer.

16.6 Formal Statement and Parallelism Predicate

The new neutral infrastructure for parallel planes is

```
def HilbertSpacePlanesParallelIncidence
  [S : HilbertSpacePrimitive Geo]
  (pi rho : S.Plane) : Prop :=
  Not
    (exists X : Geo.Point,
      S.OnPlane X pi /\
      S.OnPlane X rho)
```

Thus the meaning of

$$\pi \parallel \rho$$

in the production proof is simply: the two planes have no common point.

The target parallelism for lines is the existing spatial predicate

```
HilbertSpaceLinesParallel Geo l m
```

which packages two facts:

- the two lines lie in a common plane;
- the two lines are disjoint.

In XI.16 the common plane is exactly the cutting plane σ .

16.7 Proof Architecture of the Reconstruction

The formal proof has four stable stages.

16.7.1 Stage 1: show that the intersections are nontrivial

To apply the general intersection theorem for two planes, the proof must first show

$$\pi \neq \sigma, \quad \rho \neq \sigma.$$

These facts are not hypotheses of the theorem. They are derived from the parallel-plane assumption by contradiction.

If $\pi = \sigma$, then the point B , already known to lie in $\rho \cap \sigma$, would lie in both π and ρ , contradicting parallelism. The same argument with A gives $\rho \neq \sigma$.

16.7.2 Stage 2: construct the two section lines

With those inequalities available, the proof applies the ambient incidence theorem


```
hilbert_plane_intersection_line
```

twice:

- to π and σ , using the point A ;
- to ρ and σ , using the point B .

This yields lines l and m , together with:

- $A \in l$,
- $B \in m$,
- $l \subseteq \pi$ and $l \subseteq \sigma$,
- $m \subseteq \rho$ and $m \subseteq \sigma$,
- the pointwise equivalences

$$X \in l \iff X \in \pi \cap \sigma, \quad X \in m \iff X \in \rho \cap \sigma.$$

This is the key formal replacement for the informal geometric phrase “the common section of two planes.”

16.7.3 Stage 3: prove that the two section lines are disjoint

Assume a point P lies on both l and m . Because

$$l \subseteq \pi, \quad m \subseteq \rho,$$

the point P lies in both π and ρ . But this contradicts

```
hPlanesParallel :
  HilbertSpacePlanesParallelIncidence Geo pi rho
```

So the lines l and m are disjoint.

16.7.4 Stage 4: assemble spatial line parallelism

The final step is immediate. Both l and m lie in σ , and they are disjoint. Therefore they are parallel in the sense of

```
HilbertSpaceLinesParallel Geo l m
```

This closes the theorem.

16.8 Wyler-Style Flat Reconstruction

The independent Wyler reconstruction of XI.16 keeps exactly the same geometric content but changes the preferred formal state. Instead of storing the two plane intersections only as pointwise equivalences, it records them as exact equalities of carriers.

Write

$$C(\pi) = \text{carrier}(\pi), \quad C(l) = \text{carrier}(l).$$

For the first section line the Wyler proof obtains

$$C(\pi) \cap C(\sigma) = C(l),$$

and for the second one

$$C(\rho) \cap C(\sigma) = C(m).$$

In Lean these identities are expressed through

```
HilbertPlaneCarrier3D
HilbertLineCarrier3D
hilbertPlaneCarrier3D_inter_eq_lineCarrier
```

The last theorem is a flat-facing wrapper around the neutral three-dimensional plane-intersection theorem. Thus the Wyler version does not introduce a different incidence theorem. It changes the level at which the result is retained: the intersection is preserved as a reusable *meet identity*.

The proof first derives

$$\pi \neq \sigma, \quad \rho \neq \sigma$$

from the parallel-plane hypothesis, exactly as in the canonical proof. It then constructs l and m and records

$$C(l) = C(\pi) \cap C(\sigma), \quad C(m) = C(\rho) \cap C(\sigma).$$

These identities immediately recover all four containments

$$l \subseteq \pi, \quad l \subseteq \sigma, \quad m \subseteq \rho, \quad m \subseteq \sigma$$

whenever they are needed.

The decisive disjointness argument becomes especially transparent in this language. If a point P belonged to both section lines, then

$$P \in C(l) \cap C(m).$$

Using the two meet identities gives

$$P \in C(\pi) \cap C(\sigma) \cap C(\rho) \cap C(\sigma),$$

hence in particular

$$P \in C(\pi) \cap C(\rho),$$

contradicting the parallelism of the original planes. Therefore

$$C(l) \cap C(m) = \emptyset.$$

The final line-parallelism statement still uses the ordinary spatial predicate

```
HilbertSpaceLinesParallel Geo l m
```

because both section lines lie in the common cutting plane σ and are disjoint there.

The complete Wyler-style structure is therefore

```
pi || rho
|
sigma meets pi and rho
|
+--> C(pi) inter C(sigma) = C(l)
|
+--> C(rho) inter C(sigma) = C(m)
|
assume P in C(l) inter C(m)
|
P in C(pi) inter C(rho)
|
contradiction
|
C(l) inter C(m) = empty
|
l,m subset sigma
|
l || m
```

The production theorem is

```
euclid_proposition_11_16_wyler
```

and it coexists with the canonical theorem because the Wyler route now has its own public name.

For XI.16 the difference between the two reconstructions is therefore not a difference in mathematical strength. The canonical proof stores a pointwise description of the two intersections; the Wyler proof stores exact carrier equalities. The latter is more compositional for subsequent spatial arguments, because a later theorem may rewrite directly with the meet identity instead of reconstructing membership information point by point. This is precisely why the Wyler form becomes useful in XI.17.

16.9 Lean Formalization

The production proof is notably small because the geometry is concentrated in the two applications of the plane-intersection constructor. The theorem begins by deriving

```
have hPiSigma : Ne pi sigma := ...
have hRhoSigma : Ne rho sigma := ...
```

Then it constructs the first section line:

```

cases
  hilbert_plane_intersection_line
    (Geo := Geo)
    pi sigma
    hPiSigma
    A hApi hAsigma with
| intro l hLData =>

```

and similarly for the second one:

```

cases
  hilbert_plane_intersection_line
    (Geo := Geo)
    rho sigma
    hRhoSigma
    B hBrho hBsigma with
| intro m hMData =>

```

The contradiction argument is then expressed as:

```

have hDisjoint :
  HilbertLinesDisjoint Geo l m := by
intro hMeet
cases hMeet with
| intro P hPData =>
  have hP1 : H.OnLine P l := hPData.1
  have hPm : H.OnLine P m := hPData.2
  apply hPlanesParallel
  refine Exists.intro P ?_
  exact And.intro (hlpi P hP1) (hmrho P hPm)

```

Finally, line parallelism is assembled in the common plane σ .

16.10 Hidden Formal Data

Euclid's statement suppresses several obligations that Lean must make explicit.

- The cutting plane does not arrive with named section lines; the proof must construct them.
- The theorem needs witness points A and B showing that σ actually meets the two parallel planes.
- The inequalities

$$\pi \neq \sigma, \quad \rho \neq \sigma$$

are required before the plane-intersection theorem can be used.

- “Common section” must be encoded formally by a universal characterization of points on the line.
- Parallelism of lines is not primitive equality of directions; it is a structured statement combining coplanarity with disjointness.

These are not artificial complications. They are the formal content hidden by the classical diagram.

16.11 Relationship with Earlier Book XI Propositions

Classically, XI.16 belongs to the cluster of early Book XI results about plane sections and parallelism. In textbook language one would naturally refer back to the fact that the common section of two planes is a straight line.

The current formal reconstruction, however, does not import any earlier Book XI production theorem. Instead it uses the general ambient incidence theorem

```
hilbert_plane_intersection_line
```

from the 3D Hilbert infrastructure.

So the mathematical content agrees with Euclid, but the formal dependency DAG is shorter at the Book XI level and deeper at the infrastructure level.

16.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No proposition from Book I is used here, and there is no dependence on the Book Zero segment infrastructure.

The proof lives entirely in the 3D incidence layer:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
```

The key infrastructure items are:

```
HilbertSpacePlanesParallelIncidence
hilbert_plane_intersection_line
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel
```

This makes XI.16 one of the cleanest examples in Book XI of a theorem whose formal content is purely spatial incidence.

16.13 Dependency Summary

Classical.

$$\begin{array}{c}
 \pi \parallel \rho, \quad \sigma \text{ cuts } \pi, \rho \\
 \downarrow \\
 l = \pi \cap \sigma, \quad m = \rho \cap \sigma \\
 \downarrow \\
 \text{if } l \cap m \neq \emptyset, \text{ then } \pi \cap \rho \neq \emptyset \\
 \downarrow \\
 \perp \\
 \downarrow \\
 l \parallel m.
 \end{array}$$

Formal.

$$\begin{array}{c}
 \text{Hilbert incidence in 3-space} \\
 \downarrow \\
 \text{HilbertSpacePlanesParallelIncidence} \\
 \downarrow \\
 hPiSigma, hRhoSigma \\
 \downarrow \\
 \text{hilbert_plane_intersection_line twice} \\
 \downarrow \\
 l, m, \text{ pointwise section characterizations} \\
 \downarrow \\
 \text{HilbertLinesDisjoint} \\
 \downarrow \\
 \text{HilbertSpaceLinesParallel} \\
 \downarrow \\
 \text{euclid_proposition_11_16.}
 \end{array}$$

The conceptual difference is clear. Euclid speaks directly about the common sections. Lean first constructs them from the ambient incidence interface and then proves their disjointness.

16.14 Conclusion

Proposition XI.16 is mathematically elementary, but formally very informative. It shows that an early Book XI result can be reconstructed in a completely synthetic and purely incidence-theoretic way.

This theorem is also structurally important for later propositions. In particular, both the canonical and the Wyler reconstruction of XI.17 use XI.16 as their spatial section engine, while the more delicate ratio argument of XI.17 is handled in a separate proportion layer. XI.16 therefore marks a clean boundary: it solves the ambient intersection geometry once and for all, without importing any metric or affine machinery. The Wyler version makes that boundary explicit by storing the two sections as exact carrier meets.

Chapter 17

Proposition XI.17

17.1 Introduction

Proposition XI.17 is the first proposition in Book XI where the spatial incidence theory and the classical theory of proportion interact in an essential way. Three parallel planes cut two transversals in the orders

$$A - E - B, \quad C - F - D,$$

and Euclid proves

$$AE : EB = CF : FD.$$

The classical construction introduces a point O on AD in the middle plane. Proposition XI.16 then produces two pairs of parallel section lines, and Proposition VI.2 is applied in two different planar triangles:

$$AE : EB = AO : OD, \quad AO : OD = CF : FD.$$

Proposition V.11 gives the final equality of ratios.

The formal reconstruction keeps this classical architecture, but it makes three hidden issues explicit.

First, the point O is not assumed: it is constructed synthetically from the spatial configuration by XI.16 and an ambient Pasch argument. Second, the two applications of VI.2 occur in two explicit induced plane geometries, not in the ambient 3-space. Third, segment proportion is represented by a purely synthetic raw relation based on right-triangle witnesses and angle congruence; no numerical length, segment division, multiplication, real coordinates, or quotient segment arithmetic is used.

One limitation remains visible in the production theorem: planar VI.2 is supplied as an explicit hypothesis

```
hVI2 :  
  forall tau : S.Plane,  
    HilbertVI2Raw (PlaneGeo Geo tau)
```

rather than derived here from the primitive Hilbert axioms. Thus XI.17 closes the spatial reconstruction around VI.2 without pretending that the whole planar proportion theory has already been reconstructed from first principles.

17.2 Classical Configuration

Let π_0, π_1, π_2 be three parallel planes. Two transversals meet them at

$$A, E, B$$

and

$$C, F, D,$$

respectively, with

$$A - E - B, \quad C - F - D.$$

Thus

$$A, C \in \pi_0, \quad E, F \in \pi_1, \quad B, D \in \pi_2.$$

The conclusion is

$$AE : EB = CF : FD.$$

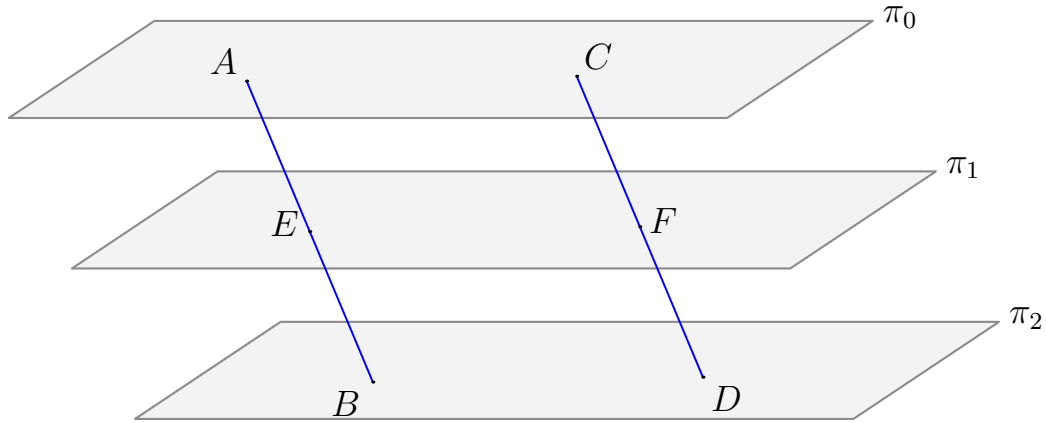


Figure 17.1: The classical input configuration of Proposition XI.17. The three light planes π_0, π_1, π_2 are pairwise parallel. The two genuinely spatial transversals are shown in blue and meet the planes in the orders $A - E - B$ and $C - F - D$.

The formal statement records pairwise parallelism explicitly:

$$\pi_0 \parallel \pi_1, \quad \pi_1 \parallel \pi_2, \quad \pi_0 \parallel \pi_2.$$

It also records the two betweenness relations directly.

17.3 Statement

Theorem 5 (Euclid XI.17). *If two straight lines are cut by parallel planes, they are cut in the same ratio.*

In the notation of Figure 17.1,

$$AE : EB = CF : FD.$$

The current production declaration is:


```

theorem euclid_proposition_11_17
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (hVI2 :
    forall tau : S.Plane,
      HilbertVI2Raw (PlaneGeo Geo tau))
  (pi0 pi1 pi2 : S.Plane)
  (A E B C F D : Geo.Point)
  (hParallel01 :
    HilbertSpacePlanesParallelIncidence Geo pi0 pi1)
  (hParallel12 :
    HilbertSpacePlanesParallelIncidence Geo pi1 pi2)
  (hParallel02 :
    HilbertSpacePlanesParallelIncidence Geo pi0 pi2)
  (hApi0 : S.OnPlane A pi0)
  (hCpi0 : S.OnPlane C pi0)
  (hEpi1 : S.OnPlane E pi1)
  (hFpi1 : S.OnPlane F pi1)
  (hBpi2 : S.OnPlane B pi2)
  (hDpi2 : S.OnPlane D pi2)
  (hAEB : Geo.Between A E B)
  (hCFD : Geo.Between C F D)
  (hBD : Ne B D)
  (hAC : Ne A C) :
  HilbertSpaceSegmentProportionRaw
    Geo
    A E
    E B
    C F
    F D

```

The hypotheses hBD and hAC are explicit nondegeneracy conditions. They guarantee that the two outer-plane connector lines BD and AC used in Euclid's construction are genuine lines determined by distinct points.

17.4 Classical Proof

Join AC , BD , and AD . Let AD meet the middle plane π_1 at O , and join EO and OF .

The plane through E, B, D, O cuts the two parallel planes π_1, π_2 . By XI.16 their common sections are parallel:

$$EO \parallel BD.$$

Similarly, the plane through A, O, F, C cuts the two parallel planes π_0, π_1 , so XI.16 gives

$$AC \parallel OF.$$

Now work first in triangle ABD . Since E lies on AB , O lies on AD , and

$$EO \parallel BD,$$

VI.2 gives

$$AE : EB = AO : OD.$$

Next work in triangle ADC . Since O lies on AD , F lies on DC , and

$$OF \parallel AC,$$

VI.2 gives

$$AO : OD = CF : FD.$$

By V.11,

$$AE : EB = CF : FD.$$

This is exactly the classical proof pattern recorded in the standard text of XI.17: XI.16 is used twice, VI.2 is used twice, and V.11 composes the two proportions.

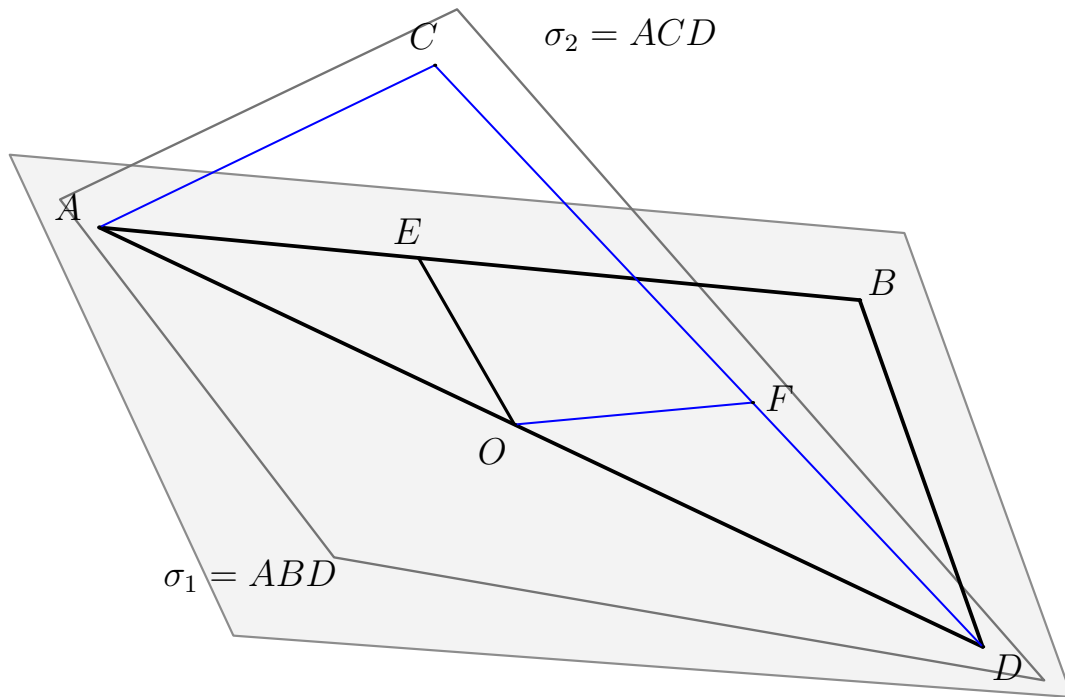


Figure 17.2: Euclid's auxiliary construction. The reference plane $\sigma_1 = ABD$ is shown as the main light plane: $E \in AB$, $O \in AD$, and $EO \parallel BD$. The second plane $\sigma_2 = ACD$ meets σ_1 along AD ; the genuinely spatial part leaving the reference plane is shown in blue, with $F \in CD$ and $OF \parallel AC$. The two VI.2 applications therefore occur in different planar slices sharing the line AD .

Diagrammatic audit. The first figure records the input state before Euclid's construction. The second records the stable proof state after the auxiliary point O and the two section-parallelisms have been obtained. No separate figure is needed for the raw ratio witnesses: they belong to the formal representation of proportion, not to a new Euclidean configuration.

17.5 The Auxiliary Point O

In Euclid's text the phrase "let AD meet the plane π_1 at O " is geometrically natural but formally nontrivial. The production theorem does not take O as input. It constructs O in

```
euclid_proposition_11_17_construct_0
```

with conclusion

```
exists O : Geo.Point,
  Geo.Between A O D /\
  S.OnPlane O pi1
```

The construction is synthetic.

First, the proof constructs the plane

$$\sigma = \text{plane}(A, B, D).$$

Because $A - E - B$, the point E also lies in σ .

Next XI.16 is applied to the parallel planes π_1 and π_2 , cut by σ . It produces the section line through E and the section line BD , and proves these two lines parallel. Hence the section through E is disjoint from BD .

The line through E in the middle-plane section meets the open side AB of triangle ABD . Ambient Pasch in σ says that it must exit through AD or BD . The BD alternative is impossible because the section line is parallel to BD . Therefore it meets AD at a point O , and the section characterization supplied by XI.16 shows

$$O \in \pi_1.$$

This is one of the main formal gains of the reconstruction: a diagrammatic intersection that Euclid takes as immediate is factored into XI.16 plus the order-theoretic Pasch mechanism.

17.6 What is genuinely three-dimensional here?

XI.17 has two distinct spatial layers.

The first is the original arrangement of three parallel planes and two transversals. The point O cannot be constructed inside one fixed pre-existing plane; its existence depends on the interaction between the middle plane and the auxiliary plane ABD .

The second is the construction of the two section-parallelisms

$$EO \parallel BD, \quad AC \parallel OF.$$

Each is produced by XI.16 from a pair of ambient parallel planes and a cutting plane.

Only after these spatial steps are complete does the proof reduce to two planar slices. The first VI.2 calculation is performed in a plane τ_1 containing EO and BD ; the second is performed in a plane τ_2 containing AC and OF . These need not be the same plane.

Thus the formal proof does not “flatten” the whole XI.17 configuration. It moves from ambient 3-space to two different local planar geometries and then transports the resulting proportions back to space.

17.7 Synthetic Representation of Ratio and Proportion

The production proof deliberately avoids segment arithmetic.

A concrete ratio $PQ : RS$ is represented by a right-triangle witness: one leg is congruent to PQ , the other to RS , and the corresponding acute angle represents the ratio. Equality of ratios is then expressed by congruence of the defining acute angles.

The planar relation is

```
HilbertSegmentProportionRaw
```

and its ambient 3D counterpart is

```
HilbertSpaceSegmentProportionRaw
```

The raw VI.2 interface is

```
def HilbertVI2Raw
  [HilbertIncidence Geo]
  [HilbertCongruence Geo] : Prop :=
  forall X E Y F Z : Geo.Point,
    Geo.Between X E Y ->
    Geo.Between X F Z ->
    Geo.Parallel E F Y Z ->
    HilbertSegmentProportionRaw
      Geo X E E Y X F F Z /\
      HilbertSegmentProportionRaw
        Geo E Y X E F Z X F
```

This is exactly the proportional consequence needed from VI.2, stated without division or numerical lengths.

The final V.11-type composition is supplied by transitivity of the raw proportion relation.

17.8 Spatial / Planar Architecture Used

The ambient layer of XI.17 uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpacePlanesParallelIncidence
HilbertSpaceLinesParallel
```

The key spatial theorem is

```
euclid_proposition_11_16
```

and the auxiliary point construction also uses ambient Pasch through

```
HilbertSpaceOrder.pasch_in_plane
```

The two VI.2 applications live in explicit induced plane geometries

```
PlaneGeo Geo tau1
PlaneGeo Geo tau2
```

The resulting planar proportions are transported back to ambient space by

```
hilbertSegmentProportionRaw_to_space
```

and finally composed by

```
hilbertSpaceSegmentProportionRaw_trans
```

This respects the permanent Book XI instance discipline: there is no global ambient planar order or planar congruence instance. Planar proportion arguments occur only after an explicit plane slice has been selected.

17.9 Proof Architecture of the Reconstruction

The production proof is organized into five mathematical stages.

17.9.1 Stage 1: construct O

The theorem

```
euclid_proposition_11_17_construct_0
```

constructs

$$A - O - D, \quad O \in \pi_1.$$

As explained above, this uses the plane ABD , XI.16, and Pasch.

17.9.2 Stage 2: obtain the two XI.16 parallelisms

The spatial core constructs four lines:

$$EO, \quad BD, \quad AC, \quad OF$$

and proves

$$EO \parallel BD, \quad AC \parallel OF.$$

The corresponding production helper is

```
euclid_proposition_11_17_spatial_core
```

This stage is purely about ambient incidence, order, and the section theorem XI.16.

17.9.3 Stage 3: first VI.2 slice

From the spatial parallelism $EO \parallel BD$, the proof opens a witness plane τ_1 containing both lines. The points A, E, B, O, D are placed in the induced geometry $\text{PlaneGeo}(\tau_1)$.

The local VI.2 interface then yields

$$AE : EB = AO : OD.$$

This planar proportion is transported back to ambient 3-space.

17.9.4 Stage 4: second VI.2 slice

The parallelism $AC \parallel OF$ similarly provides a plane τ_2 . Inside $\text{PlaneGeo}(\tau_2)$, the proof uses the reversed betweenness orientations

$$D - O - A, \quad D - F - C$$

to fit the exact argument order of the raw VI.2 interface.

After applying VI.2, the formal proportion is reversed back to the desired orientation by

```
hilbertSegmentProportionRaw_reverse_all
```

and transported to ambient space. The result is

$$AO : OD = CF : FD.$$

17.9.5 Stage 5: V.11

The two ambient raw proportions

$$AE : EB = AO : OD, \quad AO : OD = CF : FD$$

are composed by

```
hilbertSpaceSegmentProportionRaw_trans
```

to obtain

$$AE : EB = CF : FD.$$

17.10 Wyler-Style Flat Reconstruction

The independent Wyler reconstruction of XI.17 now has a genuinely affine endpoint. Its spatial layer is expressed by flat/carrier calculus, while its final proportional statement is no longer obtained by crossing into `HilbertVI2Raw`. Instead, the Wyler path introduces its own affine notion of equality of division positions on transversals.

The first generated slice is

$$\sigma_1 = \text{plane}(A, B, D).$$

Because $A - E - B$ and the auxiliary point O satisfy $A - O - D$, both E and O lie in σ_1 . Wyler XI.16 gives the exact section identities

$$C(\pi_1) \cap C(\sigma_1) = C(l_{EO}), \quad C(\pi_2) \cap C(\sigma_1) = C(l_{BD}),$$

together with

$$l_{EO} \parallel l_{BD}.$$

The incidences $O \in l_{EO}$ and $D \in l_{BD}$ are recovered by direct rewriting with these carrier equalities.

The second generated slice is

$$\sigma_2 = \text{plane}(A, C, D).$$

Likewise,

$$C(\pi_0) \cap C(\sigma_2) = C(l_{AC}), \quad C(\pi_1) \cap C(\sigma_2) = C(l_{OF}),$$

and XI.16 yields

$$l_{AC} \parallel l_{OF}.$$

Again, $C \in l_{AC}$ and $F \in l_{OF}$ follow from the meet identities.

This spatial skeleton is packaged by

```
euclid_XI17_wyler_first_sections
euclid_XI17_wyler_sections
```

The auxiliary point O is constructed in the same carrier-first spirit. In the plane

$$\sigma = \text{plane}(A, B, D),$$

the middle plane π_1 cuts σ in a line l_E , while π_2 cuts it in the line l_{BD} . Wyler XI.16 gives

$$l_E \parallel l_{BD}.$$

Since l_E meets the open side AB at E , Pasch forces it to meet AD or BD . The second branch contradicts the disjointness encoded by line parallelism. Hence there is O with

$$A - O - D, \quad O \in \pi_1.$$

This is implemented by

```
euclid_proposition_11_17_construct_O_wyler
```

17.10.1 Affine division in the Wyler path

The key additional module is

```
CGJteamLab.Wyler.HilbertWylerProportion
```

It introduces the elementary affine intercept configuration

```
HilbertWylerDivisionStep
```

with geometric data

$$X - E - Y, \quad X - F - Z, \quad EF \parallel YZ.$$

This configuration expresses that E and F occupy the same affine division position on the two transversals. No numerical lengths, segment quotients, field operations, coordinates, or angle ratios occur in this definition.

The generated relation

```
HilbertWylerSameDivision
```

acts on ordered triples of points. Informally,

$$(A, E, B) \sim (C, F, D)$$

means that E divides AB in the same affine position in which F divides CD . The relation contains the elementary parallel-section step and is closed under reflexivity, symmetry, transitivity, and simultaneous reversal of the two ordered triples.

The first section pair gives directly

$$(A, E, B) \sim (A, O, D),$$

because

$$A - E - B, \quad A - O - D, \quad EO \parallel BD.$$

For the second pair the proof reverses the two transversals:

$$D - O - A, \quad D - F - C, \quad OF \parallel AC.$$

Hence

$$(D, O, A) \sim (D, F, C).$$

Simultaneous reversal gives

$$(A, O, D) \sim (C, F, D),$$

and transitivity yields the affine XI.17 conclusion

$$(A, E, B) \sim (C, F, D).$$

The production theorem is

```
euclid_proposition_11_17_wyler_affine
```


and its conclusion is

```
HilbertWylerSameDivision Geo A E B C F D
```

This theorem is the genuine Wyler endpoint of XI.17. It does not assume `HilbertVI2Raw`, does not use `HilbertSpaceCongruence`, and does not represent ratios by congruent acute angles. The older theorem

```
euclid_proposition_11_17_wyler
```

remains available as a compatibility result whose target is the previously used synthetic relation

```
HilbertSpaceSegmentProportionRaw
```

Thus the project retains the connection with the canonical Book V/VI proportion API without making that API foundational for the Wyler proof.

The complete Wyler dependency pattern is now

```
pi0 || pi1 || pi2
|
+--> sigma1 = plane(A,B,D)
|   |
|   +--> C(pi1) inter C(sigma1) = C(1EO)
|   +--> C(pi2) inter C(sigma1) = C(1BD)
|   +--> 1EO || 1BD
|
+--> sigma2 = plane(A,C,D)
|   |
|   +--> C(pi0) inter C(sigma2) = C(1AC)
|   +--> C(pi1) inter C(sigma2) = C(1OF)
|   +--> 1AC || 1OF
|
+--> Pasch in plane(A,B,D)
|   |
|   +--> O on AD and O in pi1
|
+--> HilbertWylerDivisionStep
|   |
|   +--> (A,E,B) ~ (A,O,D)
|   +--> (A,O,D) ~ (C,F,D)
|   |
|   transitivity
|   |
|   (A,E,B) ~ (C,F,D)
```

The characteristic Wyler discipline is therefore complete in XI.17: first determine the generated flats and their exact meets, then express the remaining proportional content as an affine invariant of parallel sections. There is no remaining VI.2 boundary inside the affine theorem itself.

17.11 Lean Formalization

The public theorem is intentionally short at the top level. It first calls

```
euclid_proposition_11_17_construct_0
```

to obtain O , then delegates the complete remaining argument to

```
euclid_proposition_11_17_with_0
```

The latter handles two conceptually different tasks:

- reconstruction of Euclid’s section lines and parallelisms;
- the two plane-local VI.2 calculations and their transport back to ambient space.

This separation is mathematically useful. It isolates the genuinely spatial existence problem for O from the proportion calculation that begins once O is available.

17.12 Hidden Formal Data

Several pieces of information visible in Euclid’s diagram become explicit proof obligations.

- Pairwise parallelism of the three planes is given explicitly.
- The order relations $A - E - B$ and $C - F - D$ are hypotheses, not inferred from a picture.
- The connector points A, C and B, D are required to be distinct: $A \neq C$ and $B \neq D$.
- The point O must be proved to exist, to satisfy $A - O - D$, and to lie in the middle plane.
- The planes ABD and ACD must be constructed from verified noncollinearity.
- XI.16 supplies section lines pointwise; their incidences with E, O, B, D, A, C, F must be recovered from those characterizations.
- Each spatial line-parallelism carries a witness plane, and that plane becomes the correct `PlaneGeo` slice for VI.2.
- The second VI.2 application requires orientation normalization before the resulting proportion can be put into Euclid’s final order.

These obligations explain why the formal DAG is longer than Euclid’s visible proof even though the mathematical strategy is the same.

17.13 Relationship with Earlier Book XI Propositions

XI.16 is the decisive Book XI dependency, both classically and formally.

Classically it is invoked twice:

$$EO \parallel BD, \quad AC \parallel OF.$$

Formally it plays an additional role in the construction of O : the middle and lower parallel planes are cut by the plane ABD , and the resulting section line through E is shown parallel to BD . This disjointness is what eliminates one of the two Pasch exits.

Thus the formal use of XI.16 is slightly richer than the visible textbook proof:

- once to construct O ;
- twice to obtain the parallel section pairs used in the ratio proof.

No Wyler flat calculus is imported. XI.17 depends on the canonical synthetic XI.16 in the main Book XI route.

17.14 Relationship with Books V and VI / Hilbert Infrastructure

The classical proportion dependencies are exactly:

$$\text{VI.2}, \quad \text{V.11}.$$

In the current formal architecture they appear in synthetic raw form.

VI.2 is represented by the explicit plane-local assumption

```
forall tau : S.Plane,
  HilbertVI2Raw (PlaneGeo Geo tau)
```

and V.11 by transitivity of

```
HilbertSpaceSegmentProportionRaw
```

The right-triangle representation of ratio uses Hilbert congruence and right-angle infrastructure. The production audit also confirms that no new XI.17-specific geometric axiom was introduced. However, because VI.2 is an explicit hypothesis, the theorem should not yet be described as a derivation of Euclidean proportion theory from primitive Hilbert incidence, order, and congruence alone.

For the canonical theorem and the compatibility theorem `euclid_proposition_11_17_wyler`, this Book V/VI layer remains present. The new affine theorem `euclid_proposition_11_17_wyler_affine` bypasses it entirely: its conclusion is `HilbertWylerSameDivision`, obtained from parallel-section steps, reversal, and transitivity.

No Group IV typeclass appears in the theorem. The spatial section geometry itself remains independent of the Euclidean parallel postulate.

17.15 Dependency Summary

Classical.

$$\begin{array}{c}
 \pi_0 \parallel \pi_1 \parallel \pi_2, \quad A - E - B, \quad C - F - D \\
 \downarrow \\
 \text{join } AC, BD, AD; \quad O = AD \cap \pi_1 \\
 \downarrow \\
 \text{XI.16 : } EO \parallel BD, \quad AC \parallel OF \\
 \downarrow \\
 \text{VI.2 in } \triangle ABD : AE : EB = AO : OD \\
 \downarrow \\
 \text{VI.2 in } \triangle ADC : AO : OD = CF : FD \\
 \downarrow \\
 \text{V.11} \\
 \downarrow \\
 AE : EB = CF : FD.
 \end{array}$$

Formal.

```

Hilbert 3D incidence/order/congruence
+ HilbertVI2Raw on every PlaneGeo
|
v
Proposition11_16 + ambient Pasch
|
v
euclid_proposition_11_17_construct_0
|
v
euclid_proposition_11_17_spatial_core
|
v
PlaneGeo tau1: AE:EB = AO:OD
|
v
PlaneGeo tau2: AO:OD = CF:FD
|
v
transport to ambient raw proportions
|
v
hilbertSpaceSegmentProportionRaw_trans
|
v
euclid_proposition_11_17

```

The main conceptual difference is the treatment of O . Euclid names the intersection directly. Lean proves its existence from the spatial incidence/order infrastructure before the classical VI.2 chain can begin.

17.16 Conclusion

Proposition XI.17 is the point where the Book XI reconstruction begins to combine genuinely spatial geometry with a statement that is classically expressed as a proportion.

The canonical route follows Euclid closely: XI.16 supplies the two section parallelisms, Pasch constructs the auxiliary point O , and two plane-local applications of VI.2 are composed by a V.11-type transitivity theorem. This route ends in `HilbertSpaceSegmentProportionRaw`.

The Wyler reconstruction now separates the same mathematics more sharply. Its three-dimensional content is expressed entirely through generated planes, exact carrier meets, parallel section lines, and Pasch. Its final invariant is the affine relation `HilbertWylerSameDivision`. The elementary parallel-section configuration gives equality of division position directly, and the proof of XI.17 consists of two such steps joined through the auxiliary transversal AD .

Thus XI.17 no longer contains the previously identified foundational gap in the Wyler path. The affine theorem does not assume VI.2 and does not use metric segment-ratio machinery. The older raw-proportion theorem remains only as a compatibility interface with the canonical Book V/VI development.

This gives XI.17 a particularly useful architectural role in the project: it shows that the flat/carrier calculus is strong enough not only to reconstruct the spatial configuration but also to carry the natural affine invariant associated with three parallel planes and their transversals.

Chapter 18

Proposition XI.18

18.1 Introduction

Proposition XI.18 is a particularly useful theorem for the present project because it can now be read in two different formal reconstructions of the same three-dimensional geometry.

The project contains two production proofs:

```
CGJteamLab/Proposition11_18.lean
CGJteamLab/Wyler/Proposition11_18.lean
```

with main theorems

```
euclid_proposition_11_18
euclid_proposition_11_18_wyler
```

The first is the direct Hilbert 3D route. The second is the Wyler route, retained deliberately as an independent reconstruction and as a laboratory for the later passage to E^4 . The purpose is not to replace Hilbert's three-dimensional language by Wyler's language. Hilbert remains the primary synthetic description of E^3 ; the Wyler proof is a second realization of the same theorem, useful because it exposes flat-intersection and transport mechanisms in a form that can be compared with higher-dimensional geometry.

The two proofs share the theorem statement and the source-faithful interpretation of Euclid XI.Def.4. That common definition has therefore been moved out of the Wyler directory into the neutral module

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

The Wyler compatibility module

```
CGJteamLab/Wyler/HilbertWylerPlanePerpendicular.lean
```

no longer owns the definition; it re-exports the neutral interface together with the lower Wyler

helper layers needed by the older development.

This distinction is mathematically important. A modern textbook may define two planes to be perpendicular when one of them contains a line perpendicular to the other. Such a definition would make XI.18 almost tautological. The present reconstruction instead follows Euclid XI.Def.4 and therefore preserves the theorem content.

18.2 Euclid's Statement

Euclid states Proposition XI.18 as follows:

If a straight line is at right angles to any plane, then all the planes through it are also at right angles to the same plane.

Let l be perpendicular to the plane π at O , and let ρ be any plane containing l . The hypotheses are

$$l \perp \pi, \quad l \subset \rho.$$

The conclusion is

$$\rho \perp \pi.$$

In the formal reconstruction, $\rho \perp \pi$ is not the assertion that ρ merely contains one normal to π . It is the directed XI.Def.4 condition attached to the exact common section of the two planes.

18.3 Classical Configuration

The classical configuration contains:

- the reference plane π ;
- a line l perpendicular to π at O ;
- an arbitrary plane ρ containing l ;
- the common section

$$s = \rho \cap \pi.$$

Since $O \in l \subset \rho$ and $O \in \pi$, the point O lies on the section s .

To verify XI.Def.4, take an arbitrary point $F \in s$ and an arbitrary line $g \subset \rho$ perpendicular to s at F . The goal is

$$g \perp \pi.$$

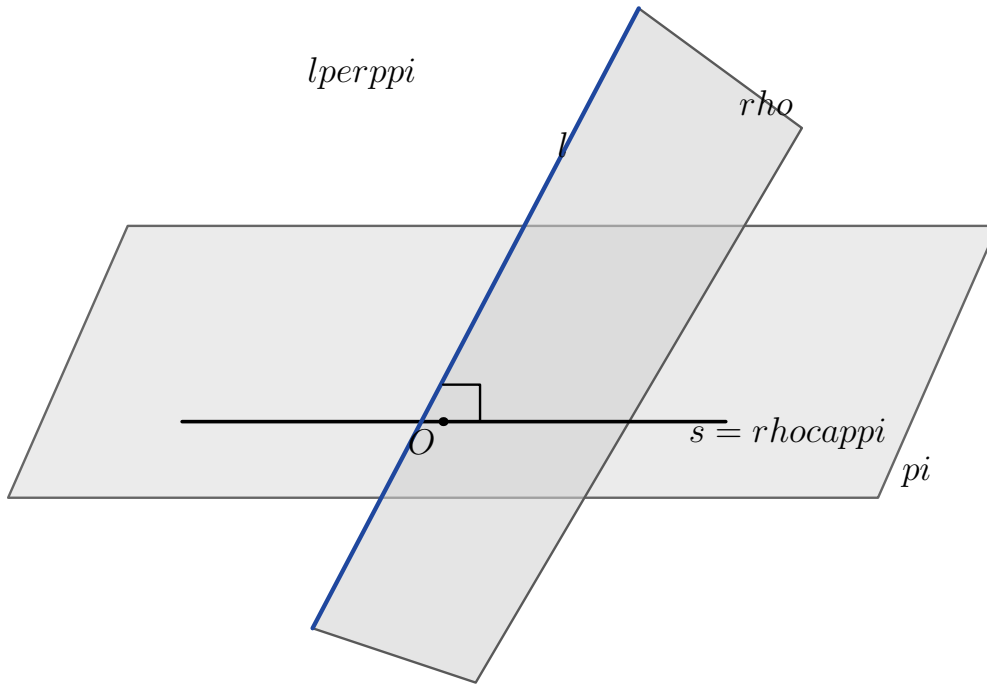


Figure 18.1: The classical configuration of Proposition XI.18. The line l is perpendicular to the reference plane π at O , and the arbitrary plane ρ contains l . Their common section is $s = \rho \cap \pi$. Proposition XI.18 proves that ρ is perpendicular to π in the sense of XI.Def.4.

18.4 Euclid's Classical Proof

Euclid begins with a straight line AB perpendicular to the reference plane and lets an arbitrary plane DE pass through AB . The line CE is the intersection of DE with the reference plane.

Choose an arbitrary point F on CE , and inside the plane DE draw FG perpendicular to CE .

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through its foot. In particular,

$$AB \perp CE.$$

Thus AB and FG lie in one plane and are both perpendicular to CE . Euclid obtains

$$AB \parallel FG.$$

Then XI.8 gives

$$FG \perp \pi.$$

Since F and FG were arbitrary, XI.Def.4 yields

$$DE \perp \pi.$$

Finally, the plane DE was arbitrary among the planes through AB .

18.5 Classical Dependency Pattern

The source proof is summarized by

$$\begin{array}{c}
 l \perp \pi, \quad l \subset \rho \\
 \Downarrow \\
 s = \rho \cap \pi \\
 \Downarrow \\
 l \perp s \\
 \Downarrow \\
 F \in s, \quad g \subset \rho, \quad g \perp s \\
 \Downarrow \\
 l \parallel g \\
 \Downarrow \\
 \text{XI.8 : } \quad g \perp \pi \\
 \Downarrow \\
 \text{XI.Def.4 : } \quad \rho \perp \pi.
 \end{array}$$

The formal proof exposes one case suppressed by the classical diagram. If $F = O$, then the two perpendiculars need not be distinct parallel lines. They may coincide. The correct formal alternative is therefore

$$l = g \quad \text{or} \quad l \parallel g.$$

18.6 The Shared XI.Def.4 Interface

The decisive architectural change in the present revision is that perpendicularity of planes is no longer owned by the Wyler layer.

The neutral module

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

defines

```
HilbertSpacePlanesPerpendicularAlong Geo rho pi s
HilbertSpacePlanesPerpendicular Geo rho pi
```

The first predicate records four facts:

- $s \subset \rho$;
- $s \subset \pi$;
- s is exactly the common point set of ρ and π ;
- every line $g \subset \rho$ perpendicular to s at F is perpendicular to π at F .

The second predicate adds

$$\rho \neq \pi$$

and existentially packages the section line.

This interface is shared because it belongs to the mathematics of Euclid XI.Def.4, not to either proof technology. Both XI.18 proofs therefore prove the same target predicate.

The relation is directed. In

$$\text{HilbertSpacePlanesPerpendicular}(\rho, \pi)$$

the first plane is the plane in which the test lines are chosen; the second is the plane to which those test lines must be perpendicular.

18.7 What Is Genuinely Three-Dimensional Here?

XI.18 contains both a planar core and a genuinely spatial shell.

The genuinely spatial ingredients are:

- line–plane perpendicularity $l \perp \pi$;
- containment $l \subset \rho$;
- the exact plane meet $s = \rho \cap \pi$;
- XI.8, which transports line–plane perpendicularity along a spatially parallel line.

The comparison of l and g is planar. Both lie in ρ , and both are perpendicular to s . This part naturally takes place in

PlaneGeo Geo rho

Thus the proposition has the conceptual form

$$\text{3D incidence} \quad + \quad \text{2D perpendicular comparison} \quad + \quad \text{3D transport}.$$

That decomposition is common to both proofs, even though the two routes package the components differently.

18.8 Formal Statements

18.8.1 Direct Hilbert 3D theorem

The direct Hilbert theorem is

```

theorem euclid_proposition_11_18
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l : Geo.Line)
  (O : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi O)
  (hlrho :
    HilbertLineInPlane Geo l rho) :
  HilbertSpacePlanesPerpendicular Geo rho pi

```

Its production file is

```
CGJteamLab/Proposition11_18.lean
```

18.8.2 Wyler theorem

The Wyler theorem has the same hypotheses and conclusion:

```

theorem euclid_proposition_11_18_wyler
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l : Geo.Line)
  (O : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi O)
  (hlrho :
    HilbertLineInPlane Geo l rho) :
  HilbertSpacePlanesPerpendicular Geo rho pi

```

Its production file is

```
CGJteamLab/Wyler/Proposition11_18.lean
```

The equality of the theorem types is deliberate. The comparison concerns the proof architecture, not two different mathematical statements.

18.9 Direct Hilbert 3D Reconstruction

The direct proof imports

```
CGJteamLab.Hilbert3DPlanePerpendicular
CGJteamLab.Proposition11_12
```

and has no direct import from the Wyler directory.

18.9.1 Stage H1: recover the common section

From

$$l \perp \pi \text{ at } O$$

the proof extracts

$$O \in l, \quad O \in \pi.$$

Since $l \subset \rho$,

$$O \in \rho.$$

The planes are shown to be distinct. If $\rho = \pi$, then $l \subset \pi$, contradicting the fact that a line perpendicular to a plane cannot lie in that plane.

The spatial incidence theorem

```
hilbert_plane_intersection_line
```

then gives a line s satisfying

$$O \in s, \quad s \subset \rho, \quad s \subset \pi,$$

together with the exactness certificate

$$X \in \rho \cap \pi \iff X \in s.$$

18.9.2 Stage H2: reduce to a planar problem

Because $s \subset \pi$ and $O \in s$,

$$l \perp s \quad \text{at } O.$$

Now take an arbitrary XI.Def.4 test line

$$g \subset \rho, \quad g \perp s \text{ at } F.$$

The direct Hilbert helper is

```
hilbert_XI18_coplanar_perpendiculars_eq_or_parallel
```

and proves

$$l = g \quad \text{or} \quad l \parallel g.$$

Its same-foot sublemma is

```
hilbert_XI18_coplanar_perpendiculars_same_foot_equal
```

The distinct-foot branch reuses the direct Hilbert XI.8 helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```

so that the planar comparison remains part of the main Hilbert Book XI development.

18.9.3 Stage H3: equality branch

If $l = g$, the proof still has to identify the chosen XI.Def.4 foot F with the original normal foot O .

Since

$$F \in g = l \quad \text{and} \quad F \in s \subset \pi,$$

the direct XI.12 helper

```
hilbert_XI12_perpendicular_foot_unique
```

gives

$$F = O.$$

The original certificate $l \perp \pi$ can then be reused.

18.9.4 Stage H4: parallel branch

If

$$l \parallel g,$$

the direct Hilbert theorem

```
euclid_proposition_11_8
```

produces a point D such that

$$g \perp \pi \quad \text{at } D.$$

But $F \in g$ and $F \in \pi$, so XI.12 foot uniqueness gives

$$F = D.$$

Hence

$$g \perp \pi \quad \text{at } F.$$

This proves the universal clause of XI.Def.4.

18.10 Wyler Reconstruction

The Wyler production proof imports

```
CGJteamLab.Wyler.HilbertWylerPerpendicularParallel
CGJteamLab.Wyler.Proposition11_8
```

The high-level geometric chain is intentionally the same as in the direct Hilbert proof:

$$\text{exact section} \longrightarrow l \perp s \longrightarrow (l = g \text{ or } l \parallel g) \longrightarrow g \perp \pi.$$

The difference lies in which lower modules realize those steps.

18.10.1 Wyler coplanar bridge

The principal Wyler helper is

```
hilbert_coplanar_perpendiculars_to_same_line_eq_or_parallel_wyler
```

in

```
CGJteamLab/Wyler/HilbertWylerPerpendicularParallel.lean
```

Its same-foot branch uses the Wyler metric core, while its distinct-foot branch packages the nonintersection argument through the Wyler XI.14 right-angle helper. The helper has no `HilbertSpaceEuclidean` Geo assumption.

18.10.2 Wyler XI.8 transport

In the parallel branch the Wyler theorem calls the Wyler XI.8 reconstruction. The mathematical role is identical:

$$l \parallel g, \quad l \perp \pi \implies g \perp \pi.$$

The transported foot is again identified with F .

18.10.3 Flat viewpoint

The Wyler route is useful because it separates the carrier configuration from the metric test. The exact common section is treated as a meet object, while right-angle and congruence information are supplied by metric helper layers.

Schematically,

$$\begin{array}{c} \text{Meet}(\rho, \pi) = s \\ \Downarrow \\ l \perp s \\ \Downarrow \\ g \subset \rho, \quad g \perp s \\ \Downarrow \\ g = l \text{ or } g \parallel l \\ \Downarrow \\ g \perp \pi. \end{array}$$

The flat language does not replace XI.Def.4. It identifies the carrier on which XI.Def.4 is tested.

18.11 The Common Formal Core

Both proofs ultimately verify the same arbitrary test configuration.

The two proofs therefore agree on the visible mathematical skeleton. They differ below that skeleton.

18.12 Hilbert versus Wyler: Proof Diff

The purpose of maintaining two proofs is to expose a mathematically useful difference rather than to duplicate code for its own sake.

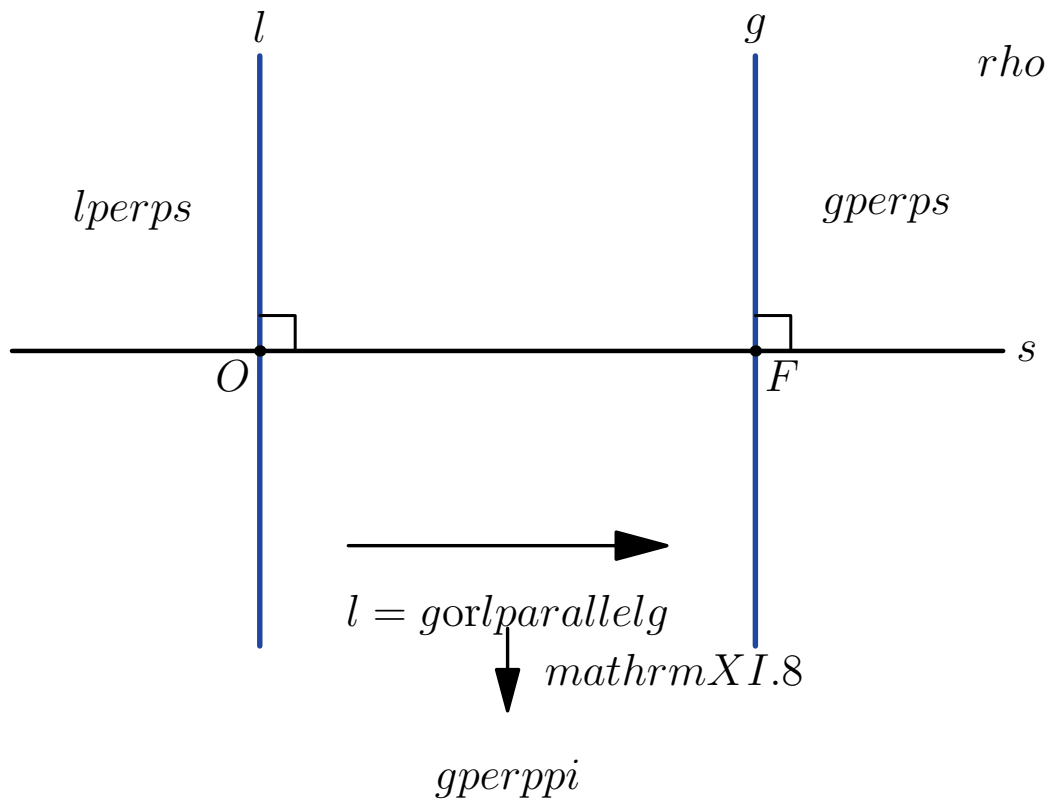


Figure 18.2: The common formal core of XI.18 inside the plane ρ . The given normal l and an arbitrary test line g are both perpendicular to the common section s . If their feet coincide, then $l = g$; if the feet are distinct, then the lines are parallel. XI.8 handles the parallel branch and foot uniqueness identifies the resulting normal foot with the prescribed point F .

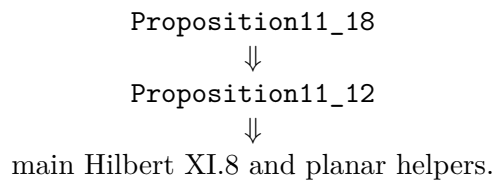
18.12.1 Shared nodes

The following nodes are common:

- the theorem statement;
- the neutral XI.Def.4 target;
- the exact section $s = \rho \cap \pi$;
- the reduction $l \perp \pi \Rightarrow l \perp s$;
- the alternative $l = g$ or $l \parallel g$;
- XI.8-type transport in the parallel branch;
- uniqueness of the line–plane perpendicular foot.

18.12.2 Direct Hilbert realization

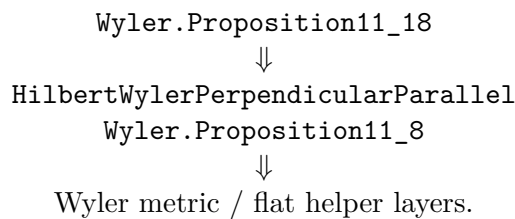
The direct route uses the main Book XI infrastructure:



The geometry is expressed directly in Hilbert’s point–line–plane language.

18.12.3 Wyler realization

The Wyler route uses



The geometry is reorganized so that flat incidence and metric transport can be inspected separately.

18.12.4 What the diff tells us

For XI.18 the mathematical proof is stable across both routes. The main difference is therefore architectural:

- Hilbert gives a natural direct proof in the native language of E^3 ;

- Wyler decomposes the same proof into carrier/flat and metric layers;
- the planar middle step survives essentially unchanged;
- the three-dimensional meet and transport interfaces are the places where higher-dimensional analogues should be investigated.

This is exactly the kind of comparison for which the double Book XI development was introduced.

18.13 Ambient Space and PlaneGeo

The global variables are ambient:

```
pi rho : S.Plane
l s g   : Geo.Line
O F     : Geo.Point
```

The induced geometry

```
PlaneGeo Geo rho
```

appears when the problem becomes intrinsically planar.

This separation is not merely a Lean implementation detail. It records the mathematical fact that the local statement

$$l \perp s, \quad g \perp s$$

is a theorem about two lines in one plane, while the interpretation of l and g as normals to π is ambient three-dimensional geometry.

18.14 Hidden Formal Data

Several facts suppressed in the classical picture are explicit in both Lean proofs.

18.14.1 The two planes are distinct

The target predicate includes

$$\rho \neq \pi.$$

This is proved from the normality hypothesis: if $\rho = \pi$, then $l \subset \pi$, impossible for a line perpendicular to π .

18.14.2 The common section is exact

The proof does not merely produce a line lying in both planes. It records

```
forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X pi) <-> H.OnLine X s
```

This exactness certificate becomes important immediately in XI.19.

18.14.3 XI.Def.4 is universal

The proof handles arbitrary

```
g : Geo.Line
F : Geo.Point
```

with $g \subset \rho$ and $g \perp s$ at F . Existence of a single normal inside ρ would not suffice.

18.14.4 The equality branch is real

The correct formal statement is

$$l = g \vee l \parallel g.$$

This is a genuine geometric distinction caused by the possible equality $F = O$.

18.14.5 The XI.8 foot is existential

XI.8 produces a foot D , while XI.Def.4 requires perpendicularity at the chosen point F . Foot uniqueness is therefore an essential final step:

$$F = D.$$

18.15 Relationship with Earlier and Later Book XI Propositions

XI.3 / plane-intersection infrastructure. Both routes need an exact common section of two distinct planes.

XI.4 metric core. The same-foot comparison ultimately depends on uniqueness of a perpendicular direction in an induced plane.

XI.8. This is the decisive Euclidean transport step in both proofs.

XI.12. The direct Hilbert route uses the production foot-uniqueness helper from XI.12.

XI.14. The Wyler coplanar bridge uses a neutral two-right-angles impossibility helper extracted in the Wyler XI.14 infrastructure.

XI.19. XI.19 consumes the XI.Def.4 interface established here. In particular, it uses exact common sections and extracts normal lines from perpendicular-plane certificates.

Thus XI.18 is the point where the perpendicular-plane interface becomes part of the reusable Book XI API.

18.16 Relationship with Hilbert's Axiom Groups

The local comparison of two perpendicular lines uses the neutral part of the Hilbert geometry:

- Group I: incidence;
- Group II: order;
- Group III: congruence and right-angle structure.

The full theorem also assumes

HilbertSpaceEuclidean Geo

because the proper parallel branch invokes XI.8.

Thus the Euclidean contribution is localized:

$$l \parallel g, \quad l \perp \pi \quad \Longrightarrow \quad g \perp \pi.$$

The exact plane meet and the same-perpendicular comparison are conceptually separate from this Group IV step.

18.17 Classical DAG and the Two Formal DAGs

18.17.1 Euclid

$$\begin{array}{c}
 l \perp \pi, \quad l \subset \rho \\
 \Downarrow \\
 s = \rho \cap \pi \\
 \Downarrow \\
 l \perp s \\
 \Downarrow \\
 g \subset \rho, \quad g \perp s \\
 \Downarrow \\
 l \parallel g \\
 \Downarrow \\
 \text{XI.8} \\
 \Downarrow \\
 g \perp \pi \\
 \Downarrow \\
 \text{XI.Def.4} \\
 \Downarrow \\
 \rho \perp \pi.
 \end{array}$$

18.17.2 Direct Hilbert 3D

$$\begin{array}{c}
 l \perp \pi \text{ at } O, \quad l \subset \rho \\
 \Downarrow \\
 O \in \rho \cap \pi, \quad \rho \neq \pi \\
 \Downarrow \\
 s = \rho \cap \pi \text{ exactly} \\
 \Downarrow \\
 l \perp s \text{ at } O \\
 \Downarrow \\
 \forall g, F: g \subset \rho, g \perp s \text{ at } F \\
 \Downarrow \\
 \text{hilbert_XI18_coplanar_perpendiculars_eq_or_parallel} \\
 \Downarrow \\
 l = g \vee l \parallel g \\
 \Downarrow \\
 \begin{cases} l = g: & \text{XI.12 foot uniqueness,} \\ l \parallel g: & \text{main XI.8 + XI.12 foot uniqueness} \end{cases} \\
 \Downarrow \\
 \rho \perp \pi.
 \end{array}$$

18.17.3 Wyler 3D

$$\begin{array}{c}
l \perp \pi \text{ at } O, \quad l \subset \rho \\
\Downarrow \\
s = \rho \cap \pi \text{ exactly} \\
\Downarrow \\
l \perp s \text{ at } O \\
\Downarrow \\
\forall g, F: g \subset \rho, g \perp s \text{ at } F \\
\Downarrow \\
\text{hilbert_coplanar_perpendiculars_to_same_line_} \\
\text{eq_or_parallel_wyler} \\
\Downarrow \\
l = g \vee l \parallel g \\
\Downarrow \\
\begin{cases} l = g: & \text{Wyler foot uniqueness,} \\ l \parallel g: & \text{Wyler XI.8 + foot uniqueness} \end{cases} \\
\Downarrow \\
\rho \perp \pi.
\end{array}$$

The visible graphs are close. The research value lies in comparing the implementation of the nodes rather than merely comparing the final theorem statements.

18.18 E^4 Transfer Audit

XI.18 is useful for the E^4 program precisely because not every part of its formulation transfers unchanged.

18.18.1 1. Purely planar step

The comparison

$$l \perp s, \quad g \perp s$$

inside one carrier plane is two-dimensional. This step is already isolated by passing to $\text{PlaneGeo}(\rho)$. It should survive any higher-dimensional development whenever the relevant configuration is restricted to a generated 2-flat.

18.18.2 2. Genuinely E^3 incidence step

The statement

$$\rho \cap \pi = s$$

uses the fact that two distinct intersecting planes in E^3 meet in a line. This is dimension-specific. In E^4 , two hyperplanes normally meet in a 2-flat, so the exact carrier type changes.

18.18.3 3. Wyler replacement

The Wyler route suggests replacing the hard-coded phrase “intersection line” by an exact flat meet whose dimension is derived from the ambient dimension and the dimensions of the two carriers.

This is the transferable architectural idea:

$$\text{typed flat meet} \quad + \quad \text{separate metric orthogonality.}$$

18.18.4 4. Expected E^4 analogue

One should not copy XI.Def.4 literally into E^4 . XI.Def.4 is formulated around a common *line* of two planes. A plausible higher-dimensional analogue would instead use:

- codimension-one flats playing the role of planes;
- their codimension-two intersection flat;
- a 2-flat normal section in which the right-angle comparison is performed;
- a dimension-independent notion of normality or flat orthogonality.

This is a programmatic target, not yet a theorem asserted by the present formalization.

18.18.5 5. Missing E^4 lemma

The main missing ingredient suggested by XI.18 is an exact dimension-aware meet theorem together with a transport principle saying that normality is preserved under the appropriate parallel translation inside the relevant normal 2-flat.

Thus the XI.18 comparison points directly to two interfaces that matter for the E^4 refactor:

$$\boxed{\text{dimension-aware flat meet}} \quad \text{and} \quad \boxed{\text{normal transport in a generated 2-flat}}.$$

18.19 Structural Lesson

XI.18 tests two independent design decisions.

First, source fidelity requires a definition of perpendicular planes strong enough to preserve the theorem. If one defined

$$\rho \perp \pi \quad :\Longleftrightarrow \quad \exists l \subset \rho, \, l \perp \pi,$$

then XI.18 would collapse from its hypotheses.

Second, the existence of two formal proofs is useful only if the routes remain conceptually distinct. The direct Hilbert proof records the theorem in the natural synthetic language of E^3 . The Wyler

proof reorganizes the same geometry so that flat incidence and metric transport can be examined independently.

The object of interest is therefore not which proof is preferable, but their difference:

$$\text{Hilbert proof} \leftrightarrow \text{Wyler proof} \implies \text{candidate } E^4 \text{ interfaces.}$$

18.20 Audit Status

The current project contains production theorems

```
Geometry.euclid_proposition_11_18
Geometry.euclid_proposition_11_18_wyler
```

and both are imported by the main project development.

During the present XI.18/XI.19 update, the direct Hilbert XI.18 file compiled cleanly and the complete project build was reported clean after restoring the explicit Wyler helper imports required by the older XI.18 dependency chain.

No new `#print axioms` result is claimed in this documentation revision.

18.21 Conclusion

Proposition XI.18 has the compact source form

$$l \perp \pi, \quad l \subset \rho \implies \rho \perp \pi.$$

The formalization now deliberately contains two proofs of this same theorem.

The direct Hilbert route shows that the proposition belongs naturally to the synthetic point–line–plane geometry of E^3 . The Wyler route exposes the same proof as a composition of exact flat incidence and metric transport. Both terminate in the same neutral XI.Def.4 predicate.

The common mathematical skeleton is

$$s = \rho \cap \pi, \quad l \perp s, \quad g \perp s, \quad l = g \text{ or } l \parallel g, \quad g \perp \pi.$$

The comparison is already informative for the planned passage to E^4 : the planar perpendicular comparison is stable, whereas the type and dimension of the common meet are not. XI.18 therefore identifies, in a small theorem, exactly the distinction that motivates the parallel Hilbert/Wyler development of Book XI.

Chapter 19

Proposition XI.19

19.1 Introduction

Proposition XI.19 is one of the most informative comparison points in the current Book XI development. The project now contains two production proofs of the same theorem:

```
CGJteamLab/Proposition11_19.lean
CGJteamLab/Wyler/Proposition11_19.lean
```

with main theorems

```
euclid_proposition_11_19
euclid_proposition_11_19_wyler
```

The first is the direct Hilbert 3D proof. The second is the Wyler reconstruction retained deliberately as a second proof route and as a laboratory for the passage from E^3 to E^4 .

The comparison is especially useful because XI.19 contains two logically different problems.

First, there is a global incidence problem: if two planes ρ, σ , both perpendicular to a third plane π , cut one another, then their traces on π must meet. The existence of this triple point is not a local right-angle fact; it is forced by the spatial parallel theory through XI.6 and XI.15.

Second, once a triple point

$$D \in \rho \cap \sigma \cap \pi$$

has been obtained, the remaining proof is local and neutral. XI.Def.4 supplies one normal to π inside each of the two perpendicular planes, XI.13 identifies the two normals, and exact plane intersection identifies that common normal with the common section of ρ and σ .

Both routes use the same neutral definition of perpendicular planes from

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

Thus, as in XI.18, the comparison concerns proof architecture rather than different theorem

statements.

19.2 Euclid's Statement

Euclid states Proposition XI.19 as follows:

If two planes cutting one another be at right angles to any plane, then their common section will also be at right angles to the same plane.

Let ρ and σ be distinct planes which cut one another, and let π be a third plane. The hypotheses are

$$\rho \perp \pi, \quad \sigma \perp \pi.$$

If

$$t = \rho \cap \sigma,$$

then the conclusion is

$$t \perp \pi.$$

The formal conclusion contains more information than the bare statement that some normal to π lies in both planes. It returns an exact common section t and proves that this very line is perpendicular to π .

19.3 Classical Configuration

The configuration consists of three planes:

- the reference plane π ;
- a plane ρ perpendicular to π ;
- a plane σ perpendicular to π .

The two cutting planes have common section

$$t = \rho \cap \sigma.$$

Their traces on π are

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi.$$

The decisive point is

$$D = r \cap s.$$

Once this point is known,

$$D \in \rho \cap \sigma \cap \pi.$$

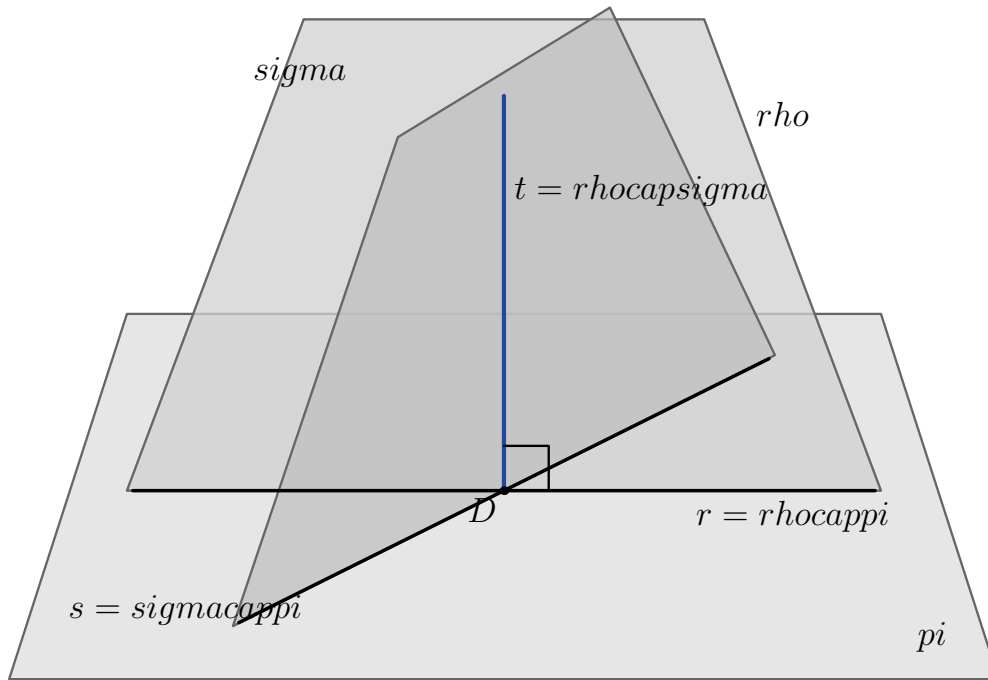


Figure 19.1: The classical configuration of Proposition XI.19. The planes ρ and σ are both perpendicular to the reference plane π . Their traces $r = \rho \cap \pi$ and $s = \sigma \cap \pi$ meet at D , and the common section $t = \rho \cap \sigma$ passes through the same point. The conclusion is $t \perp \pi$.

19.4 Euclid's Classical Proof

Euclid lets the two perpendicular planes cut one another in a straight line. At the point where the relevant traces on the reference plane meet, he constructs in each perpendicular plane a line perpendicular to the corresponding trace.

By XI.Def.4 each of those constructed lines is perpendicular to the reference plane. Since two straight lines perpendicular to the same plane at the same point must coincide, the two constructed normals are the same line.

That common normal lies in both cutting planes, hence it is their common section. Therefore the common section is perpendicular to the reference plane.

The printed proof makes the triple point visually natural. In formal geometry that point is not free. The statement

$$r \cap s \neq \emptyset$$

is the main global incidence theorem inside the proof.

19.5 Classical Dependency Pattern

At the level of the classical diagram the argument has the compact shape

$$\begin{array}{c}
 \rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \\
 \Downarrow \\
 D \in \rho \cap \sigma \cap \pi \\
 \Downarrow \\
 g \subset \rho, \quad h \subset \sigma, \quad g \perp \pi, \quad h \perp \pi \text{ at } D \\
 \Downarrow \\
 \text{uniqueness of the normal at } D \\
 \Downarrow \\
 g = h = \rho \cap \sigma \\
 \Downarrow \\
 (\rho \cap \sigma) \perp \pi.
 \end{array}$$

The formal development separates the first arrow from the remainder. That separation is mathematically significant: existence of D is the only global Euclidean part of XI.19.

19.6 Shared XI.Def.4 Structure

Both formal proofs use the neutral predicates

```
HilbertSpacePlanesPerpendicularAlong Geo rho pi r
HilbertSpacePlanesPerpendicular Geo rho pi
```

from

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

A certificate that ρ is perpendicular to π contains an exact common section r and the XI.Def.4 elimination rule:

$$g \subset \rho, \quad g \perp r \text{ at } D \quad \Longrightarrow \quad g \perp \pi \text{ at } D.$$

XI.19 uses this rule twice, once in each of the two perpendicular planes.

This is why XI.18 and XI.19 must not use the modern existential definition

$$\rho \perp \pi \quad :\Longleftrightarrow \quad \exists g \subset \rho, \quad g \perp \pi.$$

Such a definition would suppress the common-section geometry that is the actual content of both propositions.

19.7 What Is Genuinely Three-Dimensional Here?

XI.19 is spatial in several independent ways.

First, the theorem concerns perpendicularity of planes rather than merely perpendicularity of lines.

Second, three exact plane meets occur:

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi, \quad t = \rho \cap \sigma.$$

In E^3 , each of these meets is a line.

Third, the existence of the point

$$D \in \rho \cap \sigma \cap \pi$$

requires a global spatial argument. If the traces $r, s \subset \pi$ were disjoint, they would be parallel. Normals erected inside ρ and σ would then supply a second pair of parallel lines. XI.15 would force

$$\rho \parallel \sigma,$$

contradicting the given common point of the cutting planes.

Finally, after the triple point is obtained, XI.13 is a spatial uniqueness theorem: two lines perpendicular to the same plane at the same point coincide.

19.8 Formal Statements

19.8.1 Direct Hilbert 3D theorem

The direct Hilbert production theorem is

```

theorem euclid_proposition_11_19
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (rho sigma pi : S.Plane)
  (K : Geo.Point)
  (hRhoSigma : Ne rho sigma)
  (hKrho : S.OnPlane K rho)
  (hKsigma : S.OnPlane K sigma)
  (hRhoPerp :
    HilbertSpacePlanesPerpendicular Geo rho pi)
  (hSigmaPerp :
```

```

HilbertSpacePlanesPerpendicular Geo sigma pi) :
exists D : Geo.Point,
exists t : Geo.Line,
  S.OnPlane D pi /\
  H.OnLine D t /\
  HilbertLineInPlane Geo t rho /\
  HilbertLineInPlane Geo t sigma /\
  (forall X : Geo.Point,
    (S.OnPlane X rho /\ S.OnPlane X sigma) <=>
      H.OnLine X t) /\
  HilbertLinePerpendicularPlaneAt Geo t pi D

```

Its production file is

```
CGJteamLab/Proposition11_19.lean
```

19.8.2 Wyler theorem

The Wyler theorem has the same public type:

```

theorem euclid_proposition_11_19_wyler
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (rho sigma pi : S.Plane)
  (K : Geo.Point)
  (hRhoSigma : Ne rho sigma)
  (hKrho : S.OnPlane K rho)
  (hKsigma : S.OnPlane K sigma)
  (hRhoPerp :
    HilbertSpacePlanesPerpendicular Geo rho pi)
  (hSigmaPerp :
    HilbertSpacePlanesPerpendicular Geo sigma pi) :
exists D : Geo.Point,
exists t : Geo.Line,
  S.OnPlane D pi /\
  H.OnLine D t /\
  HilbertLineInPlane Geo t rho /\
  HilbertLineInPlane Geo t sigma /\
  (forall X : Geo.Point,
    (S.OnPlane X rho /\ S.OnPlane X sigma) <=>
      H.OnLine X t) /\
  HilbertLinePerpendicularPlaneAt Geo t pi D

```

Its production file is

```
CGJteamLab/Wyler/Proposition11_19.lean
```

The hypotheses

```
hRhoSigma : Ne rho sigma
hKrho      : S.OnPlane K rho
hKsigma    : S.OnPlane K sigma
```

formalize Euclid's phrase that the two planes cut one another.

19.9 Direct Hilbert 3D Reconstruction

The direct production file imports

```
CGJteamLab.Hilbert3DPlanePerpendicular
CGJteamLab.Proposition11_6
CGJteamLab.Proposition11_13
CGJteamLab.Proposition11_15
```

and contains the complete direct proof, including its own local normal construction and its own common-point core.

19.9.1 Stage H1: extract the two traces

From the two perpendicular-plane certificates the proof obtains exact section lines

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi.$$

Their containments are explicit:

$$r \subset \rho, \quad r \subset \pi, \quad s \subset \sigma, \quad s \subset \pi.$$

19.9.2 Stage H2: force the traces to meet

Assume for contradiction that r and s do not meet.

Because both lie in π ,

$$r \parallel s.$$

Choose points

$$R \in r, \quad Q \in s.$$

Since the traces are disjoint,

$$R \neq Q.$$

The direct helper

```
hilbert_XI19_planesPerpendicularAlong_normal_exists
```

uses XI.Def.4 to produce

$$g \subset \rho, \quad g \perp \pi \text{ at } R,$$

and

$$h \subset \sigma, \quad h \perp \pi \text{ at } Q.$$

Then direct XI.6 gives

$$g \parallel h.$$

The pairs (r, g) and (s, h) are intersecting line pairs inside ρ and σ , respectively. The proof verifies explicitly that the four lines cannot all lie in one plane; otherwise uniqueness of the plane through two intersecting lines would imply

$$\rho = \sigma.$$

Direct XI.15 now yields

$$\rho \parallel \sigma.$$

But the hypothesis supplies

$$K \in \rho \cap \sigma.$$

Contradiction. Therefore r and s meet.

19.9.3 Stage H3: direct common-point core

Let D be the intersection point of r and s . Then

$$D \in \rho, \quad D \in \sigma, \quad D \in \pi.$$

The direct theorem

```
euclid_proposition_11_19_common_point
```

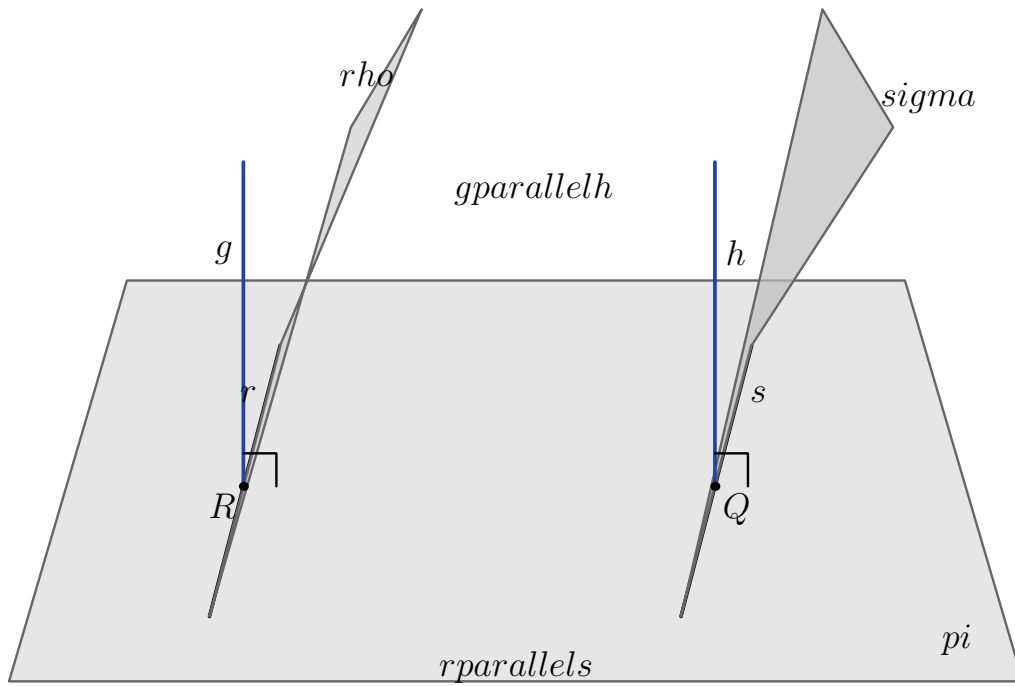



Figure 19.2: The incidence step hidden by the classical picture. If the traces $r = \rho \cap \pi$ and $s = \sigma \cap \pi$ were disjoint, then $r \parallel s$. Normals $g \subset \rho$ and $h \subset \sigma$ to π are parallel by XI.6. XI.15 would then force $\rho \parallel \sigma$, contradicting the common point K . Hence r and s meet at the required triple point D .

now handles the local part.

First, XI.Def.4 produces

$$g \subset \rho, \quad g \perp \pi \text{ at } D,$$

and

$$h \subset \sigma, \quad h \perp \pi \text{ at } D.$$

Direct XI.13 gives

$$g = h.$$

Next, exact plane intersection gives

$$t = \rho \cap \sigma.$$

Since the common normal lies in both ρ and σ , a second point on that normal lies on the exact meet t . Line uniqueness then gives

$$g = t.$$

Therefore

$$t \perp \pi \quad \text{at } D.$$

19.9.4 Stage H4: local planar construction

The lowest direct helper is

```
hilbert_XI19_perpendicular_line_through_point_in_plane
```

It enters

```
PlaneGeo Geo rho
```

and erects a perpendicular to the section line at a specified section point. The result is transported back to the ambient geometry and consumed by the XI.Def.4 elimination rule.

Thus even the direct Hilbert proof has a clear local 2-dimensional core inside an ambient 3-dimensional argument.

19.10 Wyler Reconstruction

The Wyler production theorem imports

```
CGJteamLab.Wyler.Proposition11_19_common_point_wyler
CGJteamLab.Wyler.Proposition11_6
CGJteamLab.Wyler.Proposition11_15
```

The visible proof architecture is intentionally parallel to the Hilbert route.

19.10.1 Stage W1: global triple-point layer

The Wyler proof extracts the exact traces

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi$$

from the same neutral XI.Def.4 certificates.

Assuming the traces disjoint gives

$$r \parallel s.$$

At points $R \in r$, $Q \in s$, the helper

```
hilbertSpacePlanesPerpendicularAlong_normal_exists_wyler
```

supplies normals

$$g \subset \rho, \quad h \subset \sigma$$

to π .

Wyler XI.6 gives

$$g \parallel h,$$

and Wyler XI.15 forces

$$\rho \parallel \sigma,$$

contradicting the common point K .

Hence the triple point D exists.

19.10.2 Stage W2: isolated common-point core

The local result is isolated in

```
CGJteamLab/Wyler/Proposition11_19_common_point_wyler.lean
```

with theorem

```
euclid_proposition_11_19_common_point_wyler
```

At the given triple point, XI.Def.4 supplies one normal in each cutting plane; XI.13 identifies the normals; exact flat intersection identifies that normal with the common section of ρ and σ .

The lower normal-extraction machinery belongs to

```
CGJteamLab/Wyler/HilbertWylerPlaneNormal.lean
```

The Wyler route therefore makes the separation between

global existence of D

and

local identification of the common normal

an explicit module boundary.

19.11 Hilbert versus Wyler: Proof Diff

XI.19 is particularly useful because the two proofs are mathematically close but architecturally distinct.

19.11.1 Shared skeleton

Both proofs have the same high-level graph:

$$\begin{array}{c}
 r = \rho \cap \pi, \quad s = \sigma \cap \pi \\
 \Downarrow \\
 \text{assume } r \cap s = \emptyset \\
 \Downarrow \\
 r \parallel s \\
 \Downarrow \\
 \text{construct normals } g, h \text{ to } \pi \\
 \Downarrow \\
 \text{XI.6 : } g \parallel h \\
 \Downarrow \\
 \text{XI.15 : } \rho \parallel \sigma \\
 \Downarrow \\
 \perp \\
 \Downarrow \\
 D \in \rho \cap \sigma \cap \pi \\
 \Downarrow \\
 \text{XI.Def.4 + XI.13} \\
 \Downarrow \\
 (\rho \cap \sigma) \perp \pi.
 \end{array}$$

19.11.2 Direct Hilbert realization

The direct route keeps the complete argument in the main Book XI hierarchy:

```

Proposition11_19
-> Proposition11_6
-> Proposition11_13
-> Proposition11_15

```

and defines direct local helpers inside `Proposition11_19.lean`.

This is the natural proof in Hilbert's E^3 language.

19.11.3 Wyler realization

The Wyler route factors the proof more aggressively:

```

Wyler.Proposition11_19
-> Wyler.Proposition11_19_common_point_wyler
-> Wyler.HilbertWylerPlaneNormal

```

```
-> Wyler.Proposition11_6
-> Wyler.Proposition11_15
```

The global incidence stage remains visible in the final theorem, while the local common-point argument is promoted to a separate neutral layer.

19.11.4 The mathematical difference

The difference is not the final theorem and not even the high-level proof. The difference is where the geometry is cut into reusable interfaces.

Hilbert says, in effect:

solve XI.19 directly in the native E^3 language.

Wyler says:

separate exact flat configuration from metric identification.

This is precisely the comparison needed before moving to E^4 .

19.12 Ambient Space and Plane_{GEO}

The public theorem is entirely ambient:

```
rho sigma pi : S.Plane
r s t       : Geo.Line
D K         : Geo.Point
```

The induced plane geometry appears only in the local normal construction. Given a section line $r \subset \rho$ and a point $D \in r$, the proof enters

```
PlaneGeo Geo rho
```

to erect a perpendicular to r at D .

The following steps remain genuinely ambient:

- exact intersection of two planes;
- spatial parallelism of lines;
- XI.6 for normals to a plane;
- XI.15 for parallel planes;
- XI.13 for uniqueness of a normal to a plane;
- identification of a normal line with an exact common section.

19.13 Hidden Formal Data

19.13.1 The cutting planes are distinct

The hypothesis

```
hRhoSigma : Ne rho sigma
```

is required because an exact common section line of two identical planes would not be determined.

19.13.2 The cutting planes have a common point

The data

```
K : Geo.Point
hKrho   : S.OnPlane K rho
hKsigma : S.OnPlane K sigma
```

formalize Euclid's phrase that ρ and σ cut one another.

19.13.3 The triple point is derived

The point D is not an input. The proof derives

$$D \in \rho \cap \sigma \cap \pi$$

from the trace argument using XI.6 and XI.15.

This is the most important incidence fact hidden by the classical picture.

19.13.4 The common section is exact

The result includes

```
forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X sigma) <->
    H.OnLine X t
```

Thus t is not merely a line contained in both planes. It is the full point-set intersection.

19.13.5 Normal extraction is constructive

At a specified point of the XI.Def.4 section, the proof first constructs a line inside the carrier plane perpendicular to that section, then applies the XI.Def.4 elimination rule to upgrade the line to a normal of the reference plane.

This local construction is explicit in both proof routes.

19.14 Relationship with Earlier Book XI Propositions

XI.6. Two lines perpendicular to the same plane at distinct points are parallel. This creates the second parallel pair needed in the contradiction argument.

XI.13. The perpendicular to a plane at a prescribed point is unique. This is the central local theorem after the triple point has been obtained.

XI.15. Two carrier planes containing corresponding pairs of parallel lines are parallel under the noncoplanarity hypothesis. This is the decisive global Euclidean step.

XI.18 and XI.Def.4. XI.18 established the source-faithful perpendicular-plane interface now shared by both routes. XI.19 uses that interface as an elimination rule rather than introducing a new definition.

XI.19 therefore closes a short chain:

$$\text{XI.18} \longrightarrow \text{usable XI.Def.4 plane perpendicularity} \longrightarrow \text{XI.19.}$$

19.15 Relationship with Hilbert's Axiom Groups

The local common-point core uses the neutral synthetic geometry:

- Group I incidence;
- Group II order in the planar perpendicular construction;
- Group III congruence and right-angle structure.

The full theorem additionally requires

HilbertSpaceEuclidean Geo

because the global triple-point argument invokes XI.15.

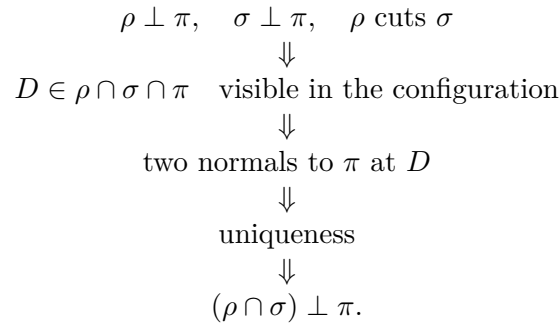
Thus Group IV is localized to the existence layer:

$$\text{if the traces do not meet} \implies \rho \parallel \sigma.$$

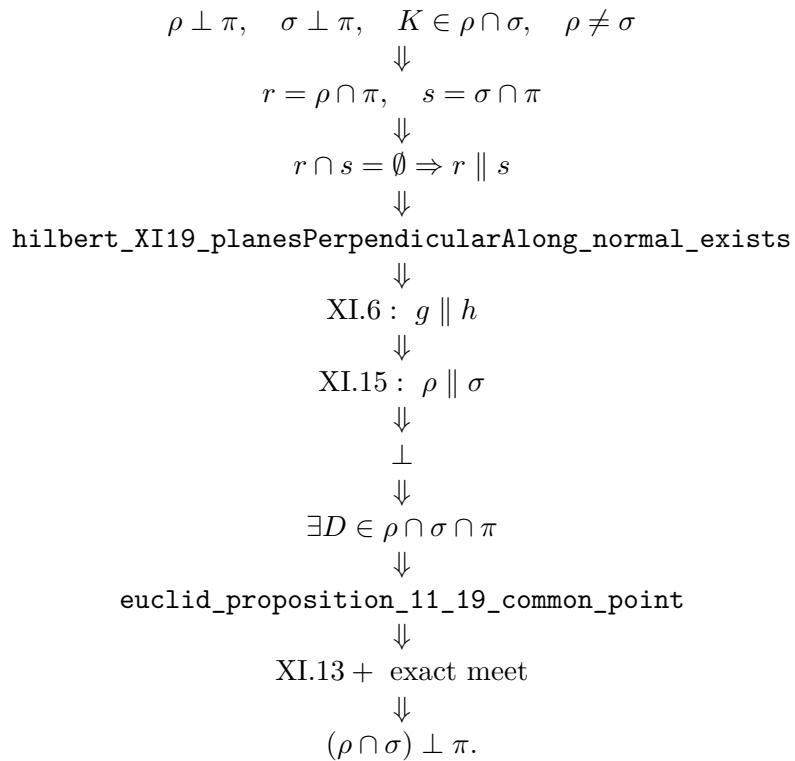
Once D exists, the identification of the common section with the normal to π is neutral.

19.16 Classical DAG and the Two Formal DAGs

19.16.1 Euclid



19.16.2 Direct Hilbert 3D



19.16.3 Wyler 3D

$$\begin{array}{c}
\rho \perp \pi, \quad \sigma \perp \pi, \quad K \in \rho \cap \sigma, \quad \rho \neq \sigma \\
\Downarrow \\
r = \rho \cap \pi, \quad s = \sigma \cap \pi \\
\Downarrow \\
r \cap s = \emptyset \Rightarrow r \parallel s \\
\Downarrow \\
\text{hilbertSpacePlanesPerpendicularAlong_} \\
\text{normal_exists_wyler} \\
\Downarrow \\
\text{Wyler XI.6 : } g \parallel h \\
\Downarrow \\
\text{Wyler XI.15 : } \rho \parallel \sigma \\
\Downarrow \\
\perp \\
\Downarrow \\
\exists D \in \rho \cap \sigma \cap \pi \\
\Downarrow \\
\text{euclid_proposition_11_19_common_point_wyler} \\
\Downarrow \\
\text{XI.13 + exact flat meet} \\
\Downarrow \\
(\rho \cap \sigma) \perp \pi.
\end{array}$$

The two DAGs are close at proposition level. Their difference lies in the modularization of the lower geometry.

19.17 E^4 Transfer Audit

XI.19 reveals a stronger dimension dependence than XI.18.

19.17.1 1. Purely local step

Once a point D lies in all relevant carriers, the construction of a normal direction can be reduced to a suitable 2-dimensional normal section.

The local pattern

$$\text{construct normal directions at } D \quad + \quad \text{use uniqueness}$$

is a good candidate for dimension-independent packaging.

19.17.2 2. Genuinely E^3 incidence step

In E^3 ,

$$\rho \cap \pi, \quad \sigma \cap \pi, \quad \rho \cap \sigma$$

are lines when the corresponding planes are distinct and intersect.

This fact is built deeply into the statement of XI.19. In E^4 , two 2-planes need not meet in a line, and two hyperplanes meet generically in a 2-flat. Therefore the carrier dimensions in the theorem cannot simply be copied.

19.17.3 3. Wyler replacement

The Wyler organization suggests replacing fixed words such as “intersection line” by typed flat meets with explicit dimension data.

The transferable pattern is

flat meet $\longrightarrow$ common lower-dimensional carrier $\longrightarrow$ metric normal identification.

This is more promising for E^4 than a direct transcription of the three-dimensional statement.

19.17.4 4. Expected E^4 analogue

A useful higher-dimensional analogue should likely be formulated in terms of codimensions rather than the names “line” and “plane”.

The XI.19 pattern suggests investigating statements of the form:

if two suitable flats are both orthogonal to a reference flat and meet with the required dimension, then their common meet inherits an orthogonality property determined in an appropriate normal 2-flat.

The exact formulation depends on the chosen E^4 flat and orthogonality interfaces and is not asserted here as an already formalized theorem.

19.17.5 5. Missing E^4 lemmas

XI.19 points to at least three higher-dimensional interfaces:

dimension formula for flat meets,

normal-space / normal-2-flat construction,

and

higher-dimensional analogue of XI.15.

The last item is particularly important. In the E^3 proof, XI.15 turns two pairs of parallel lines into parallelism of the carrier planes. In E^4 , the correct statement must know the dimensions and spanning relations of the corresponding flats.

Thus XI.19 is a stronger diagnostic for the E^4 project than XI.18: the main difficulty is no longer merely replacing one intersection line by a higher-dimensional meet; the global parallel-carrier argument itself must be reformulated dimensionally.

19.18 Structural Lesson

XI.19 shows why the project should retain the direct Hilbert and Wyler routes side by side.

The direct Hilbert proof demonstrates that the theorem is natural and manageable in Hilbert's synthetic E^3 language. Nothing about the three-dimensional argument requires abandoning that language.

The Wyler proof, however, makes a different feature visible: the theorem can be decomposed into an exact incidence layer and a local metric layer. That decomposition is precisely what becomes valuable when the dimensions of intersections change in E^4 .

The research object is therefore the proof difference:

$$\text{Hilbert } E^3 \leftrightarrow \text{Wyler } E^3 \implies \text{candidate dimension-free interfaces.}$$

For XI.19 the most important candidate is the replacement of the rigid three-dimensional pattern

$$\text{two line traces} + \text{two parallel normals} \Rightarrow \text{parallel planes}$$

by a dimension-aware statement about flats and their spanning directions.

19.19 Audit Status

The current project contains the production theorems

```
Geometry.euclid_proposition_11_19
Geometry.euclid_proposition_11_19_wyler
```

together with the neutral XI.Def.4 module

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

and both Book XI routes are included through XI.19 in the main project imports.

The direct XI.19 theorem and the complete project were reported to compile cleanly after the XI.18/XI.19 integration.

No new `#print axioms` result is claimed in this documentation revision.

19.20 Conclusion

Proposition XI.19 has the compact classical form

$$\rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \implies (\rho \cap \sigma) \perp \pi.$$

The current formalization now contains two independent reconstructions.

Both first solve the global incidence problem of producing the triple point D . Both then solve the neutral local problem by extracting two normals to π , identifying them by XI.13, and identifying the resulting line with the exact common section of ρ and σ .

The direct Hilbert route keeps this argument in the native synthetic language of E^3 . The Wyler route factors the same mathematics into explicit flat and metric layers.

For the E^4 program, XI.19 therefore supplies a particularly useful lesson: the local normal argument appears transferable, while the dimensions of the meets and the XI.15-style global parallelism step are genuinely three-dimensional and must be reformulated rather than copied.

Chapter 20

Proposition XI.20

20.1 Purpose and scope

Proposition XI.20 is the first proposition in the present reconstruction where Euclid’s verbal comparison of a *sum of two plane angles* with a third angle must be represented without introducing a numerical angle measure. The formal development therefore separates three layers:

1. a neutral language for a proper trihedral configuration and for the relation “two angles together are greater than a third”;
2. a planar metric core, based on congruence, SAS, Euclid I.20, and Euclid I.25;
3. one small spatial incidence step which prevents a certain auxiliary triangle from degenerating.

Two production proofs are provided. The direct Hilbert proof uses the three-dimensional incidence interface. The Hilbert–Wyler proof replaces the single spatial step by the dimension-free point–line–plane incidence fragment of `HilbertWylerAxioms`. The two proofs deliberately share the neutral metric core but not the final theorem itself.

20.2 Classical configuration

Let O be the vertex of a solid angle bounded by the three rays OA , OB , and OC . The three face angles are

$$\angle AOB, \quad \angle BOC, \quad \angle COA.$$

Euclid XI.20 asserts the three cyclic strict inequalities

$$\begin{aligned}\angle AOB + \angle BOC &> \angle AOC, \\ \angle BOC + \angle COA &> \angle BOA, \\ \angle COA + \angle AOB &> \angle COB.\end{aligned}$$

The formal theorem assumes that each face angle is proper and that O, A, B, C do not lie in one plane. This is exactly the content of the neutral predicate `HilbertTrihedralConfiguration`.

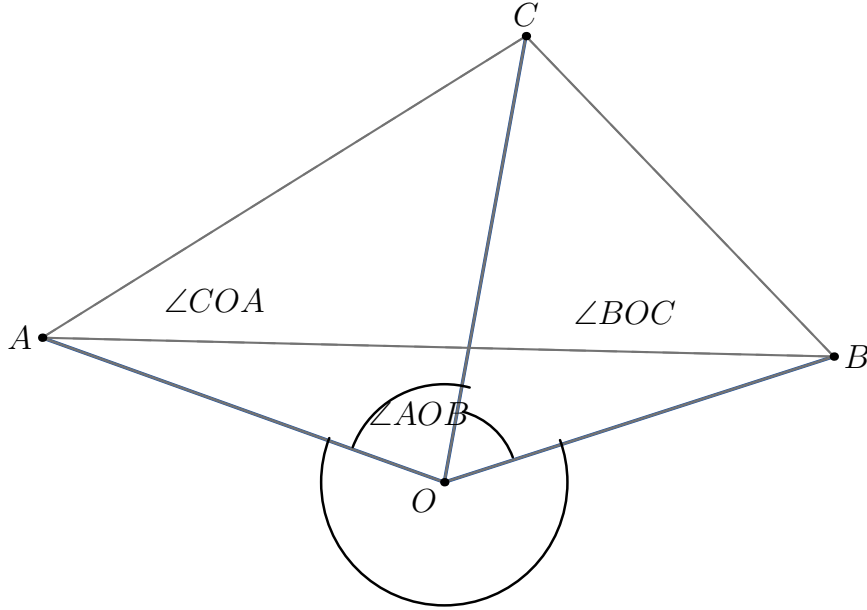


Figure 20.1: A proper trihedral configuration with vertex O and face angles $\angle AOB$, $\angle BOC$, and $\angle COA$. The drawing is schematic: the essential spatial condition is that O, A, B, C are not coplanar.

20.3 Neutral synthetic vocabulary

The file `CGJteamLab/HilbertTrihedralAngle.lean` introduces the vocabulary needed by both formal routes.

20.3.1 Trihedral configuration

The definition

```
def HilbertTrihedralConfiguration
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (O A B C : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A O B) /\
  Not (PrimCollinear Geo B O C) /\
  Not (PrimCollinear Geo C O A) /\
  Not (exists pi : S.Plane,
    S.OnPlane O pi /\
    S.OnPlane A pi /\
    S.OnPlane B pi /\
    S.OnPlane C pi)
```

records exactly the three proper face angles and the noncoplanarity of the four points. No orientation of space and no numerical angle are involved.

20.3.2 What “two angles together are greater” means

The relation `HilbertTwoAnglesGreaterThanAngle` is the central design choice of XI.20. It represents

$$\alpha + \beta > \gamma$$

without defining an addition operation on angles. The target angle γ is handled by one of three mutually sufficient geometric situations:

1. $\gamma < \alpha$ already;
2. $\gamma \cong \alpha$, while β is a proper angle;
3. an interior ray decomposes γ into two components, the first congruent to α , and the remaining component strictly smaller than β .

Thus the third case captures Euclid’s actual construction: copy the first summand into the target angle and compare the remaining part with the second summand. The public predicate `HilbertTrihedralAngleInequalities` is simply the conjunction of the three cyclic instances of this relation.

This representation is important mathematically. XI.20 is proved entirely in synthetic angle order; no real-valued angle measure, no angle addition, and no trigonometry are introduced merely to state the proposition.

20.4 The classical constructive core

It is enough to prove one cyclic inequality, for example

$$\angle ZOX + \angle ZOY > \angle XOY.$$

Compare the first summand $\angle ZOX$ with the target $\angle XOY$. There are three cases.

If $\angle XOY < \angle ZOX$, the first summand is already larger than the target. If the two angles are congruent, the proper second summand makes the sum strictly larger. The only nontrivial case is

$$\angle ZOX < \angle XOY.$$

By the synthetic definition of angle order there is an interior ray OW of $\angle XOY$ such that

$$\angle ZOX \cong \angle XOW.$$

Let the ray OW meet XY at H . On the third ray OZ choose D so that

$$OD \cong OH.$$

Transporting the copied angle from the rays OW and OZ to the same rays represented by OH and OD gives

$$\angle DOX \cong \angle XOH.$$

Together with $OD \cong OH$ and the common segment OX , SAS yields

$$DX \cong HX.$$

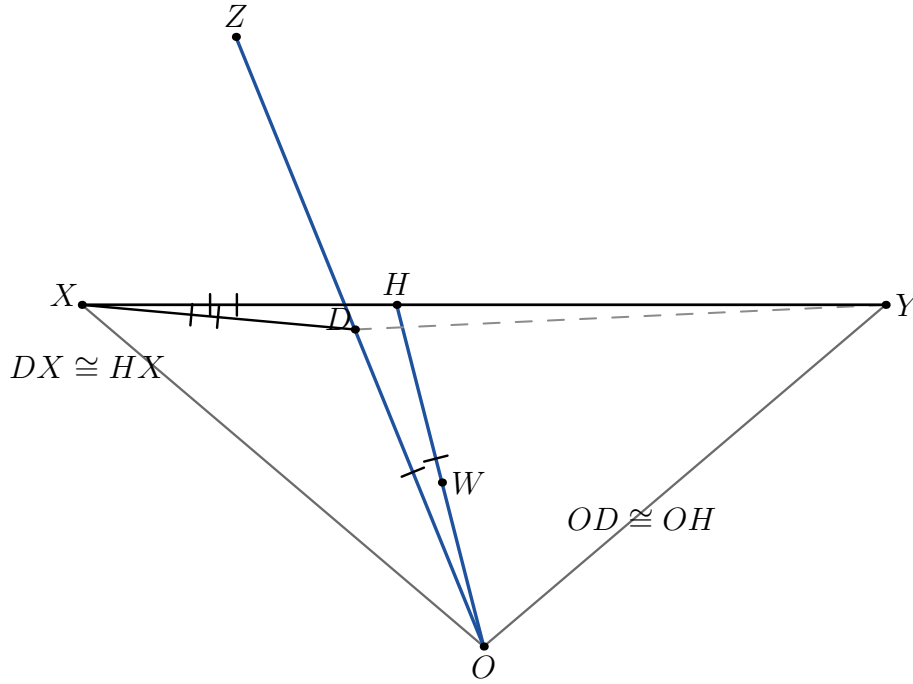


Figure 20.2: The auxiliary construction used in the nontrivial comparison case. H is the intersection of the interior ray with XY ; D is laid off on OZ with $OD \cong OH$. SAS gives $DX \cong HX$.

Now consider triangle DXY . Since H lies between X and Y and $DX \cong HX$, Euclid I.20 together with segment cancellation gives

$$HY < DY.$$

Applying Euclid I.25 to triangles OHY and ODY , with $OH \cong OD$ and OY common, gives

$$\angle HOY < \angle DOY.$$

Finally, same-ray transport returns from H to W and from D to Z :

$$\angle WOY < \angle ZOY.$$

This is precisely the strict estimate required for the remainder of the target angle.

20.5 Shared planar metric core

The file `CGJteamLab/HilbertTrihedralAngleMetric.lean` contains the parts of the proof that are independent of the ambient spatial incidence system. Its principal lemmas are:

- `hilbert_XI20_two_angles_greater_of_first_less`: extracts the interior-ray decomposition from the hypothesis that the first summand is smaller than the target;
- `hilbert_XI20_sas_base_congruent`: the SAS step yielding the congruence of the auxiliary base segments;
- `hilbert_XI20_remainder_side_less`: Euclid I.20 plus cancellation, producing the strict side inequality for the remainder;

- `hilbert_XI20_remainder_angle_less`: Euclid I.25, converting that side inequality into the strict angle inequality;
- `hilbertTwoAnglesGreaterThanAngle_swap_first`: symmetry needed to present the final cyclic inequalities in the conventional order of rays.

This factorization is not merely a code refactor. It identifies the genuinely planar metric content of XI.20 and makes the spatial dependency visible as a single separate obligation.

20.6 The genuinely spatial obligation

The auxiliary application of Euclid I.20 requires a proper triangle DXY . Thus one must prove

$$D, X, Y \text{ are not collinear.}$$

This is the only point in the core construction where the ambient point–line–plane structure is used.

Assume, for contradiction, that D, X, Y are collinear. Since O, X, Y are noncollinear, they determine a plane π . The line XY lies in π ; therefore $D \in \pi$. Since $O, D \in \pi$ and $O \neq D$, the whole line OD lies in π . But O, D, Z are collinear because D lies on the ray OZ . Hence $Z \in \pi$, contradicting the noncoplanarity of O, X, Y, Z .

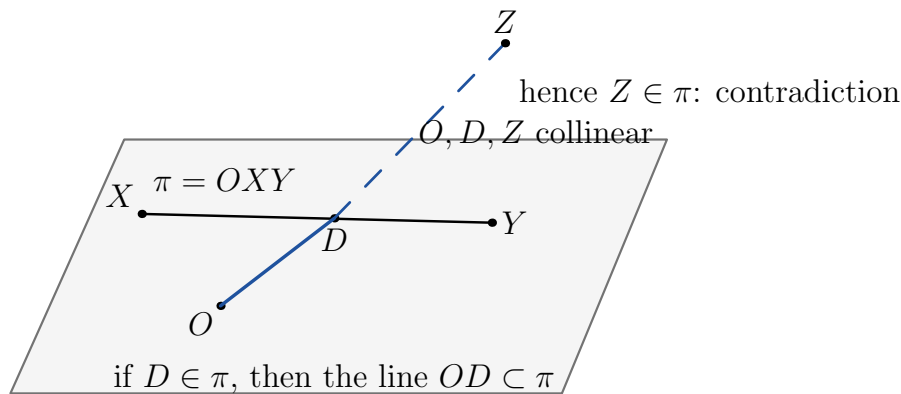


Figure 20.3: The sole spatial contradiction in the XI.20 construction. If D were on XY , then D would lie in the plane $\pi = OXY$. The line $OD = OZ$ would then lie in π , forcing $Z \in \pi$ and contradicting the trihedral noncoplanarity.

20.7 Direct Hilbert 3D reconstruction

The direct production theorem is

```
theorem euclid_proposition_11_20
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertCongruence Geo]
  (O A B C : Geo.Point)
  (hTri : HilbertTrihedralConfiguration Geo O A B C) :
  HilbertTrihedralAngleInequalities Geo O A B C
```

from CGJteamLab/Proposition11_20.lean.

Its local nontrivial theorem is `hilbert_XI20_triangular_remainder_less`. In the spatial block the proof uses

```
HilbertSpaceIncidence.plane_through
HilbertSpaceIncidence.line_in_plane
```

to carry out the contradiction described above. Once the auxiliary triangle DXY is known to be proper, the proof returns entirely to the shared metric core.

The theorem `hilbert_XI20_one_cyclic_inequality` then removes the comparison hypothesis by trichotomy between the first summand and the target. The final theorem obtains the remaining two inequalities by permuting the three rays of the triangular configuration.

20.8 Hilbert–Wyler reconstruction

The independent production theorem is

```
theorem euclid_proposition_11_20_wyler
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertCongruence Geo]
  [A : HilbertWylerAxioms Geo]
  (O A B C : Geo.Point)
  (hTri : HilbertTriangularConfiguration Geo O A B C) :
  HilbertTriangularAngleInequalities Geo O A B C
```

from CGJteamLab/Wyler/Proposition11_20.lean.

The Wyler proof does not call the direct Hilbert theorem. It has its own remainder lemma `hilbert_XI20_triangular_remainder_less_wyler` and its own cyclic theorem `hilbert_XI20_one_cyclic_inequality_wyler`. Both reuse only the neutral metric lemmas.

The decisive observation is that the spatial contradiction does not require Wyler I.7. It uses only the two fields

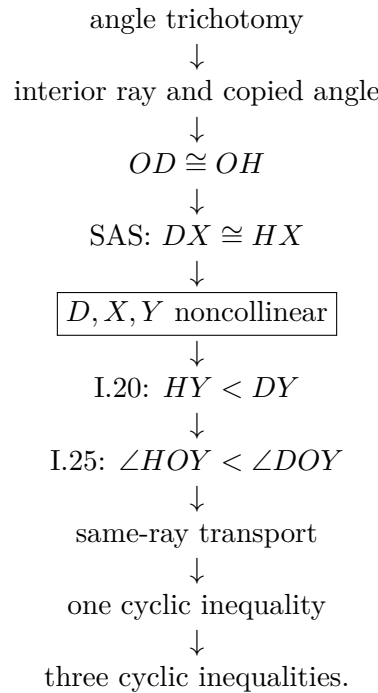
```
A.plane_through
A.line_in_plane
```

of the common Hilbert–Wyler incidence core. No use is made of Smith I.5, span exchange, or an ambient dimension bound.

A small local compatibility instance exposes the `HilbertPlaneIncidence` parent already contained in `HilbertCongruence`; this is only a typeclass-resolution adapter and adds no geometric assumption.

20.9 Hilbert versus Hilbert–Wyler proof DAG

The two production DAGs coincide after the spatial nondegeneracy step:



The boxed node is implemented differently:

Hilbert 3D	Hilbert–Wyler
HilbertSpaceIncidence.plane_through	A.plane_through
HilbertSpaceIncidence.line_in_plane	A.line_in_plane

Everything below that node is common metric geometry.

20.10 E^n transfer audit

XI.20 gives an unusually clean answer to the dimensional-transfer question.

1. Purely planar step. The copied-angle construction, SAS, Euclid I.20, Euclid I.25, angle order, and same-ray transport are planar metric geometry. They are already isolated in `HilbertTrihedralAngleMetric.lean`.

2. Genuine spatial incidence step. Only the proof that D, X, Y are noncollinear leaves the planar metric layer. Its logical form is: construct the plane OXY , place D in it from collinearity with X, Y , then place Z in it from collinearity with O, D .

3. Wyler replacement. The replacement needs only “three noncollinear points lie in a plane” and “a plane containing two distinct points of a line contains that line.” These are already fields of `HilbertWylerAxioms`.

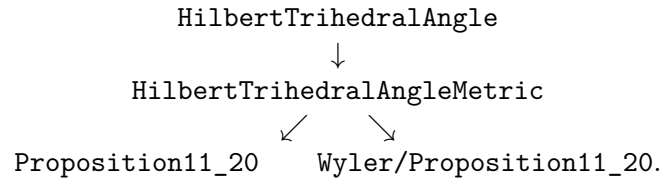
4. Expected E^4 and E^n analogue. No local 3-flat is required. The contradiction takes place inside a single ordinary plane. Therefore the same proof is meaningful in any ambient E^n satisfying the common incidence core and the same planar metric theory.

5. Missing higher-dimensional lemma. For XI.20 there is none. Neither Wyler I.7, Smith I.5, span exchange, nor a special E^4 dimension axiom is needed by the proof.

This is stronger than a reduction of XI.20 to a reconstructed local three-dimensional subspace. The proposition is, in the formal sense exposed by the proof, *dimension-free above the plane level*.

20.11 Dependency summary

The logical architecture of the formalization is



The shared metric module depends on the existing planar theory, in particular the formalizations of Euclid I.20 and I.25. The direct branch adds `HilbertSpaceIncidence`; the Wyler branch adds `HilbertWylerAxioms`. The final statements are intentionally separate, so the Wyler theorem is not a wrapper around the Hilbert theorem.

20.12 Conclusion

The formal reconstruction of XI.20 separates Euclid's argument into an unexpectedly small spatial shell around a planar metric proof. The decisive formal device is the synthetic relation `HilbertTwoAnglesGreaterThanAngle`, which expresses Euclid's comparison without numerical angle addition. Once the target angle is decomposed by an interior ray, the classical chain $\text{SAS} \rightarrow \text{I.20} \rightarrow \text{I.25}$ is reproduced directly.

The comparison of the two production routes yields a particularly useful structural result for the wider project: XI.20 does not depend on a specifically three-dimensional theorem. Its only ambient-space requirement is the common point–line–plane incidence core. Consequently XI.20 should transfer to E^4 and, more generally, to E^n without introducing a local 3-flat or additional dimension axioms.

Chapter 21

Proposition XI.21

21.1 Purpose and Scope

Proposition XI.21 is the first proposition in the present reconstruction whose classical wording appears to demand a global arithmetic of several angles:

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

A literal transcription of Euclid’s proof would combine three applications of XI.20 with four applications of I.32 and then cancel six intermediate angles from a total of six right angles.

The formal development deliberately does not introduce a numerical angle measure, a general addition operation on angles, or a special “straight angle” object merely to reproduce that calculation. Instead it uses the same complement principle already familiar from the synthetic treatment of I.17: a bound by four right angles can be represented by comparing two supplementary angles with the third angle.

This leads to a considerably shorter proof than Euclid’s printed one. The final formal argument uses only:

- two ray extensions;
- Proposition XI.20;
- vertical angles;
- congruence transport;
- one line-in-plane closure argument preserving noncoplanarity.

Two independent production routes are retained. The direct Hilbert route uses `HilbertSpaceIncidence`; the Hilbert–Wyler route replaces the single spatial closure step by the corresponding field of `HilbertWylerAxioms`. Both routes share only the neutral target language from

```
CGJteamLab/HilbertThreeAnglesFourRight.lean
```

21.2 Euclid's Statement and the Trihedral Scope

The classical statement is usually given in the unrestricted form:

Any solid angle is contained by plane angles whose sum is less than four right angles.

The proof, however, treats a solid angle bounded by three plane angles. Let A be the vertex and let the three edges be AB, AC, AD . The face angles are

$$\angle BAC, \quad \angle CAD, \quad \angle DAB.$$

The conclusion proved in the classical argument is

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

The Greek–English source used in the project explicitly notes that the printed proposition is proved only for the trihedral case; the extension to more than three face angles is stated to be straightforward. The current Lean theorem therefore formalizes exactly the trihedral case. It does not claim a general polyhedral-angle theorem.

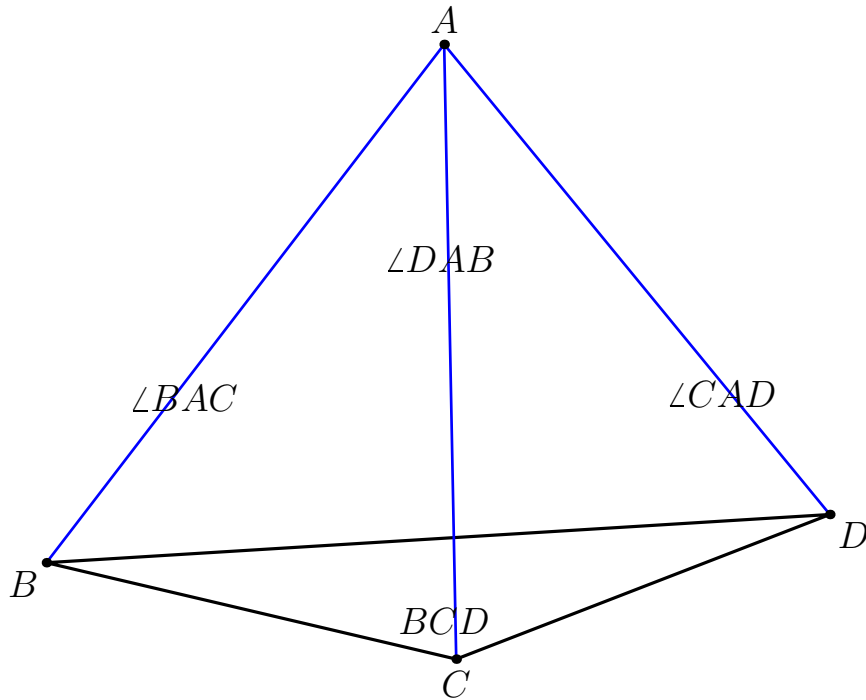


Figure 21.1: The trihedral configuration used in Proposition XI.21. The base points B, C, D are joined, while the three spatial edges from the vertex A determine the face angles $\angle BAC$, $\angle CAD$, and $\angle DAB$. The essential hypothesis is that A, B, C, D are not coplanar.

21.3 Euclid's Classical Proof

Euclid joins the three base points B, C, D . Proposition XI.20 is then applied at the three base vertices:

$$\begin{aligned} \angle CBA + \angle ABD &> \angle CBD, \\ \angle BCA + \angle ACD &> \angle BCD, \\ \angle CDA + \angle ADB &> \angle CDB. \end{aligned}$$

Hence the sum of the six angles

$$\angle CBA, \angle ABD, \angle BCA, \angle ACD, \angle CDA, \angle ADB$$

is greater than the sum of the three angles of triangle BCD .

By I.32,

$$\angle CBD + \angle BCD + \angle CDB = 2R.$$

Therefore those six angles are together greater than $2R$.

Euclid next applies I.32 to the three triangles

$$ABC, \quad ACD, \quad ADB.$$

Their nine angles are together equal to $6R$. Removing the six angles already shown to exceed $2R$, the remaining three angles at the vertex A are together less than $4R$:

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

The classical dependency graph is therefore

$$\begin{array}{c}
 \text{trihedral configuration } A; B, C, D \\
 \Downarrow \\
 \text{XI.20 at } B, C, D \\
 \Downarrow \\
 \text{six lateral/base angles exceed the three angles of } BCD \\
 \Downarrow \\
 \text{I.32 on } BCD : \quad \text{base sum} = 2R \\
 \Downarrow \\
 \text{I.32 on } ABC, ACD, ADB : \quad \text{nine-angle sum} = 6R \\
 \Downarrow \\
 \angle BAC + \angle CAD + \angle DAB < 4R.
 \end{array}$$

21.4 Why the Arithmetic Proof Is Not Formalized Literally

The project treats an angle synthetically as a proper ray-pair configuration. A straight configuration is represented through betweenness and supplementary constructions rather than by enlarging the ordinary angle type to include a numerical angle of measure $2R$.

This is already visible in the formal treatment of I.17. A statement of the form

$$\alpha + \beta < 2R$$

is encoded by comparing α with a supplement of β , not by forming a numerical sum.

Likewise, I.32 in the present library records an exterior-angle decomposition. It is not a theorem about an additive monoid of angle measures. A literal formalization of Euclid's XI.21 bookkeeping would therefore require substantial new infrastructure only to support a single cancellation argument.

During the XI.21 development such an infrastructure was deliberately postponed. The successful proof shows that it is not needed.

21.5 Neutral Synthetic Meaning of “Less Than Four Right Angles”

The common module

```
CGJteamLab/HilbertThreeAnglesFourRight.lean
```

introduces

```
def HilbertThreeAnglesLessThanFourRightAngles
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A O B C P D E Q F : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A O B) /\
  Not (PrimCollinear Geo C P D) /\
  Not (PrimCollinear Geo E Q F) /\
  exists X Y : Geo.Point,
    Geo.Between B O X /\
    Geo.Between D P Y /\
    HilbertTwoAnglesGreaterThanAngle
      Geo
      A O X
      C P Y
      E Q F
```

If the three original angles are denoted by

$$\alpha, \quad \beta, \quad \gamma,$$

then the two extensions construct their supplements

$$2R - \alpha, \quad 2R - \beta.$$

The formal certificate is

$$(2R - \alpha) + (2R - \beta) > \gamma,$$

which is the synthetic complement form of

$$\alpha + \beta + \gamma < 4R.$$

No angle-valued addition operation occurs in the definition. The only comparison object is the already established XI.20 relation

```
HilbertTwoAnglesGreaterThanAngle
```

For the trihedral proposition the public target is

```
def HilbertTrihedralAnglesLessThanFourRightAngles
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point) : Prop :=
  HilbertThreeAnglesLessThanFourRightAngles
```


Geo
 C A D
 B A C
 D A B

The order CAD, BAC, DAB is intentional: it matches the comparison delivered directly by XI.20 after replacing one ray by its opposite.

21.6 The Direct Synthetic Shortcut

Extend the rays AC and AD through the vertex A :

$$C - A - X, \quad D - A - Y.$$

Thus

$$\angle BAX$$

is supplementary to $\angle BAC$, while

$$\angle CAY$$

is supplementary to $\angle CAD$.

Now replace the ray AC by its opposite ray AX and reorder the trihedral configuration as

$$A; D, X, B.$$

Proposition XI.20 applied to this new trihedral configuration gives

$$\angle DAX + \angle XAB > \angle DAB.$$

Because $C - A - X$ and $D - A - Y$,

$$\angle DAX \cong \angle CAY$$

by vertical angles. Since the project uses unoriented ordinary angles,

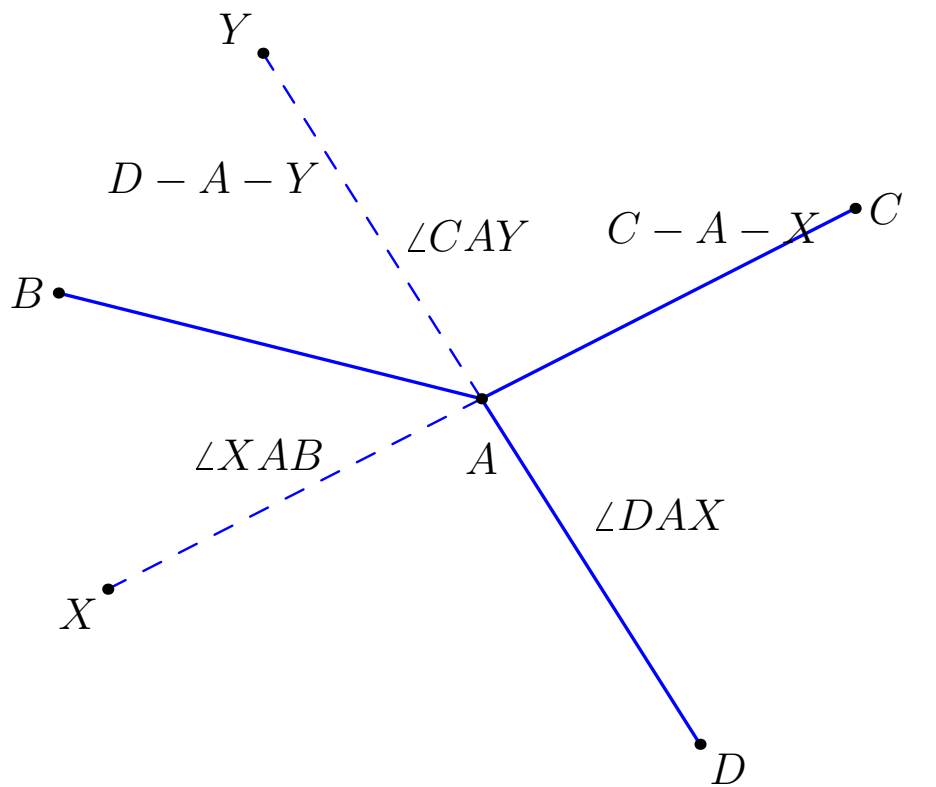
$$\angle XAB \cong \angle BAX.$$

Congruence transport therefore yields

$$\angle CAY + \angle BAX > \angle DAB.$$

This is exactly the neutral complement certificate for

$$\angle CAD + \angle BAC + \angle DAB < 4R.$$



XI.20 on $A; D, X, B$: $\angle DAX + \angle XAB > \angle DAB$

Figure 21.2: The formal reduction of XI.21. The rays AC and AD are extended through A to X and Y . XI.20 is applied to the trihedral configuration $A; D, X, B$, giving $\angle DAX + \angle XAB > \angle DAB$. Vertical-angle and ray-order symmetries transport this to $\angle CAY + \angle BAX > \angle DAB$, the complement form of the desired four-right-angle bound.

The Lean proof graph is therefore much shorter than the classical angle-sum graph:

$$\begin{array}{c}
 C - A - X, \quad D - A - Y \\
 \Downarrow \\
 A; D, X, B \text{ is trihedral} \\
 \Downarrow \\
 \text{XI.20} \\
 \Downarrow \\
 \angle DAX + \angle XAB > \angle DAB \\
 \Downarrow \\
 \angle DAX \cong \angle CAY, \quad \angle XAB \cong \angle BAX \\
 \Downarrow \\
 \angle CAY + \angle BAX > \angle DAB \\
 \Downarrow \\
 \angle BAC + \angle CAD + \angle DAB < 4R.
 \end{array}$$

21.7 The Genuinely Spatial Obligation

The only nontrivial ambient-space obligation is to prove that replacing AC by the opposite ray AX preserves noncoplanarity.

Assume for contradiction that

$$A, B, X, D$$

lie in one plane π . Since $A \neq X$, the line AX lies in π . But C, A, X are collinear, so C lies on that same line. Consequently

$$C \in \pi.$$

Thus

$$A, B, C, D$$

would all lie in π , contradicting the original trihedral noncoplanarity.

$$C, A, X \text{ collinear} \implies C \in \pi$$

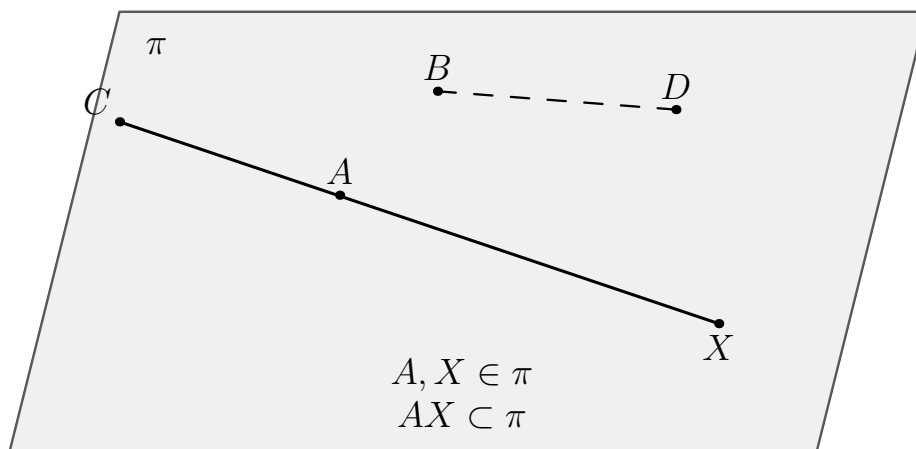


Figure 21.3: The sole spatial closure step in XI.21. If A, B, X, D lay in a plane π , then the line through the distinct points A, X would lie in π . Since C, A, X are collinear, this forces $C \in \pi$, contradicting the noncoplanarity of the original points A, B, C, D .

This is even weaker spatially than the corresponding XI.20 argument. XI.20 first had to construct a plane through three noncollinear points. XI.21 starts its contradiction with an already hypothesized plane and needs only line-in-plane closure.

21.8 Direct Hilbert 3D Reconstruction

After minimization of the inherited test binder, the direct production theorem is

```
theorem euclid_proposition_11_21
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hTri :
    HilbertTrihedralConfiguration
      Geo A B C D) :
  HilbertTrihedralAnglesLessThanFourRightAngles
    Geo A B C D
```

The key proposition-local lemmas are:

- hilbert_XI21_supplement_rays_exist;
- hilbert_XI21_trihedral_opposite_middle;
- hilbert_XI21_trihedral_reorder;
- hilbert_XI21_vertical_DAX_CAY;
- hilbert_XI21_angle_XAB_BAX;
- hilbert_XI21_direct_comparison.

The only Hilbert-space-specific call in the geometric core is

```
HilbertSpaceIncidence.line_in_plane
```

inside the proof that the opposite-ray trihedral configuration remains noncoplanar.

No use is made of I.32 in the production proof.

21.9 Hilbert–Wyler Reconstruction

The independent Wyler theorem is

```
theorem euclid_proposition_11_21_wyler
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertCongruence Geo]
```

```
[W : HilbertWylerAxioms Geo]
(A B C D : Geo.Point)
(hTri :
  HilbertTrihedralConfiguration
    Geo A B C D) :
HilbertTrihedralAnglesLessThanFourRightAngles
  Geo A B C D
```

Its production file is

```
CGJteamLab/Wyler/Proposition11_21.lean
```

The Wyler proof does not import or call `euclid_proposition_11_21`. It reconstructs the opposite-ray noncoplanarity lemma independently and calls the already independent Wyler version of XI.20:

```
euclid_proposition_11_20_wyler
```

In the sole spatial closure step the Wyler proof uses

```
W.line_in_plane
```

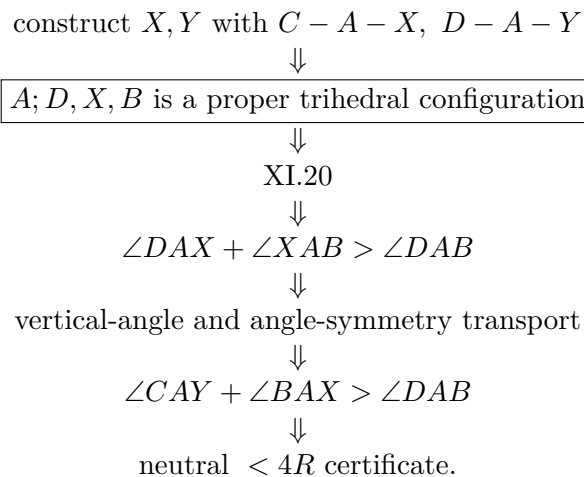
instead of

```
HilbertSpaceIncidence.line_in_plane
```

No `plane_through` field is needed in XI.21 itself. In particular, the proof does not use Wyler I.7, Smith I.5, span exchange, or an ambient dimension bound.

21.10 Hilbert versus Hilbert–Wyler Proof DAG

The two production routes are identical after the noncoplanarity-preservation step:



Only the boxed node differs. The direct route uses

```
HilbertSpaceIncidence.line_in_plane
```

whereas the Hilbert–Wyler route uses

```
HilbertWylerAxioms.line_in_plane
```

The common target and congruence-transport lemmas live in

```
HilbertThreeAnglesFourRight.lean
```

and are independent of the ambient spatial route.

21.11 E^n Transfer Audit

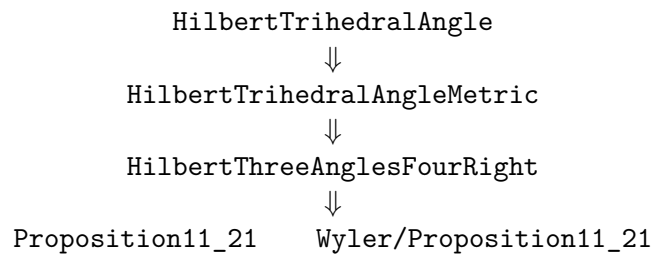
XI.21 gives an even cleaner dimensional-transfer result than XI.20.

1. **Purely planar/order step.** Ray extension, vertical angles, ordinary angle symmetry, congruence transport, and the complement representation are all dimension-free.
2. **Spatial step.** One must show that replacing one ray of a noncoplanar trihedral configuration by its opposite ray preserves noncoplanarity.
3. **Wyler replacement.** The proof needs only the principle that a plane containing two distinct points of a line contains the whole line.
4. **Expected E^4 and E^n analogue.** No local 3-flat is required. The same trihedral theorem applies to any ambient dimension in which the common point–line–plane incidence and planar metric theory are available.
5. **Missing higher-dimensional lemma.** For XI.21 there is none. The proposition is not an E^3 -specific theorem disguised by notation; the proof exposes it as a local dimension-free theorem about three rays.

This does not mean that all of Book XI is dimension-free. Rather, XI.20 and XI.21 now form a useful test pair showing where classical “solid geometry” can be promoted directly into the reusable synthetic library for later E^4 work.

21.12 Dependency Summary

The production architecture is



The two XI.21 branches additionally depend on their corresponding XI.20 production theorems:

euclid_proposition_11_20 euclid_proposition_11_20_wyler

The dependency classification is:

New proposition-specific geometric axiom	no
Numerical angle measure	no
General angle addition	no
I.32 in the production proof	no
Continuity	no new use
Hilbert Group IV	no, after binder minimization
Wyler I.7	no
Smith I.5 / exchange	no
Ambient dimension bound	no
Coordinates / vectors / scalar product	no
sorry/admit	no
Full repository build	successful, 2224 jobs

21.13 Conclusion

Proposition XI.21 initially appears to be the point where the project must introduce an arithmetic of finite angle sums. The formal reconstruction shows that this is unnecessary.

Euclid's printed proof remains mathematically important: it explains the classical relation between XI.20, the triangle angle theorem I.32, and the four-right-angle bound. But the synthetic complement representation reveals a shorter route. Two opposite-ray extensions transform the target into a single XI.20 comparison; vertical angles and unoriented-angle symmetry then transport that comparison directly to the required certificate.

The result has a second architectural consequence. The independent Wyler proof needs only line-in-plane closure at the ambient level. XI.21 therefore joins XI.20 as a reusable dimension-free theorem above the planar metric layer. In later E^4 work the natural task is not to reprove XI.21, but to show that the local Coxeter configuration satisfies the already available trihedral hypotheses.

Chapter 22

Proposition XI.22

22.1 Purpose and Scope

Proposition XI.22 is a striking boundary case in Book XI. It occurs in the solid-geometry book and prepares the construction of a solid angle in the next proposition, but its own mathematical content is completely planar.

The proposition begins with three proper plane angles. Each angle is contained by two segments of one common length. If the sum of any two angles is greater than the third, Euclid joins the endpoints of the equal radial segments and proves that the three resulting chords satisfy the triangle inequalities. Proposition I.22 can then construct a triangle with those three chord lengths.

The formal reconstruction preserves that structure, but makes two points explicit.

First, the project does not introduce a numerical angle measure or a general addition operation on angles. The phrase “the sum of two angles is greater than a third angle” is represented by the neutral synthetic relation `HilbertTwoAnglesGreaterThanAngle`.

Second, the final appeal to I.22 is not logically free. In the present Hilbert development, constructing a triangle from three side lengths ultimately uses circle–circle intersection. The axiom audit for XI.22 therefore exposes `hilbert_circle_circle_intersection`, even though every preceding angle-to-chord comparison is neutral and dimension-free.

Two public production entry points are retained:

```
Geometry.euclid_proposition_11_22
Geometry.euclid_proposition_11_22_wyler
```

They do not call one another. Both are proved from the same neutral planar core, and neither requires a spatial incidence class.

22.2 Euclid’s Statement

Let the three plane angles be

$$\angle AOB, \quad \angle CPD, \quad \angle EQF,$$

and assume that the six radial segments have one common length:

$$OA = OB = PC = PD = QE = QF.$$

Assume also the three strict angle inequalities

$$\begin{aligned}\angle AOB + \angle CPD &> \angle EQF, \\ \angle CPD + \angle EQF &> \angle AOB, \\ \angle EQF + \angle AOB &> \angle CPD.\end{aligned}$$

Joining the outer endpoints gives the three chords

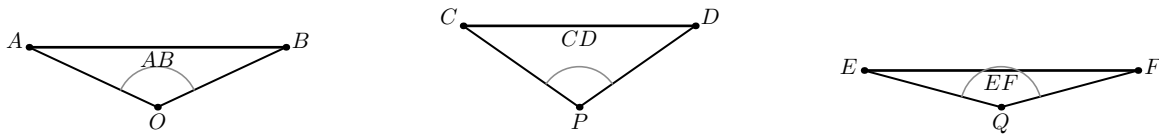
$$AB, \quad CD, \quad EF.$$

The proposition asserts that these three segments satisfy the triangle inequalities and hence can be used as the three sides of a triangle.

The historical statement is therefore not intrinsically spatial. Its role in Book XI is preparatory: XI.23 uses the triangle supplied by XI.22 in the construction of a solid angle.

22.3 Classical Configuration

Figure 22.1 displays the configuration used throughout the formal reconstruction. The three angle vertices are unrelated points in a plane-level theory; no common ambient plane needs to be named because the theorem is formulated only through point, order, congruence, and angle relations.



$$OA = OB = PC = PD = QE = QF$$

$$\angle AOB + \angle CPD > \angle EQF$$

$$\angle CPD + \angle EQF > \angle AOB$$

$$\angle EQF + \angle AOB > \angle CPD$$

Figure 22.1: Three equal-radius angle configurations for Proposition XI.22. The radial segments have one common length. The chords AB , CD , and EF are the three candidate side lengths.

The geometric mechanism is the familiar equal-radius chord principle: for two isosceles triangles with corresponding radial sides equal, equality of included angles gives equality of the opposite chords, while strict inequality of the included angles gives strict inequality of the opposite chords.

In the formal library these two facts are isolated as

```
hilbert_equalRadiusChord_congruent
hilbert_equalRadiusChord_less
```

in `HilbertEqualRadiusChord.lean`.

22.4 Euclid's Classical Proof

Euclid proves one triangle inequality and obtains the other two cyclically. Using the historical lettering, suppose the three angles are

$$\angle ABC, \quad \angle DEF, \quad \angle GHK,$$

with

$$AB = BC = DE = EF = GH = HK.$$

The corresponding chords are

$$AC, \quad DF, \quad GK.$$

To prove, for example,

$$AC + GK > DF,$$

Euclid constructs at H a ray HL such that

$$\angle KHL = \angle ABC$$

and lays off

$$HL = AB.$$

By SAS,

$$KL = AC.$$

The angle hypothesis

$$\angle ABC + \angle GHK > \angle DEF$$

and the placement of HL give

$$\angle GHL > \angle DEF.$$

The equal-radius comparison then yields

$$GL > DF.$$

Finally the triangle inequality in triangle GKL gives

$$GK + KL > GL,$$

hence

$$GK + AC > DF.$$

Cycling the construction gives

$$AC + DF > GK, \quad DF + GK > AC.$$

The classical proof then invokes I.22 to construct a triangle from AC, DF, GK .

The local classical dependency spine is

copy one angle on an adjacent ray and lay off an equal radius

↓

I.4 (SAS)

↓

one auxiliary chord equals the first source chord

↓

I.24 (larger included angle gives larger base)

↓

I.20 (triangle inequality)

↓

one chord-sum inequality

↓

cyclic repetition

↓

I.22.

22.5 Why Ordinary Angle Addition Is Not Used

The expression

$$\alpha + \beta > \gamma$$

is potentially misleading if ordinary angles are treated only as the smaller angles between two rays. A numerical sum can pass through two right angles, while the project angle object remains an ordinary proper angle.

For this reason XI.20 introduced a witness-based relation rather than a numerical operation:

```
def HilbertTwoAnglesGreaterThanAngle
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A O B C P D E Q F : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A O B) /\
  Not (PrimCollinear Geo C P D) /\
  Not (PrimCollinear Geo E Q F) /\
  (
    HilbertAngleLess Geo E Q F A O B
    /\
    Geo.AngleCongruent E Q F A O B
    /\
    exists X,
      HilbertRayMeetsSegment Geo Q X E F /\
      Geo.AngleCongruent A O B E Q X /\
      HilbertAngleLess Geo X Q F C P D
  )
```

The three cases have a direct geometric meaning.

1. The target angle is already smaller than the first source angle.
2. The target angle equals the first source angle.

3. The target angle decomposes into a copy of the first source angle plus a remainder smaller than the second source angle.

This relation was moved to the neutral module `HilbertAngleSumComparison.lean`. It is shared by XI.20, XI.21, and XI.22 and contains no spatial hypothesis.

22.6 The Equal-Radius Chord Bridge

The central reusable theorem of the XI.22 development is

```

theorem
  hilbert_twoAnglesGreaterThanAngle_equalRadius_chordSumGreater
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    ...
    (hSum :
      HilbertTwoAnglesGreaterThanAngle
        Geo A O B C P D E Q F)
    ...
    :
    HilbertSegmentSumGreater
      Geo A B C D E F

```

Its mathematical content is

$$\angle AOB + \angle CPD > \angle EQF \implies AB + CD > EF,$$

under the common-radius hypotheses.

This theorem is the genuine neutral core of XI.22. Once it is available, the public proposition is only a cyclic threefold application followed by I.22.

The proof is split according to the three constructors of `HilbertTwoAnglesGreaterThanAngle`.

22.6.1 First-angle-greater branch

If

$$\angle EQF < \angle AOB,$$

then equal-radius chord comparison gives

$$EF < AB.$$

Since CD is non-null,

$$EF < AB < AB + CD.$$

The corresponding formal lemma is

```

hilbert_equalRadiusChordSum_of_first_greater

```

22.6.2 First-angle-equal branch

If

$$\angle EQF \cong \angle AOB,$$

then SAS gives

$$EF = AB.$$

Again the non-null segment CD makes the sum strictly larger:

$$EF = AB < AB + CD.$$

The formal lemma is

hilbert_equalRadiusChordSum_of_first_equal

22.6.3 Decomposition branch

The third branch is the formal counterpart of Euclid's auxiliary construction. A ray QX meets the open segment EF , the angle EQX is congruent to the first source angle, and the remainder XQF is smaller than the second source angle.

A point R is laid off on the same ray with

$$QR = OA.$$

Then

$$AB \cong ER$$

by equal-radius SAS and

$$RF < CD$$

by the equal-radius strict chord comparison.

Figure 22.2 shows this branch.

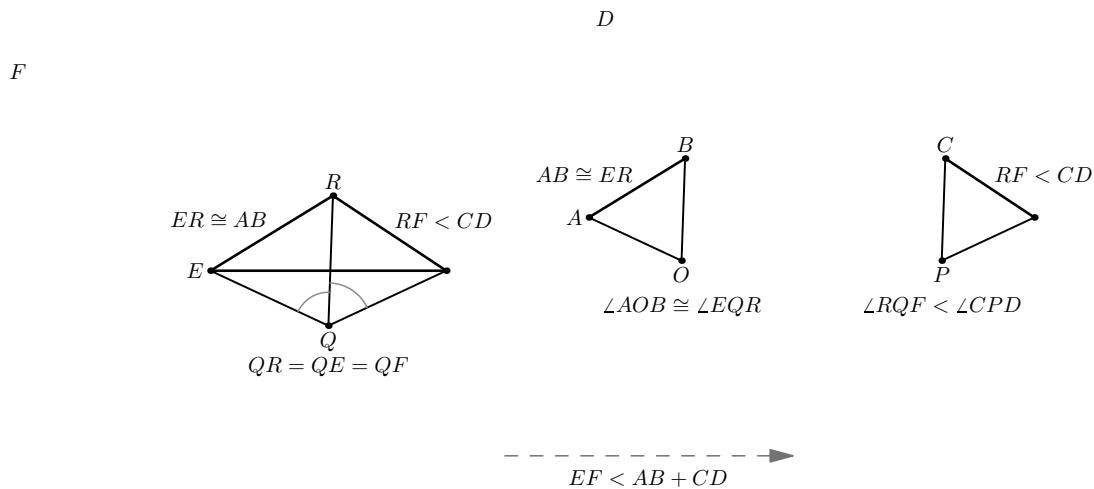


Figure 22.2: The decomposition branch. The ray QR carries a copy of the first source angle and the common radius. Thus $ER \cong AB$, while the remainder comparison gives $RF < CD$. The proof then converts these two facts into $EF < AB + CD$.

The formal proof deliberately handles both possibilities for E, R, F .

If they are noncollinear, I.20 applied to triangle ERF gives a segment representing $ER + RF$ and strictly exceeding EF . Replacing ER by AB and RF by the longer segment CD yields

$$EF < AB + CD.$$

If they are collinear, the interior-ray crossing forces the laid-off point R to coincide with the crossing point on the open segment EF . Hence

$$E - R - F,$$

so

$$EF = ER + RF < AB + CD.$$

This second branch avoids introducing a separate theorem saying that a line meets an equal-radius circle in at most two points.

The proposition-local lemmas recording this split are

```
hilbert_equalRadiusChordSum_of_decomposition_core
hilbert_chordSum_of_collinear_decomposition
hilbert_equalRadiusChordSum_of_decomposition
```

22.7 Synthetic Segment Addition

The project does not use numerical segment lengths. A strict inequality

$$AB + CD > EF$$

is represented by `HilbertSegmentSumGreater`: one lays off a copy of CD beyond B , obtaining a representative segment for the sum, and then compares EF with that representative.

The neutral helper module `HilbertSegmentSum.lean` isolates the corresponding operations:

```
hilbert_segmentSum_witness
hilbertSegmentSumGreater_of_less_first
hilbertSegmentSumGreater_of_congruent_first
hilbertSegmentSumGreater_swap
```

This layer is important for XI.22 because the final I.22 interface expects exactly the three strict segment-sum inequalities, not a numerical length equation.

22.8 The Formal Proof DAG

After the neutral chord theorem has been proved, XI.22 has a very small public proof graph:

$$\begin{array}{ccc} \angle AOB + \angle CPD > \angle EQF & \angle CPD + \angle EQF > \angle AOB & \angle EQF + \angle AOB > \angle CPD \\ \Downarrow & \Downarrow & \Downarrow \\ AB + CD > EF & CD + EF > AB & EF + AB > CD \end{array}$$

followed by

$$\begin{array}{c}
 AB + CD > EF, \quad CD + EF > AB, \quad EF + AB > CD \\
 \Downarrow \\
 \text{I.22} \\
 \Downarrow \\
 \text{a triangle with side lengths } AB, CD, EF.
 \end{array}$$

There is no spatial node in this graph.

22.9 Direct Hilbert Reconstruction

The production theorem is

```

theorem euclid_proposition_11_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A O B C P D E Q F : Geo.Point)
  (h12_3 :
    HilbertTwoAnglesGreaterThanAngle
      Geo A O B C P D E Q F)
  (h23_1 :
    HilbertTwoAnglesGreaterThanAngle
      Geo C P D E Q F A O B)
  (h31_2 :
    HilbertTwoAnglesGreaterThanAngle
      Geo E Q F A O B C P D)
  ...
  :
  exists K : Geo.Point,
    Geo.Congruent C K A B /\
    Geo.Congruent D K E F

```

The third side of the constructed triangle is CD , while the two new sides are congruent to AB and EF . This is exactly the form supplied by the existing production theorem `euclid_proposition_22`.

The proof has four stages:

1. extract properness of the three angles and hence non-nullity of the three chords;
2. apply the neutral angle-to-chord-sum theorem to the first cyclic inequality;
3. rebase the common-radius congruences twice and apply the same neutral theorem to the remaining two cyclic inequalities;
4. normalize the order of one segment sum and call I.22.

No earlier proposition of Book XI is called by the production theorem.

22.10 Hilbert–Wyler Reconstruction

The independent Wyler entry point is

```

theorem euclid_proposition_11_22_wyler
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  ...
  :
  exists K : Geo.Point,
    Geo.Congruent C K A B /\
    Geo.Congruent D K E F

```

Its production file is

```
CGJteamLab/Wyler/Proposition11_22.lean
```

This file does not import or call `euclid_proposition_11_22`. It independently performs the three cyclic applications of the common neutral chord theorem and then calls I.22.

Unlike XI.20 and XI.21, there is no point at which the Hilbert and Wyler routes need different spatial infrastructure. In particular XI.22 requires neither

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertWylerAxioms

```

nor any local flat, exchange, or dimension assumption.

Thus the two public entry points differ only as architectural endpoints. Their mathematical proof DAG is identical.

22.11 Planar and Dimension-Free Status

XI.22 is therefore stronger architecturally than its location in Book XI might suggest.

1. The statement mentions only points, proper angles, congruent segments, synthetic angle comparison, and segment-sum comparison.
2. The equal-radius chord lemmas are ordinary planar SAS/I.24 facts.
3. The decomposition branch uses only order, angle comparison, congruence, and I.20.
4. The final construction is I.22.
5. No proposition-specific spatial incidence statement occurs.

Consequently the same theorem can be used unchanged inside E^3 , E^4 , or a general E^n development whenever the common planar Hilbert metric layer is available. No local 3-flat reduction is required.

This is an instructive contrast with XI.20 and XI.21. Those propositions are dimension-free after a small ambient incidence obligation has been discharged. XI.22 has no such obligation at all.

22.12 Continuity Boundary and Proposition I.22

The final call is

```
euclid_proposition_22
```

with the three chords AB, CD, EF as the prescribed side lengths.

The existing I.22 theorem takes the three strict segment-sum inequalities and constructs a point K satisfying

$$CK \cong AB, \quad DK \cong EF,$$

with CD as the third side.

This is precisely where circle continuity enters. The angle-to-chord reduction itself is neutral; the existence of the final triangle is not. The formal dependency therefore clarifies a point hidden in the classical presentation: proving the three triangle inequalities and constructing the triangle from them are distinct logical steps.

The historical proof simply cites I.22 at this point. The formal theorem makes the construction boundary explicit through the already audited implementation of I.22.

22.13 Axiom Audit

The audit commands are

```
#print axioms Geometry.euclid_proposition_11_22
#print axioms Geometry.euclid_proposition_11_22_wyler
```

and Lean reports the same dependency set for both:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

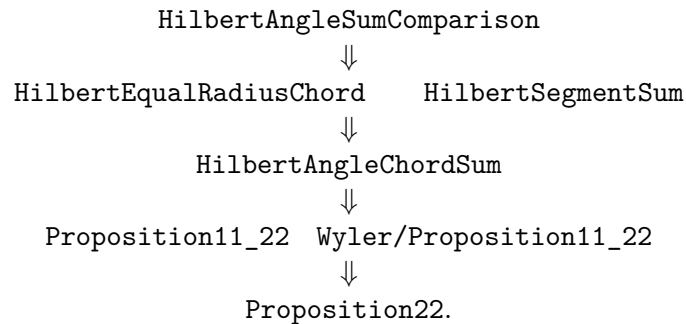
The equality of the two audit results confirms that the Wyler entry point adds no Wyler-specific axiom and that the direct entry point adds no spatial Hilbert axiom.

The occurrence of `hilbert_circle_circle_intersection` comes from the final I.22 construction. It should not be attributed to the angle comparison or equal-radius chord machinery.

The two `bookZero_nullSegment` declarations are inherited elementary segment infrastructure. No new XI.22-specific axiom constant is introduced.

22.14 Dependency Summary

The production architecture is



The status can be summarized as follows:

New proposition-specific geometric axiom	no,
Numerical angle measure	no,
General angle addition operation	no,
Spatial incidence	no,
HilbertSpacePrimitive	no,
HilbertWylerAxioms	no,
Wyler I.7 / Smith I.5 / exchange	no,
Ambient dimension bound	no,
Coordinates / vectors / scalar product	no,
Circle continuity	yes, through I.22,
sorry/admit	no,
Full repository build	clean.

22.15 Conclusion

Proposition XI.22 is formally valuable for a reason that is easy to miss in the classical organization of the *Elements*. Although it belongs to Book XI and prepares a solid-angle construction, the proposition itself is a pure theorem of planar metric geometry.

The main formal work is not spatial. It is the conversion

$$\text{two-angle comparison} \implies \text{two-chord sum comparison.}$$

The witness-based relation `HilbertTwoAnglesGreaterThanAngle` makes this conversion possible without numerical angle measure. Equal-radius SAS and I.24 handle the simple branches, while the decomposition branch reconstructs Euclid's auxiliary ray synthetically and covers both the collinear and noncollinear placements of the laid-off point.

After that neutral theorem is isolated, XI.22 itself collapses to three cyclic applications and one invocation of I.22. The identical Hilbert and Hilbert–Wyler axiom audits show that there is no residual spatial content.

The only non-neutral boundary is constructive: I.22 uses circle–circle intersection. Thus XI.22 provides a particularly clean example of the architectural distinction between neutral comparison geometry and continuity needed for existence constructions.

Chapter 23

Proposition XI.23

23.1 Purpose and Scope

Proposition XI.23 closes the introductory trihedral block of Book XI. The preceding propositions provide the two necessary conditions for three plane angles to occur as the faces of a proper trihedral angle:

1. any two of the three angles are together greater than the third (XI.20);
2. the three angles are together less than four right angles (XI.21).

XI.23 proves the converse construction. Starting from three proper plane angles satisfying exactly these two conditions, one constructs a proper solid angle whose three face angles are respectively congruent to the given ones.

The production Lean theorem is

```
theorem euclid_proposition_11_23
  [Hinc : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [HSO : HilbertSpaceOrder
    (Geo := Geo) (H := Hinc) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := Hinc) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (A O B C P D E Q F : Geo.Point)
  (h12_3 :
    HilbertTwoAnglesGreaterThanAngle
      Geo A O B C P D E Q F)
  (h23_1 :
    HilbertTwoAnglesGreaterThanAngle
      Geo C P D E Q F A O B)
  (h31_2 :
    HilbertTwoAnglesGreaterThanAngle
      Geo E Q F A O B C P D)
  (hFour :
    HilbertThreeAnglesLessThanFourRightAngles
      Geo A O B C P D E Q F) :
```

HilbertThreeAnglesHaveTrihedralRealization
Geo A O B C P D E Q F

The final target is deliberately neutral. It does not expose the auxiliary circumcenter, the chosen base plane, or the normal construction. It asserts only the existence of a proper trihedral configuration whose three face angles realize the three prescribed angles.

23.2 Euclid's Statement

In the classical statement one is given three plane angles

$$\angle ABC, \quad \angle DEF, \quad \angle GHK,$$

such that every two together are greater than the remaining one and such that their total is less than four right angles. The task is to construct a solid angle bounded by three plane angles equal to these.

Thus XI.23 is the sufficiency theorem corresponding to XI.20 and XI.21. The logical pattern is

$$\begin{array}{c} \text{pairwise angle inequalities} \\ + \text{ three-angle bound } < 4R \\ \Downarrow \\ \text{existence of a trihedral realization.} \end{array}$$

No numerical angle measure is introduced in the formal theorem. The two source conditions are represented synthetically by `HilbertTwoAnglesGreaterThanAngle` and `HilbertThreeAnglesLessThanFourRightAngles`.

23.3 Euclid's Classical Construction

Euclid first cuts six equal radii on the sides of the three given angles and joins the corresponding chord endpoints. By XI.22 the three resulting chord lengths form a triangle. Let this triangle be LMN .

A circle is described through L, M, N , with center O . The decisive metric claim is that the common source radius, say AB , is greater than the circumradius LO .

Euclid excludes equality by SSS: if $AB = LO$, then the three given angles would equal the three central angles

$$\angle LOM, \quad \angle MON, \quad \angle NOL,$$

whose sum is four right angles, contradicting XI.21.

He then excludes $AB < LO$ by laying off equal shorter radii on OL and OM , producing a smaller similar triangle. The printed proof uses VI.2, I.29, VI.4, V.16 and I.25 to compare its chord with LM , and hence to obtain strict inequalities between all three source angles and the three central angles. Their sum would then be greater than four right angles, again contradicting the hypothesis.

Having proved $AB > LO$, Euclid erects OR perpendicular to the plane of the circumcircle by XI.12 and chooses its length so that

$$OR^2 = AB^2 - LO^2.$$

The auxiliary lemma constructs this length by a semicircle argument using IV.1, III.31 and I.47. Finally one joins R to L, M, N . Right-triangle arguments give

$$RL = RM = RN = AB,$$

and three applications of SSS recover

$$\angle LRM \cong \angle ABC, \quad \angle MRN \cong \angle DEF, \quad \angle NRL \cong \angle GHK.$$

Thus the solid angle at R has the required three faces.

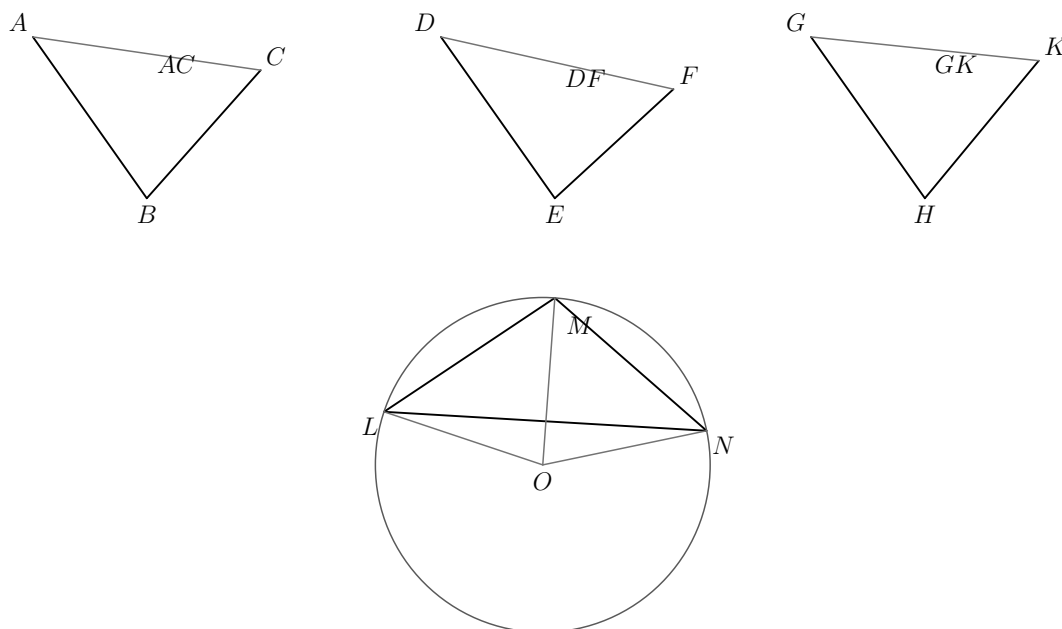
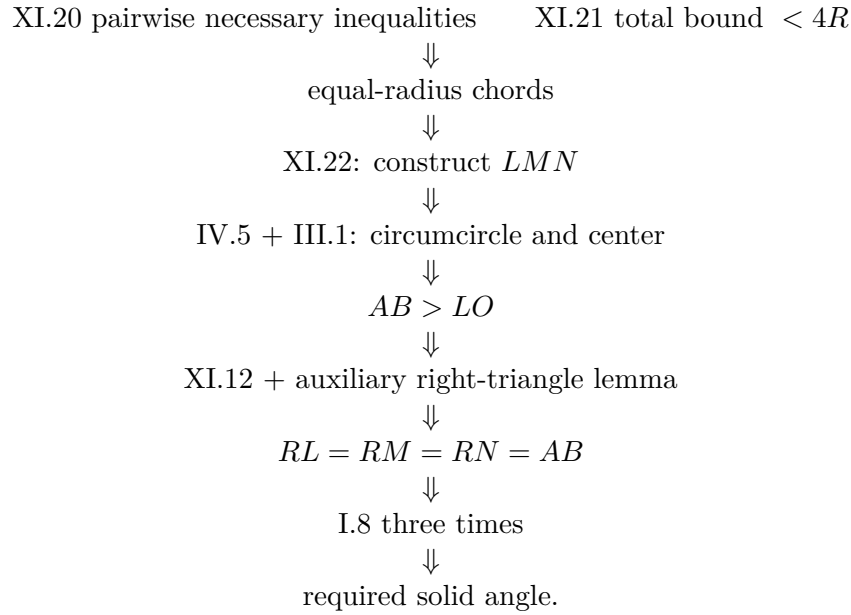


Figure 23.1: The first classical stage of XI.23. Equal radii are cut on the three prescribed angles; their three chords become the side lengths of the auxiliary triangle LMN by XI.22. The right panel shows its circumcircle and center O .

23.4 Classical Dependency Pattern

At proposition level the source DAG is



The long middle step $AB > LO$ is where Euclid uses the largest collection of earlier plane propositions, including the Book V–VI proportional route in the strict case.

23.5 The Formal Target in HilbertInterfaceXI

The Book XI interface contains the proposition-independent language

```

def HilbertTrihedralRealizesThreeAngles
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (R L M N : Geo.Point)
  (A O B C P D E Q F : Geo.Point) : Prop :=
HilbertTrihedralConfiguration Geo R L M N /\
Geo.AngleCongruent L R M A O B /\
Geo.AngleCongruent M R N C P D /\
Geo.AngleCongruent N R L E Q F

def HilbertThreeAnglesHaveTrihedralRealization
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (A O B C P D E Q F : Geo.Point) : Prop :=
exists R L M N : Geo.Point,
  HilbertTrihedralRealizesThreeAngles
    Geo R L M N A O B C P D E Q F

```

This is an important architectural boundary. The target records the mathematical output of XI.23 but none of its construction witnesses. In particular it does not require a global ambient `HilbertCongruence` instance merely to state the result.

23.6 Normalization into One Explicit Plane

The production proof does not assume that the three prescribed angles already live in a common plane. It chooses the plane π containing the second source angle and copies all three angles into that same plane with one common initial ray.

Schematically the copied angles are

$$\angle CPR_1, \quad \angle CPR_2, \quad \angle CPR_3,$$

with

$$PC \cong PR_1 \cong PR_2 \cong PR_3.$$

The helper

```
hilbert_XI23_copy_angle_equal_radius_in_plane_XI
```

performs one such copy. The general spatial transport layer then carries the three XI.20-type inequalities and the XI.21-type four-right-angle certificate to the copied angles:

```
hilbert_space_twoAnglesGreater_transport_all_in_plane
hilbert_space_threeAnglesLessThanFourRight_transport_all_in_plane
```

Only after this transport are the points repackaged as `PlanePoint` values and the four hypotheses converted to statements in `PlaneGeo( $\pi$ )`.

This separation is deliberate:

$$\text{ambient space} \longrightarrow \text{explicit plane } \pi \longrightarrow \text{PlaneGeo}(\pi).$$

No global planar geometry is installed on the ambient three-space.

23.7 XI.22 Inside `PlaneGeo( $\pi$ )`

Once the three angles have been normalized into the fixed plane, XI.22 is called entirely inside `PlaneGeo( $\pi$ )`. The result is a chord triangle, represented in the final normalized pipeline by

$$C, \quad R_2, \quad K.$$

The two nontrivial side data exported from XI.22 are congruences of the constructed sides with the appropriate equal-radius chords. The proof then establishes

$$\neg \text{Collinear}(C, R_2, K),$$

so the triangle has a circumcenter U in the same plane.

23.8 The Circumradius Inequality

The central planar theorem is

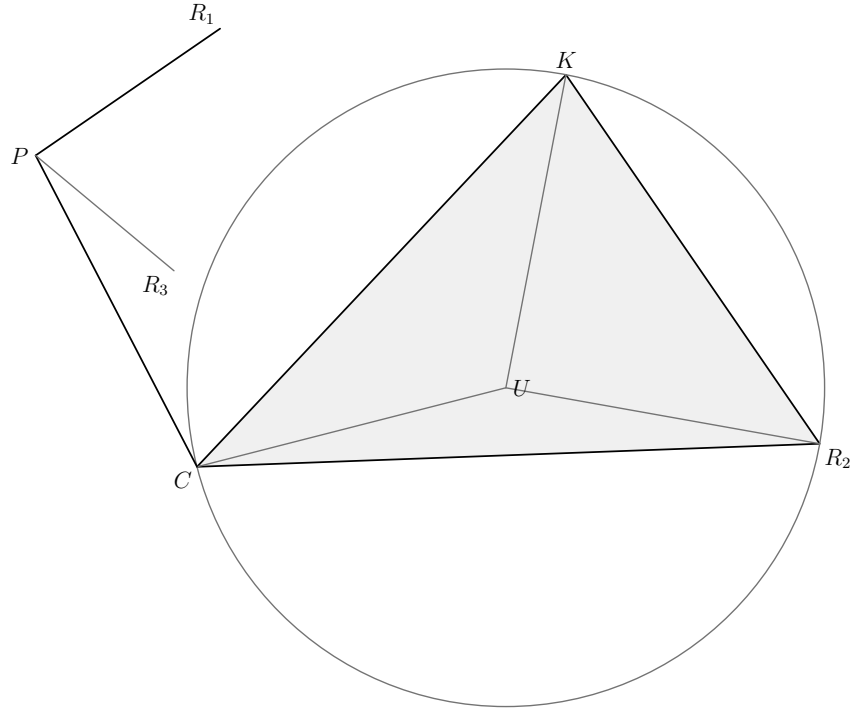


Figure 23.2: The normalized planar core. The three prescribed angles have been copied to one explicit plane. XI.22 produces the base triangle CR_2K ; its circumcenter U and equal radii remain entirely inside $\text{PlaneGeo}(\pi)$.

```
hilbert_XI23_circumradius_less_source_radius_XI
```

which proves, in the planar setting,

$$UC < OA.$$

The formal proof follows the same high-level trichotomy as Euclid but replaces the proportional Book V–VI argument by synthetic angle-order arguments.

First, equality $UC \cong OA$ is impossible. Equal radii would identify the three source angles with the three central angles by SSS, contradicting the strict four-right-angle certificate.

Second, the opposite strict inequality $OA < UC$ is impossible. The proof compares the corresponding equal-radius chords and central angles. Its main difficulty is not numerical computation but the possible cyclic order of the three rays from the circumcenter.

Finally, segment trichotomy leaves only

$$UC < OA.$$

A substantial part of the proposition-local proof exists to make this argument configuration-complete. The three-ray classifier separates the three “one ray enters the opposite chord” cases from the cyclic case, and each branch is discharged synthetically.

This is stronger as a proof audit than relying on a representative diagram: the conclusion is proved for every permitted incidence/order configuration covered by the theorem hypotheses.

23.9 The Auxiliary Height Segment

The theorem

```
hilbert_XI23_radius_height_in_plane_XI
```

packages the radius comparison together with the planar construction needed for the spatial apex.

From $UC < OA$ it constructs points E_0, K_0, H_0 such that the relevant right triangle has one leg congruent to the circumradius and hypotenuse congruent to the source radius. In particular the output includes

$$OE_0 \cong UC, \quad OH_0 \cong OA, \quad \angle OE_0H_0 \text{ right.}$$

The entire statement is made in $\text{PlaneGeo}(\pi)$. The ambient Euclidean-space structure supplies the Euclidean-plane instance only for this explicit slice. There is still no global `HilbertEuclideanPlaneGeo` assumption on the ambient space.

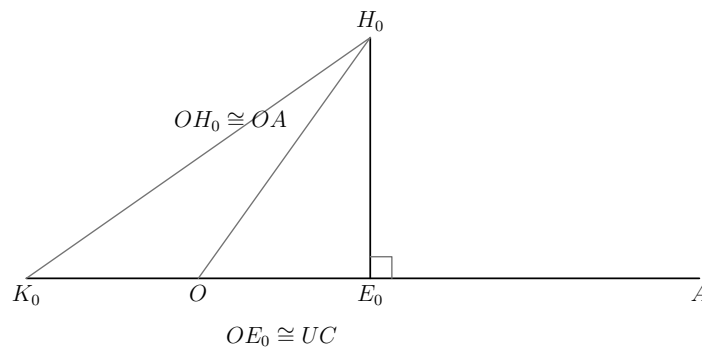


Figure 23.3: The synthetic replacement for Euclid's square-difference lemma. The planar construction encodes the source radius as the hypotenuse and the circumradius as one leg; the remaining leg is transported to the spatial normal.

23.10 The Spatial Apex

The normalized spatial package is

```
hilbert_XI23_normalized_spatial_apex_XI
```

Starting from the base plane π and its circumcenter U , XI.12 erects a line through U perpendicular to π . The height obtained from the auxiliary right triangle is laid off on this normal, producing an apex R with

$$R \notin \pi.$$

Spatial right-angle congruence and SAS then yield

$$RC \cong RR_2 \cong RK \cong PC.$$

The key point is that the final lateral edges are obtained without introducing coordinates, vector lengths, or a numerical Pythagorean equation. The role played classically by

$$OR^2 = AB^2 - LO^2$$

is represented by an explicitly constructed right triangle plus spatial SAS.

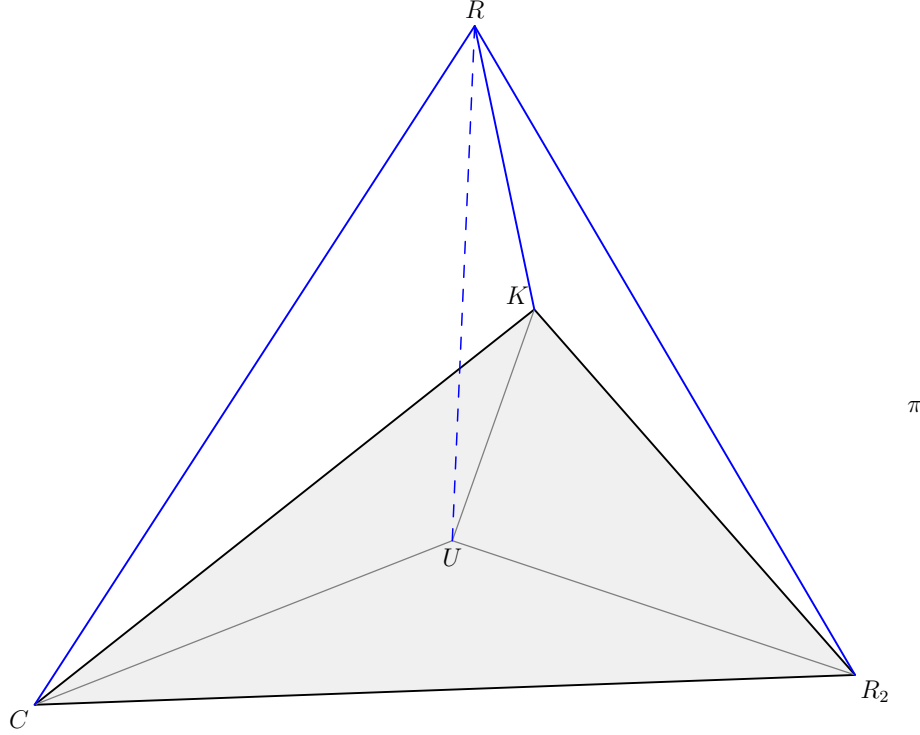


Figure 23.4: The final spatial construction. The base triangle CR_2K lies in π , U is its circumcenter, and R lies on the normal through U . The three blue lateral edges are proved congruent to the common source radius.

23.11 Recovery of the Three Face Angles

The final theorem now has three spatial triangles:

$$RKC, \quad RCR_2, \quad RR_2K.$$

Each is compared by SSS with one of the normalized planar equal-radius triangles:

$$PCR_1, \quad PCR_2, \quad PCR_3.$$

The common spatial theorem

```
hilbert_space_sss_angleA_in_plane
```

returns the required angle congruences across the ambient/plane boundary. They are then composed with the original angle-copy congruences to obtain

$$\begin{aligned} \angle KRC &\cong \angle AOB, \\ \angle CRR_2 &\cong \angle CPD, \\ \angle R_2RK &\cong \angle EQF. \end{aligned}$$

The cyclic order

$$R; K, C, R_2$$

is chosen because it aligns exactly with the three copied source angles.

23.12 Noncoplanarity and the Final Trihedral Package

It remains to prove that the three face triangles form a proper solid angle, not merely three planar triangles sharing a vertex.

Since C, R_2, K are noncollinear, they determine the base plane π . If another plane contained

$$R, K, C, R_2,$$

plane uniqueness would force it to equal π . But the constructed apex satisfies $R \notin \pi$. Hence the four points are not coplanar.

This supplies the `HilbertTrihedralConfiguration` part of the target. The final packaging is then purely structural:

```
hilbertTrihedralRealizesThreeAngles_intro
hilbertThreeAnglesHaveTrihedralRealization_intro
```

23.13 Hidden Formal Data

XI.23 exposes several pieces of geometric information that the classical diagram suppresses.

23.13.1 The source angles need not share a plane

The formal inputs are three ambient angles. Before XI.22 can be reused, the proof must construct congruent copies in one explicit plane and transport all four hypotheses to those copies.

23.13.2 The circumcenter must be certified to lie in the base plane

The spatial normal in XI.12 must be erected through a point already known to lie in the plane. The formal proof therefore reconstructs the circumcenter inside $\text{PlaneGeo}(\pi)$, rather than using an ambient circumcenter without an incidence certificate.

23.13.3 The circumradius comparison has genuine order cases

A single picture cannot encode all cyclic orders of the three radii. The Lean proof contains a three-ray classifier and treats the cyclic branch explicitly. This removes the diagram-dependent gap noted in modern commentary on the printed proof.

23.13.4 The final noncoplanarity is not automatic

Three correct face angles do not by themselves guarantee a proper solid angle. The proof uses $R \notin \pi$, noncollinearity of the base triangle, and plane uniqueness to establish the required trihedral configuration.

23.14 Relationship with XI.20, XI.21, and XI.22

XI.23 is best read as the converse synthesis of the previous three results.

XI.20	:	each pair of face angles is greater than the third,
XI.21	:	the three face angles are less than $4R$,
XI.22	:	the three equal-radius chords form a triangle,
XI.23	:	these data construct the trihedral angle.

At API level the same structure appears as

```

HilbertTwoAnglesGreaterThanAngle
      +
HilbertThreeAnglesLessThanFourRightAngles
      ↓↓
euclid_proposition_11_22
      ↓↓
euclid_proposition_11_23

```

The production proof therefore reuses the synthetic language developed for XI.20–XI.22 instead of introducing a new numerical theory of angle sums.

23.15 Book XI Interface and the New 3D Transport Layer

The final refactor separates three levels.

```

Hilbert3DAngleComparisonTransport
      ↓↓
HilbertInterfaceXI
      ↓↓
Proposition11_23

```

The first file contains proposition-independent spatial transport machinery: betweenness transport, ray/segment transport, subangle transport, transport of angle order, transport of the XI.20 comparison relation, transport of the four-right-angle certificate, and ambient-to-PlaneGeo bridges.

`HilbertInterfaceXI` remains the public Book XI facade. It owns the neutral trihedral-realization language and imports the generic transport layer.

`Proposition11_23` contains only the proposition-specific construction: the radius contradiction, the auxiliary height, the spatial apex, and the final assembly.

23.16 Comparison with Euclid's Metric Route

The classical and formal proofs have the same large-scale geometry:

chord triangle $\longrightarrow$ circumcenter $\longrightarrow$ radius comparison
 $\longrightarrow$ normal apex $\longrightarrow$ SSS face recovery.

The main difference is the middle metric technology.

Euclid's strict-radius contradiction uses similarity and proportion:

VI.2 + VI.4 + V.16 + I.25.

The production Lean proof instead uses the already developed synthetic angle-order API, equal-radius chord comparison, ray-order classification and the four-right-angle certificate. It therefore does not require a general Book V/VI ratio calculus merely to prove XI.23.

The second difference is the construction of the height. Euclid represents it by a difference of squares and verifies it by I.47. The formal proof constructs an auxiliary right triangle in the plane and transports its leg to the spatial normal. The result is geometrically equivalent for the purpose of the construction, but no numerical square operation is introduced into the Lean statement.

23.16.1 The Hilbert–Wyler route for XI.23

XI.23 now also has a production Hilbert–Wyler route in

```
CGJteamLab/Wyler/Proposition11_23.lean
```

with two final entry points:

```
euclid_proposition_11_23_wyler_core
euclid_proposition_11_23_wyler
```

The distinction between them is structural. The core theorem assumes an explicit

```
[W : HilbertWylerAxioms Geo]
```

and routes the spatial incidence part of the construction through the Wyler interface. In particular, the proof uses the Wyler fields

```
W.plane_through
W.line_in_plane
W.plane_unique
```

for the choice of the normalization plane, preservation of points on lines in that plane, and the final noncoplanarity argument.

The public theorem does not add `HilbertWylerAxioms` as a new three-dimensional axiom. Instead it derives the package from the established Hilbert 3D environment by

```
hilbertWylerAxioms_of_hilbert3D
```

proved in

```
CGJteamLab/HilbertWyler3DCompatibility.lean
```

Thus, in the present XI.23 theorem, the dependency direction is

$$\text{Hilbert 3D} \implies \text{Hilbert–Wyler incidence} \implies \text{XI.23}.$$

Wyler is therefore derived incidence theory in the three-dimensional Hilbert environment, not a competing additional foundation.

Only the incidence routing is changed. The metric and order core remains the same synthetic machinery used by the direct proof: `PlaneGeo`, XI.22, the circumcenter construction, the radius comparison, the auxiliary right triangle, XI.12, spatial SAS, and the final SSS recovery of the three face angles. This is intentional. The Wyler theorem tests the incidence organization of XI.23 without duplicating an already established metric library.

The resulting theorem is genuinely different from merely aliasing the direct Hilbert theorem: the normalization and noncoplanarity steps are reconstructed through the Wyler incidence API. At the same time it is not yet a theorem of arbitrary-dimensional Euclidean geometry. No E^4 structure occurs in the production proof, and no local metric 3-flat inside E^n is assumed.

23.17 E^n Transfer Audit

XI.23 is not as dimension-free as XI.20 and XI.21.

The planar core—angle transport, XI.22, circumcenter geometry, chord comparison, and the radius contradiction—lives in one explicit `PlaneGeo` slice and is therefore structurally portable.

The new Hilbert–Wyler production theorem sharpens the dimensional audit. It shows that the incidence operations used by the 3D proof can be routed through `HilbertWylerAxioms`: construction of the base plane, line-in-plane closure, and final plane uniqueness. In genuine Hilbert 3D this package is itself derived by `hilbertWylerAxioms_of_hilbert3D`.

The final construction nevertheless still uses genuinely spatial metric data:

- a line through the circumcenter perpendicular to the base plane;
- a point on this normal outside the plane;
- spatial congruence and SAS across different plane slices;
- SSS recovery of the three trihedral face angles.

Consequently the project now has a Wyler-routed XI.23 theorem in Hilbert 3D, but not yet an arbitrary-dimensional E^n version. A higher-dimensional transfer would still need an explicit local three-dimensional carrier equipped with the required perpendicularity and congruence theory, or another theorem showing that the metric apex construction can be confined to a suitable 3-flat.

23.18 Axiomatic Status

The kernel audit was run on both the direct Hilbert theorem and the Hilbert–Wyler production route:

```
import CGJteamLab.Proposition11_23
import CGJteamLab.Wyler.Proposition11_23

#print axioms Geometry.euclid_proposition_11_23
#print axioms Geometry.euclid_proposition_11_23_wyler
#print axioms Geometry.euclid_proposition_11_23_wyler_core
```

Lean reports the same dependency set in all three cases:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

This equality of kernel dependency sets is important. The Wyler proof is not an alias of the direct theorem: its spatial incidence front end is reconstructed through `HilbertWylerAxioms`. Nevertheless, neither the explicit `wyler_core` theorem nor the public theorem obtained from `hilbertWylerAxioms_of_hilbert3D` introduces any additional axiom. The architectural refactor changes the route of the proof, not its axiomatic cost.

The logical axioms `propext`, `Classical.choice`, and `Quot.sound` are part of the standard project baseline.

Two Book Zero convenience axioms occur:

```
Geometry.bookZero_nullSegment1
Geometry.bookZero_nullSegment2
```

The first expresses that a segment congruent to a null segment is itself null. The second expresses congruence of null segments. Both are already tracked project-level interface debt; they are not proposition-specific axioms introduced by XI.23.

The genuinely stronger geometric dependency is

```
Geometry.hilbert_circle_circle_intersection
```

This is the project’s circle–circle continuity axiom: under the usual inside/outside hypotheses, two circles meet. XI.23 therefore does depend, through its planar circle/circumcircle construction machinery, on a continuity principle.

This is an important distinction from XI.20 and XI.21. Those propositions could be expressed in the neutral angle-comparison and spatial-incidence layers, while XI.23 must actually construct the planar circle geometry used to build the base configuration and its circumcenter.

Accordingly, the precise status is:

New proposition-specific axiom:	no
New spatial axiom:	no
Book Zero null-segment axioms used:	yes, two
Circle–circle continuity used:	yes
Numerical angle measure used:	no
Coordinate or vector structure used:	no
Global ambient HilbertEuclideanPlane Geo:	no
Explicit PlaneGeo Euclidean slice:	yes
sorry / admit:	no
Full lake build:	clean

The continuity dependency is inherited from established circle-construction infrastructure. XI.23 does not introduce a new local continuity axiom of its own.

23.19 Diagrammatic Audit

Four geometric states are retained.

1. The equal-radius chord construction together with the XI.22 base triangle and circumcircle.
2. The normalized planar Lean core C, R_2, K, U .
3. The auxiliary right triangle representing the height construction.
4. The final spatial apex over the base plane.

No separate figure is introduced for:

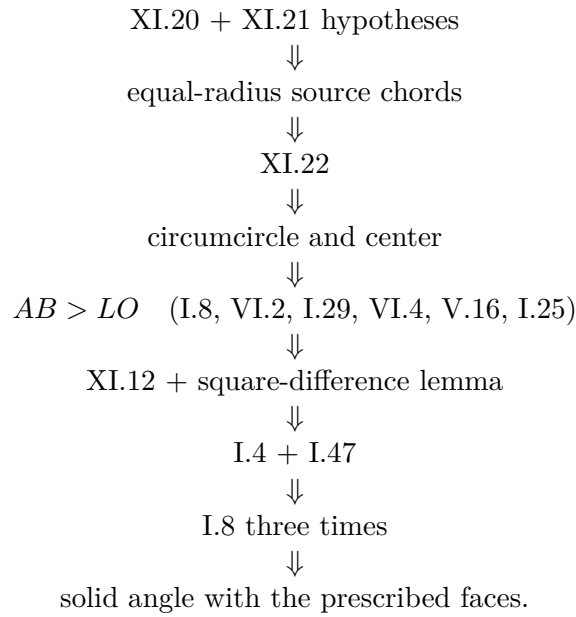
- each of the three angle-copy operations;
- each transport of an XI.20 inequality;
- each SSS call;
- the final plane-uniqueness contradiction.

Those operations alter logical information but do not create a new stable geometric state.

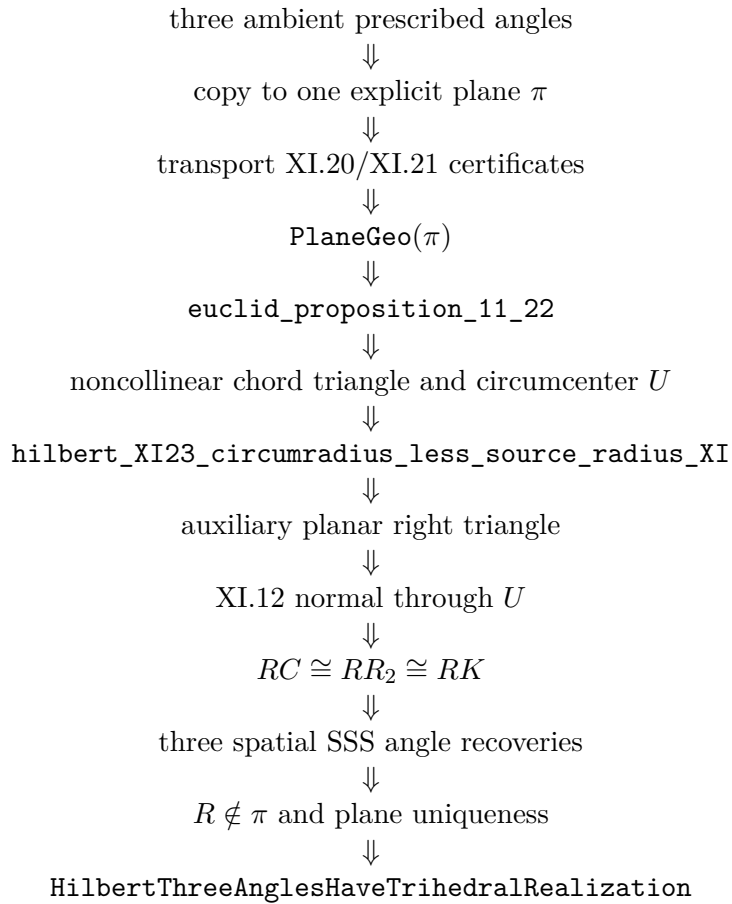
The figure convention follows the established Book XI style: planar geometry is black or gray; genuinely spatial edges are blue; a hidden spatial continuation may be blue dashed; plane fills are light.

23.20 Dependency Summary

The classical local chain is



The production Lean chain is



23.21 Conclusion

Proposition XI.23 is the constructive converse of XI.20 and XI.21 and the point at which the opening theory of solid angles becomes complete.

The final formal proof preserves Euclid’s global architecture remarkably well: construct the chord triangle, find its circumcenter, prove the common edge length exceeds the circumradius, erect a normal through the center, and recover the three required face angles by SSS.

At the same time, the formalization changes two important local mechanisms. The circumradius comparison is proved synthetically through angle order rather than through a Book V–VI proportional calculation, and the height is encoded by an auxiliary right triangle rather than by a numerical difference of squares.

The most important formal gain is control of hidden configuration data. The three source angles are explicitly normalized into one plane; the circumcenter is certified to lie in that plane; all cyclic ray-order cases are closed; the apex is proved to lie outside the base plane; and noncoplanarity is derived from plane uniqueness.

Thus XI.23 is not merely a long terminal construction. It is a useful integration test for the mature Book XI architecture:

$$\begin{aligned} &\text{synthetic angle comparison} + \text{PlaneGeo} + \text{spatial congruence} + \text{XI.12} \\ &\quad \longrightarrow \text{proper trihedral realization.} \end{aligned}$$

Chapter 24

Proposition XI.24

24.1 Purpose and Scope

Proposition XI.24 begins a new phase of Book XI. The preceding propositions develop the geometry of perpendicular and parallel lines and planes, solid angles, and the realization of trihedral data. XI.24 is the first proposition in which this infrastructure is assembled into the geometry of a complete six-faced solid.

In modern language the configuration is a parallelepiped determined by three pairs of parallel planes. Euclid’s conclusion has two parts:

1. every face of the solid is a parallelogram;
2. each pair of opposite faces is equal.

The formal reconstruction keeps these two tasks separate. The first is primarily an incidence theorem and is driven by XI.16. The second is metric: I.34 provides corresponding side equalities, XI.10 provides equality of the included spatial angles, and spatial SAS identifies corresponding diagonal triangles.

XI.24 is also particularly useful for comparing the direct Hilbert and Hilbert–Wyler routes. Both routes have the same input configuration, the same metric core, the same final result type, and the same audited kernel dependencies. They differ only in the incidence mechanism used to construct the section lines that form the six parallelogram faces.

24.2 Euclid’s Statement

Euclid’s statement is:

If a solid be contained by parallel planes, the opposite planes in it are equal and parallelogrammic.

The phrase “contained by parallel planes” is geometrically compressed. For XI.24 it means that the solid is bounded by three pairs of opposite parallel planes. The formal statement therefore makes this six-plane configuration explicit rather than leaving the combinatorics of the solid implicit.

24.3 Classical Configuration

Let

$$\pi_0 \parallel \pi_1, \quad \rho_0 \parallel \rho_1, \quad \sigma_0 \parallel \sigma_1$$

be the three pairs of parallel planes.

The eight vertices are indexed by choosing one plane from each pair:

	π	ρ	σ
A	π_0	ρ_0	σ_1
B	π_0	ρ_0	σ_0
C	π_0	ρ_1	σ_0
D	π_0	ρ_1	σ_1
E	π_1	ρ_0	σ_1
F	π_1	ρ_0	σ_0
G	π_1	ρ_1	σ_0
H	π_1	ρ_1	σ_1 .

Thus the six faces are

$$\begin{aligned} \pi_0 : ABCD, \quad \pi_1 : EFGH, \\ \rho_0 : ABFE, \quad \rho_1 : DCGH, \\ \sigma_0 : CBFG, \quad \sigma_1 : DAEH. \end{aligned}$$

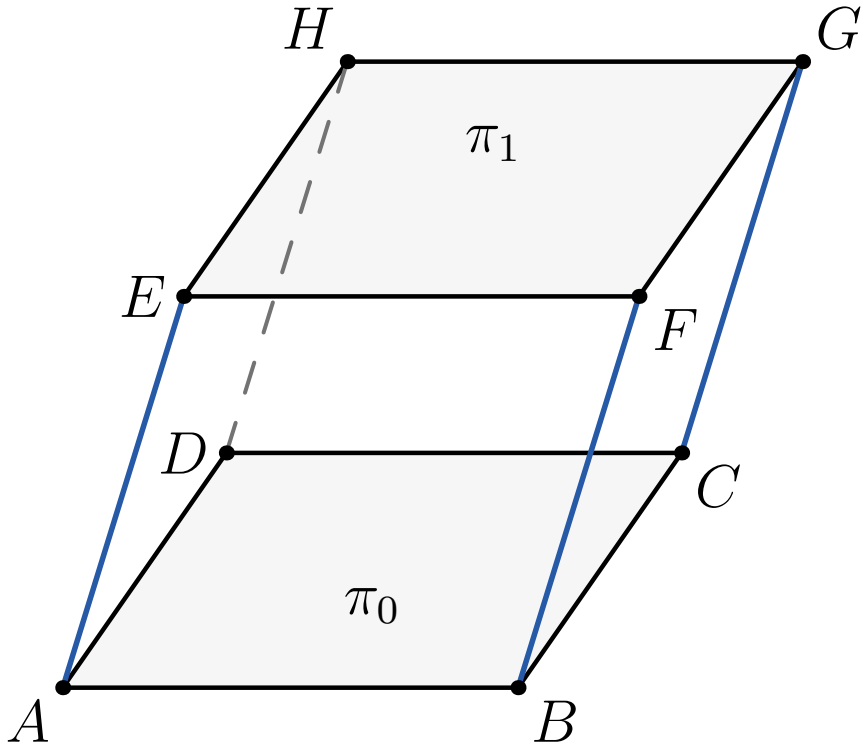


Figure 24.1: Parallelepiped configuration.

The indexing is chosen so that every face is obtained by fixing one of the six planes and allowing the other two plane families to vary. This symmetry is exploited heavily in the Lean proof: only one face and one pair of opposite faces need a proposition-specific proof; the remaining cases follow by permutation of the three plane directions.

24.4 Euclid's Classical Proof

24.4.1 Every face is a parallelogram

Consider the face $ABCD$ in the plane π_0 .

The parallel planes ρ_0 and ρ_1 are cut by the plane π_0 . Proposition XI.16 says that the two section lines are parallel. These section lines are AB and CD , hence

$$AB \parallel CD.$$

Likewise, cutting the parallel planes σ_0 and σ_1 by π_0 gives

$$BC \parallel DA.$$

Therefore $ABCD$ is a parallelogram. The same argument applies to every other face after permuting the three pairs of parallel planes.

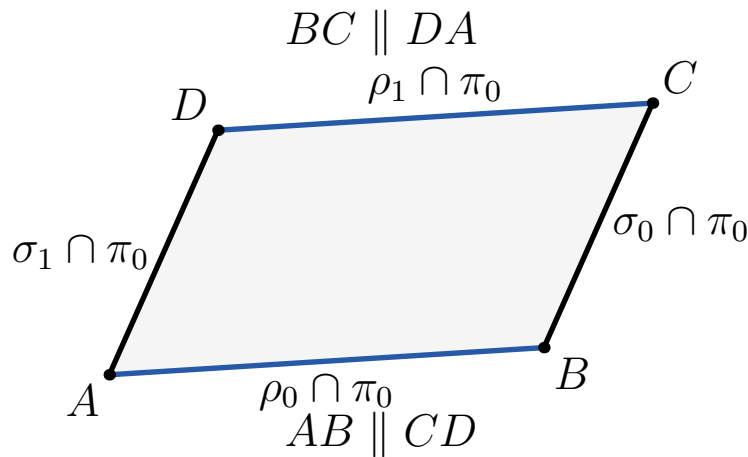


Figure 24.2: XI.16 on one face.

24.4.2 Opposite faces are equal

Consider the opposite faces $ABCD$ and $EFGH$. Euclid compares diagonal triangles in these faces.

The side face $ABFE$ is a parallelogram, so by I.34

$$AB \cong EF.$$

The side face $CBFG$ is also a parallelogram, so

$$CB \cong GF.$$

The lines forming the angle at B are parallel, in the appropriate directions, to the lines forming the angle at F . Proposition XI.10 therefore gives

$$\angle ABC \cong \angle EFG.$$

By I.4, the triangles

$$BAC \quad \text{and} \quad FEG$$

are congruent.

Euclid then passes from the diagonal triangles to the complete parallelograms: by I.34 each diagonal bisects a parallelogram into two equal triangles, so the two opposite parallelogram faces are equal. The same argument applies to the other two pairs of opposite faces.

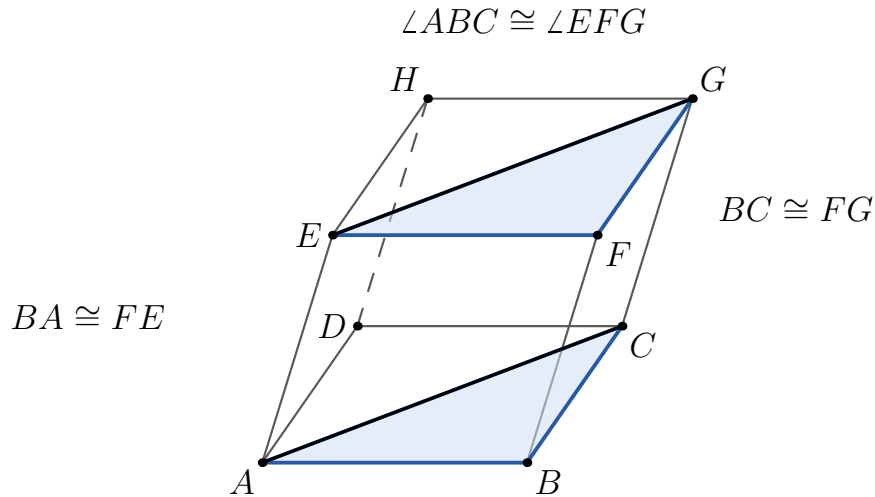


Figure 24.3: Opposite-face SAS.

24.5 Classical Dependency Pattern

The classical proof has a particularly transparent dependency structure.

First:

$$\boxed{\text{XI.16}} \longrightarrow \boxed{\text{six parallelogram faces}}.$$

Then, for one pair of opposite faces:

$$\begin{aligned} \text{I.34} &\implies \text{two corresponding side equalities,} \\ \text{XI.10} &\implies \text{included spatial angle equality.} \end{aligned}$$

Finally:

$$\boxed{\text{I.4 (SAS)}} \longrightarrow \boxed{\text{congruent diagonal triangles}} \longrightarrow \boxed{\text{equal opposite parallelograms}}.$$

No proportion theory is needed. No volume notion is needed. The theorem is a synthesis of parallel-plane incidence with ordinary congruence geometry.

24.6 What Is Genuinely Three-Dimensional Here?

The conclusion concerns six planar parallelograms, but XI.24 is not a planar theorem disguised as a solid theorem.

There are two genuinely spatial steps.

First, XI.16 converts parallelism of planes into parallelism of their section lines. The face parallelograms therefore arise from three-dimensional incidence, not from an independent planar assumption.

Second, XI.10 compares angles formed by pairs of corresponding parallel lines situated in different parts of the solid. This is the spatial bridge needed for the SAS comparison of opposite-face triangles.

By contrast, I.34 is used entirely within one face plane. The formal proof therefore treats I.34 through an induced **PlaneGeo** slice rather than installing planar Euclidean structure globally on the ambient three-dimensional geometry.

XI.24 is thus a useful example of the project's basic architecture:

spatial incidence + local planar geometry + spatial congruence.

24.7 Formal Target and Statement

The input configuration is represented by

```
HilbertParallelepipedConfiguration
```

It stores:

- the three parallel-plane hypotheses;
- the twenty-four point-plane incidences locating the eight vertices.

The conclusion is represented by

```
HilbertXI24Conclusion
```

It contains six parallelogram fields:

```
face_pi0
face_pi1
face_rho0
face_rho1
face_sigma0
face_sigma1
```

and three opposite-face triangle-congruence witnesses:

```
opposite_pi
opposite_rho
opposite_sigma
```

The direct public theorem is

```
Geometry.euclid_proposition_11_24
```

and the Hilbert–Wyler public theorem is

```
Geometry.euclid_proposition_11_24_wyler
```

Both have the same hypothesis type and the same conclusion type.

24.8 Spatial / Planar Architecture Used

The direct theorem lives in the established Book XI three-dimensional environment:

```
HilbertIncidence
HilbertOrder
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

The proof deliberately avoids a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient geometry.

Whenever a planar theorem is required, the relevant carrier plane is turned into an induced plane geometry. In particular, I.34 is used inside

```
PlaneGeo
```

and its congruence conclusions are transported back to the ambient space.

This separation is essential. It prevents a theorem about a three-dimensional configuration from silently inheriting planar Euclidean structure at the ambient level.

24.9 Proof Architecture of the Direct Reconstruction

24.9.1 Step 1: one face from XI.16

The reusable one-face theorem is

```
hilbert_XI24_face_parallelagram
```


For $ABCD \subset \pi_0$, it applies

```
euclid_proposition_11_16
```

twice.

The first application uses ρ_0, ρ_1 and obtains the two section lines carrying AB and CD . Their spatial parallelism is converted into the point-defined relation

```
Geo.Parallel A B C D
```

The second application uses σ_0, σ_1 and obtains

```
Geo.Parallel B C D A
```

Together these are exactly the two opposite-side parallelisms required for

```
IsParallelogram Geo A B C D
```

24.9.2 Step 2: deriving the endpoint inequalities

The point-defined relation `Geo.Parallel` contains nondegeneracy data. Thus the proof must establish

$$A \neq B, \quad C \neq D, \quad B \neq C, \quad D \neq A.$$

These inequalities are not added as hypotheses. Each follows directly from the disjointness of one of the relevant pairs of parallel planes. For example, if $A = B$, then the same point would lie on both σ_0 and σ_1 , contradicting $\sigma_0 \parallel \sigma_1$.

This is a typical example of hidden formal data that Euclid leaves implicit because the diagram already displays a nondegenerate solid.

24.9.3 Step 3: all six faces

The theorem

```
hilbert_XI24_all_faces_parallelograms
```

applies the one-face theorem six times.

The results are:

$$\begin{aligned} &\text{Par}(A, B, C, D), \quad \text{Par}(E, F, G, H), \\ &\text{Par}(A, B, F, E), \quad \text{Par}(D, C, G, H), \\ &\text{Par}(C, B, F, G), \quad \text{Par}(D, A, E, H). \end{aligned}$$

No new geometry occurs here. The proof is a symmetry layer over one incidence argument.

24.9.4 Step 4: I.34 inside a face plane

The helper

```
hilbert_XI24_face_opposite_sides_congruent
```

takes an ambient parallelogram known to lie in a plane.

It performs three tasks:

1. realize the four vertices in the induced **PlaneGeo**;
2. invoke the planar I.34 theorem there;
3. transport the resulting segment congruences back to the ambient geometry.

For the first pair of opposite faces this gives, from the two side faces,

$$BA \cong FE, \quad BC \cong FG.$$

This is exactly the side data needed for SAS.

24.9.5 Step 5: directed parallelism for XI.10

XI.10 does not consume merely unordered parallel lines. It needs directed parallel-segment data.

The helper

```
hilbert_XI24_parallelogram_directed_opposite_sides
```

extracts the required direction from a parallelogram.

For the side faces $ABFE$ and $CBFG$, the proof obtains

$$AB \parallel_{\text{dir}} EF, \quad CB \parallel_{\text{dir}} GF.$$

This directed information is an important formal refinement of the classical phrase “the sides are parallel respectively.”

24.9.6 Step 6: the noncoplanarity hypothesis of XI.10

The six points used in the angle comparison must not all lie in one plane.

For the first opposite-face pair, suppose that

$$A, B, C, E, F, G$$

were contained in some plane ω .

Since A, B, C are three noncollinear vertices of the parallelogram $ABCD$, plane uniqueness forces

$$\omega = \pi_0.$$

Likewise the noncollinear triple E, F, G forces

$$\omega = \pi_1.$$

Hence $\pi_0 = \pi_1$, contradicting their parallelism.

Thus the noncoplanarity required by XI.10 is derived from the solid configuration rather than assumed independently.

24.9.7 Step 7: XI.10 gives the included angle

The directed side data and the noncoplanarity proof are passed to

```
euclid_proposition_11_10
```

The result is the ambient angle congruence

$$\angle ABC \cong \angle EFG.$$

This is the genuinely spatial metric step in the opposite-face comparison.

24.9.8 Step 8: spatial SAS

The theorem

```
hilbert_XI24_opposite_pi_faces_triangle_congruent
```

combines

$$BA \cong FE, \quad BC \cong FG, \quad \angle ABC \cong \angle EFG.$$

The ambient three-dimensional geometry does not use a global planar HilbertCongruence instance. Instead the proof invokes the spatial SAS consequences

```
hilbert_space_sas_third_side
hilbert_space_sas_remaining_angles
```

The result is

```
TriangleCongruenceResult Geo B A C F E G
```

Thus the entire congruence of the two diagonal triangles is made explicit.

24.9.9 Step 9: the other two opposite-face pairs

Only the π_0/π_1 pair is proved metrically from scratch.

The theorem

```
hilbert_XI24_all_opposite_faces_triangle_congruent
```

constructs two relabeled parallelepiped configurations and reuses the same theorem for the ρ - and σ -directions.

The three witnesses are

opposite faces	triangle witness
π_0/π_1	$BAC \cong FEG,$
ρ_0/ρ_1	$BAF \cong CDG,$
σ_0/σ_1	$BCF \cong ADE.$

This permutation strategy is preferable to copying the metric proof three times: it exposes the actual symmetry of the solid.

24.9.10 Step 10: final assembly

The public theorem

```
euclid_proposition_11_24
```

has a deliberately short final body.

It obtains the six faces from

```
hilbert_XI24_all_faces_parallelograms
```

and the three opposite-face witnesses from

```
hilbert_XI24_all_opposite_faces_triangle_congruent
```

and then fills the fields of

```
HilbertXI24Conclusion.
```

This separation keeps proposition-specific construction out of the public wrapper.

24.10 Why the Formal Conclusion Uses Diagonal Triangles

Euclid's final conclusion is equality of opposite parallelogram faces. Classically this is obtained by saying that each parallelogram is twice one of its diagonal triangles.

The present library does not introduce a general theory of area or equality of arbitrary plane figures merely to formalize this one sentence. Instead, the conclusion stores a stronger local synthetic witness: the corresponding diagonal triangles are congruent.

For the first pair this witness is

```
TriangleCongruenceResult Geo B A C F E G
```

which contains all three side congruences and all three angle congruences.

This choice has three advantages.

1. It stays inside the existing synthetic congruence language.
2. It avoids introducing numerical area or measure.
3. A future theory of equal plane figures or area can recover Euclid’s final doubling argument from the stored triangle congruence.

Thus the formal proof does not replace Euclid by coordinates or arithmetic. It stops one structural step before the classical area statement.

24.11 Hilbert–Wyler Reconstruction

XI.24 has a second production theorem:

```
Geometry.euclid_proposition_11_24_wyler
```

The important point is that this is not an alias of the direct theorem.

24.11.1 Wyler XI.16 as the incidence front end

The Wyler one-face theorem is

```
hilbert_XI24_face_parallelogram_wyler
```

Instead of the direct XI.16 theorem, it invokes

```
euclid_proposition_11_16_wyler
```

The Wyler XI.16 interface returns exact carrier identities for the section lines. Schematically,

$$\text{carrier}(\rho_0) \cap \text{carrier}(\pi) = \text{carrier}(l_{AB}),$$

and similarly for the opposite section.

The incidence $B \in l_{AB}$, for example, is recovered from

$$B \in \rho_0, \quad B \in \pi$$

by rewriting with the carrier equality.

Only after these exact meet identities are used does the proof convert the section-line result to the project’s point-defined `Geo.Parallel` relation.

Thus the Wyler route changes the incidence language at precisely the place where XI.24 depends on XI.16.

24.11.2 Six faces through the Wyler route

The theorem

```
hilbert_XI24_all_faces_parallellograms_wyler
```

repeats the same carrier-based one-face construction under the six symmetric relabelings.

The result is the same six parallelogram faces as in the direct proof.

24.11.3 The metric core is intentionally shared

After the six faces have been established, the direct and Wyler proofs rejoin.

The Wyler route uses the same helpers for:

- I.34 in PlaneGeo;
- directed opposite sides of a parallelogram;
- XI.10;
- spatial SAS.

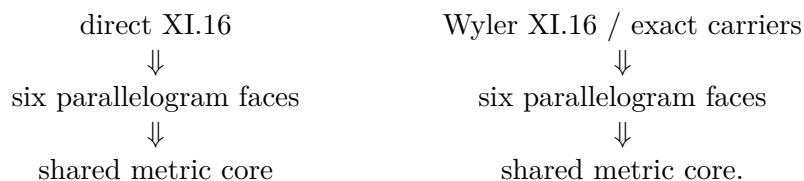
Its one-pair theorem is

```
hilbert_XI24_opposite_pi_faces_triangle_congruent_wyler
```

and the all-pairs theorem is

```
hilbert_XI24_all_opposite_faces_triangle_congruent_wyler
```

The two proof spines are therefore:



This is exactly the intended role of the Hilbert–Wyler program: alter the incidence interface where it matters without duplicating a successful metric library.

24.12 Lean Formalization

The production direct file is

```
CGJteamLab/Proposition11_24.lean
```

Its principal declarations are:

```
hilbert_XI24_face_parallelogram
HilbertParallelepipedConfiguration
hilbert_XI24_all_faces_parallelograms
hilbert_XI24_parallelogram_directed_opposite_sides
hilbert_XI24_face_opposite_sides_congruent
hilbert_XI24_opposite_pi_faces_triangle_congruent
hilbert_XI24_all_opposite_faces_triangle_congruent
HilbertXI24Conclusion
euclid_proposition_11_24
```

The production Wyler file is

```
CGJteamLab/Wyler/Proposition11_24.lean
```

Its principal declarations are:

```
hilbert_XI24_face_parallelogram_wyler
hilbert_XI24_all_faces_parallelograms_wyler
hilbert_XI24_opposite_pi_faces_triangle_congruent_wyler
hilbert_XI24_all_opposite_faces_triangle_congruent_wyler
euclid_proposition_11_24_wyler
```

The final full project build is clean.

24.13 Hidden Formal Data

XI.24 is visually simple but formally dense. Several pieces of information that are immediate from Euclid’s diagram must be made explicit.

24.13.1 The solid is not determined by the phrase “parallel planes”

The formal theorem needs six named planes, grouped into three parallel pairs, and eight named vertices with explicit incidence data. Without this combinatorial information there is no formal notion of which four vertices form each face.

24.13.2 Face edges must be nondegenerate

The relation `Geo.Parallel A B C D` requires $A \neq B$ and $C \neq D$. These facts are derived from disjoint parallel planes rather than assumed separately.

24.13.3 Section-line membership must be reconstructed

XI.16 returns line-level section data. XI.24 must prove that the other vertices of a face lie on those same section lines. In the direct route this uses the section characterization supplied by XI.16. In the Wyler route it uses exact carrier equalities.

24.13.4 Planar and ambient congruence are different layers

I.34 is a planar theorem. XI.24 must enter `PlaneGeo`, apply I.34, and transport the resulting segment congruences back to the ambient three-space.

24.13.5 Parallelism for XI.10 has direction

The classical wording “parallel respectively” suppresses orientation. The formal XI.10 interface requires directed parallel segments, so the orientation of the opposite sides of each parallelogram must be recovered explicitly.

24.13.6 Noncoplanarity is a theorem, not a picture

The two opposite face triangles cannot simply be declared to occupy different planes. Their noncoplanarity is proved using noncollinearity of parallelogram vertices, plane uniqueness, and the parallelism of the two opposite face planes.

24.13.7 Equality of faces has no primitive area meaning

The current library has no general primitive relation saying that two parallelograms have equal area. XI.24 therefore stores congruent diagonal triangles as the synthetic equality witness.

24.14 Relationship with XI.10 and XI.16

XI.24 is the first proposition that combines XI.16 and XI.10 in a single solid-geometry theorem.

XI.16 provides the face structure:

$$\text{parallel planes} \implies \text{parallel section lines} \implies \text{parallelogram faces.}$$

XI.10 provides the metric comparison between different parts of the solid:

$$\text{corresponding directed parallel lines} \implies \text{equal included spatial angles.}$$

Thus the two earlier propositions have complementary roles. XI.16 is the incidence engine; XI.10 is the angle-transport engine.

I.34 and I.4 then supply the planar and metric closure needed to identify opposite faces.

24.15 Relationship with the Book XI Architecture

XI.24 is an integration theorem.

It consumes established infrastructure rather than adding a new primitive geometric notion:

- parallel-plane incidence from XI.16;
- spatial angle transfer from XI.10;
- local `PlaneGeo` geometry;
- ambient segment and angle congruence;
- plane uniqueness and noncollinearity infrastructure.

It also introduces a reusable explicit model of a parallelepipedal configuration:

```
HilbertParallelepipedConfiguration
```

which can serve as a stable interface for later propositions involving parallelepipeds and parallelepipedal solids.

In this sense XI.24 marks a transition from local line–plane results to structured solid configurations.

24.16 Relationship with the Foundations

No coordinate model is used.

No vector space is used.

No numerical length function is used.

No numerical angle measure is used.

No area or volume function is used.

The direct proof relies on the project’s established Hilbert incidence, order, congruence, and Euclidean layers. Planar Euclidean reasoning is localized to explicit `PlaneGeo` slices.

The Wyler proof does not introduce a second metric foundation. It changes only the incidence route used to obtain face section lines. The downstream metric argument remains the same synthetic Hilbert argument.

24.17 E^n Transfer Audit

XI.24 is not yet formalized as a theorem of arbitrary-dimensional Euclidean geometry.

Nevertheless, its structure separates the dimension-sensitive and dimension-local pieces quite clearly.

The face construction requires an incidence theorem saying that suitable parallel flats cut by another flat produce parallel section lines. The Wyler XI.16 route already expresses this step through carrier intersections, which is closer to the language needed for dimension-independent transfer.

The metric part of the proof is more strongly three-dimensional in the current library:

- XI.10 is presently formulated in the three-dimensional Hilbert environment;
- the noncoplanarity argument uses uniqueness of the two opposite face planes inside that environment;
- spatial SAS is provided by the established 3D congruence layer.

Accordingly, XI.24 currently proves:

Hilbert 3D $\implies$ direct XI.24 and Wyler-routed XI.24.

It does not yet prove:

$E^n \implies$ XI.24 inside an arbitrary ambient dimension.

A future E^n transfer would naturally seek a local three-dimensional flat containing the parallelepipedal configuration and then reuse the present metric proof inside that flat.

24.18 Axiomatic Status

The kernel audit of the direct theorem

```
Geometry.euclid_proposition_11_24
```

reports:

```
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The kernel audit of the Wyler theorem

```
Geometry.euclid_proposition_11_24_wyler
```

reports exactly the same set:

```
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Thus:

New proposition-specific axiom:	no
New spatial axiom:	no
<code>bookZero_nullSegment1</code> :	yes
<code>bookZero_nullSegment2</code> :	no
Circle–circle continuity:	no
Numerical angle measure:	no
Coordinates or vectors:	no
Global ambient planar Euclidean instance:	no
Explicit <code>PlaneGeo</code> slices:	yes
<code>sorry / admit</code> :	no
Full <code>lake build</code> :	clean

The equality of the two audit profiles is significant. The direct and Wyler proofs are genuinely different at the incidence layer, but the change in proof organization has no additional kernel cost.

XI.24 also differs sharply from XI.23: it does not depend on

```
Geometry.bookZero_nullSegment2
Geometry.hilbert_circle_circle_intersection
```

and hence does not cross the circle–circle continuity boundary.

24.19 Diagrammatic Audit

Three stable geometric states are retained.

1. The complete parallelepipedal configuration with the eight vertices and the two displayed opposite face planes.
2. A single face together with the four section lines obtained from the two relevant parallel-plane pairs.
3. The two opposite diagonal triangles used in the SAS comparison.

No separate figure is introduced for:

- each of the remaining five face-parallelogram constructions;
- each application of I.34;
- the directed-parallel extraction;
- the plane-uniqueness contradiction used for noncoplanarity;
- the two relabelings used for the remaining opposite-face pairs.

Those operations change logical information but do not create a new stable geometric state.

The figure convention follows the established Book XI style: planar geometry is black or gray; genuinely spatial connectors may be emphasized; hidden spatial continuation may be dashed; plane or face fills are light.

24.20 Dependency Summary

The classical local chain is:

```

three pairs of parallel planes
  ↓↓
XI.16
  ↓↓
six parallelogram faces
  ↓↓
I.34 + XI.10
  ↓↓
I.4 (SAS)
  ↓↓
congruent diagonal triangles
  ↓↓
equal opposite parallelograms.

```

The production direct Lean chain is:

```

HilbertParallelepipedConfiguration
  ↓↓
hilbert_XI24_face_parallelogram
  ↓↓
hilbert_XI24_all_faces_parallelograms
  ↓↓
PlaneGeo / I.34 + directed parallel segments + XI.10
  ↓↓
spatial SAS
  ↓↓
hilbert_XI24_all_opposite_faces_triangle_congruent
  ↓↓
euclid_proposition_11_24.

```

The production Wyler chain differs only at the incidence front end:

```

HilbertParallelepipedConfiguration
  ↓↓
euclid_proposition_11_16_wyler
  ↓↓
exact plane-meet / line-carrier identities
  ↓↓
hilbert_XI24_all_faces_parallelograms_wyler
  ↓↓
shared metric core
  ↓↓
euclid_proposition_11_24_wyler.

```

24.21 Conclusion

Proposition XI.24 is the point at which the theory of parallel planes becomes the geometry of a structured solid.

The formal reconstruction preserves Euclid’s global proof architecture very closely. XI.16 produces the six parallelogram faces; I.34 provides corresponding side equalities; XI.10 transfers the included spatial angle; and SAS identifies the diagonal triangles of opposite faces.

The main formal refinements concern data that the classical diagram suppresses: the six-plane combinatorics of the solid, endpoint distinctness, exact section-line membership, the distinction between planar and ambient congruence, direction on parallel sides, and the proof that opposite face triangles are not coplanar.

The Hilbert–Wyler version makes the architectural separation especially clear. Its incidence front end is genuinely different: XI.16 is routed through exact carrier intersections. Once the six parallelogram faces are available, however, the proof rejoins the direct route and uses exactly the same metric core.

The final audit therefore gives a particularly clean result:

different incidence organization + shared synthetic metric core $\implies$ same theorem and same axiom pro

XI.24 is consequently not only a reconstruction of one Euclidean proposition. It is a compact integration test for the mature Book XI architecture and one of the clearest production examples of the project’s Hilbert–Wyler strategy.

Chapter 25

Proposition XI.25

25.1 Purpose and Scope

Proposition XI.25 is the point at which Book XI passes from the geometry of individual parallelepipeds to a genuine theory of magnitude and proportion. XI.24 established the geometric structure of a parallelepiped bounded by three pairs of parallel planes. XI.25 asks for substantially more: when such a solid is cut by a plane parallel to a pair of opposite faces, the ratio of the two corresponding base parallelograms is the same as the ratio of the two resulting solids.

The classical proof is Eudoxian. Euclid does not assign numerical areas or volumes. Instead he constructs arbitrary equal multiples of the two base magnitudes and the two solid magnitudes, proves that their three-way comparisons agree, and invokes Book V, Definition 5.

That proof architecture creates the main formal difficulty of XI.25. It is not enough to show that one pair of adjacent slabs is equal when their widths are equal. The reconstruction must build a coherent magnitude system in which

- equal bases and equal solids are genuine geometric equivalence classes;
- strict comparison means a genuine proper-part relation;
- arbitrary positive multiples are represented geometrically by repeated adjacent slabs;
- comparison of base multiples and solid multiples is synchronized; and
- the final appeal to Eudoxus' definition is not made true by definition.

The current production source therefore has two large layers. The first is the local geometry of adjacent parallelepipedal slabs. The second turns that geometry into quotient magnitudes, strict comparison, repeated multiples, and finally a faithful Book V proportion statement.

No coordinates, numerical area, or numerical volume are introduced.

25.2 Euclid's Statement

Euclid's proposition states that if a parallelepipedal solid is cut by a plane parallel to a pair of opposite faces, then the two base faces are in the same ratio as the two corresponding solids.

Schematically, if a solid is divided into adjacent parts

$$S_1 \quad \text{and} \quad S_2$$

by a parallel section, with corresponding base parallelograms

$$B_1 \quad \text{and} \quad B_2,$$

then the conclusion is

$$B_1 : B_2 = S_1 : S_2.$$

The assertion is not numerical. The equality of ratios is to be understood in the Eudoxian sense of Book V, Definition 5: for arbitrary positive multiples, the comparison pattern of the two base multiples must agree with the comparison pattern of the corresponding solid multiples.

25.3 Classical Configuration

Euclid begins with one parallelepiped cut by a plane parallel to two opposite faces. The cut produces two adjacent parallelepipeds sharing the section plane.

For the formal reconstruction it is convenient to represent the same local geometry by three parallel section planes

$$\pi_0 \parallel \pi_1 \parallel \pi_2$$

and two fixed pairs of side planes

$$\rho_0 \parallel \rho_1, \quad \sigma_0 \parallel \sigma_1.$$

Each section plane contains four vertices:

	π	ρ	σ
A_i	π_i	ρ_0	σ_1
B_i	π_i	ρ_0	σ_0
C_i	π_i	ρ_1	σ_0
D_i	π_i	ρ_1	σ_1 .

Thus the first slab lies between π_0 and π_1 , the second between π_1 and π_2 , and the whole lies between π_0 and π_2 .

The ordered version of the configuration also records

$$\begin{aligned} A_0 - A_1 - A_2, & \quad B_0 - B_1 - B_2, \\ C_0 - C_1 - C_2, & \quad D_0 - D_1 - D_2, \end{aligned}$$

so that the middle section is genuinely internal to the whole solid.

25.4 Euclid's Classical Proof

25.4.1 Equal repeated widths

Euclid extends one longitudinal direction on both sides of the original cut. He marks off any number of intervals equal to the width of the first slab and any number of intervals equal to

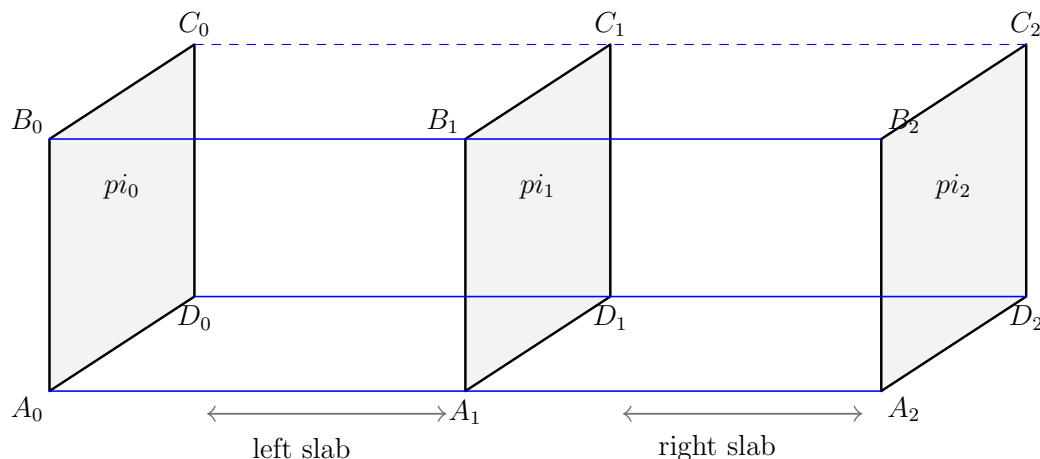


Figure 25.1: Three parallel sections determine two adjacent parallelepipedal slabs and their whole. Section edges are shown in black; the longitudinal spatial edges are shown in blue.

the width of the second slab. Through the new division points he completes the corresponding parallelograms and parallelepipeds.

The purpose of the construction is not merely to obtain a larger solid. It constructs arbitrary positive multiples of four magnitudes simultaneously:

$$mB_1, \quad nB_2, \quad mS_1, \quad nS_2.$$

25.4.2 XI.24 supplies equal cells

For a chain of equal successive widths, XI.24 implies that the corresponding faces of successive elementary parallelepipeds are equal. Euclid then uses XI.Def.10 to conclude that the elementary solids are equal.

Thus a chain of m equal base cells corresponds to a chain of m equal solid cells, and similarly for the second family.

25.4.3 The comparison step

The decisive sentence in Euclid's proof is the three-way comparison:

base multiple less than / equal to / greater than base multiple

if and only if the corresponding solid multiple is less than / equal to / greater than the other solid multiple.

The equality case is supported by XI.Def.10. The strict cases use the geometric principle behind Common Notion 5: when one magnitude is represented as a genuine proper part of another, the whole is greater than the part. Classically this proper-part geometry is largely left to the diagram.

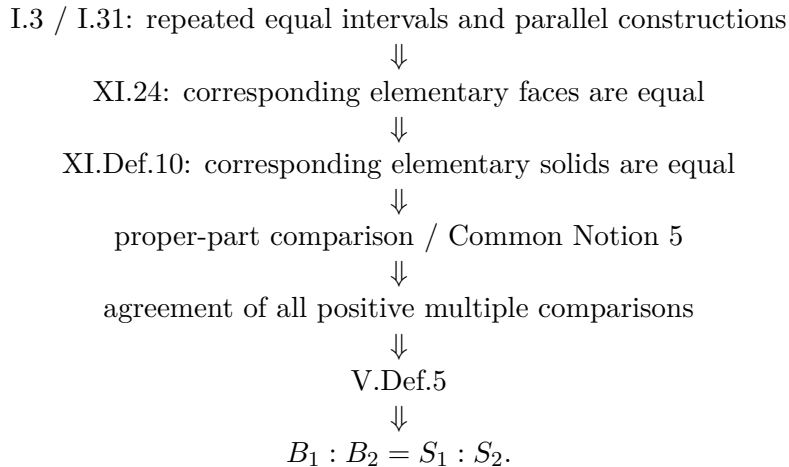
This is the point at which the formal reconstruction must do substantially more work than Euclid's text. It must produce an actual parallel cut that witnesses the smaller base and the smaller solid as proper parts of the larger ones; it cannot merely postulate a compatible order.

25.4.4 Book V, Definition 5

Once comparison agreement has been established for arbitrary positive multiples, V.Def.5 gives

$$B_1 : B_2 = S_1 : S_2.$$

The classical dependency chain is therefore

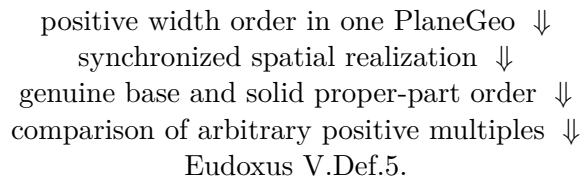


25.5 Why XI.25 Is Structurally Difficult

XI.25 is the first proposition in the present Book XI reconstruction whose main difficulty is not a single incidence or congruence argument. The proof must connect three different mathematical levels without identifying them: concrete three-dimensional parallelepipeds, geometric magnitude classes of bases and solids, and Eudoxian comparison of arbitrary positive multiples. The distinction is essential. A width parameter may organize the construction, but it must not define the very base and solid orders whose agreement XI.25 is supposed to prove.

This is also why the genuinely three-dimensional content cannot be hidden inside Book V. The proof needs three parallel section planes, two pairs of parallel side planes, four longitudinal edges, synchronized layoff of widths on those edges, construction of a section plane through three realized points, proof that the fourth point lies in that plane, and finally an actual cut exhibiting a smaller solid as a proper part of a larger one. The planar geometry **PlaneGeo** enters only where it should: widths are positive segment classes in one fixed carrier plane and their Book V order is handled there. These planar magnitude data are then realized back in ambient space.

The production development therefore has the following conceptual shape:



25.6 Formal Reconstruction: From One Cut to Eudoxus

25.6.1 The geometric core: two adjacent slabs and equal cells

The basic spatial object is

```
HilbertXI25TwoSlabConfiguration
```

with three parallel section planes, two pairs of parallel side planes, and twelve vertices. Its ordered refinement

```
HilbertXI25OrderedTwoSlabConfiguration
```

adds the four strict betweenness relations along the longitudinal edges. Thus the middle section is not merely another parallel plane: it is required to lie genuinely between the two outer sections on every longitudinal edge.

The direct route recognizes the two elementary slabs and their union through XI.24:

```
hilbert_XI25_twoSlab_XI24
hilbert_XI25_whole_XI24
```

If two consecutive widths are equal on one longitudinal edge, the parallelogram-face information supplied by XI.24 propagates that equality to the other three longitudinal edges. This is recorded by

```
hilbert_XI25_equal_width_B
hilbert_XI25_equal_width_C
hilbert_XI25_equal_width_D
hilbert_XI25_equal_width_all_edges
```

The corresponding adjacent side faces are then equal, and XI.Def.10 packages the resulting equality of elementary parallelepipeds:

```
hilbert_XI25_adjacent_rho_face_equal
hilbert_XI25_adjacent_sigma_face_equal
hilbert_XI25_adjacent_elementary_solids_equal
```

This is the formal version of Euclid's repeated use of equal successive intervals together with XI.24.

25.6.2 Magnitudes, proper parts, and positive multiples

The next step is not to attach numbers to bases or solids, but to form their geometric magnitude classes. A concrete parallelepiped is represented by

```
HilbertXI25Parallelepiped
```

and XI.Def.10 equality is encoded by

```
HilbertXI25SolidEquivalent
HilbertXI25SolidClass
```

Bases are treated analogously by

```
HilbertXI25BaseEquivalent
HilbertXI25BaseClass
```

so that equality of magnitudes is quotient equality rather than literal equality of records.

Strict comparison is geometric as well. The relations

```
HilbertXI25ProperBasePart
HilbertXI25BaseLess
HilbertXI25ProperSolidPart
HilbertXI25SolidLess
```

are generated by an actual proper-part decomposition with a nonempty complement. The synchronized object

```
HilbertXI25AlignedProperPartWitness
```

records one cut that proves both the base comparison and the corresponding solid comparison. This is the precise formal content of the classical “whole greater than part” step.

Euclid’s positive multiples are then represented by actual chains of equal slabs rather than by an external volume function. The structures

```
HilbertXI25AlignedCutSum
HilbertXI25RepeatedSlabMultiple
HilbertXI25PositiveMultipleSystem
HilbertXI25PairedPositiveMultipleSystems
```

keep base and solid multiples synchronized. For every pair of indices m, n , a comparison certificate records the common trichotomy: equality, left less than right, or right less than left, simultaneously for base classes and solid classes.

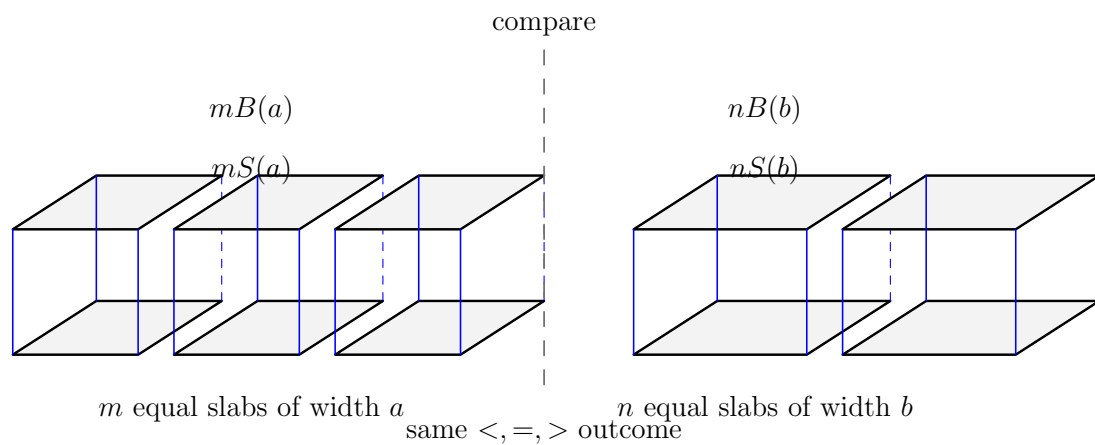


Figure 25.2: Positive multiples are represented by chains of adjacent equal-width slabs. The same geometric repetition generates the corresponding base and solid multiples.

25.6.3 The noncircular comparison layer

This is the conceptual center of the reconstruction. A tempting formal shortcut would define both base order and solid order directly from the same width order. The final comparison agreement would then be true by construction and would no longer express Euclid's theorem.

The production structure

```
HilbertXI25ComparisonFaithfulFamily
```

reverses that dependency. Widths retain the existing positive-segment order in one fixed `PlaneGeo`. Bases retain their geometric quotient order `HilbertXI25BaseClassLess`; solids retain `HilbertXI25SolidClassLess`. The family is accepted only after one proves that a strict width comparison produces genuine proper-part comparisons in both geometric systems. Trichotomy and asymmetry then show that base and solid comparison are faithfully reflected by the width source, not defined by it.

This distinction is what makes the final V.Def.5 argument substantive. The common width system supplies synchronized positive multiples, while the two target orders remain independently geometric.

25.6.4 Realizing arbitrary positive widths in space

A single nontrivial seed cut fixes the longitudinal direction and the four side planes. Every positive width class is first laid off on a fixed ray in `PlaneGeo`; Book Zero segment construction gives a canonical endpoint and the strict-less/betweenness theorem identifies the magnitude order with point order on that ray. The same width is then copied to the other three longitudinal rays by ambient segment construction.

For $x < y$ the result is four synchronized betweenness relations

$$\begin{aligned} A_0 - A(x) - A(y), & \quad B_0 - B(x) - B(y), \\ C_0 - C(x) - C(y), & \quad D_0 - D(x) - D(y). \end{aligned}$$

The first three realized endpoints determine the section plane $\pi(x)$. The fourth point is not definitionally coplanar: its incidence with $\pi(x)$ is proved using the spatial parallel-line machinery. XI.9 transfers the required line parallelism, XI.16 identifies the relevant section lines of parallel planes, and XI.15 proves that the resulting section plane is parallel to the original base plane. For $x < y$, XI.15 also yields $\pi(x) \parallel \pi(y)$.

The theorem

```
orderedTwoSlabConfiguration_of_width_less
```

turns these data into a complete ordered two-slab configuration, and

```
solidCutWitness_of_width_less
```

packages it as a genuine cut. The canonical direct-route representatives are built by

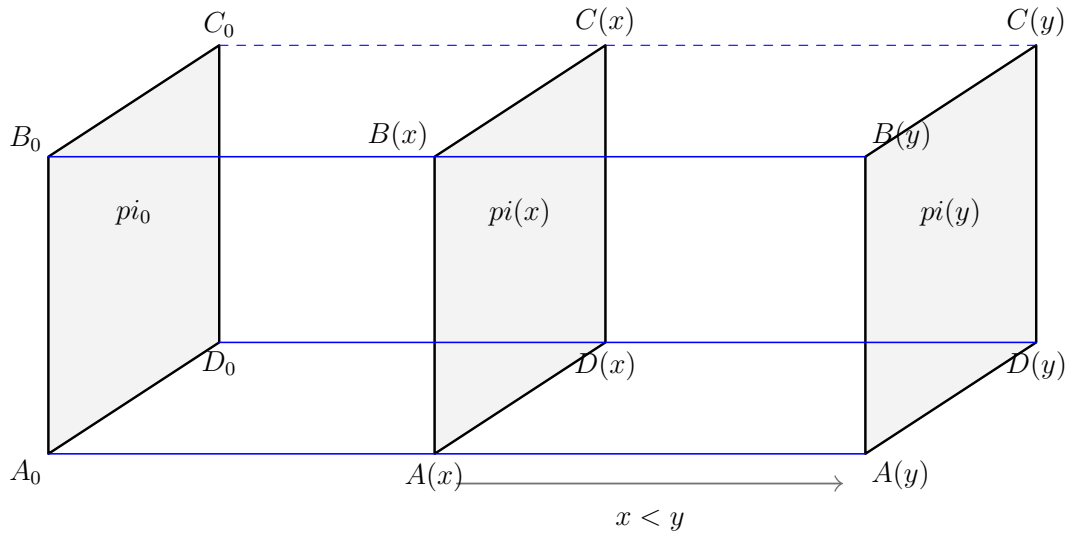


Figure 25.3: For $x < y$, one normalized width comparison in the carrier plane is transported to all four longitudinal edges. The realized endpoints determine two parallel section planes and an actual ordered cut.

```
widthXI24
widthSolid
widthBase
```

and are proved equivalent to the left and whole pieces of the cut. Hence

```
alignedProperPartWitness_of_width_less
```

supplies exactly the geometric strict-comparison field required by the faithful family.

25.6.5 Closing the Eudoxus argument

The definition

```
HilbertXI25SolidCutWitness.toWidthRealization
```

packages the canonical base and solid representatives, the exact width specification, and the proper-part witness for every strict pair. From this, `toComparisonFaithfulFamily` provides the two geometric magnitude systems with their common positive-width source. The general theorem

```
eudoxusProportionBetween_of_common_source
```

then proves V.Def.5 without introducing numerical volume.

The terminal theorem of the direct production file

```
CGJteamLab/Proposition11_25.lean
```

is

```
Geometry.HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut
```

with conclusion

```
EudoxusProportionBetween
  ((X.toWidthRealization (Geo := Geo)).toComparisonFaithfulFamily
    (Geo := Geo)).baseEudoxusMagnitude
  ((X.toWidthRealization (Geo := Geo)).toComparisonFaithfulFamily
    (Geo := Geo)).solidEudoxusMagnitude
a b a b
```

for arbitrary positive width classes a, b in the seed carrier plane. Thus one nontrivial cut generates not merely one ratio but a complete local family in which every positive pair satisfies the required base-to-base and solid-to-solid proportion.

25.7 The Hilbert–Wyler Route for XI.25

The separate production file

```
CGJteamLab/Wyler/Proposition11_25.lean
```

implements XI.25 through the Hilbert–Wyler route. It is important that this is not a second copy of the fifteen-thousand-line direct development. The neutral XI.25 structures, quotient magnitudes, proper-part relations, positive-multiple systems, faithful-family mechanism, section-plane construction, and final Book V argument are shared. The genuinely different part is the XI.24 front end and every XI.25 lemma whose proof depends on that front end.

The first replacement occurs at the level of the two adjacent slabs and their whole:

```
hilbert_XI25_twoSlab_XI24_wyler
hilbert_XI25_whole_XI24_wyler
```

These theorems use

```
euclid_proposition_11_24_wyler
```

instead of the direct XI.24 theorem. Consequently the six parallelogram faces are obtained by the Wyler XI.16 carrier-intersection argument developed for Proposition XI.24. The metric consequences remain neutral once the XI.24 face package has been produced.

The XI.25 geometry that actually consumes XI.24 is then rebuilt over this Wyler front end. Equal-width propagation is provided by

```

hilbert_XI25_equal_width_B_wyler
hilbert_XI25_equal_width_C_wyler
hilbert_XI25_equal_width_D_wyler
hilbert_XI25_equal_width_all_edges_wyler

```

and the equality of adjacent elementary solids by

```

hilbert_XI25_adjacent_elementary_solids_equal_wyler
hilbert_XI25_adjacent_solids_equal_of_base_equal_wyler

```

The quotient comparison layer is correspondingly connected by

```

hilbert_XI25_classComparisonCertificate_of_twoSlab_wyler

```

while the strict proper-part branches themselves remain the shared geometric ones.

The second essential replacement occurs in the canonical realization of an arbitrary width. The neutral construction of the points $A(x)$, $B(x)$, $C(x)$, $D(x)$, their section plane, and all parallelism results is reused unchanged. What changes is the XI.24 package assigned to the resulting slab:

```

HilbertXI25SolidCutWitness.widthXI24_wyler
HilbertXI25SolidCutWitness.widthSolid_wyler
HilbertXI25SolidCutWitness.widthBase_wyler

```

The width specification is then re-established for these representatives by

```

HilbertXI25SolidCutWitness.widthBase_widthClass_wyler

```

and four equivalence lemmas identify the Wyler representatives with the left and whole pieces of the actual cut generated by $x < y$.

This yields the Wyler proper-part bridge

```

HilbertXI25SolidCutWitness.alignedProperPartWitness_of_width_less_wyler

```

and therefore the complete realization

```

HilbertXI25SolidCutWitness.toWidthRealization_wyler

```

At this point the two routes rejoin completely. The comparison-faithful family and the Eudoxus theorem do not need a second Wyler-specific magnitude theory. They consume the same abstract `HilbertXI25WidthRealization` interface.

The terminal Wyler theorem is

```

Geometry.HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut_wyler

```

It has the same mathematical conclusion as the direct theorem, but its canonical solids and bases are generated through the Wyler XI.24 carrier route. This separation is architecturally significant: the Wyler method changes the spatial incidence proof where it genuinely differs, while Book V, quotient magnitudes, proper-part order, and the final Eudoxian logic remain shared.

25.8 Proof Architecture and Formal Obligations

The whole reconstruction is easier to read as one continuous argument than as a ladder of tiny formal layers. Starting from one nontrivial seed cut, the proof proceeds as follows.

1. XI.24 recognizes the two adjacent slabs and their whole and supplies the parallelogram-face data needed to transport equal consecutive widths.
2. Equal widths determine equal elementary faces and then equal elementary solids in the sense of XI.Def.10. Concrete bases and solids are passed to quotient magnitude classes, while strict comparison is defined by actual proper-part decompositions.
3. Repeated equal slabs realize the arbitrary positive multiples required by Book V. Their base and solid comparisons are synchronized by one geometric comparison certificate.
4. The comparison-faithful family prevents circularity: width order is a source order, whereas base and solid orders remain genuine geometric proper-part orders.
5. Every positive width class is laid off canonically on one carrier-plane ray and copied to the other three longitudinal rays. XI.9, XI.16 and XI.15 turn the synchronized endpoints into an actual parallel section.
6. For $x < y$, the two realized sections produce a genuine ordered cut. The smaller canonical base and solid are equivalent to the left piece, and the larger ones to the whole, so strict width comparison generates actual base and solid proper-part comparison.
7. The resulting `HilbertXI25WidthRealization` feeds the shared comparison-faithful family. Since all positive multiple comparisons agree, the common-source Eudoxus theorem yields V.Def.5.

Several facts that are visually implicit in Euclid's diagram must therefore be explicit formal obligations. The middle section must be internal on all four longitudinal edges; equal widths must be synchronized on four rays; the fourth realized endpoint must be proved coplanar with the first three; two section planes obtained from $x < y$ must be proved distinct before their parallelism is used; a proper part must carry a nonempty complementary piece; and magnitude equality must be quotient equality rather than equality of chosen records. Finally, V.Def.5 quantifies over arbitrary positive multiples, so the proof cannot stop at one pair of adjacent slabs.

These are not implementation accidents. They are precisely the pieces of mathematical information compressed by the classical diagram and by Euclid's short comparison argument.

25.9 Dependencies and Reuse

The most visible dependency is XI.24: it supplies the complete face package for every elementary or canonical parallelepiped used by XI.25. In the direct route this is `euclid_proposition_11_24`;

in the Wyler route it is `euclid_proposition_11_24_wyler`. Earlier Book XI results enter the arbitrary-width construction more locally: XI.9 transports parallelism of spatial lines, XI.16 identifies section lines of parallel planes, and XI.15 proves the realized section planes parallel to the base and to one another.

Book V supplies the magnitude-theoretic target: positive segment classes, positive multiples, trichotomy, comparison of multiples, and Eudoxus' Definition 5. XI.25 does not replace this by a numerical area or volume model; it proves that the geometric base and solid magnitude systems have the same comparison pattern.

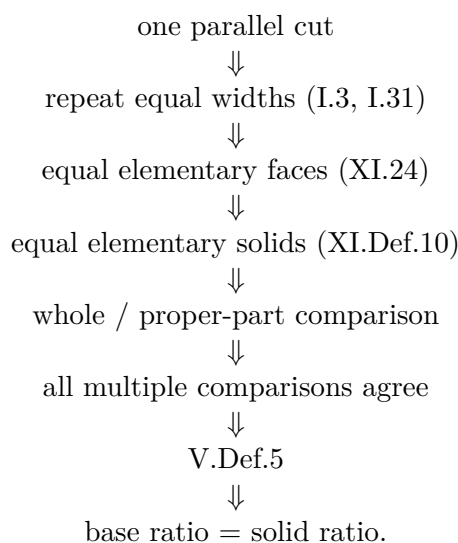
Book I contributes the planar parallelogram and congruence results inherited through XI.24. Book Zero supplies the canonical segment-layoff machinery and the strict-less/betweenness bridge used to realize arbitrary positive widths on a fixed ray. The ambient spatial proof uses the established

```
HilbertIncidence
HilbertOrder
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
```

interfaces, while planar magnitude reasoning is localized to explicit `PlaneGeo` slices.

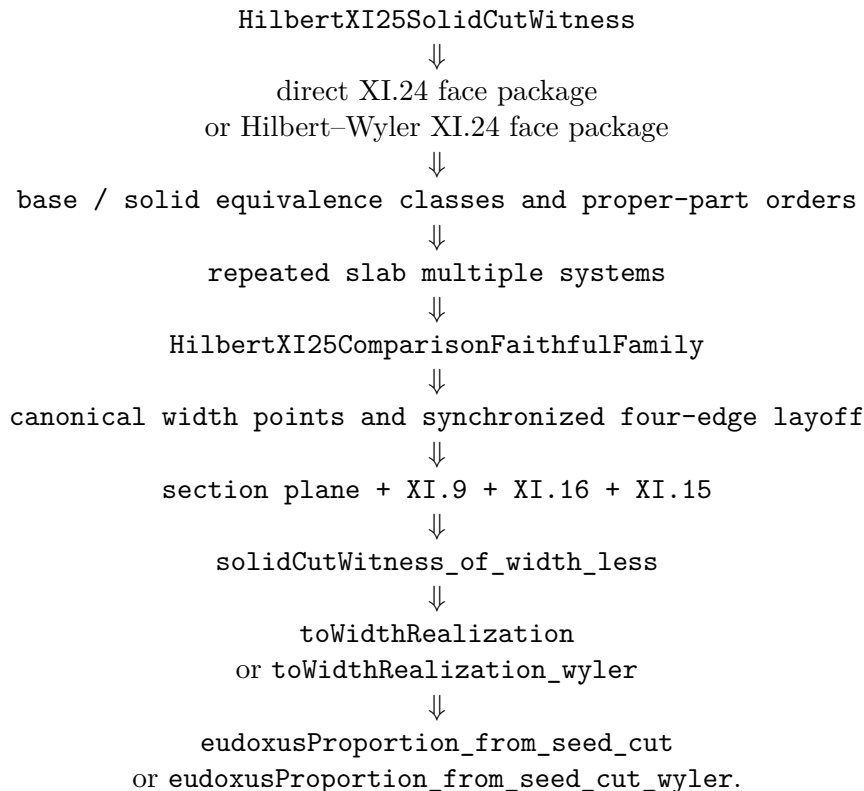
25.10 Classical and Formal Dependency DAGs

The classical chain is



The production Lean chain is longer because every implicit comparison must be exposed. The direct and Hilbert–Wyler routes differ only at the XI.24 front end and at the canonical

representatives built from that front end:



The additional formal length is not numerical overhead. It is the explicit geometry of the comparison statement that Euclid leaves largely implicit. The Wyler reconstruction confirms that this long comparison layer is not tied to one incidence proof of XI.24: once the alternative carrier-based face package has been supplied, the same magnitude and Eudoxus architecture can be reused.

25.11 Comparison with Euclid

The reconstruction preserves Euclid’s essential architecture unusually closely at the highest level:

1. construct repeated equal geometric cells;
2. identify equal elementary solids through XI.Def.10;
3. compare arbitrary multiples;
4. apply V.Def.5.

The major difference is the treatment of strict comparison. Euclid states that if one extended base exceeds another, then the corresponding extended solid exceeds the other. The formal proof cannot accept this as a diagrammatic fact. It must build a real section plane, exhibit the smaller solid as a proper part of the larger solid, and prove that this proper part is not equivalent to the whole.

A second difference is that the production theorem develops a complete local width family from one seed cut. This gives a reusable formal structure in which arbitrary positive widths, not only the original two pieces, have canonical base and solid representatives.

A third difference is organizational. Euclid moves directly from repeated segments to repeated bases and solids. Lean inserts quotient layers so that “equal base” and “equal solid” become well-defined magnitude equalities independent of representatives.

These changes strengthen the logical transparency of the argument without replacing its synthetic content by coordinates or measure.

25.12 Diagrammatic Audit

Three figures are retained because they show three genuinely different stable states.

1. Figure 25.1 shows the local geometric cut: two adjacent slabs and their whole.
2. Figure 25.2 shows the repeated-cell mechanism behind arbitrary positive multiples and V.Def.5.
3. Figure 25.3 shows the later canonical realization of arbitrary ordered widths $x < y$ on all four longitudinal edges.

No separate figure is introduced for quotient construction, setoid transport, asymmetry lemmas, or the final Eudoxus wrapper. Those steps change the logical organization of the magnitudes but do not create a new stable geometric state.

25.13 Audit Status

This documentation has been checked against both current production sources:

```
CGJteamLab/Proposition11_25.lean
CGJteamLab/Wyler/Proposition11_25.lean
```

The direct terminal theorem is `HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut`; the Hilbert–Wyler terminal theorem is `HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut_wyler`. Both production modules compile cleanly in the completed XI.25 development.

The separate kernel audit of the direct theorem reports no new proposition-specific geometric axiom. It uses the established Book Zero null-segment infrastructure required by the positive-segment magnitude theory, together with the standard logical quotient and choice infrastructure. The Wyler route changes the spatial XI.24 derivation and the representatives built from it; it does not introduce a second magnitude or Eudoxus theory.

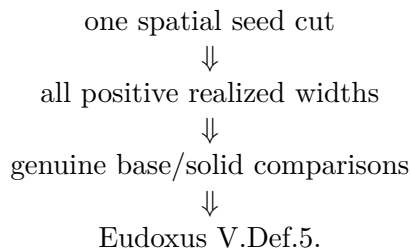
25.14 Conclusion

Proposition XI.25 is the first full volume-proportion theorem of Book XI and one of the most structurally revealing propositions in the reconstruction. It shows that Euclid’s proportion of bases and solids can be recovered without introducing numerical volume.

The proof proceeds instead through geometric equality, geometric proper parts, repeated equal slabs, and Eudoxus' comparison of arbitrary multiples. The formal development makes explicit the step that is most compressed in the classical text: strict comparison of bases and solids must be supported by a genuine cut and a nonempty remainder.

The decisive architectural achievement is the noncircular faithful family. Width classes organize the construction and provide a common multiple operation, but they do not define the target base and solid orders. Those orders remain geometric and are connected to width only by proved proper-part transfer.

Thus the final chain is genuinely synthetic:



XI.25 therefore marks the transition from the incidence and congruence theory of parallelepipeds to a formal synthetic theory of solid magnitude.

Chapter 26

Proposition XI.26

26.1 Purpose and Scope

Proposition XI.26 is the constructive transition from the theory of spatial incidence, perpendicularity, and congruence developed earlier in Book XI to the explicit copying of a complete trihedral angle. The input consists of a given trihedral angle at a point D , together with a prescribed straight line AB and a prescribed point A on that line. The goal is to construct at A a new trihedral angle whose three plane face angles are respectively congruent to the three face angles of the source trihedron.

Classically the construction is short. Euclid projects a point on one edge of the source trihedron perpendicularly to the plane of the other two edges, copies the resulting planar data at the prescribed point, erects a perpendicular to the copied base plane, and recovers the remaining face angles by triangle congruence [1].

The formal reconstruction reveals that the short classical proof suppresses a substantial order problem. If G is the perpendicular projection of the third source point F to the base plane through D, E, C , then the ray DG need not lie in the interior of the base angle EDC . It may coincide with the vertex, lie on either boundary ray, continue beyond a boundary ray, lie in either adjacent exterior sector, lie in the interior sector, or lie in the remaining cyclic sector. A complete theorem must construct the copied point in every one of these positions.

The production proof therefore has two conceptual layers:

1. a proposition-specific planar construction and complete order classification of the projection foot;
2. a shared spatial metric layer using perpendicular-to-plane data, right triangles, SAS, SSS, and the spatial form of Hilbert's angle-addition theorem.

The direct production theorem is

```
Geometry.euclid_proposition_11_26
```

in

```
CGJteamLab/Proposition11_26.lean
```

A second production route is provided in

```
CGJteamLab/Wyler/Proposition11_26.lean
```

by

```
Geometry.euclid_proposition_11_26_wyler_core
Geometry.euclid_proposition_11_26_wyler
```

The Hilbert–Wyler route does not duplicate the long metric and order proof. Instead it reroutes the common spatial incidence fields through `HilbertWylerAxioms` and reuses the same XI.26 reconstruction.

No coordinates, vectors, scalar products, numerical angle measures, or trigonometric functions occur in either production route.

26.2 Euclid’s Statement

Euclid’s Proposition XI.26 states:

To construct a solid angle equal to a given solid angle on a given straight line, and at a given point on it.

Let AB be the prescribed straight line and A the prescribed point on it. Let the source solid angle have vertex D and be bounded by the three plane angles

$$\angle EDC, \quad \angle EDF, \quad \angle FDC.$$

The required construction is a solid angle at A , one of whose edges lies on the prescribed ray from A through B , and whose three face angles are congruent to the three source face angles.

The formal theorem uses the project predicate

```
HilbertTrihedralConfiguration Geo D E C F
```

for the source trihedron and concludes the existence of points L, H such that

```
HilbertTrihedralRealizesThreeAngles
Geo A B L H E D C C D F F D E
```

holds. Thus the three target face angles are

$$\angle BAL, \quad \angle LAH, \quad \angle HAB,$$

and they correspond respectively to

$$\angle EDC, \quad \angle CDF, \quad \angle FDE.$$

26.3 Classical Configuration

The source configuration has vertex D and three rays DE, DC, DF . Euclid chooses a point F on the third ray and drops the perpendicular FG to the plane determined by ED and DC . The foot is G , so

$$G \in \text{plane}(E, D, C), \quad FG \perp \text{plane}(E, D, C).$$

The segment DG is then joined.

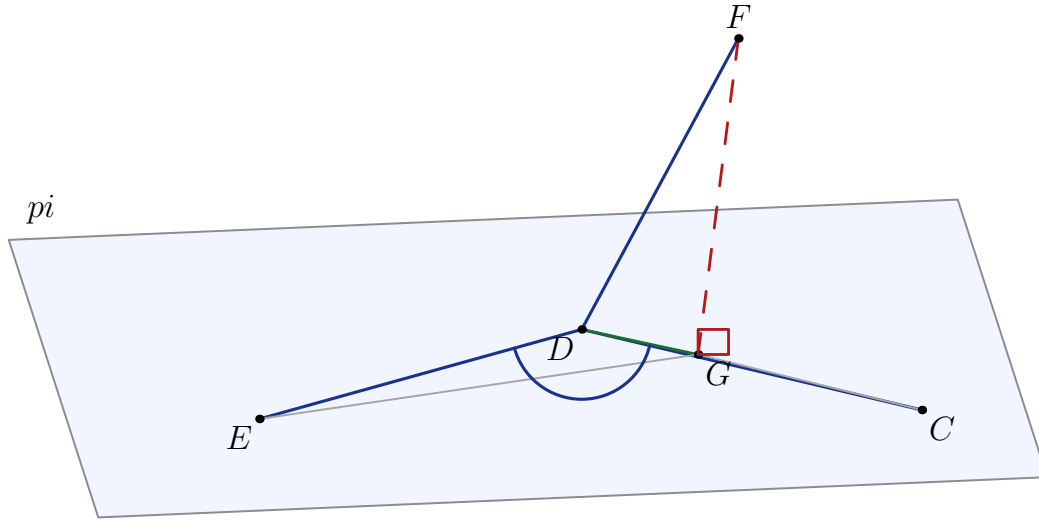


Figure 26.1: The source trihedral angle at D . The point F on the third spatial edge is projected perpendicularly to the base plane π through E, D, C , with foot G . The formal proof does not assume any particular order position of G inside that plane.

At the prescribed point A on the line AB , Euclid constructs a plane angle equal to $\angle EDC$ and a second plane angle equal to $\angle EDG$. On the second constructed ray he chooses K with

$$AK = DG.$$

Through K he erects a line perpendicular to the target base plane and chooses H on that line so that

$$KH = GF.$$

The metric verification is then driven by right triangles. Euclid cuts off on the first target ray a segment equal to DE , joins the necessary segments, and repeatedly applies planar triangle congruence. The right triangles based on DG, GE, GC are matched with the corresponding target right triangles based on AK and the copied planar distances. This ultimately gives

$$\angle BAH = \angle EDF, \quad \angle HAL = \angle FDC,$$

while the copied base construction already gives

$$\angle BAL = \angle EDC.$$

The classical diagram depicts one generic position of G . The formal theorem cannot use that diagrammatic convention as a hidden hypothesis.

26.4 Euclid's Classical Proof

Euclid's proof can be organized into five steps.

26.4.1 Project the third edge to the base plane

A point F is chosen on the third source ray and FG is drawn perpendicular to the plane through E, D, C , using Proposition XI.11. The point G is the foot of this perpendicular.

The defining property of a line perpendicular to a plane implies that FG is perpendicular to every relevant line in the base plane through G .

26.4.2 Copy the base angle and the projected direction

At A , on the prescribed line AB , Euclid constructs

$$\angle BAL = \angle EDC$$

and

$$\angle BAK = \angle EDG.$$

He then lays off

$$AK = DG.$$

In the classical text this is presented as a direct application of planar angle and segment construction. The production Lean proof packages the same information differently: it first copies a nondegenerate base triangle and then constructs the point corresponding to G by its position relative to the copied base angle.

26.4.3 Erect the target normal

At K Euclid erects a line perpendicular to the plane through B, A, L , using XI.12, and chooses H on this normal such that

$$KH = GF.$$

Thus the source and target configurations have equal perpendicular heights above their respective base planes.

26.4.4 Recover the spatial distances

Because FG and KH are perpendicular to their respective base planes, the relevant triangles are right triangles. Euclid first establishes the copied base distances and then applies I.4 repeatedly to obtain equal spatial distances from H and F to the corresponding base vertices.

Schematically,

$$DG = AK, \quad GF = KH$$

and equality of the appropriate base distances give the equal hypotenuses needed for the final face comparisons.

26.4.5 Recover the remaining face angles

The final use of I.8 identifies the remaining plane angles by SSS. Hence all three face angles of the constructed trihedron at A are equal to the corresponding face angles of the source trihedron at D .

The classical proof is therefore fundamentally synthetic: orthogonal projection, planar copying, perpendicular erection, SAS/SSS triangle congruence, and no numerical representation of solid angles.

26.5 Why XI.26 Is Structurally Difficult

The main formal difficulty is not the final metric verification. Once a correct target copy of the projected base point G has been constructed, the perpendicular lift and SSS recovery are systematic.

The difficult point is that Euclid's diagram suppresses the complete order type of G in the base plane.

The source projection theorem produces

$$G \in \pi, \quad FG \perp \pi, \quad D, E, C \in \pi,$$

where π is the base plane. It does *not* say that DG lies inside $\angle EDC$. A complete synthetic proof must therefore distinguish all possible ray positions.

The production theorem handles:

- $G = D$;
- G on the ray DE ;
- $E - D - G$, the ray opposite DE ;
- G on the ray DC ;
- $C - D - G$, the ray opposite DC ;
- the first adjacent exterior sector;
- the second adjacent exterior sector;
- the interior sector;
- the cyclic/opposite sector.

The noncollinear part of this classification is expressed by the reusable interface theorem

```
hilbert_space_three_rays_order_or_cyclic_in_plane
```

while collinear boundary positions are handled by

```
hilbert_space_collinear_ray_position_in_plane
```

This is the central formal refinement of Euclid XI.26: the classical picture shows a representative configuration; the Lean theorem proves the construction independently of where the perpendicular foot happens to lie.

26.6 Formal Reconstruction: Projection, Base Copy, and Lift

26.6.1 The copied base triangle

The first production structure is

```
HilbertXI26BaseCopy
```

For a source base triangle D, E, C and prescribed target ray AB , it records a target plane σ and points B_0, L such that:

- $A, B, B_0, L \in \sigma$;
- B_0 lies on the same ray from A as B ;
- $AB_0 \cong DE$;
- $AL \cong DC$;
- $B_0L \cong EC$;
- the copied base triangle is nondegenerate;
- $\angle B_0AL \cong \angle EDC$;
- the same angle equality is available with the prescribed ray AB .

The local theorem

```
hilbert_XI26_copy_base_triangle_in_plane
```

performs the construction in an already chosen target plane. The wrapper

```
hilbert_XI26_copy_base_triangle_on_ray
```

chooses such a plane through the prescribed ray.

The use of the auxiliary point B_0 is deliberate. The theorem must keep the original prescribed point B as the direction of the first target ray, but there is no reason for the original segment AB itself to have length DE . The construction therefore lays off the correct length on the same ray.

26.6.2 The source projection

The theorem

```
hilbert_XI26_source_projection
```

constructs the base plane π , a normal line, and the foot G . Its output includes

$$D, E, C, G \in \pi, \quad F \notin \pi, \quad F \in \text{normal}, \quad F \neq G,$$

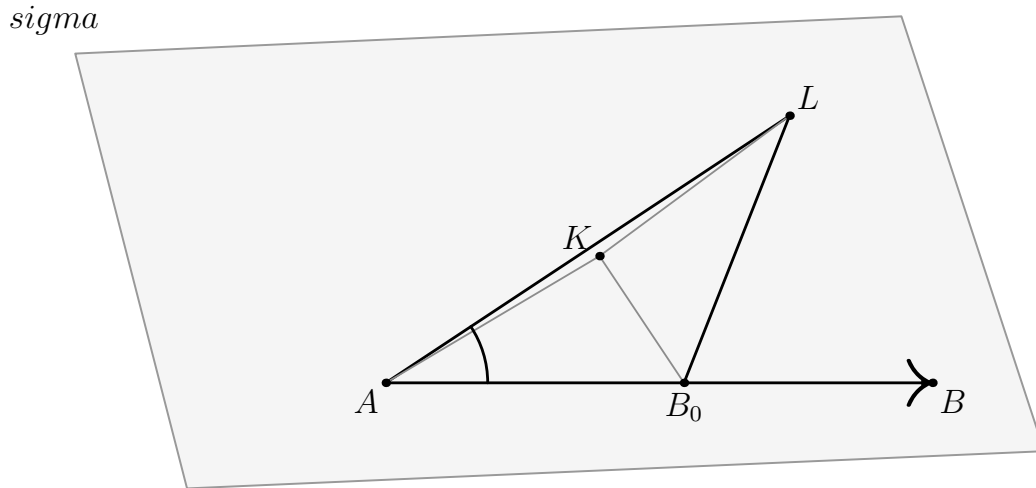


Figure 26.2: The copied planar base at the prescribed point A . The point B_0 lies on the prescribed ray AB , L completes the copied base triangle, and K represents the copied projection point. Metric equalities and angle congruences are stated in the text rather than written on the diagram.

together with

$$\text{HilbertLinePerpendicularPlaneAt}(\text{normal}, \pi, G).$$

This stage uses the already established Proposition XI.11.

26.6.3 The interior model

When the ray DG meets the open segment EC , the target point can be constructed inside the copied base angle. The structure

```
HilbertXI26InteriorPointCopy
```

records a target point K with

$$DG \cong AK, \quad EG \cong B_0K, \quad CG \cong LK,$$

together with the corresponding two component angle congruences.

The combined theorem

```
hilbert_XI26_copy_base_with_interior_point
```

packages the copied base and the copied interior projection point.

The full interior branch is

```
euclid_proposition_11_26_of_interior_projection
```

26.6.4 Boundary and exterior positions

The remaining projection positions cannot be forced into the interior construction without losing orientation information. The production proof therefore contains separate theorems for the boundary rays, opposite rays, two adjacent exterior sectors, and the cyclic sector.

The terminal branch theorems are:

```
euclid_proposition_11_26_of_projection_foot_eq_vertex
euclid_proposition_11_26_of_projection_on_first_ray
euclid_proposition_11_26_of_projection_on_second_ray
euclid_proposition_11_26_of_first_exterior_projection
euclid_proposition_11_26_of_second_exterior_projection
euclid_proposition_11_26_of_cyclic_projection
euclid_proposition_11_26_of_projection_opposite_first_ray
euclid_proposition_11_26_of_projection_opposite_second_ray
```

The exterior cases are where the reusable spatial Hilbert-T15 infrastructure is needed. The proposition-specific file builds the local source and target extension data; the proposition-independent angle-addition and same-side machinery now lives in

```
CGJteamLab/HilbertInterfaceXI.lean
```

26.6.5 The spatial lift

Once a target base point K has been constructed with the three required base distances, the remaining step is common in spirit to all branches. XI.12 erects a target normal at K , the source height GF is copied to that normal, and right-triangle congruence recovers the spatial distances.

The two principal lifting interfaces in the proposition file are

```
hilbert_XI26_lift_base_point_copy
hilbert_XI26_lift_base_point_copy_ne
```

They culminate in a point H for which

```
HilbertTrihedralRealizesThreeAngles
Geo A B L H E D C C D F F D E
```

holds.

26.6.6 The complete dispatcher

The public direct theorem is:

```
theorem euclid_proposition_11_26
  [Hinc : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
```

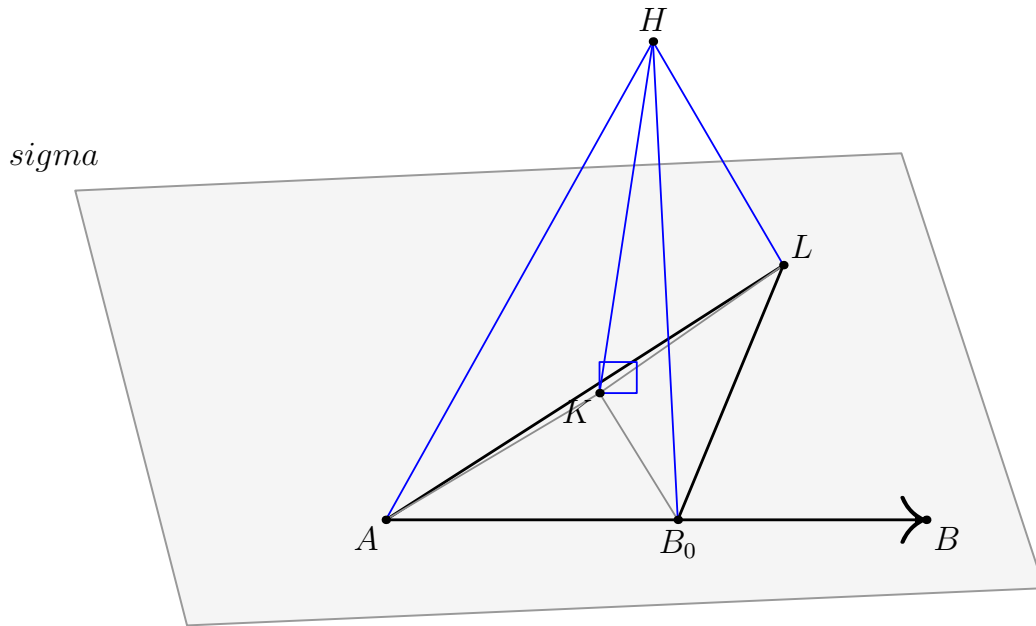


Figure 26.3: The final spatial lift. The points A, B_0, L, K lie in the target base plane, while H is obtained on the perpendicular through K . The three face-angle congruences with the source trihedron are recovered by the shared synthetic metric argument.

```
[S : HilbertSpacePrimitive Geo]
[HSI : HilbertSpaceIncidence Geo]
[HSO : HilbertSpaceOrder
  (Geo := Geo) (H := Hinc) (S := S)]
[HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := Hinc) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(D E C F A B : Geo.Point)
(hTri : HilbertTrihedralConfiguration Geo D E C F)
(hAB : Ne A B) :
exists L H : Geo.Point,
  HilbertTrihedralRealizesThreeAngles
    Geo A B L H E D C C D F F D E
```

The final proof first constructs the source projection and then performs a complete order dispatch. The dispatcher introduces no new geometric construction; every branch has already been solved by a dedicated theorem.

26.7 The Hilbert–Wyler Route for XI.26

The separate production file

```
CGJteamLab/Wyler/Proposition11_26.lean
```

implements XI.26 through a Hilbert–Wyler incidence route.

The architectural situation is notably cleaner than in XI.24 and XI.25. There is no second copy of the approximately five-thousand-line XI.26 development. The audited direct proof is already

parametric in

```
HilbertSpaceIncidence Geo
```

and proposition-independent geometry has already been moved to `HilbertInterfaceXI`. The Wyler route can therefore change the incidence package supplied to the existing theorem while leaving the whole order, congruence, `PlaneGeo`, right-angle, SAS, SSS, and T15 core unchanged.

26.7.1 The routed incidence package

The theorem

```
hilbertSpaceIncidence_routed_through_wyler
```

constructs a new `HilbertSpaceIncidence` value from:

- the existing Hilbert 3D spatial incidence package;
- an explicit `HilbertWylerAxioms Geo` package.

The common incidence fields are taken from Wyler:

```
two_points_on_each_line
plane_through
plane_unique
line_in_plane
```

The compatibility facade also requires a point on each plane. The Wyler package provides the stronger fact that every plane contains three noncollinear points, so

```
point_on_each_plane
```

is derived from

```
W.three_noncollinear_on_plane
```

Two genuinely three-dimensional Hilbert fields are retained from the ambient 3D package:

```
plane_second_common_point
four_noncoplanar
```

Thus the routing has the conceptual form

Hilbert–Wyler common incidence I.3–I.6 level data	+	Hilbert 3D residual incidence I.7–I.8 level data	⇒	XI.26 incidence package.
--	---	---	---	--------------------------

This is an incidence refactor, not a claim that the present XI.26 theorem is already dimension-free.

26.7.2 The explicit Wyler core

The theorem

```
euclid_proposition_11_26_wyler_core
```

assumes an explicit

```
[W : HilbertWylerAxioms Geo]
```

It constructs the routed incidence value

```
HSIw : HilbertSpaceIncidence Geo
```

and invokes the already proved direct theorem with

```
(HSI := HSIw)
```

This point is mathematically significant. The Wyler theorem is not an alias of the final direct conclusion: the direct XI.26 proof is re-instantiated with a different spatial incidence record. Consequently the common incidence operations used by the proof are definitionally supplied through Wyler.

At the same time the theorem deliberately retains the established spatial order and congruence classes. It does not yet construct a local metric 3-flat in an arbitrary-dimensional ambient geometry.

26.7.3 The public Hilbert 3D corollary

The public theorem

```
euclid_proposition_11_26_wyler
```

does not add `HilbertWylerAxioms` as an extra public geometric assumption. It first defines

```
let W : HilbertWylerAxioms Geo :=
  hilbertWylerAxioms_of_hilbert3D
  (Geo := Geo)
```

and then applies the explicit Wyler core.

Thus, in the present three-dimensional library,

$$\text{Hilbert 3D} \implies \text{Hilbert–Wyler incidence} \implies \text{XI.26.}$$

The result is parallel to the architecture already used for XI.23, but XI.26 requires dramatically less duplicated proposition code because its direct proof has already been refactored around

reusable interfaces.

26.8 Proof Architecture and Formal Obligations

The complete reconstruction is most naturally read as one continuous construction.

1. The source trihedron determines the nondegenerate base angle EDC and guarantees that F lies outside the base plane.
2. XI.11 constructs the perpendicular projection $F \rightarrow G$ onto the source base plane.
3. The base triangle DEC is copied to a target plane through the prescribed ray AB , producing B_0, L .
4. The position of G relative to DE and DC is classified completely.
5. In the corresponding target sector a point K is constructed so that the three distances from G to D, E, C are copied by the three distances from K to A, B_0, L .
6. XI.12 erects a normal through K to the target base plane.
7. The height GF is copied to the target normal.
8. Spatial right-triangle congruence gives the three lateral distances.
9. Spatial SSS recovers the two remaining face-angle congruences.
10. The target noncoplanarity and the three angle congruences are packaged as `HilbertTrihedralRealizesTh`

Several obligations that are invisible in Euclid's diagram are explicit in Lean:

- every use of planar order occurs in an explicit `PlaneGeo`;
- every target point must be proved to remain in the chosen target plane;
- ray orientation must be preserved under each segment construction;
- opposite-ray and exterior-sector cases require explicit same-side and opposite-side information;
- angle addition requires a synthetic T15 interface rather than numerical angle measures;
- the final four points of the target trihedron must be proved noncoplanar;
- the prescribed point B fixes a ray, not the target length, hence the auxiliary point B_0 .

These obligations explain the formal length of XI.26. They are not numerical overhead; they are the explicit incidence and order content hidden by the classical drawing.

26.9 Dependencies and Reuse

The production direct file imports

```
CGJteamLab.HilbertInterfaceXI
CGJteamLab.Proposition11_11
CGJteamLab.Proposition11_12
```

XI.11 supplies the source perpendicular from F to the base plane. XI.12 supplies the target perpendicular through the copied projection point K .

The remaining substantial machinery is reusable interface mathematics rather than proposition-local dependence on earlier Book XI theorem files. In particular, the post-audit `HilbertInterfaceXI` contains the general results for:

- points off lines inside planes;
- spatial right-angle recovery from a plane normal;
- lifted-distance comparison;
- spatial SSS angle recovery;
- ray-position classification;
- opposite-extension and same-side transport;
- spatial segment-construction uniqueness;
- the spatial Hilbert-T15 angle-addition chain.

A notable consequence of the audit is that XI.26 no longer imports `Proposition11_23`. The planar three-ray classification originally developed during XI.23 has been promoted to the common interface.

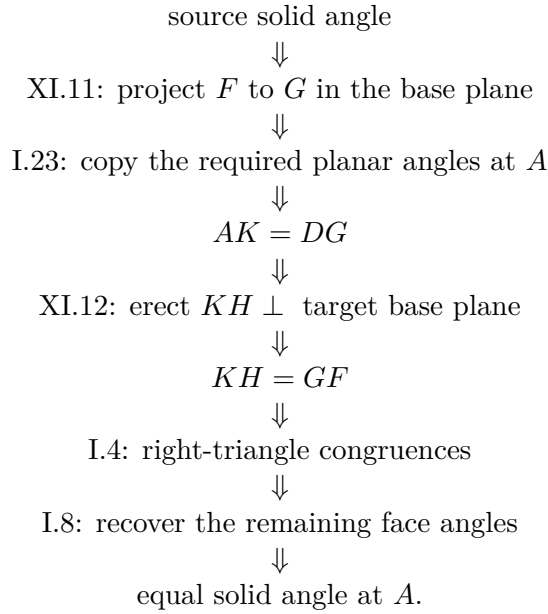
The Wyler production file imports

```
CGJteamLab.Proposition11_26
CGJteamLab.HilbertWyler3DCompatibility
```

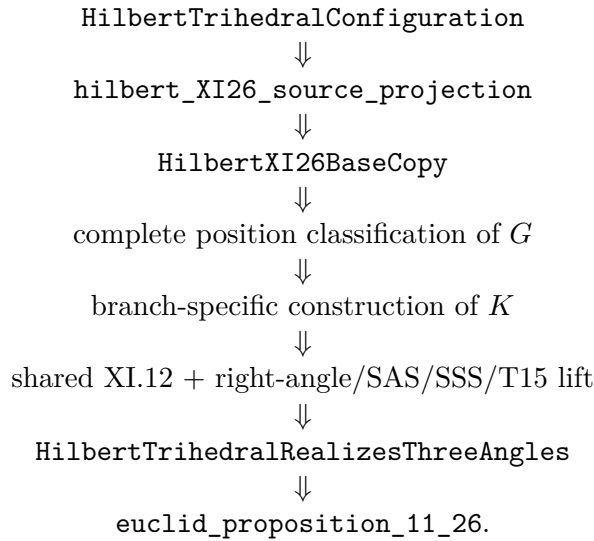
and reuses the complete direct metric/order proof after changing the supplied incidence record.

26.10 Classical and Formal Dependency DAGs

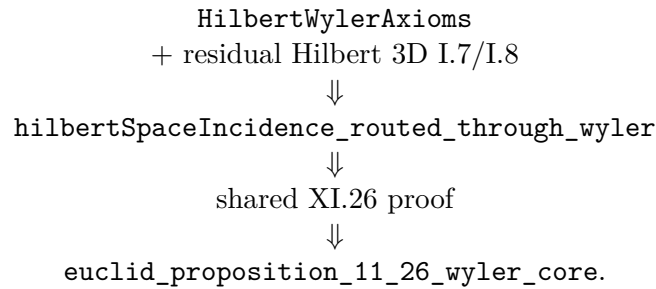
The classical construction has the compact chain



The formal direct chain is longer:



The Hilbert–Wyler production route inserts an incidence factorization before the same direct theorem:



For the public 3D theorem,

$$\text{Hilbert 3D} \implies \text{hilbertWylerAxioms_of_hilbert3D} \implies \text{euclid_proposition_11_26_wyler.}$$

26.11 Comparison with Euclid

At the highest level the formal reconstruction follows Euclid closely:

1. choose a base plane of the source solid angle;
2. project the third edge orthogonally to that plane;
3. reproduce the projected planar geometry at the prescribed ray;
4. erect a perpendicular at the copied projection point;
5. copy the perpendicular height;
6. recover the remaining face angles from congruent triangles.

The main difference is not metric but order-theoretic. Euclid draws one position of G and proceeds as if the required planar construction has a single visible orientation. The formal theorem must be valid for every trihedral configuration satisfying its hypotheses. Consequently it proves a complete classification of G and supplies a construction in every sector.

A second difference is representational. Euclid directly says that the required plane angles are constructed at A . The formal proof often represents that construction by a copied triangle or by three copied distances. SSS then recovers the angle statement. This is not a change in geometry; it is a more stable synthetic interface for a proof assistant.

A third difference is architectural. The completed audit separates proposition-specific construction from reusable spatial mathematics. General order and T15 lemmas no longer live inside the XI.26 file merely because they were discovered while proving XI.26.

The Hilbert–Wyler route adds a further structural observation: much of the spatial incidence needed by XI.26 belongs to the common Wyler-compatible incidence layer, while the present theorem still retains specifically three-dimensional residual incidence and metric structure.

26.12 Diagrammatic Audit

Three figures are mathematically justified for the final documentation.

1. **Source projection.** The source trihedron at D , its base plane through E, D, C , the point F , and the perpendicular projection $F \rightarrow G$. This figure should make clear that the order position of G is not fixed by the projection theorem.
2. **Copied planar configuration.** The prescribed ray AB , the auxiliary point B_0 on the same ray, the copied base point L , and a representative copied projection point K . The figure should distinguish the prescribed direction AB from the laid-off segment AB_0 .
3. **Final spatial lift.** The target base plane through A, B_0, L, K , the perpendicular through K , the lifted point H , and the three target face angles corresponding to the source trihedron.

No separate figure is required for each of the nine order branches. Those branches represent different logical positions of the same planar datum, not nine geometrically stable constructions. A small schematic ray-order diagram may be useful inside the second figure, but the documentation should avoid a page of near-duplicate branch pictures.

26.13 Audit Status

This documentation is based on the current production sources

```
CGJteamLab/Proposition11_26.lean
CGJteamLab/Wyler/Proposition11_26.lean
```

Both production modules have been compiled successfully after the XI.26 architecture audit.

The direct proposition file was reduced by moving proposition-independent mathematics to `HilbertInterfaceXI`. In particular, the spatial Hilbert-T15 chain and the three-ray order classification are now shared interface results. Redundant and superseded local helpers were removed, and XI.26 no longer imports Proposition XI.23 merely to obtain a general planar order theorem.

The Hilbert–Wyler production theorem was then checked by constructing a routed `HilbertSpaceIncidence` package and running the already compiled XI.26 theorem against it. This confirms that the common incidence portion of the current proof can be supplied through `HilbertWylerAxioms` without duplicating the metric/order development.

A separate kernel `#print axioms` comparison for

```
Geometry.euclid_proposition_11_26
Geometry.euclid_proposition_11_26_wyler_core
Geometry.euclid_proposition_11_26_wyler
```

is not recorded here until that audit is run explicitly. The source architecture already shows that the public Wyler theorem adds no new *typeclass assumption*: its Wyler package is derived from the established Hilbert 3D environment.

26.14 Conclusion

Proposition XI.26 is a constructive theorem about transporting a complete trihedral angle to a prescribed spatial location and direction. Euclid’s classical argument is short because the diagram fixes an apparently generic projection geometry. Formal reconstruction shows that the decisive hidden content is the order type of the projection foot.

The production proof therefore separates the problem into three layers:

source projection $\implies$ complete planar position analysis $\implies$ common spatial lift and face recovery.

This organization preserves Euclid’s synthetic mechanism while making every incidence, order, orientation, and nondegeneracy obligation explicit.

The Hilbert–Wyler route gives a second structural conclusion. XI.26 does not require a second proposition-scale proof in order to expose its Wyler-compatible incidence content. After the direct development is properly modularized, the same theorem can be run with a spatial incidence record whose common fields are supplied by `HilbertWylerAxioms`, while the residual 3D Hilbert incidence and metric layers remain unchanged.

Thus XI.26 provides a particularly clean example of the role of the Wyler program in the project:

change the incidence route where it genuinely changes + reuse the established synthetic metric core.

It also prepares the next proposition structurally. XI.27 begins by invoking XI.26 to reproduce the solid angle of a given parallelepiped on a prescribed straight line; the completed formal XI.26 theorem now supplies that construction without any hidden assumption on the position of the projected foot.

Appendix A

Euclid in Heath's Edition

A.1 Scope

This appendix records the Book XI source audit against Euclid's *Elements* in Heath's edition through Proposition XI.25 [1]. The purpose is to control the classical statement, construction, and local proposition dependencies. Lean-specific incidence, subtype, and plane-transport obligations are documented in the proposition chapters and are not attributed to Euclid or Heath.

A.2 Proposition XI.4

Proposition XI.4 states that if a straight line is set up at right angles to each of two straight lines at their point of intersection, then it is also at right angles to the plane through those two lines.

The classical proof chooses equal segments on the two given lines and introduces an arbitrary line of the plane through the common point. Triangle-congruence arguments are then used to prove that the candidate spatial line makes a right angle with that arbitrary transversal.

For the formal reconstruction the significant classical features are:

- the target is universal over every line of the plane through the common point;
- the arbitrary transversal is part of the proposition's logical content, not merely a feature of the diagram;
- the proof is synthetic and based on incidence, right angles, equal segments, and triangle congruence.

A.3 Proposition XI.5

Proposition XI.5 says that if one straight line is perpendicular at a common point to three straight lines meeting there, then those three straight lines lie in one plane.

Euclid's contradiction argument uses the plane through two of the three candidate coplanar lines and another plane through the common perpendicular and the remaining line. The intersection line of the two planes produces, inside one plane, two distinct lines through the same point both perpendicular to the common perpendicular.

The local classical dependency is therefore governed by the preceding spatial propositions: the intersection-of-planes configuration and Proposition XI.4 supply the contradiction. The current Lean proof organizes this material differently, but preserves the same spatial core: perpendicularity to a plane forces the required intersection line to be perpendicular to the common normal.

A.4 Proposition XI.6

Heath gives the statement:

If two straight lines be at right angles to the same plane, the straight lines will be parallel.

Let AB and CD be perpendicular to the reference plane at B and D . Euclid joins BD , draws DE in the reference plane perpendicular to BD , makes

$$DE = AB,$$

and joins BE , AE , and AD .

Since AB is perpendicular to the reference plane,

$$\angle ABD, \quad \angle ABE$$

are right. Since CD is also perpendicular to the same plane, the corresponding lines of the plane through D are perpendicular to CD .

The first triangle comparison uses

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles. Proposition I.4 gives

$$AD = BE.$$

The second comparison uses

$$DE = BA, \quad EA = AE, \quad DA = BE.$$

Proposition I.8 therefore gives

$$\angle EDA = \angle ABE.$$

Hence $\angle EDA$ is right and

$$ED \perp DA.$$

At D , the line ED is now perpendicular to

$$DB, \quad DA, \quad DC.$$

Proposition XI.5 places these three lines in one plane. Proposition XI.2 supplies the corresponding triangle-plane incidence, bringing AB into that same plane. The two right angles made with the transversal BD then give the final parallelism by Proposition I.28.

The local classical DAG can be recorded as

$$\begin{array}{c}
 \text{construct } DE \perp BD, \quad DE = AB \\
 \Downarrow \\
 \text{I.4} \\
 \Downarrow \\
 AD = BE \\
 \Downarrow \\
 \text{I.8} \\
 \Downarrow \\
 ED \perp DA \\
 \Downarrow \\
 \text{XI.5} \\
 \Downarrow \\
 DB, DA, DC \text{ coplanar} \\
 \Downarrow \\
 \text{XI.2} \\
 \Downarrow \\
 AB, BD, DC \text{ coplanar} \\
 \Downarrow \\
 \text{I.28} \\
 \Downarrow \\
 AB \parallel CD.
 \end{array}$$

A.5 Comparison with the Lean reconstruction at XI.6

The Lean proof preserves the central metric sequence of the classical argument:

$$\text{auxiliary } DE \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

The formal expansion is concentrated in the data suppressed by the classical diagram:

- each planar construction is performed in an explicit induced plane geometry;
- cross-plane segment and angle comparisons use the spatial congruence layer;
- noncollinearity and point-in-plane obligations are explicit;
- the common plane of the two final normals is produced as a witness;
- spatial parallelism records both coplanarity and disjointness, so skew lines are excluded;
- the last planar disjointness proof uses the established neutral alternate-angle infrastructure rather than a direct call to the public I.28 wrapper.

The last point is a dependency-graph difference, not a change in the geometric target. Euclid closes the configuration by I.28; the current Lean source imports the Proposition I.27 module and invokes the lower alternate-angle criterion in the induced common plane.

A.6 Checkpoint XI.6

Through XI.6 the reconstruction remains synthetic. No coordinate, vector, scalar-product, trigonometric, or numerical-angle representation is used in the proposition proofs.

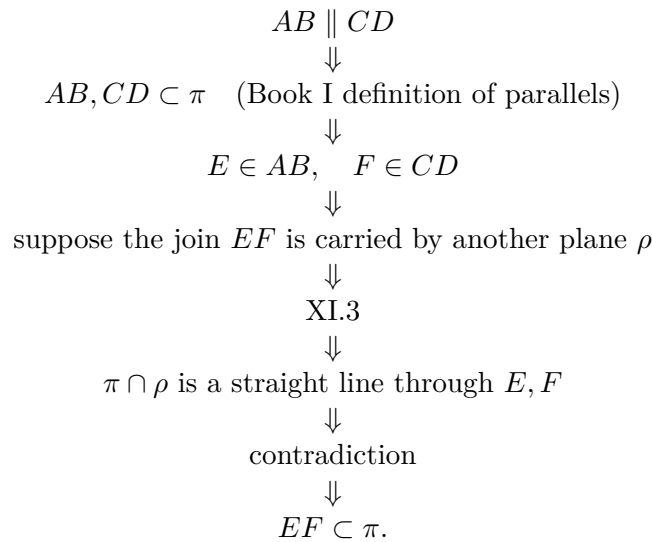
The source comparison also isolates the new structural point of XI.6: in three-space the proof of parallelism has to establish coplanarity before planar nonintersection can be invoked. Euclid obtains this by XI.5 and XI.2. The Lean theorem records the same distinction directly in the target relation `HilbertSpaceLinesParallel`, which consists of an explicit common plane together with disjointness of the two ambient carriers.

A.7 Proposition XI.7

Proposition XI.7 concerns the plane determined by two parallel straight lines. In Book I the definition of parallel lines already requires the two lines to be in one plane. Thus the reference plane is part of the hypothesis, not an additional conclusion of XI.7.

Choose a point on each of the two parallel lines and join the chosen points. Euclid argues by contradiction. He supposes that the joining straight line is carried by another, more elevated plane. A plane through that alleged join is then intersected with the reference plane. Proposition XI.3 makes the common section a straight line. Since both chosen points lie in both planes, that section passes through both of them. The contradiction is expressed in the classical text by the impossibility of two straight lines enclosing an area. The join must therefore lie in the original plane of the parallels.

The local classical DAG is



No Proposition XI.4, XI.5, or XI.6 occurs in this local proof. The explicit Book XI dependency is XI.3.

A.8 Comparison with the Lean reconstruction at XI.7

The Lean proof has a shorter incidence DAG. The formal relation `HilbertSpaceLinesParallel` already contains a witness plane σ for the two parallel carriers, together with their disjointness.

The selected points therefore lie in σ . Their distinctness follows from disjointness of the parallel carriers. After the joining carrier is constructed, `HilbertSpaceIncidence.line_in_plane` places the whole join in σ .

This final call represents the Hilbert incidence principle that two distinct points of a line lying in a plane force the entire line to lie in that plane. Consequently the formal proof does not reproduce Euclid's auxiliary second plane or his use of XI.3. This is a difference in foundational packaging, not in the geometric conclusion.

A.9 Checkpoint XI.7

Through XI.7 the reconstruction remains entirely synthetic. Proposition XI.7 uses only incidence: no order, congruence, angle theory, continuity, coordinate representation, vector formalism, or scalar product enters its proof.

The source comparison now exhibits two distinct ways to express the same closure principle. Euclid derives it by a plane-intersection contradiction; the Hilbert-style formalization uses an explicit common-plane witness for spatial parallelism and a general line-in-plane incidence axiom.

A.10 Proposition XI.8

Proposition XI.8 states that if two straight lines are parallel and one of them is perpendicular to a plane, then the other is perpendicular to the same plane.

Let $AB \parallel CD$, with AB perpendicular to the reference plane at B . Euclid lets CD meet the plane at D , joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

The local classical dependencies are unusually informative because they cross the Euclidean parallel boundary.

- XI.7 places AB , CD , and BD in one plane.
- I.11 and I.3 supply the auxiliary perpendicular and equal segment.
- The definition of line-plane perpendicularity makes $\angle ABD$ and $\angle ABE$ right.
- I.29 converts the given parallelism into the right angle $\angle CDB$, hence $CD \perp BD$.
- I.4 and I.8 give $AD = BE$ and then show $\angle EDA$ right.
- XI.4 applied in the plane through BD, DA gives $ED \perp CD$.
- XI.4 applied again in the reference plane, to DB, DE , gives $CD \perp \pi$.

The incidence step placing CD in the plane through BD, DA uses the planarity of triangle ABD together with the common plane of AB, CD, BD . The printed diagram suppresses this plane identification.

The local classical DAG is

$$\begin{array}{c}
AB \parallel CD, \quad AB \perp \pi \\
\Downarrow \\
\text{XI.7} \\
\Downarrow \\
DE \perp BD, \quad DE = AB \\
\Downarrow \\
\text{I.29} \\
\Downarrow \\
CD \perp BD \\
\Downarrow \\
\text{I.4, I.8} \\
\Downarrow \\
ED \perp AD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
ED \perp CD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
CD \perp \pi.
\end{array}$$

A.11 Comparison with the Lean reconstruction at XI.8

The Lean proof preserves the synthetic metric core but reorganizes two classical steps.

First, the point D where the second parallel meets the reference plane is not assumed. It is derived from the common carrier plane of the parallels, the intersection of that plane with the reference plane, and Group IV uniqueness.

Second, the proof does not call I.29 directly. It constructs through D a line perpendicular to BD , proves by the neutral I.27 alternate-angle criterion that this line is disjoint from the first parallel, and then uses Group IV uniqueness to identify it with the second parallel.

Thus the formal factorization is

$$\text{neutral I.27} + \text{Euclidean uniqueness of parallels} \implies \text{the I.29-type conclusion needed in XI.8.}$$

The two applications of XI.4 remain visible in the production proof. The first is made in the common plane of the parallel lines; the second is made in the reference plane.

A.12 Checkpoint XI.8

Through XI.8 the reconstruction remains synthetic: no coordinates, vectors, scalar product, trigonometric functions, or numerical angle measure are introduced.

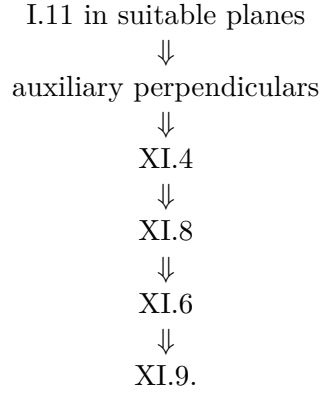
Unlike XI.4–XI.7, however, XI.8 is genuinely Euclidean in the axiomatic sense. The formal proof requires plane-local parallel uniqueness, represented by `HilbertSpaceEuclidean`. No new continuity assumption is used.

A.13 Proposition XI.9

Heath gives the proposition in the form:

If two straight lines are parallel to the same straight line, then they are parallel to one another, if they do not lie in the same plane with it.

Euclid's printed proof is organized through the perpendicularity theory built in the immediately preceding propositions rather than through an affine plane-intersection argument. Its characteristic local dependency pattern is



The important source-level point is that XI.9 is not simply planar I.30 with “space” substituted for “plane”. Euclid explicitly uses the preceding line–plane perpendicularity results to bridge between the distinct carrier planes of the given parallels.

A.14 Comparison with the Lean reconstruction at XI.9

The production Lean proof reaches the same conclusion by a different synthetic route. From

$$l \parallel n, \quad m \parallel n,$$

it extracts the two carrier planes. It then chooses a point $B \in m$ not on l , constructs a plane ω through l and B , and intersects ω with the carrier plane τ of m, n .

The resulting line r passes through B . Inside τ , both r and m are parallels to n through the same point. Group IV uniqueness therefore gives $r = m$. Hence $l, m \subset \omega$, and a final disjointness argument yields

$$l \parallel m.$$

Thus the formal proof replaces Euclid's perpendicularity chain by

$$\text{spatial incidence} + \text{plane intersection} + \text{plane-local parallel uniqueness}.$$

It remains fully synthetic and uses no coordinates, vectors, or scalar product.

A.15 Checkpoint XI.9

The classical and Lean proofs establish the same spatial transitivity statement but expose different local DAGs. Euclid packages the theorem through XI.4, XI.8, and XI.6; the current formalization packages it through explicit carrier planes and one Group IV uniqueness step.

This is a difference in proof architecture, not in theorem content. No continuity principle is introduced by the Lean route.

A.16 Proposition XI.10

Heath gives the statement:

If two straight lines meeting one another be parallel to two straight lines meeting one another not in the same plane, they will contain equal angles.

Let AB, BC meet at B , and let DE, EF meet at E , with the corresponding pairs parallel. Euclid first uses I.3 to cut equal segments so that

$$BA = BC = ED = EF.$$

He then joins

$$AD, \quad CF, \quad BE, \quad AC, \quad DF.$$

The first two I.33 applications give

$$AD \parallel BE, \quad AD = BE,$$

and

$$CF \parallel BE, \quad CF = BE.$$

Then XI.9 gives

$$AD \parallel CF,$$

while transitivity of equality through BE gives $AD = CF$. A third application of I.33 yields

$$AC \parallel DF, \quad AC = DF.$$

Finally I.8 applied to the two triangles gives

$$\angle ABC = \angle DEF.$$

The exact local classical DAG is therefore

$$\begin{array}{c} \text{I.3} \\ \Downarrow \\ \text{I.33 twice} \\ \Downarrow \\ \text{XI.9} \\ \Downarrow \\ \text{I.33} \\ \Downarrow \\ \text{I.8} \\ \Downarrow \\ \text{XI.10.} \end{array}$$

A subtle point is carried by the classical diagram: the selected parallel segments must be taken in corresponding directions. Reversing exactly one ray changes the relevant angle to its supplement. Euclid does not isolate this orientation datum as a separate predicate.

A.17 Comparison with the Lean reconstruction at XI.10

The Lean proof preserves the same I.33–XI.9–I.33–SSS spine but makes the orientation information explicit through

```
HilbertSpaceDirectedParallelSegments
```

Ordinary spatial parallelism is sufficient for XI.9 but not for the third I.33 step.

The normalized formal core has the following actual DAG:

```
BA ~= ED, BC ~= EF
directed parallels
|
I.33 twice
|
AD ~= BE, CF ~= BE
|
XI.9
|
AD || CF
|
Borsuk-Szmielew 52
|
AD directed-parallel CF
|
I.33
|
AC ~= DF
|
spatial SSS
|
angle ABC ~= angle DEF
```

The Borsuk–Szmielew 52 line in this DAG is not an additional Euclidean proposition. It records the formal section-line/plane-separation bridge used to recover the direction of AD relative to CF .

The public theorem also reconstructs Euclid's opening I.3 normalization by segment construction. It lays off points D_1, F_1 on the original rays ED, EF , applies the normalized theorem, and then transports the resulting angle back along SameRay in the explicit plane of D, E, F .

The final SameRay step is deliberately planar. The proof does not install a global ambient HilbertOrder Geo; it works inside the induced PlaneGeo of the target angle plane and maps the resulting angle data back to ambient space.

A.18 Checkpoint XI.10

Through XI.10 the source comparison shows two kinds of information which classical diagrams suppress but the formalization must expose:

- explicit carrier planes and noncoplanarity witnesses;

- orientation of selected segments on parallel carriers.

The Lean theorem remains synthetic. Its extra machinery is plane-side and ray-order bookkeeping, not coordinates or numerical angle measure. XI.9 is a genuine direct dependency of XI.10, whereas XI.11–XI.13 below have their own local DAGs and do not consume XI.9 or XI.10.

A.19 Proposition XI.11

Heath gives the statement:

From a given elevated point to draw a straight line perpendicular to a given plane.

Let A be the elevated point and let π be the plane of reference. Euclid draws an arbitrary line $BC \subset \pi$, then draws AD from A perpendicular to BC by I.12. If $AD \perp \pi$, the problem is finished.

In the nontrivial branch he constructs

$$DE \perp BC \quad (\text{I.11}), \quad AF \perp DE \quad (\text{I.12}), \quad GH \parallel BC \quad (\text{I.31}).$$

Since BC is perpendicular to both DA and DE , XI.4 gives

$$BC \perp \text{plane}(ED, DA).$$

Then XI.8 transfers this perpendicularity to the parallel line GH :

$$GH \perp \text{plane}(ED, DA).$$

By XI.Def.3, GH is perpendicular to AF , which meets it and lies in that plane. Hence AF is perpendicular to both GH and DE . A second application of XI.4 gives

$$AF \perp \text{plane}(ED, GH) = \pi.$$

Thus the exact local classical dependency chain is

$$\begin{array}{c} \text{I.12} \\ \Downarrow \\ \text{I.11, I.12, I.31} \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ \text{XI.8} \\ \Downarrow \\ \text{XI.Def.3} \\ \Downarrow \\ \text{XI.4} \\ \Downarrow \\ \text{XI.11.} \end{array}$$

Neither XI.9 nor XI.10 is cited in Heath's proof, and neither is needed transitively for the local construction described there. This verifies that the current reconstruction may pass directly from XI.8 to XI.11 without silently skipping a classical dependency.

One incidence point is implicit in the classical prose. Before the first I.12 construction can be interpreted as a planar construction, the elevated point A and the chosen line BC must be placed in a common plane. The formal reconstruction supplies this plane explicitly.

A.20 Comparison with the Lean reconstruction at XI.11

The production proof follows the same construction and retains Euclid's initial case split. Its additional work is spatial bookkeeping rather than a change of method.

The principal formal expansions are:

- construction of the plane σ through A and BC before the first planar I.12;
- construction of the plane τ through AD and DE before the second planar I.12;
- proof that the second foot F does not lie on BC , needed before I.31 is applied in the reference plane;
- conversion of planar I.31 parallelism into the ambient `HilbertSpaceLinesParallel` relation;
- proof that the foot K returned existentially by XI.8 equals the already constructed point F ;
- proof that $DE \neq GH$ before the final XI.4 application.

The two XI.4 applications and the single XI.8 application remain exactly visible in the formal proof architecture.

A.21 Checkpoint XI.11

Through XI.11 the reconstruction remains synthetic: no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure are introduced.

The classical proof of XI.11 explicitly uses XI.8 and therefore inherits the Euclidean parallel content of that proposition. No continuity principle is cited. XI.9 and XI.10 are absent from the local source DAG.

A.22 Proposition XI.12

Heath gives the statement:

To set up a straight line at right angles to a given plane from a given point in it.

Let A be the given point in the plane of reference. Euclid lets any elevated point B be conceived and applies XI.11 to draw BC perpendicular to the reference plane. Through A he then draws

$$AD \parallel BC$$

by I.31. Proposition XI.8 immediately transfers the line–plane perpendicularity:

$$BC \perp \pi, \quad AD \parallel BC \implies AD \perp \pi.$$

The complete local classical dependency chain is therefore

$$\begin{array}{c}
 A \in \pi \\
 \Downarrow \\
 B \text{ elevated} \\
 \Downarrow \\
 \text{XI.11} \\
 \Downarrow \\
 BC \perp \pi \\
 \Downarrow \\
 \text{I.31} \\
 \Downarrow \\
 AD \parallel BC \\
 \Downarrow \\
 \text{XI.8} \\
 \Downarrow \\
 AD \perp \pi.
 \end{array}$$

Neither XI.9 nor XI.10 is cited or required in this local proof.

There is one exceptional incidence configuration that the classical prose does not discuss. If the foot C of the XI.11 perpendicular happens to equal the prescribed point A , then the line BC itself already solves XI.12. In that configuration I.31 should not be read as producing a distinct parallel through a point lying on BC . The formal proof therefore separates this completed case before carrying out the generic parallel construction.

A.23 Comparison with the Lean reconstruction at XI.12

The production proof preserves the source-level chain

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8},$$

but expands three pieces of hidden incidence data.

First, it explicitly tests whether the prescribed point A already lies on the perpendicular returned by XI.11. A proposition-local foot-uniqueness theorem then identifies A with the XI.11 foot and closes that branch.

Second, in the nontrivial branch the spatial line and the point A must be placed in a common plane σ before I.31 can be used. The parallel construction is performed inside $\text{PlaneGeo}(\sigma)$, after which its planar disjointness data are packaged as ambient `HilbertSpaceLinesParallel`.

Third, XI.8 returns the new perpendicular foot existentially. The same foot-uniqueness theorem proves that this returned point is exactly the prescribed point A .

These are formal expansions of incidence and carrier-plane information; they do not replace Euclid's geometric method.

A.24 Checkpoint XI.12

Through XI.12 the reconstruction remains synthetic: no coordinates, vectors, scalar products, trigonometric functions, or numerical angle measure enter the construction.

The classical proof explicitly uses XI.11, I.31, and XI.8. Hence the present Euclid-faithful route inherits the Euclidean parallel content already present in XI.8 and XI.11. No continuity principle is cited, and XI.9 and XI.10 remain absent from the local source DAG.

A.25 Proposition XI.13

Heath gives the statement:

From the same point two straight lines cannot be set up at right angles to the same plane on the same side.

Assume that AB and AC are two such straight lines erected from the common point A . Euclid draws the plane through BA and AC . By XI.3 its section with the reference plane is a straight line through A , written DAE .

Definition XI.3 then applies to each supposed perpendicular. Since DAE lies in the reference plane and meets both lines at A ,

$$AC \perp DAE, \quad AB \perp DAE.$$

Hence $\angle CAE$ and $\angle BAE$ are both right and therefore equal. They lie in the same plane, share the side AE , and the other two sides are assumed to be erected on the same side. Euclid concludes that this is impossible.

The complete local source DAG is therefore

$$\begin{array}{c}
 AB \perp \pi, \quad AC \perp \pi \\
 \Downarrow \\
 \text{plane through } AB, AC \\
 \Downarrow \\
 \text{XI.3} \\
 \Downarrow \\
 DAE = \pi \cap \text{plane}(AB, AC) \\
 \Downarrow \\
 \text{XI.Def.3} \\
 \Downarrow \\
 AB \perp DAE, \quad AC \perp DAE \\
 \Downarrow \\
 \text{two equal right angles in one plane on the same side} \\
 \Downarrow \\
 \text{contradiction.}
 \end{array}$$

Neither XI.9 nor XI.10 occurs in the proof. XI.11 and XI.12 are also not source dependencies: they are existence constructions, while XI.13 proves uniqueness.

A.26 Comparison with the Lean reconstruction at XI.13

The formal proof preserves Euclid's two genuinely spatial steps:

1. construct the plane σ through the two supposed distinct perpendicular carriers;
2. construct the intersection line $q = \pi \cap \sigma$ through the common foot.

The production proof makes one hidden condition explicit before the second step: it proves $\sigma \neq \pi$. If the planes were equal, one of the perpendicular lines would lie in the reference plane, which is impossible.

The terminal planar contradiction is then packaged differently. From Definition XI.3, represented by the universal clause of `HilbertLinePerpendicularPlaneAt`, the proof gets

$$l \perp q, \quad m \perp q.$$

After symmetry, XI.4 is applied in the auxiliary plane σ . If $l \neq m$, it yields $q \perp \sigma$. But the plane-intersection construction already gives $q \subset \sigma$, producing a contradiction.

Thus the Lean DAG replaces Euclid's final same-side angle-uniqueness sentence by an earlier Book XI theorem, but it does not replace the synthetic method by coordinates or metric formulas.

A.27 Checkpoint XI.13

Through XI.13 the source DAG remains independent of XI.9 and XI.10. The classical dependencies specific to the proposition are XI.3 and Definition XI.3.

The proposition is also independent of XI.11 and XI.12 as source dependencies: those propositions establish the two existence cases, whereas XI.13 supplies the corresponding uniqueness statement at a point of the plane.

The current reconstruction remains synthetic and introduces no coordinate, vector, scalar-product, trigonometric, or numerical-angle argument.

A.28 Proposition XI.14

Heath gives the statement:

Planes to which the same straight line is at right angles are parallel.

Let a line l be perpendicular to two planes π and ρ , meeting them at A and B , respectively. Euclid argues by contradiction. If the two planes meet, choose a common point K . The line l together with K determines an auxiliary plane. In that plane the triangle ABK has two right angles:

$$\angle BAK \text{ is right,} \quad \angle ABK \text{ is right.}$$

This contradicts Proposition I.17.

The classical proof is therefore governed by the compact pattern

$$\begin{array}{c}
 l \perp \pi, \quad l \perp \rho \\
 \Downarrow \\
 \text{suppose } K \in \pi \cap \rho \\
 \Downarrow \\
 \triangle ABK \text{ in the plane through } l \text{ and } K \\
 \Downarrow \\
 \text{two right angles} \\
 \Downarrow \\
 \text{I.17} \\
 \Downarrow \\
 \pi \parallel \rho.
 \end{array}$$

The formal reconstruction exposes an exceptional configuration suppressed by the printed diagram. If the two feet coincide, the triangle ABK degenerates. The production proof therefore separates the equal-foot case and proves that two planes perpendicular to the same line at the same point must coincide. The distinct-foot branch then follows Euclid's two-right-angle contradiction.

This is a formal completion of the classical argument, not a change of its geometric core.

A.29 Proposition XI.15

Heath states:

If two straight lines meeting one another be parallel to two straight lines meeting one another, not being in the same plane, the planes through them are parallel.

Let AB, BC meet at B , and let DE, EF be the corresponding pair in the second plane. Euclid assumes

$$AB \parallel DE, \quad BC \parallel EF.$$

From B , XI.11 constructs a normal BG to the second plane. Through G , Proposition I.31 produces

$$GH \parallel DE, \quad GK \parallel EF.$$

Since BG is perpendicular to the second plane, XI.Def.3 gives

$$BG \perp GH, \quad BG \perp GK.$$

By XI.9,

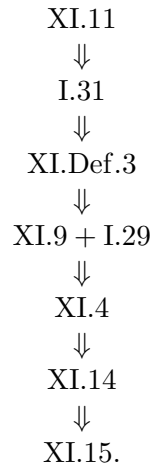
$$BA \parallel GH, \quad BC \parallel GK.$$

Proposition I.29 transfers the two right angles to BA and BC . Since those two lines meet, XI.4 yields

$$BG \perp \text{plane}(AB, BC).$$

The same line is already perpendicular to plane(DE, EF), so XI.14 gives the final plane parallelism.

The local source DAG is



The formal reconstruction keeps this outer structure but makes explicit an exception hidden in the instruction “draw through G a parallel.” The point G may already lie on one of the reference lines. In that branch the correct line is the reference line itself; only in the generic branch is a new parallel constructed.

A.30 Proposition XI.16

Proposition XI.16 is the section theorem for parallel planes:

If two parallel planes be cut by any plane, their common sections are parallel.

Let

$$l = \pi \cap \sigma, \quad m = \rho \cap \sigma,$$

where $\pi \parallel \rho$ and the cutting plane σ meets both reference planes.

The classical argument is incidence-theoretic. If l and m met at a point P , then

$$P \in \pi \quad \text{and} \quad P \in \rho,$$

contradicting the parallelism of π and ρ . Since both section lines lie in σ , they are parallel.

Thus the essential proof is

$$\text{exact plane sections} \quad + \quad \text{disjointness of the reference planes} \quad \implies \quad \text{parallel section lines.}$$

The current formal theorem makes the two section lines explicit rather than treating the notation $\pi \cap \sigma$ and $\rho \cap \sigma$ as primitive.

A.31 Proposition XI.17

Heath’s statement is:

If two straight lines be cut by parallel planes, they shall be cut in the same ratio.

Let three parallel planes π_0, π_1, π_2 cut two transversals in the orders

$$A - E - B, \quad C - F - D.$$

The conclusion is

$$AE : EB = CF : FD.$$

Euclid introduces an auxiliary point O on AD in the middle plane. Proposition XI.16 gives the two parallel-section relations

$$EO \parallel BD, \quad AC \parallel OF.$$

The remaining argument is planar proportion theory. Proposition VI.2 applied in the two relevant triangles gives

$$AE : EB = AO : OD, \quad AO : OD = CF : FD,$$

and V.11 composes the two equalities.

The classical dependency pattern is therefore

$$\begin{array}{c} \text{three parallel planes and two transversals} \\ \Downarrow \\ \text{XI.16 twice} \\ \Downarrow \\ \text{VI.2 twice} \\ \Downarrow \\ \text{V.11} \\ \Downarrow \\ AE : EB = CF : FD. \end{array}$$

The formal reconstruction must additionally prove the existence and location of the auxiliary point O . Its canonical route then reproduces the two plane-local VI.2 steps without introducing numerical lengths or coordinates. A separate Wyler-oriented affine reconstruction is discussed in the Wyler appendix.

A.32 Proposition XI.18

Heath states:

If a straight line is at right angles to any plane, then all the planes through it are also at right angles to the same plane.

Let $l \perp \pi$ and let ρ be a plane containing l . Let

$$s = \rho \cap \pi.$$

Euclid chooses an arbitrary point $F \in s$ and, in ρ , draws a line g perpendicular to s at F .

The original normal l is also perpendicular to s , since $s \subset \pi$. Hence, inside ρ , the lines l and g are both perpendicular to the same line. They are therefore parallel. Proposition XI.8 transfers the line-plane perpendicularity:

$$l \perp \pi, \quad g \parallel l \implies g \perp \pi.$$

Since F and g were arbitrary in the configuration required by XI.Def.4, the plane ρ is perpendicular to π .

The proof is therefore a combination of:

exact common section + planar perpendicular comparison + XI.8.

The formal reconstruction retains Euclid's XI.Def.4 content rather than replacing it by the modern shortcut "a plane is perpendicular to another if it contains one normal to it."

A.33 Proposition XI.19

Heath gives:

If two planes cutting one another be at right angles to any plane, then their common section will also be at right angles to the same plane.

Let

$$\rho \perp \pi, \quad \sigma \perp \pi,$$

and suppose ρ and σ cut one another. Write

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi, \quad t = \rho \cap \sigma.$$

The classical diagram uses the point

$$D = r \cap s.$$

At D , XI.Def.4 supplies in each perpendicular plane a line perpendicular to the corresponding trace and hence perpendicular to π . The two normals to π at the same point must coincide. The common normal lies in both ρ and σ , so it is the common section t . Therefore

$$t \perp \pi.$$

The formal reconstruction exposes the principal hidden incidence obligation: the existence of the triple point D cannot simply be read from the diagram. It must be proved that the traces r and s meet. Once that point exists, the remainder of the classical proof is local.

A.34 Proposition XI.20

Euclid XI.20 states:

If a solid angle is contained by three plane angles, any two of them are together greater than the remaining one.

Let the solid angle have vertex O and edges OA, OB, OC . Its three face angles are

$$\angle AOB, \quad \angle BOC, \quad \angle COA.$$

The proposition asserts the three cyclic strict inequalities, for example

$$\angle AOB + \angle AOC > \angle BOC.$$

The nontrivial classical case is constructive. Inside the target angle choose a ray carrying a copy of one summand. Let it meet the opposite segment and lay off an equal radial segment on the third edge. SAS gives equality of the corresponding auxiliary base segments. Proposition I.20 then gives the strict inequality between the two remaining side segments, and Proposition I.25 converts that side inequality into a strict angle inequality. Adding back the copied component gives the desired comparison.

The proof spine is

copy one summand inside the target angle
 $\Downarrow$
 lay off an equal radial segment
 $\Downarrow$
 SAS
 $\Downarrow$
 I.20
 $\Downarrow$
 I.25
 $\Downarrow$
 angle decomposition
 $\Downarrow$
 XI.20.

The modern formal statement must make explicit that the three rays form a genuinely spatial configuration. In the current reconstruction this is recorded by noncoplanarity of O, A, B, C . No numerical angle measure is needed: the comparison is represented by synthetic angle order and decomposition.

A.35 Checkpoint XI.20

Through XI.20 the source audit reveals three distinct regions in Book XI.

First, XI.4–XI.13 build the line–plane perpendicularity and spatial parallelism machinery. Second, XI.14–XI.19 organize relations among planes, their sections, normals, and perpendicularity. Third, XI.20 begins the systematic study of a genuinely spatial vertex configuration through the three plane angles of a solid angle.

The formal reconstruction remains synthetic throughout this checkpoint. Coordinates, vectors, scalar products, trigonometric functions, and real-valued angle measure are not used to replace Euclid's arguments.

A.36 Proposition XI.21

Euclid XI.21 states that any solid angle is contained by plane angles whose sum is less than four right angles.

The proof actually treats the trihedral case. Let A be the vertex and let the three edges be AB, AC, AD . The three face angles are

$$\angle BAC, \quad \angle CAD, \quad \angle DAB.$$

The conclusion proved is

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

Euclid joins BC, CD, DB and applies XI.20 at the three base vertices:

$$\begin{aligned}\angle CBA + \angle ABD &> \angle CBD, \\ \angle BCA + \angle ACD &> \angle BCD, \\ \angle CDA + \angle ADB &> \angle CDB.\end{aligned}$$

Hence the six lateral angles in the three triangles ABC, ACD, ADB are together greater than the three angles of triangle BCD .

By I.32 the angles of BCD sum to two right angles. Therefore those six lateral angles are together greater than $2R$. Again by I.32, the nine angles of the three triangles ABC, ACD, ADB are together equal to $6R$. The remaining three angles at A are consequently together less than $4R$.

The classical dependency spine is

$$\begin{aligned}&\text{XI.20 at } B, C, D \\&\quad \Downarrow \\&\text{six lateral angles} > \text{angles of } BCD \\&\quad \Downarrow \\&\text{I.32 on } BCD : 2R \\&\quad \Downarrow \\&\text{I.32 on } ABC, ACD, ADB : 6R \\&\quad \Downarrow \\&\angle BAC + \angle CAD + \angle DAB < 4R.\end{aligned}$$

The formal reconstruction proves exactly this trihedral case. It does not formalize the broader polygonal solid-angle generalization implicit in the wording of the proposition.

The Lean proof deliberately does not reproduce Euclid's global angle-sum bookkeeping. Instead it extends two rays through the vertex and converts the $< 4R$ target into one XI.20 comparison between two supplementary angles and the third face angle. Thus I.32 is a classical dependency of Euclid's printed proof, but not a dependency of the production Lean proof.

A.37 Checkpoint XI.21

Through XI.21 the source audit now separates two successive roles of the trihedral theory.

XI.20 is a local comparison theorem: any two face angles of a proper trihedral configuration are together greater than the third. XI.21 uses that comparison classically to obtain a global bound by four right angles.

The formal reconstruction preserves the same mathematical bound but exposes a shorter synthetic route through opposite rays and complements. This is an important source/formalization distinction: the formal theorem is not a numerical reformulation of Euclid's angle arithmetic, and no real-valued angle measure is introduced.

A.38 Proposition XI.22

Euclid XI.22 starts with three plane angles such that any two together are greater than the third. The two straight lines containing each angle are taken equal, and the endpoints of each

equal pair are joined. The proposition asserts that the three joining segments satisfy the triangle inequalities and therefore can be used as the three sides of a triangle.

Let the three angles be

$$\angle AOB, \quad \angle CPD, \quad \angle EQF,$$

with

$$OA = OB = PC = PD = QE = QF.$$

Assume

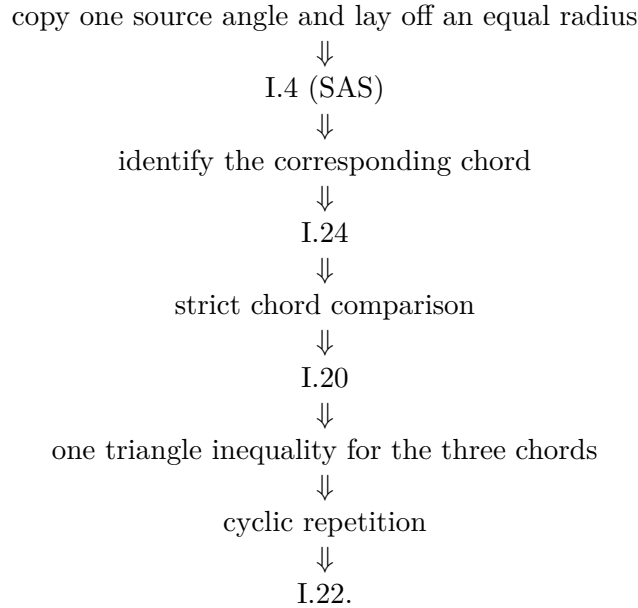
$$\begin{aligned} \angle AOB + \angle CPD &> \angle EQF, \\ \angle CPD + \angle EQF &> \angle AOB, \\ \angle EQF + \angle AOB &> \angle CPD. \end{aligned}$$

Joining the corresponding endpoints gives the three chords

$$AB, \quad CD, \quad EF.$$

Euclid proves one of the required side inequalities and obtains the other two cyclically. In the nontrivial case he copies one of the source angles inside the comparison configuration and lays off an equal radial segment. SAS gives equality of the corresponding auxiliary chord with one of the original chords. Proposition I.24 converts the strict included-angle comparison into a strict comparison of the opposite chords. Proposition I.20 then supplies the triangle inequality needed to obtain the corresponding chord-sum inequality.

The classical proof spine is therefore



The present Lean reconstruction isolates the angle-to-chord step in the dimension-free theorem

```
hilbert_twoAnglesGreaterThanAngle_equalRadius_chordSumGreater
```

which converts

$$\angle AOB + \angle CPD > \angle EQF$$

into

$$AB + CD > EF.$$

The angle-sum hypothesis is represented by `HilbertTwoAnglesGreaterThanAngle`; no numerical angle measure and no general addition operation on angles are introduced.

Applying the neutral chord theorem cyclically gives

$$AB + CD > EF, \quad CD + EF > AB, \quad EF + AB > CD.$$

The production proof then calls the already formalized Proposition I.22.

XI.22 is therefore entirely planar in the current architecture. Neither the direct theorem nor the Hilbert–Wyler theorem requires `HilbertSpacePrimitive`, `HilbertSpaceIncidence`, or `HilbertWylerAxioms`. The two public entry points

```
Geometry.euclid_proposition_11_22
Geometry.euclid_proposition_11_22_wyler
```

have identical axiom audits:

```
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

The occurrence of `hilbert_circle_circle_intersection` belongs to the final I.22 construction, not to the angle-to-chord comparison. Thus XI.22 exposes a clean logical boundary: all comparison geometry is neutral and dimension-free, while the final existence of a triangle with the three prescribed side lengths uses the circle-continuity machinery already present in Proposition I.22.

A.39 Checkpoint XI.22

Through XI.22 the source audit reveals a change of role in Book XI.

XI.20 and XI.21 are statements about a genuinely spatial trihedral configuration, even though their final metric content can be transported dimension-freely once the relevant incidence obligation is available. XI.22 is different: it contains no spatial obligation at all. Its entire formal content lies in the common planar Hilbert metric layer.

This makes XI.22 a useful architectural checkpoint. Its placement in the solid-geometry book reflects its use in the following construction rather than a need for three-dimensional hypotheses in the proposition itself.

A.40 Proposition XI.23

Proposition XI.23 proves that the two restrictions obtained in XI.20 and XI.21 are not only necessary but sufficient for a trihedral solid angle.

The three given plane angles may be denoted

$$\angle ABC, \quad \angle DEF, \quad \angle GHK.$$

The hypotheses are that every two, taken together, are greater than the remaining one and that the three together are less than four right angles. The problem is to construct a solid angle contained by three plane angles respectively equal to the given ones.

Euclid first cuts

$$AB = BC = DE = EF = GH = HK$$

and joins AC, DF, GK . Proposition XI.22 then gives a triangle LMN with

$$LM = AC, \quad MN = DF, \quad NL = GK.$$

A circle is described about LMN , its center is denoted by O , and LO, MO, NO are joined.

The decisive claim is

$$AB > LO.$$

Euclid excludes equality first. If $AB = LO$, then SSS gives

$$\angle ABC = \angle LOM, \quad \angle DEF = \angle MON, \quad \angle GHK = \angle NOL.$$

The three central angles make four right angles, contradicting the XI.21 hypothesis.

He next excludes $AB < LO$. Points P, Q are chosen on OL, OM so that

$$OP = AB, \quad OQ = BC.$$

Since $AB = BC$, one has $OP = OQ$, hence the remainders LP, QM are equal. The classical route then uses

$$\text{VI.2, I.29, VI.4, V.16}$$

to compare LM with PQ . Proposition I.25 converts the strict chord comparison into a strict angle comparison. Repeating cyclically would make the three given angles greater than the three central angles, although the former are assumed less than four right angles and the latter are exactly four right angles. This contradiction proves $AB > LO$.

The spatial stage now begins. By XI.12 Euclid erects OR perpendicular to the plane of the circumcircle and chooses it so that

$$OR^2 = AB^2 - LO^2.$$

The appended lemma constructs such a length by placing LO as a chord in a semicircle on AB and applying III.31 and I.47.

Since OR is perpendicular to the plane of the circumcircle, it is perpendicular to LO, MO, NO . Euclid obtains

$$RL = RM = RN = AB.$$

Finally three applications of I.8 give

$$\angle LRM = \angle ABC, \quad \angle MRN = \angle DEF, \quad \angle NRL = \angle GHK.$$

Thus the solid angle at R realizes the three prescribed plane angles.

The local source DAG is therefore

$$\begin{array}{c}
 \text{XI.20} + \text{XI.21} \\
 \Downarrow \\
 \text{XI.22} \\
 \Downarrow \\
 \text{IV.5} + \text{III.1} \\
 \Downarrow \\
 AB > LO \\
 \Downarrow \\
 \text{XI.12} + \text{lemma using III.31 and I.47} \\
 \Downarrow \\
 RL = RM = RN = AB \\
 \Downarrow \\
 \text{I.8 three times} \\
 \Downarrow \\
 \text{XI.23.}
 \end{array}$$

A.41 Comparison with the Lean reconstruction at XI.23

The production Lean proof retains the same large-scale construction:

chord triangle $\longrightarrow$ circumcenter $\longrightarrow$ radius comparison $\longrightarrow$ normal apex $\longrightarrow$ SSS recovery of the three face angles

The main difference lies in the radius comparison. The formal proof does not reproduce the VI.2–VI.4–V.16 proportional calculation. Instead it uses the synthetic angle-order infrastructure already developed for XI.20–XI.22, together with an explicit classification of the possible cyclic orders of the three radii from the circumcenter.

This explicit case control matters. Modern commentary on XI.23 notes that the printed argument for the radius comparison is drawn in the case where the circumcenter lies inside the base triangle; the outside and boundary positions require separate treatment. The formal theorem does not infer the missing cases from the picture: its ray-order analysis closes them inside the proof.

The formal proof also makes explicit a plane-normalization step absent from Euclid’s prose. The three source angles are first copied into one specified plane, the XI.20 and XI.21 certificates are transported to those copies, and XI.22 is then run inside the induced planar geometry. This supplies a certified base plane and a circumcenter lying in that plane before XI.12 is used.

At kernel level XI.23 has the audited dependency

```
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

The circle–circle intersection axiom enters through the established planar circle-construction infrastructure. Thus the comparison stage itself remains synthetic angle/order geometry, but the complete construction is not continuity-free.

A.42 Checkpoint XI.23

XI.23 completes the opening trihedral sequence of Book XI:

- XI.20 : pairwise necessary inequalities
- XI.21 : total bound below $4R$
- XI.22 : chord triangle exists
- XI.23 : the three angles are realizable as a solid angle.

The source audit also exposes a sharp formal boundary. XI.20 and XI.21 can be kept in neutral angle-comparison language; XI.22 introduces the circle continuity inherited from I.22; XI.23 combines that planar existence machinery with a genuinely spatial perpendicular-to-plane construction.

For the current project, XI.23 is therefore both the converse synthesis of XI.20–XI.21 and the first point in this trihedral block where the complete formal theorem necessarily spans neutral angle geometry, circle construction, and spatial Euclidean infrastructure.

A.43 Propositions XI.24–XI.25: from parallelepiped faces to Eudoxian proportion

A.43.1 Proposition XI.24

Proposition XI.24 begins with a solid contained by three pairs of parallel planes and proves that each pair of opposite faces is both parallelogrammic and equal.

The source proof has two clearly separated parts. First, Proposition XI.16 is applied to the intersections of the parallel planes with each face plane. This proves the required pairs of opposite edges parallel and hence shows that all six faces are parallelograms. Second, Euclid compares opposite faces metrically. Opposite sides of the relevant parallelograms are equal by I.34, the included spatial angles are equal by XI.10, and I.4 gives congruent diagonal triangles. Since each parallelogram is double the corresponding diagonal triangle, the opposite faces are equal.

Thus XI.24 is not a volume theorem. It produces exactly the face data later used in XI.Def.10: corresponding triples of faces of suitable parallelepipeds can be compared without assigning a numerical measure to the solids.

A.43.2 Proposition XI.25

Proposition XI.25 is the first proposition in this part of Book XI whose conclusion is explicitly a proportion between plane and solid magnitudes. If a parallelepiped is cut by a plane parallel to a pair of opposite faces, Euclid asserts

$$\text{base}_1 : \text{base}_2 = \text{solid}_1 : \text{solid}_2.$$

The proof is Eudoxian. Euclid prolongs one longitudinal direction on both sides of the given cut and lays off arbitrarily many equal intervals of the two cut widths. The associated parallelograms and parallelepipeds are then completed. Proposition XI.24 supplies equality of the corresponding opposite faces, and XI.Def.10 converts equality of the appropriate triples of faces into equality of the repeated elementary solids.

Consequently equal multiples of the two base magnitudes and equal multiples of the two solid magnitudes are synchronized. Euclid then compares an arbitrary multiple on the first side with an arbitrary multiple on the second side: equality, excess, and deficiency of the bases are accompanied by equality, excess, and deficiency of the corresponding solids. This is exactly the comparison scheme required by V.Def.5, and the desired proportion follows.

One point is logically compressed in the classical text. After the repeated equal slabs have been constructed, the implications

$$B_m > B_n \implies S_m > S_n, \quad B_m < B_n \implies S_m < S_n$$

are used without a separate geometric development of the strict comparison. This omission is harmless on the classical page, where “the whole is greater than the part” is read directly from the construction, but it becomes a genuine obligation in a formal proof.

The production Lean reconstruction makes precisely this step explicit. Proper-part witnesses are built for bases and solids, their strict orders are proved asymmetric, and the resulting comparison certificates are fed into the common Eudoxus magnitude interface. The formal endpoint therefore preserves Euclid’s V.Def.5 architecture rather than replacing XI.25 by a numerical volume calculation.

A.44 Checkpoint through XI.25

XI.24 and XI.25 form a compact transition in the classical source:

$$\begin{array}{c}
 \text{XI.16 + I.34 + XI.10 + I.4} \\
 \Downarrow \\
 \text{XI.24 : opposite faces are equal parallelograms} \\
 \Downarrow \\
 \text{XI.Def.10 : equality of elementary parallelepipeds} \\
 \Downarrow \\
 \text{arbitrary repeated slabs} \\
 \Downarrow \\
 \text{V.Def.5} \\
 \Downarrow \\
 \text{XI.25.}
 \end{array}$$

The important source-level point is that XI.25 still belongs to Euclid’s theory of magnitudes. No numerical volume is introduced. The ratio of the two solids is established by the behavior of all positive integral multiples, exactly as required by Book V.

Appendix B

Hilbert: Spatial Axiomatic Architecture

B.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book XI at the current checkpoint, Proposition XI.25. The purpose is architectural: identify which layers of incidence, order, congruence, and Euclidean parallelism enter at each theorem, and distinguish them from continuity assumptions.

B.2 Spatial architecture after the Hilbert refactor

The project now distinguishes a source-faithful Hilbert foundation from the compatibility interface still consumed by the existing Book XI theorem signatures.

At source level, `HilbertAxioms.lean` contains

```
HilbertGrundlagenPrimitive
HilbertGrundlagenGroupI
HilbertGrundlagenGroupII
HilbertGrundlagenGroupIII
HilbertGrundlagenGroupIV
HilbertGrundlagenGroupV
```

in Hilbert’s source order. The spatial incidence clauses I.4–I.8 therefore belong to the same Group I foundation as the planar incidence clauses. There is no separate source-level `Hilbert3DAxioms.lean` foundation.

The module `HilbertSpaceIncidence.lean` now has two roles. It proves spatial incidence consequences directly from `HilbertGrundlagenPrimitive` and `HilbertGrundlagenGroupI`, and it also preserves the old-shaped downstream records

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
```



```
HilbertSpaceEuclidean
```

as a migration facade. The current Proposition XI.1–XI.17 production theorems still use these compatibility names in their signatures. Their presence must therefore not be read as evidence for a second Hilbert foundation.

Above that boundary, `Hilbert3DInterface.lean` contains derived spatial definitions, induced-plane constructions, and plane/ambient bridge theorems. Ordinary planar Hilbert geometry on each fixed plane is recovered through

```
PlanePoint
PlaneLine
PlaneGeo
```

with explicit bridges for incidence, betweenness, congruence, same rays, angles, right angles, perpendicularity, and parallelism.

This separation is essential at XI.4. There is deliberately no global `HilbertCongruence Geo` or `HilbertOrder Geo` instance installed on the ambient three-space. Planar order arguments are carried out inside a specified `PlaneGeo`; spatial congruence transports only the data needed between such plane slices.

B.3 Neutral status of XI.4 in the current hierarchy

The public theorem assumes the spatial incidence/order/congruence hierarchy but no `HilbertEuclideanPlane` instance. In particular, Hilbert Group IV – the Euclidean parallel axiom – does not occur in the proof of XI.4.

The final axiom audit reported by Lean is

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this checkpoint. XI.4 introduces no proposition-specific geometric axiom.

B.4 What XI.4 adds to the interface

The theorem forced one useful correction to the general 3D API: `HilbertLinesPerpendicularAt` now carries an explicit noncollinearity witness. Consequently distinctness of perpendicular carriers is derived by `hilbert_linesPerpendicularAt_ne` rather than supplied as a separate hypothesis.

The universal target is represented by

```
HilbertLinePerpendicularPlaneAt
```

which quantifies over every incidence line of the plane through the foot. This is a direct synthetic representation and introduces no metric structure.

B.5 Checkpoint XI.4

At XI.4 Hilbert’s role is therefore not a one-to-one proposition-number match. It is the axiomatic architecture in which Euclid’s spatial proof can be made explicit. The most significant result of the audit is the boundary: XI.4 is completed in the current incidence/order/congruence 3D hierarchy without Group IV and without a new continuity assumption.

B.6 Incidence axiom I.6 and Proposition XI.7

Proposition XI.7 isolates a much smaller part of the spatial hierarchy than XI.4–XI.6. The decisive Hilbert incidence principle is source-level Hilbert I.6, represented in `HilbertAxioms.lean` by

```
HilbertGrundlagenGroupI.I6_line_in_plane
```

and exposed to the current Book XI compatibility API by

```
HilbertSpaceIncidence.line_in_plane
```

Its geometric content is: if two distinct points of a line lie in a plane, then the whole line lies in that plane.

The public XI.7 theorem assumes only

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
```

It does not assume `HilbertSpaceOrder`, `HilbertSpaceCongruence`, or an induced `PlaneGeo` instance. The Euclidean parallel axiom, Group IV, and continuity are therefore absent from the proof architecture.

B.7 Spatial parallelism as reusable incidence data

The relation

```
HilbertSpaceLinesParallel
```

is now part of `Hilbert3DInterface`. It packages two assertions: a common carrier plane and disjointness of the two ambient lines. The common plane is essential because ambient disjointness alone would also admit skew lines.

This notion was introduced during the XI.6 development and promoted to the general 3D interface before XI.7. Proposition XI.7 is the first subsequent Book XI theorem that consumes the common-plane witness directly.

B.8 Axiomatic status of XI.7

The final audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This result concerns global Lean axiom constants. It does not remove the explicit Hilbert incidence structures from the theorem signature: those structures remain hypotheses of the theorem. What the audit establishes is that no additional globally declared axiom constant occurs in the proof term.

B.9 Checkpoint XI.7

At XI.7 the spatial reconstruction reaches a pure incidence theorem. The formal proof needs neither Group IV nor continuity, and it introduces no new proposition-specific geometric or spatial axiom. The decisive spatial content is exactly the combination of a common-plane witness for parallel lines and Hilbert's line-in-plane incidence principle.

B.10 Group IV becomes active at Proposition XI.8

Proposition XI.8 is the first theorem in the present Book XI sequence whose production signature assumes

```
[HSE : HilbertSpaceEuclidean Geo]
```

At source level, Group IV is part of `HilbertGrundlagenGroupIV` in `HilbertAxioms.lean`. The class `HilbertSpaceEuclidean` is the retained Book XI compatibility representation of that Euclidean layer. Its parallel-uniqueness field is explicitly restricted to one ambient plane: through a point of that plane outside a given line, two lines in the same plane that are disjoint from the given line must coincide.

This formulation is essential in three-space. The project does not install a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient spatial geometry. Instead, every fixed plane inherits the planar Euclidean interface through

```
planeGeo_linesDisjoint_iff_ambient
planeGeoHilbertEuclidean
```

The first theorem identifies planar disjointness in a fixed `PlaneGeo` with ambient disjointness of the underlying spatial lines. The second packages the plane-local Group IV field as the existing planar `HilbertEuclideanPlane` instance.

B.11 Neutral I.27 and Euclidean uniqueness in XI.8

The formal replacement for Euclid’s I.29 step exposes the logical boundary between neutral and Euclidean geometry.

Inside the common plane σ of the parallel lines, a line p is constructed through D perpendicular to the transversal $d = BD$. The established I.27 alternate-angle theorem shows that p is disjoint from the first parallel l . This part uses order and congruence but not Group IV.

The given second parallel m is also disjoint from l . Both p and m pass through D and lie in σ . Only now is Group IV used: uniqueness forces $p = m$. Therefore $m \perp d$.

The formal proof consequently separates

existence/nonintersection from Euclidean uniqueness.

B.12 Axiomatic status of XI.8

The final audit command is

```
import CGJteamLab.Proposition11_8

#print axioms Geometry.euclid_proposition_11_8
```

and Lean reports

```
'Geometry.euclid_proposition_11_8' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited global Lean/project dependencies. No new proposition-specific global axiom constant is introduced.

This audit output must be distinguished from the theorem’s explicit geometric hypotheses. The typeclass `HilbertSpaceEuclidean Geo` is a genuine mathematical assumption of XI.8: it represents Group IV. It is passed as a theorem parameter, rather than appearing as an additional global Lean axiom constant.

No continuity assumption is used.

B.13 Checkpoint XI.8

At XI.8 the spatial hierarchy first crosses from the neutral incidence/order/congruence layers into the Euclidean parallel layer. The transition is plane-local: Group IV is applied only inside a specified ambient plane and is inherited by its induced **PlaneGeo** geometry.

The proof remains synthetic. Its Euclidean content is exactly the uniqueness step required to turn neutral nonintersection into the I.29-type perpendicularity conclusion.

B.14 Proposition XI.9 in the current Hilbert architecture

The production XI.9 theorem assumes

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

although the mathematical proof uses the order/congruence layers only through inherited spatial infrastructure. The decisive Euclidean call is

```
HilbertSpaceEuclidean.parallel_unique_in_plane
```

in one explicitly chosen carrier plane.

The formal proof does not reconstruct Euclid's perpendicularity chain. It uses two proposition-local incidence helpers,

```
hilbert_XI9_point_on_second_line_off_first
hilbert_XI9_planes_eq_of_common_line_and_external_point
```

together with the general spatial theorems

```
hilbert_plane_through_line_and_external_point
hilbert_plane_intersection_line
```

to create a common carrier plane for the two target lines.

No global ambient **HilbertOrder Geo** or **HilbertCongruence Geo** instance is installed. In fact, the public proof does not need to enter an induced **PlaneGeo**; Group IV is already available in its ambient plane-indexed form.

B.15 Axiomatic status of XI.9

The audit reports

```
'Geometry.euclid_proposition_11_9' depends on axioms:
[proptext, Classical.choice, Quot.sound]
```

The theorem therefore introduces no proposition-specific global axiom and does not inherit `Geometry.bookZero_nullSegment1`.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

remains a genuine geometric hypothesis. As elsewhere, an explicit theorem parameter is not listed as a hidden global constant by `#print axioms`. No continuity assumption is used.

B.16 Checkpoint XI.9

At XI.9 the current hierarchy isolates a particularly clean decomposition:

$$\begin{array}{c} \text{spatial incidence} \\ + \text{one plane-local Group IV uniqueness step} \\ \Downarrow \\ \text{spatial transitivity of parallelism} \end{array}$$

The theorem adds no new reusable axiom class and no analytic representation.

B.17 Proposition XI.10 in the current Hilbert architecture

The production XI.10 theorem assumes the full spatial hierarchy

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

and concludes an ambient angle-congruence statement.

XI.10 introduces a new proposition-local orientation relation:

```
HilbertSpaceDirectedParallelSegments
```

This relation refines ordinary

```
HilbertSpaceLinesParallel
```

by adding a connector line through the terminal endpoints and a same-side condition for the initial endpoints in the common carrier plane. It is not a new axiom class; it is a definition built from incidence, disjointness, and plane-local order.

The proof also defines proposition-local plane-separation language

```
HilbertSegmentMeetsPlane
HilbertSameSideOfPlane
```

and proves the section bridge

```
hilbert_XI10_borsuk_szmielew_52
```

from the existing spatial incidence/order infrastructure. Thus no primitive half-space axiom is added to the Hilbert hierarchy.

The normalized proof spine is

```
BA ~= ED, BC ~= EF
directed parallels
|
I.33 twice
|
AD ~= BE, CF ~= BE
|
XI.9
|
AD || CF
|
Borsuk-Szmielew 52
|
AD directed-parallel CF
|
I.33
|
AC ~= DF
|
spatial SSS
|
angle ABC ~= angle DEF
```

The I.33 steps and XI.9 consume the Euclidean parallel layer. The final SSS step is synthetic congruence. The Borsuk–Szmielew 52 bridge consumes spatial order but no continuity.

The public theorem removes the normalized equal-length assumptions by spatial segment construction. The constructed points D_1, F_1 lie on the original rays ED, EF . After the normalized angle theorem is applied, the final SameRay angle transport is performed inside the explicit induced geometry $\text{PlaneGeo}(\text{plane}(D, E, F))$.

This last point is architecturally important. XI.10 does not install ambient

```
[HilbertOrder Geo]
```

merely to reuse planar ray lemmas. The order-theoretic angle transport stays inside a genuine plane slice and is transferred back by the existing PlaneGeo/ambient bridge.

B.18 Axiomatic status of XI.10

The validated audit is

```
'Geometry.euclid_proposition_11_10' depends on axioms:
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited global Lean/project dependencies. XI.10 introduces no proposition-specific global axiom constant, no new spatial axiom class, and no continuity principle.

The appearance of `Geometry.bookZero_nullSegment1` distinguishes the public XI.10 wrapper from XI.9 at the level of global audit output. It is inherited through existing construction/congruence infrastructure rather than introduced by the new directed-parallel or plane-side definitions.

B.19 Checkpoint XI.10

At XI.10 the Hilbert architecture acquires no new primitive layer, but the formal proof exposes a new interface problem: orientation data must move between several plane slices and ambient three-space.

The solution is entirely synthetic:

plane-local same side $\longleftrightarrow$ ambient same side of a plane $\longrightarrow$ directed spatial parallelism.

This is the principal new structural lesson of XI.10. Group IV is reused, spatial order becomes genuinely active, and no continuity or analytic geometry is introduced.

B.20 Proposition XI.11 in the current Hilbert architecture

The production theorem XI.11 assumes the same full spatial hierarchy as XI.8:

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```


The proof does not install ambient `HilbertOrder Geo` or ambient `HilbertCongruence Geo`. Instead, Book I perpendicular and parallel constructions are applied inside explicit induced planes $\text{PlaneGeo}(\sigma)$, $\text{PlaneGeo}(\pi)$, and $\text{PlaneGeo}(\tau)$.

The Group IV hypothesis enters through the call to

```
euclid_proposition_11_8
```

which transfers perpendicularity from BC to the constructed parallel GH . No independent Group IV argument is added by XI.11 itself.

The two calls to

```
euclid_proposition_11_4
```

remain in the incidence/order/congruence spatial layer. The first proves $BC \perp \tau$; the second proves the final $AF \perp \pi$.

B.21 Axiomatic status of XI.11

The final audit command is

```
import CGJteamLab.Proposition11_11
#print axioms Geometry.euclid_proposition_11_11
```

and Lean reports

```
'Geometry.euclid_proposition_11_11' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited global Lean/project dependencies. XI.11 introduces no proposition-specific global axiom constant, no new spatial axiom declaration, and no continuity assumption.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

is nevertheless a genuine hypothesis of the current theorem because XI.8 is used in the proof. As at XI.8, this theorem parameter does not appear as an extra global constant in the output of `#print axioms`.

B.22 Checkpoint XI.11

At XI.11 the existing plane-local Group IV layer is consumed but not extended. The proposition introduces no new general 3D interface theorem. Its main architectural role is to coordinate three induced plane slices and to return a spatial line–plane perpendicularity statement.

The current formal proof is therefore Euclid-faithful with respect to its parallel-theoretic step: it inherits the Euclidean strength of XI.8. The audit does not claim that this hypothesis is minimal for the abstract existence statement.

B.23 Proposition XI.12 in the current Hilbert architecture

The public XI.12 theorem assumes the same full spatial hierarchy as XI.11 and XI.8:

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceEuclidean Geo]
```

The proof again avoids global ambient `HilbertOrder Geo` and `HilbertCongruence Geo`. The only new planar construction is the parallel through the prescribed point. It is carried out in an explicit auxiliary plane σ via `PlaneGeo( $\sigma$ )`.

XI.12 introduces two proposition-local theorems.

The first,

```
hilbert_XI12_perpendicular_foot_unique
```

is an incidence consequence of the universal definition of line–plane perpendicularity. If two distinct points of the perpendicular line lay in the plane, Hilbert spatial incidence would force the whole line into the plane; the universal perpendicularity clause would then make the line perpendicular to itself.

The second,

```
hilbert_XI12_parallel_through_point_in_plane
```

performs the I.31 construction inside a fixed induced plane and packages the resulting planar disjointness as `HilbertSpaceLinesParallel`. Its signature does not assume `HilbertSpaceEuclidean`; the helper itself remains below Group IV.

The public theorem nevertheless assumes Group IV because it consumes XI.11 and XI.8. No additional Euclidean uniqueness argument is introduced by the two new XI.12 helpers.

B.24 Axiomatic status of XI.12

The public audit command is

```
import CGJteamLab.Proposition11_12

#print axioms Geometry.euclid_proposition_11_12
```

and Lean reports

```
'Geometry.euclid_proposition_11_12' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

The two local helpers were audited separately:

```
'Geometry.hilbert_XI12_perpendicular_foot_unique'
depends on axioms: [propext, Classical.choice, Quot.sound]

'Geometry.hilbert_XI12_parallel_through_point_in_plane'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Thus the XI.12-specific incidence and plane-slice packaging steps add no new global geometric axiom constant and do not themselves depend on `bookZero_nullSegment1`. The public theorem inherits that existing project dependency through the established earlier construction chain.

The explicit parameter

```
[HSE : HilbertSpaceEuclidean Geo]
```

remains a genuine geometric hypothesis of the current theorem. It enters through XI.11 and XI.8, not through a new XI.12-specific Group IV step.

No new continuity assumption, spatial axiom class, or `sorry/admit` is introduced.

B.25 Checkpoint XI.12

XI.12 adds no new primitive 3D structure and no new plane-space bridge declaration. Its two local helpers expose reusable consequences of the existing architecture: uniqueness of the perpendicular foot and the packaging of a planar parallel construction as spatial parallelism.

The current public proof is Euclid-faithful and therefore retains the Group IV strength inherited from XI.11 and XI.8. The axiom audit distinguishes that explicit theorem hypothesis from the global Lean/project axiom constants appearing in the proof term.

B.26 Proposition XI.13 in the current Hilbert architecture

The production XI.13 theorem assumes

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder (Geo := Geo) (H := H) (S := S)]
[HilbertSpaceCongruence (Geo := Geo) (H := H) (S := S)]
```

and, unlike XI.8, XI.11, and XI.12, does not assume

```
HilbertSpaceEuclidean Geo
```

Thus XI.13 returns to the neutral spatial incidence/order/congruence layer of the current hierarchy.

The proof needs no new induced-plane construction theorem and no new plane–space bridge. Its principal spatial ingredients are:

- the plane through two distinct intersecting lines;
- the line of intersection of two distinct planes;
- the universal clause of line–plane perpendicularity;
- symmetry of line–line perpendicularity;
- XI.4;
- the fact that a line perpendicular to a plane cannot itself lie in that plane.

The explicit proof that the auxiliary plane differs from the reference plane is an incidence consequence of the existing line–plane perpendicularity interface. No parallel uniqueness principle is needed.

B.27 Axiomatic status of XI.13

The audit command is

```
import CGJteamLab.Proposition11_13

#print axioms Geometry.euclid_proposition_11_13
```

and Lean reports

```
'Geometry.euclid_proposition_11_13' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
```

Quot.sound]

These are the accepted inherited global Lean/project dependencies. XI.13 introduces no proposition-specific global axiom constant.

The absence of `HilbertSpaceEuclidean` from the theorem signature is a genuine geometric distinction from XI.8, XI.11, and XI.12. Group IV is not used in the current XI.13 proof. No continuity assumption, new spatial axiom declaration, or `sorry/admit` is introduced.

B.28 Checkpoint XI.13

At XI.13 the current reconstruction isolates uniqueness of the normal at a point without crossing the Euclidean parallel boundary. The proof uses the existing spatial incidence/order/congruence hierarchy and reuses XI.4 as the synthetic closure theorem.

No global ambient `HilbertOrder Geo` or `HilbertCongruence Geo` instance is installed, and no new `Hilbert3DInterface` declaration is required.

B.29 Proposition XI.14: common normals and plane parallelism

XI.14 returns to the neutral spatial layer. Its mathematical content is: if one line is perpendicular to two distinct planes, then the planes are parallel.

The direct proof has two branches.

In the generic branch the two perpendicular feet are distinct. If the planes had a common point K , the line together with K would determine an auxiliary plane containing a nondegenerate triangle whose two angles at the feet are right. Proposition I.17 rules this out.

The equal-foot branch is logically separate. If the same line is perpendicular to two planes at the same point, the two planes are proved equal by the spatial incidence/congruence infrastructure. This repairs the degenerate case hidden by the classical picture.

Thus XI.14 has the neutral architecture

$$\text{spatial incidence} + \text{line-plane perpendicularity} + \text{planar I.17} \implies \text{plane disjointness.}$$

No Group IV uniqueness of parallels is required by the theorem itself, and no continuity principle is introduced.

B.30 Proposition XI.15: the Euclidean plane-parallel closure theorem

XI.15 is the principal theorem in this range that genuinely consumes the Euclidean parallel layer.

The production proof starts with two intersecting lines in a plane π and two corresponding lines in a plane ρ , with the corresponding carriers spatially parallel. XI.11 constructs a normal from the first intersection point to ρ . Lines through its foot are chosen equal to or parallel to the two reference lines of ρ . Definition XI.3 gives the local perpendicularities.

The transport back to the original lines is the exact point at which Group IV enters. Inside the carrier plane of a parallel pair, the proof:

1. constructs a perpendicular to the transversal;
2. uses the neutral I.27 mechanism to prove nonintersection;
3. uses plane-local Group IV uniqueness to identify that constructed line with the given parallel.

After this transport has been performed twice, XI.4 makes the normal perpendicular to π , while it is already perpendicular to ρ . XI.14 then yields

$$\pi \parallel \rho.$$

The foundation boundary is therefore sharp:

neutral incidence/order/congruence + one plane-local Euclidean uniqueness mechanism.

No continuity assumption enters.

B.31 Proposition XI.16: an incidence theorem implemented through carriers

XI.16 states that a cutting plane intersects two parallel planes in parallel section lines.

Mathematically this is an incidence theorem. Once the exact section lines

$$l = \pi \cap \sigma, \quad m = \rho \cap \sigma$$

have been obtained, a common point of l and m would be a common point of the parallel planes π and ρ , which is impossible.

The current production implementation is noteworthy for architectural reasons. It is written through the carrier/intersection API developed in the Wyler-facing interface. Its public mathematical content, however, remains pure spatial incidence: no order, congruence, perpendicularity, or Group IV step is used in the core theorem.

This should not be read as a new source-level Hilbert axiom. The carrier language packages consequences of the existing incidence structure. XI.16 therefore marks a point where the Hilbert source audit and the Wyler organizational vocabulary meet at theorem level.

B.32 Proposition XI.17: spatial sections followed by planar proportion

The canonical XI.17 theorem combines spatial incidence/order with two plane-local applications of Euclid VI.2.

Three pairwise parallel planes cut two transversals in the orders

$$A - E - B, \quad C - F - D.$$

The formal proof first constructs Euclid’s auxiliary point O in the middle plane. XI.16 then supplies the two section-line parallelisms

$$EO \parallel BD, \quad AC \parallel OF.$$

Only after those genuinely spatial steps are complete does the proof enter two induced plane geometries. In those plane slices the raw VI.2 interface gives the two proportions, which are transported back to ambient space and composed by the spatial V.11-type transitivity theorem.

The important Hilbert-side lesson is that the theorem is not implemented by segment arithmetic or coordinates. The canonical route is synthetic, but it does assume the required VI.2 proposition as explicit plane-local infrastructure.

A separate Wyler affine endpoint removes this proportion layer; that comparison belongs to Appendix E.

B.33 Proposition XI.18: direct Hilbert 3D plane perpendicularity

XI.18 now has an independent direct Hilbert production proof in

```
CGJteamLab/Proposition11_18.lean
```

and the neutral definition of plane perpendicularity lives in

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

The direct theorem does not import the Wyler Book XI route.

Let $l \perp \pi$ and let ρ contain l . The proof:

1. identifies the common section $s = \rho \cap \pi$;
2. proves $l \perp s$;
3. takes an arbitrary line $g \subset \rho$ perpendicular to s ;
4. handles the same-foot case by planar uniqueness of a perpendicular;
5. in the distinct-foot case proves $g \parallel l$;
6. applies XI.8 to transport $l \perp \pi$ to $g \perp \pi$;
7. uses uniqueness of the perpendicular foot to identify the returned foot with the point prescribed by XI.Def.4.

Thus XI.18 mixes neutral 3D incidence and planar perpendicular comparison with the Euclidean strength already present in XI.8. The theorem remains fully synthetic.

B.34 Proposition XI.19: global incidence plus a neutral local core

XI.19 is naturally split into two logical layers.

Let

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi,$$

where both ρ and σ are perpendicular to π , and ρ, σ cut one another.

The first layer proves that the two traces r, s meet. If they did not, they would be parallel in π . XI.Def.4 supplies one normal to π inside each perpendicular plane. XI.6 makes those normals parallel. The two parallel pairs then satisfy XI.15, which would force

$$\rho \parallel \sigma,$$

contradicting the given common point of the cutting planes.

The second layer is neutral once the triple point

$$D \in \rho \cap \sigma \cap \pi$$

is known. XI.Def.4 supplies normals to π in both planes at D . XI.13 identifies them, and exact plane intersection identifies that common normal with the section $\rho \cap \sigma$.

Hence the Euclidean hypothesis enters only through the global existence stage, via XI.15. The common-point core itself is an incidence/congruence theorem.

B.35 Proposition XI.20: the spatial shell around a planar metric proof

XI.20 changes the character of the development. The object is now a proper trihedral configuration rather than a relation among whole planes.

The direct Hilbert theorem is

```
euclid_proposition_11_20
```

and uses the neutral vocabulary

```
HilbertTrihedralConfiguration
HilbertTrihedralAngleInequalities
```

The metric core has been factored into

```
HilbertTrihedralAngleMetric.lean
```

and consists of planar congruence, SAS, Euclid I.20, Euclid I.25, angle order, and same-ray transport.

Only one proof obligation is genuinely spatial: the auxiliary triangle used for I.20 must be nondegenerate. The direct Hilbert proof handles this by:

1. constructing the plane through three noncollinear points;
2. using line-in-plane closure to propagate an assumed collinearity;
3. deriving a contradiction with the noncoplanarity of the four points defining the trihedral configuration.

This is an unusually small spatial dependency. It explains why the independent Hilbert–Wyler proof can replace the direct 3D incidence calls by the common dimension-free point–line–plane core without changing the metric part of the theorem.

B.36 Proposition XI.21: complement geometry below Group IV

The direct Hilbert XI.21 theorem is

```
euclid_proposition_11_21
```

with the minimized ambient assumptions

```
[HilbertIncidence Geo]
[S : HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertCongruence Geo]
```

An earlier test-stage version still carried

```
[HilbertEuclideanPlane Geo]
```

because the first attempted proof followed Euclid’s I.32 angle-sum route. After the proof was replaced by the opposite-ray argument, this hypothesis was removed. The production theorem therefore does not use Hilbert Group IV through the planar Euclidean interface.

The neutral target lives in

```
HilbertThreeAnglesFourRight.lean
```

and expresses the four-right-angle bound by supplements rather than by a general numerical or algebraic sum of angles.

The direct proof extends AC and AD through the vertex A , obtaining

$$C - A - X, \quad D - A - Y.$$

It then replaces AC by the opposite ray AX , reorders the trihedral configuration as $A; D, X, B$, and applies XI.20:

$$\angle DAX + \angle XAB > \angle DAB.$$

Vertical angles and ordinary unoriented-angle symmetry give

$$\angle DAX \cong \angle CAY, \quad \angle XAB \cong \angle BAX,$$

hence

$$\angle CAY + \angle BAX > \angle DAB.$$

This is the complement certificate for

$$\angle CAD + \angle BAC + \angle DAB < 4R.$$

Only one proof obligation uses the ambient spatial interface. If A, B, X, D were coplanar, then the plane containing A and X would contain the whole line AX , hence also C , contradicting the original noncoplanarity of A, B, C, D . The only genuinely spatial operation needed in this step is line-in-plane closure.

Thus the direct Hilbert proof of XI.21 is strictly weaker than its original test signature suggested: it uses incidence and congruence, but no Euclidean parallel axiom, no I.32 call, no continuity principle, and no numerical angle measure.

B.37 Hilbert checkpoint through XI.21

The addition of XI.21 sharpens the classification already visible at XI.20.

1. **Neutral spatial incidence/congruence:** XI.13–XI.14 and the local cores of XI.18–XI.21.
2. **Plane-local Euclidean parallel transport:** XI.15 and the XI.8-dependent part of XI.18; XI.19 inherits the same strength through XI.15.
3. **Pure incidence once the objects are packaged:** XI.16.
4. **Spatial setup followed by planar or local metric theory:** XI.17, XI.20, and XI.21. XI.21 is especially economical because its new spatial step is only line-in-plane closure.

XI.20 and XI.21 now form a useful pair for dimensional transfer. Both are stated for a trihedral configuration, but neither final proof needs a principle whose content is intrinsically “ambient dimension equals three.” The Hilbert route remains a direct three-dimensional presentation, while the proof-term audit identifies exactly which pieces are capable of surviving in the dimension-free Wyler route.

B.38 Proposition XI.22: planar comparison plus circle continuity

The direct production theorem

```
euclid_proposition_11_22
```

has the minimized signature

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

and is therefore dimension-free at theorem-statement level.

Its proof separates into two layers. The new XI.22 comparison theorem turns the three pairwise angle inequalities into the three triangle inequalities for the corresponding equal-radius chords. This part uses only the neutral planar incidence/order/congruence machinery.

The final existence step calls Euclid I.22. Consequently the kernel audit contains

```
Geometry.hilbert_circle_circle_intersection
```

as well as the two Book Zero null-segment axioms. Thus XI.22 is a useful example of a theorem whose statement and comparison core are neutral, while the chosen ruler-and-compass existence proof uses continuity.

No spatial Hilbert class and no Group IV parallel argument is needed by the public XI.22 theorem itself.

B.39 Proposition XI.23: explicit plane slices and a genuine spatial apex

XI.23 changes the architecture again. The final theorem is

```
euclid_proposition_11_23
```

with explicit assumptions

```
[Hinc : HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[S : HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
[HilbertSpaceOrder
  (Geo := Geo) (H := Hinc) (S := S)]
[HilbertSpaceCongruence
  (Geo := Geo) (H := Hinc) (S := S)]
[HilbertSpaceEuclidean Geo]
```

The proof deliberately does not install a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient space. Instead it chooses an actual spatial plane π , copies the three prescribed angles into that plane, transports the XI.20-type and XI.21-type hypotheses to the copies, and then repackages the points as

```
PlanePoint Geo pi
```

so that the planar core can run in

```
PlaneGeo Geo pi
```

This is the exact architectural boundary between ambient space and plane geometry. The long circumradius argument is proved once in planar language and then reused in the selected plane.

The central planar theorem is

```
hilbert_XI23_circumradius_less_source_radius_XI
```

which proves that the circumradius of the XI.22 chord triangle is strictly less than the common source radius. The proof eliminates both the equality case and the opposite strict inequality. Its proposition-local ray-order analysis is what prevents the proof from depending on one representative position of the circumcenter.

The theorem

```
hilbert_XI23_radius_height_in_plane_XI
```

then constructs, still inside `PlaneGeo`, an auxiliary right triangle whose hypotenuse has the source radius and whose one leg has the circumradius. The remaining leg is the length needed for the spatial apex.

XI.12 erects the normal through the circumcenter. Spatial right-angle congruence and SAS yield three equal lateral edges. Spatial SSS then recovers the three prescribed face angles. Finally $R \notin \pi$, base noncollinearity, and plane uniqueness establish the proper trihedral configuration.

B.40 Axiomatic status of XI.23

The final audit command is

```
import CGJteamLab.Proposition11_23

#print axioms Geometry.euclid_proposition_11_23
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

This profile should be read carefully.

The two declarations

```
Geometry.bookZero_nullSegment1
Geometry.bookZero_nullSegment2
```

are the already tracked Book Zero interface axioms concerning null segments.

The declaration

```
Geometry.hilbert_circle_circle_intersection
```

is the project-level circle–circle continuity axiom. Its occurrence is genuine: XI.23 consumes the circle-construction infrastructure needed for XI.22 and for the planar circumcircle stage.

Therefore XI.23 is not continuity-free. This is the principal axiomatic difference from XI.20 and XI.21.

At the same time, no proposition-specific XI.23 axiom is introduced, there is no `sorryAx`, and no global ambient planar-Euclidean instance is added. The user-reported full

```
lake build
```

is clean after deletion of the complete XI.23 workshop/test chain.

B.41 Hilbert checkpoint through XI.23

The Hilbert classification can now be extended as follows.

1. **Neutral spatial incidence/congruence:** the local cores of XI.13–XI.14 and XI.18–XI.21.
2. **Plane-local Euclidean parallel transport:** XI.15 and the XI.8-dependent part of XI.18; XI.19 inherits the same strength through XI.15.
3. **Pure incidence once the objects are packaged:** XI.16.
4. **Spatial setup followed by neutral local metric theory:** XI.17, XI.20, and XI.21.
5. **Dimension-free statement with continuity in the construction path:** XI.22.
6. **Explicit planar slice plus genuine spatial construction:** XI.23. The planar radius calculation is confined to `PlaneGeo`; XI.12 and the off-plane apex are genuinely spatial.

XI.20–XI.23 therefore form a particularly informative four-proposition sequence. They distinguish neutral necessary inequalities, a planar circle-based existence theorem, and the final spatial realization. The kernel audit makes the continuity boundary visible exactly where the construction begins to require circles rather than only angle comparison.

B.42 Proposition XI.25: proportion without a numerical theory of volume

XI.25 introduces a new issue for the Hilbert-side audit. The proposition is not merely another spatial incidence or congruence theorem: its conclusion is a proportion between two plane magnitudes and two solid magnitudes.

Hilbert’s *Grundlagen* contains two source-level observations that are especially relevant here.

First, Chapter III is devoted to the theory of proportion. In §14 Hilbert states explicitly that Euclid’s theory of proportion is to be developed from the plane axioms I.1–I.3 and Groups II–IV, independently of the Archimedean axiom; §§15–16 then introduce segment arithmetic and express proportion through that arithmetic. This is not the route chosen in the present Lean development: XI.25 deliberately retains the Eudoxian V.Def.5 comparison pattern rather than replacing it by multiplication and division of numerical segment measures. Nevertheless Hilbert’s treatment is an important source comparison because it confirms that proportion is a separate logical layer, not a synonym for continuity or for volume.

Second, at the end of §21, after completing the theory of plane area, Hilbert warns that an analogous development of spatial volume by the same equidecomposition methods is impossible. He points to Dehn’s result and says that other means, such as the Cavalieri principle, are required for a theory of volume in space.

This warning is directly relevant to the formal interpretation of XI.25. The types

```
HilbertXI25BaseClass
HilbertXI25SolidClass
```

must not be read as numerical area and numerical volume. The solid class is an XI.Def.10 equivalence class of concrete parallelepipeds; strict comparison is represented by geometric proper-part witnesses. Repeated slabs provide positive integral multiples, and V.Def.5 is proved by synchronized comparison of those multiples.

Thus the formal theorem remains intentionally pre-measure-theoretic:

geometric magnitude classes+ proper-part order+ positive multiples $\implies$ Eudoxian proportion.

It does not formalize Hilbert–Suss Cavalieri volume, Dehn invariants, or a real-valued volume function.

At kernel level the completed direct XI.25 theorem has the audited dependency

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Quot.sound]
```

and therefore does not inherit the circle–circle continuity axiom that appeared at XI.22–XI.23. The new difficulty at XI.25 is instead the construction of a noncircular comparison theory for the base and solid magnitudes.

B.43 Hilbert checkpoint through XI.25

The Hilbert-side classification now gains one genuinely new category after the trihedral block:

1. **XI.24:** spatial parallel-plane incidence followed by the established congruence machinery for opposite faces;

2. **XI.25:** spatial cut geometry coupled to an explicit magnitude/order layer and a Book V comparison argument.

The decisive distinction is conceptual. XI.25 is not a theorem about a previously defined numerical volume. It proves a proportion relation at the level of synthetic magnitudes. Hilbert's own separation between proportion theory and the much more delicate theory of spatial volume explains why this boundary should remain explicit in the formal architecture.

Appendix C

Borsuk–Szmielw: Axiomatic Layers and Spatial Geometry

C.1 Scope

This appendix records the Book XI audit against Karol Borsuk and Wanda Szmielw, *Foundations of Geometry*, through Proposition XI.23. Their primitive language includes points, lines, planes, betweenness, and equidistance. Their axioms are organized into incidence, order, congruence, continuity, and the Euclidean axiom.

C.2 Congruence results relevant to XI.4

The Book I audit already identifies the triangle-congruence results used in the XI.4 argument within the pre-Euclidean congruence layer. In particular:

- the SSS theorem corresponding to Euclid I.8 occurs as Borsuk–Szmielw Theorem 21;
- right-angle and perpendicular constructions belong to their synthetic congruence development and are not formulated by numerical angle measure.

The formal XI.4 proof likewise treats SAS/SSS and right-angle transport as congruence infrastructure rather than as consequences of the Euclidean parallel axiom.

C.3 Spatial geometry

Borsuk–Szmielw develop three-dimensional geometry with points, lines, planes, perpendicular lines, perpendicular projection, and later an absolute coordinate system in space. In that spatial development a line perpendicular to a plane is part of the geometric language before coordinates are introduced as a later representation.

The source comparison at XI.4 is now exact. In Chapter II, Section 31, Borsuk–Szmielw define a line K to be perpendicular to a plane P when it meets P at a point a and is perpendicular to every line of P through a . Their Theorem 87 then proves that it is sufficient for K to be perpendicular to two distinct lines of P through a . This is precisely the theorem pattern of Euclid XI.4.

C.4 Axiomatic comparison

For XI.4 the comparison is:

$$\begin{array}{c}
 \text{incidence and order in planes} \\
 + \\
 \text{synthetic segment and angle congruence} \\
 + \\
 \text{spatial incidence/congruence transport} \\
 \Downarrow \\
 \text{line-plane perpendicularity.}
 \end{array}$$

The formal proof does not use the Euclidean parallel axiom. This agrees with the placement of the triangle-congruence ingredients below the Euclidean layer in the Borsuk–Szmielew organization.

C.5 Checkpoint XI.4: Borsuk–Szmielew Theorem 87

Theorem 87 is an exact structural counterpart of XI.4. Its universal definition of line–plane perpendicularity matches the target relation used in the formal project, while the theorem reduces that universal condition to two line–line perpendicularities in the plane.

The placement is also significant. The result occurs in the synthetic congruence development, before coordinates are introduced as a later representation. It therefore supports the current classification of XI.4 as a synthetic incidence/order/congruence theorem rather than an analytic fact.

C.6 Proposition XI.6 and Borsuk–Szmielew Theorem 92

The source comparison at XI.6 is also stronger than the older audit recorded. In Chapter II, Section 32, Borsuk–Szmielew prove Theorem 92:

If two distinct lines are both perpendicular to a plane, then they lie in one plane and are disjoint.

This theorem packages the two ingredients that spatial parallelism must contain:

$$\text{coplanarity} \quad + \quad \text{nonintersection.}$$

Thus Theorem 92 is a direct strengthened counterpart of Euclid XI.6. In the formal library the same distinction is made explicit by

```
HilbertSpaceLinesParallel
```

which stores a common carrier plane together with disjointness.

Borsuk–Szmielew’s proof uses their perpendicular-plane theory and uniqueness of a perpendicular to a plane. The important architectural point is the conclusion: the theorem explicitly rules out the skew-line alternative which would remain possible if one recorded only ambient nonintersection.

C.7 Checkpoint XI.6

Theorem 92 confirms that XI.6 belongs naturally to the synthetic spatial perpendicularity theory. No numerical vector representation is required for the theorem itself, and its modern formulation already has the stronger structure needed by the Lean definition of spatial parallelism.

C.8 Proposition XI.7 and incidence Axiom I7

For XI.7 the Borsuk–Szmielw source contains an exact structural counterpart of the decisive Hilbert incidence clause.

Their Axiom I7 states that for a line L and a plane P , if there exist two distinct points a, b belonging both to L and to P , then

$$L \subset P.$$

This is exactly the line-in-plane principle used in the formal proof of XI.7. Borsuk–Szmielw also classify Axioms I1–I4 as plane axioms and Axioms I5–I9 as space axioms. Hence I7 belongs to the spatial incidence layer.

Their general organization is especially transparent here: the first group consists of incidence axioms concerning points, lines, and planes; order, congruence, continuity, and the Euclidean axiom are introduced only in later groups. Proposition XI.7 therefore lies strictly below all of those stronger layers.

The correspondence with the Lean proof is

$$\begin{array}{c} E, F \in \sigma, \quad E \neq F \\ \Downarrow \\ \text{line } n \text{ through } E, F \\ \Downarrow \\ \text{Borsuk–Szmielw I7 / Hilbert I.6} \\ \Downarrow \\ n \subset \sigma. \end{array}$$

No Borsuk–Szmielw theorem number is needed here: the relevant source item is the incidence axiom itself.

C.9 Checkpoint XI.7

At XI.7 the comparison becomes sharper than at XI.4. The source contains an explicit spatial-incidence axiom whose content matches the decisive formal step exactly.

Thus XI.7 requires only incidence structure in the Borsuk–Szmielw hierarchy. It does not require their axioms of order, congruence, continuity, or the Euclidean axiom. This agrees with the Lean theorem, which assumes only the ambient incidence classes and whose axiom audit reports no additional global axiom constants.

C.10 Proposition XI.8 and the Euclidean parallel boundary

No exact Borsuk–Szmielw theorem number matching the full statement of Euclid XI.8 has been identified in the audited source. The source does, however, give a precise axiomatic placement for the parallel-theoretic step on which XI.8 depends.

The Book I source audit places the I.27-type implication

$$\text{appropriate angle equality} \implies \text{nonintersection}$$

in the congruence development before continuity and before the Euclidean axiom.

After the Euclidean axiom E is introduced, the source obtains uniqueness of the parallel and develops the converse Euclidean parallel-angle theory. In particular, the material corresponding to the I.29 direction lies on the Euclidean side of this boundary.

This is exactly the split exposed by the Lean proof of XI.8. The proposition-local helper first uses the neutral alternate-angle theorem to construct a nonintersecting line. The spatial Group IV class then supplies uniqueness in the fixed plane and identifies that line with the given parallel.

Thus the source-layer comparison is

$$\begin{array}{c} \text{incidence + order + congruence} \\ \Downarrow \\ \text{I.27-type nonintersection} \\ \Downarrow \\ + \text{ Euclidean axiom } E / \text{ Group IV} \\ \Downarrow \\ \text{unique parallel and I.29-type converse} \\ \Downarrow \\ \text{XI.8 perpendicularity transport.} \end{array}$$

The metric SAS/SSS and right-angle ingredients used around this step remain in the pre-Euclidean congruence layer. The new axiomatic input is parallel uniqueness, not a new metric or continuity principle.

C.11 Checkpoint XI.8

At XI.8 the Borsuk–Szmielw comparison records the first Book XI checkpoint in this reconstruction that requires the Euclidean axiom layer. No new continuity assumption is needed.

The source does not justify assigning a one-to-one Borsuk–Szmielw theorem number to XI.8 itself. The durable source observation is instead the same boundary made explicit by the formal API: neutral nonintersection precedes E , while the converse parallel-angle information requires Euclidean uniqueness.

C.12 Proposition XI.9 and the spatial parallel-transitivity boundary

No exact Borsuk–Szmielw theorem number matching Euclid XI.9 has been identified in the audited source. The axiomatic placement of the theorem is, however, clear enough to compare

with the current Lean proof.

The formal XI.9 reconstruction combines two ingredients:

spatial incidence + uniqueness of a parallel in a fixed plane.

The first ingredient belongs to the spatial incidence layer: planes through a line and an exterior point, intersections of distinct planes, and uniqueness of a plane from a line plus an exterior point. The second ingredient belongs to the Euclidean parallel layer.

This agrees with the Borsuk–Szmielew organization. Their incidence axioms supply the line–plane and plane–plane closure principles, while uniqueness of a parallel is obtained only after the Euclidean axiom is active. Thus the source comparison does not suggest a new metric or continuity assumption for XI.9.

The current Lean proof is more affine than Euclid’s printed perpendicularity argument. It constructs a plane through one target line and a point of the second target line, intersects this plane with the carrier plane of the second parallel pair, and uses plane-local parallel uniqueness to identify the intersection line. The resulting theorem is therefore naturally classified as

3D incidence + Group IV $\implies$ spatial transitivity of parallelism under Euclid’s proviso.

C.13 Checkpoint XI.9

At XI.9 the Borsuk–Szmielew comparison confirms the same Euclidean boundary already visible at XI.8. The theorem requires no continuity principle and no new proposition-specific spatial axiom. The decisive non-incidence content is plane-local uniqueness of a parallel, not a new theorem of congruence.

No numbered Borsuk–Szmielew theorem is assigned to XI.9 at this checkpoint. That attribution should be added only if an exact source counterpart is located.

C.14 Proposition XI.10 and Borsuk–Szmielew Theorem 52

Proposition XI.10 adds a new issue which is not visible in XI.9: orientation of parallel segments. Ordinary carrier parallelism records coplanarity and disjointness, but it does not determine which of the two possible directions on each carrier corresponds to the intended angle.

The present Lean reconstruction makes this information explicit through the relation

```
HilbertSpaceDirectedParallelSegments
```

whose geometric content is expressed by a same-side condition with respect to a connector line inside the common carrier plane.

The key source comparison is exact. In Chapter I, Section 19, Borsuk–Szmielew prove Theorem 52: if a plane Q intersects a plane P along a line K , then for points $p, q \in Q$, the plane P separates p, q if and only if the line K separates them inside Q . Equivalently, under the corresponding off-boundary hypotheses, same-side in a plane slice with respect to the section line is the same as same-half-space in ambient space with respect to the cutting plane.

This is exactly the structural bridge used by

```
hilbert_XI10_borsuk_szmielew_52
```

and by the orientation theorem

```
hilbert_XI10_directed_AD_CF_from_common_BE
```

The formal proof first lifts two plane-local same-side statements to same-side-of-plane information, composes them in ambient space, and then cuts the result back down to the plane containing the final pair of parallel segments.

The comparison is important for axiomatic reasons. The Lean development does *not* postulate a new global plane-separation axiom. The required bridge is reconstructed from the existing spatial incidence and order structure, in the same synthetic spirit as Borsuk–Szmielew Theorem 52.

The remaining ingredients of XI.10 lie in already identified layers:

- I.33-type equal-parallel-segment transport uses the Euclidean parallel layer together with congruence;
- XI.9 supplies the unoriented spatial parallel conclusion;
- SSS belongs to the synthetic congruence layer;
- no numerical angle measure or coordinate representation is used.

Thus the source-level architecture of XI.10 can be summarized as

$$\begin{array}{c}
 \text{spatial incidence + order} \\
 \Downarrow \\
 \text{Borsuk–Szmielew 52 type section bridge} \\
 \Downarrow \\
 \text{directed parallel transport} \\
 \Downarrow \\
 + \text{Euclidean parallel uniqueness and congruence} \\
 \Downarrow \\
 \angle ABC \cong \angle DEF.
 \end{array}$$

C.15 Checkpoint XI.10

At XI.10 the Borsuk–Szmielew audit identifies an exact numbered theorem for the new formal ingredient: Theorem 52 is the plane-section separation bridge needed to transport orientation between plane slices and ambient space.

This does not introduce a continuity dependency. Nor does it justify a new primitive half-space axiom in the Lean hierarchy. The production theorem continues to use the existing Hilbert spatial incidence/order/congruence architecture plus the Euclidean parallel layer already active at XI.8–XI.9.

The source comparison therefore sharpens the interpretation of the formal proof: the extra XI.10 machinery is not analytic geometry and not topology added from outside the Hilbert system; it is synthetic order geometry made explicit at the plane–space interface.

C.16 Proposition XI.11 and Borsuk–Szmielew Theorem 89

At XI.11 the source comparison becomes stronger than a general architectural analogy. In Chapter II, Section 31, Borsuk–Szmielew prove Theorem 89: through a given point there exists exactly one line perpendicular to a given plane.

Their theorem covers both cases:

- the given point lies in the plane;
- the given point does not lie in the plane.

The second case contains the existence statement corresponding directly to Euclid XI.11, while the full theorem also includes uniqueness and the on-plane case later treated by Euclid XI.12.

The construction in their off-plane case is recognizably related to Euclid’s synthetic strategy. Starting from a line in the plane, they use perpendicular projections and a second in-plane perpendicular, then invoke their two-line criterion for perpendicularity to a plane (Theorem 87). Their proof is organized inside the congruence chapter, before the later continuity and Euclidean-axiom layers.

This yields an important comparison with the current Lean theorem. The production proof of XI.11 deliberately follows Euclid’s route through XI.8, and therefore assumes `HilbertSpaceEuclidean`. Borsuk–Szmielew Theorem 89 shows that the existence-and-uniqueness result itself admits a synthetic treatment below the Euclidean parallel layer in their axiomatic organization.

Consequently the present Group IV hypothesis should be read as a property of the chosen Euclid-faithful Lean proof, not as evidence that line–plane perpendicular construction is intrinsically Euclidean.

C.17 Comparison of the two proof architectures at XI.11

The comparison can be summarized as

Euclid / current Lean route	Borsuk–Szmielew route
two successive line perpendiculars	successive perpendicular projections
parallel $GH \parallel BC$	no Euclidean parallel step required
XI.8 / Group IV	Theorem 87 in congruence layer
XI.4	Theorem 87
existence	existence + uniqueness

This does not invalidate the current formalization. It identifies a possible later strengthening: a neutral theorem proving the existence (and potentially uniqueness) of the perpendicular through a point could be added independently, after its proof is reconstructed in the project’s current Hilbert spatial API.

C.18 Checkpoint XI.11

Borsuk–Szmielew Theorem 89 is the first exact numbered source result in this appendix whose statement directly subsumes the target of Euclid XI.11. It also sharpens the axiomatic audit: the

current Lean proof uses the Euclidean layer because it reuses XI.8, whereas the target theorem has a pre-Euclidean synthetic proof in the Borsuk–Szmielew development.

No continuity assumption is suggested by this comparison, and no new axiom should be added to the Lean project on its basis.

C.19 Proposition XI.12 and Theorem 89: the on-plane case

Theorem 89 already identified at the XI.11 checkpoint is stronger than the separate Euclidean construction problems XI.11 and XI.12. It states that through a given point there exists exactly one line perpendicular to a given plane, and its proof treats both positions of the point.

For XI.12 the relevant branch is precisely

$$A \in \pi.$$

Thus the on-plane existence statement proved by Euclid XI.12 is contained directly in Borsuk–Szmielew Theorem 89, while the source also supplies uniqueness. The theorem remains in Chapter II, Section 31, in the synthetic congruence development before the later continuity and Euclidean axiom layers.

This sharpens the comparison with the current Lean proof. The production XI.12 deliberately follows Euclid:

$$\text{XI.11} \longrightarrow \text{I.31} \longrightarrow \text{XI.8}.$$

Consequently it assumes `HilbertSpaceEuclidean`, because both XI.11 and XI.8 in the present library use the plane-local Group IV layer. Borsuk–Szmielew Theorem 89 shows that this Euclidean strength belongs to the chosen proof route rather than to the abstract existence-and-uniqueness result.

The source also clarifies the status of the new Lean foot-uniqueness helper. Euclid XI.12 asks only for existence, while Theorem 89 packages existence and uniqueness together. The formal helper is therefore not an ad hoc strengthening foreign to the synthetic theory; it isolates one incidence consequence needed to make the Euclid-faithful construction total over all configurations.

C.20 Checkpoint XI.12

At XI.12 the Borsuk–Szmielew comparison is now exact for both Euclidean perpendicular-construction problems: the off-plane case of Theorem 89 subsumes XI.11, and the on-plane case subsumes XI.12.

The source therefore provides a precise future strengthening target for the formal library: a neutral existence-and-uniqueness theorem for a perpendicular through an arbitrary point. No new axiom should be added to obtain such a theorem; its reconstruction would belong to the existing synthetic incidence/order/congruence architecture.

No continuity assumption is suggested by this source comparison.

C.21 Proposition XI.13 and the uniqueness part of Theorem 89

Theorem 89, already identified at the XI.11–XI.12 checkpoints, states that through a given point there exists exactly one line perpendicular to a given plane. Proposition XI.13 corresponds directly to the uniqueness part of this result when the given point lies in the plane.

Thus the relationship among the three Euclidean propositions and Borsuk–Szmielew Theorem 89 is

Euclid	Theorem 89
XI.11	existence, point outside the plane
XI.12	existence, point in the plane
XI.13	uniqueness of the perpendicular carrier

Borsuk–Szmielew formulate the theorem for an arbitrary given point, so their uniqueness statement is not restricted to the on-plane position of XI.13.

Theorem 89 belongs to Chapter II, Section 31, within the synthetic congruence development before the later continuity and Euclidean-axiom layers. This source placement agrees with the current Lean XI.13 signature, which uses spatial incidence/order/congruence but no `HilbertSpaceEuclidean` parameter.

The two proof architectures are not identical. The current Lean XI.13 follows Euclid’s auxiliary-plane geometry and closes the contradiction through XI.4. Borsuk–Szmielew package existence and uniqueness as one structural perpendicular-to-plane theorem. The source comparison therefore identifies an exact theorem-level counterpart without forcing a line-by-line proof correspondence.

C.22 Checkpoint XI.13

At XI.13 the comparison with Borsuk–Szmielew completes the three-part Euclidean sequence XI.11–XI.13. Theorem 89 subsumes both existence cases and the uniqueness conclusion.

The source also confirms the axiomatic placement exposed by the final Lean theorem: uniqueness of the perpendicular carrier is available below continuity and below the Euclidean parallel axiom. No new axiom should be introduced for XI.13.

C.23 Propositions XI.14–XI.17: what the audited source does and does not supply

For XI.14–XI.17 the present audit has not located a clean numbered Borsuk–Szmielew theorem sequence that should be identified proposition by proposition with Euclid.

This absence should not be confused with absence of the underlying mathematics. By this point the source has already developed:

- spatial incidence of points, lines, and planes;
- line–plane perpendicularity and its uniqueness theorems;
- perpendicular planes;

- orthogonal projection;
- the Euclidean parallel layer later in the book;
- synthetic segment and angle comparison.

Nevertheless, no one-to-one theorem-number attribution is made here for:

XI.14, XI.15, XI.16, XI.17.

The formal project should therefore continue to treat these propositions by their own source DAGs rather than retrofitting a Borsuk–Szmielw number where the audited text does not support one.

There is, however, a useful layer comparison.

XI.14 is naturally adjacent to the uniqueness of perpendiculars and the perpendicular-plane theory. XI.15 crosses the Euclidean parallel boundary in the current Euclid-faithful formal proof. XI.16 is fundamentally incidence once plane parallelism has been packaged. XI.17 adds plane order and proportion or, in the Wyler affine route, an affine division invariant.

These classifications are compatible with the layered Borsuk–Szmielw organization even when an exact numbered counterpart is not assigned.

C.24 Perpendicular planes: Definition and Theorems 90–91

Chapter II, Section 32 introduces perpendicular planes explicitly. Borsuk–Szmielw say that a plane P is perpendicular to a plane Q when there exists in P a line perpendicular to Q .

Their Theorem 90 proves symmetry:

$$P \perp Q \implies Q \perp P.$$

Theorem 91 then gives a useful section criterion. If $P \perp Q$, if $L = P \cap Q$, and if a line $K \subset P$ is perpendicular to L , then K is perpendicular to Q .

This block occupies the same mathematical territory as Euclid XI.Def.4 and the later Propositions XI.18–XI.19. The definitions are not identical, however, and that difference matters.

Euclid XI.Def.4, as represented in the project, is a directed statement along the exact common section: every appropriate perpendicular to the section in one plane must be perpendicular to the other plane. Borsuk–Szmielw use the more modern existential definition “one plane contains a normal to the other” and then derive the corresponding structural theorems.

The source is therefore a strong comparison for the theory of perpendicular planes, but it should not be cited as though its definition were literally Euclid XI.Def.4.

C.25 Proposition XI.18 in the Borsuk–Szmielw organization

XI.18 says that every plane containing a line perpendicular to a reference plane is perpendicular to that reference plane.

Under the Borsuk–Szmielw definition this becomes essentially immediate: if $l \perp Q$ and $l \subset P$, then $P \perp Q$ by the very definition of perpendicular planes.

This is an instructive contrast with the Euclid-faithful reconstruction. Because the project preserves the stronger directed content of XI.Def.4, XI.18 remains a substantive theorem: it must recover the exact common section and prove the universal perpendicularity behavior along that section.

Thus the comparison is not that one proof is shorter. The two developments have chosen different definitions:

Euclid / current project	Borsuk–Szmielw
directed condition along the section	existence of a contained normal

and XI.18 measures the gap between those formulations.

C.26 Proposition XI.19: structural support from Theorems 90–91

No exact Borsuk–Szmielw theorem number matching the full statement of Euclid XI.19 has been identified in the audited source.

Theorems 90–91 nevertheless provide the correct local structural language. Symmetry of plane perpendicularity allows either plane to be used as the carrier of a normal, and Theorem 91 links a line perpendicular to the common section with perpendicularity to the opposite plane.

The present Lean proof of XI.19 has an additional global incidence problem: the two traces on the reference plane must first be shown to meet. That existence step is not supplied merely by Theorems 90–91; in the current formal reconstruction it is forced by the spatial parallel theory.

Accordingly the durable Borsuk–Szmielw comparison for XI.19 is local rather than proposition-number exact:

$$\text{perpendicular-plane structure} \quad + \quad \text{normal uniqueness}$$

matches the neutral common-point core after the triple point has been constructed.

C.27 Proposition XI.20 and the planar metric ingredients

The audited Borsuk–Szmielw text has not yielded an exact theorem on a solid or trihedral angle corresponding to Euclid XI.20. Searches for an explicit “solid angle” or “trihedral angle” theorem in the available text did not identify such a result, so none is attributed here.

The planar metric ingredients of XI.20 are, however, already present in the Borsuk–Szmielw development.

Their theory of free segments includes a strong triangle-inequality theorem: for distinct points the sum of two consecutive segments is greater than the third, with the collinear equality case treated explicitly. Their angle-order theory also contains the strict side–angle comparison used in the Euclid I.18/I.19/I.25 family.

This supports the architectural decomposition exposed by the Lean proof:

$$\text{one small spatial noncoplanarity argument} \quad + \quad \text{planar metric comparison.}$$

The Borsuk–Szmielew source should therefore be cited for the layered synthetic infrastructure, not for an unverified one-to-one XI.20 theorem.

C.28 Proposition XI.21: synthetic angle calculus without a direct source match

No exact Borsuk–Szmielew theorem number corresponding to Euclid XI.21 has been identified in the audited source, and none is assigned here.

The source nevertheless contains two pieces of infrastructure directly relevant to the formal proof.

First, the Book I audit records their explicit calculus of free angles, adjacent angles, supplements, and right angles. This is an important source precedent for treating finite angle combinations synthetically rather than by introducing numerical degree measure.

Second, vertical-angle congruence is already a basic congruence theorem in their development. The current Lean XI.21 proof uses exactly such a vertical-angle transport after extending two rays through the vertex.

Thus the Borsuk–Szmielew comparison supports the local ingredients

supplement/adjacency calculus + vertical-angle congruence,

but the proposition-level four-right-angle theorem remains attributed to Euclid rather than to an unverified Borsuk–Szmielew counterpart.

The current formal result is also stronger architecturally than a literal angle-sum transcription would suggest: it avoids I.32 and general finite angle addition by reducing the target to one XI.20-type comparison between two supplements and the remaining face angle.

C.29 Updated Borsuk–Szmielew correspondence through XI.23

The strongest exact correspondences currently identified remain:

Euclid Book XI	Borsuk–Szmielew
XI.4	Theorem 87
XI.6	Theorem 92
XI.7 incidence core	Axiom I7
XI.10 orientation bridge	Theorem 52
XI.11 – –XI.13	Theorem 89
XI.18 modern-definition counterpart	definition of perpendicular planes
XI.19 local perpendicular-plane core	Theorems 90–91

For XI.20–XI.23 the source is used structurally rather than through an exact theorem number. XI.20–XI.21 draw on triangle inequality, side–angle comparison, free-angle calculus, and vertical-angle congruence. XI.22–XI.23 add the separate issues of circle-based existence and final spatial realization; no unverified Borsuk–Szmielew proposition number is assigned to either construction.

C.30 Checkpoint XI.21

By XI.21 the Borsuk–Szmielw audit continues to serve primarily as a layering control. It confirms that the angle-order and supplement machinery needed by the current proof belongs naturally to synthetic congruence geometry, while no unlocated trihedral theorem is invented for the sake of a proposition-number match.

No new continuity dependency is inferred from this comparison. The direct formal XI.21 proof uses the already developed neutral angle machinery plus a small spatial line-in-plane closure argument.

C.31 Proposition XI.22: structural support without an exact source match

No exact Borsuk–Szmielw theorem corresponding to Euclid XI.22 has been identified in the audited source, and none is assigned here.

The proposition nevertheless fits the layering already visible in their development. Its new comparison content is neutral: equal-radius angle comparisons are converted into strict inequalities among the corresponding chords. This uses the same kind of synthetic segment and angle order that the Borsuk–Szmielw comparison has already supplied for XI.20.

The final construction of a triangle from the three chord lengths is a different issue. In the current Lean development it is delegated to Euclid I.22, whose axiom audit contains the circle–circle continuity principle

```
Geometry.hilbert_circle_circle_intersection
```

The Borsuk–Szmielw appendix does not infer a numbered source counterpart for that construction merely from the presence of continuity in their global axiomatic organization. The safe comparison is architectural: neutral metric comparison and circle-based existence are separate layers.

C.32 Proposition XI.23: trihedral realization and the continuity boundary

No exact Borsuk–Szmielw existence theorem matching Euclid XI.23 has been located in the audited text.

Several established Borsuk–Szmielw ingredients nevertheless match local parts of the formal proof:

- SSS congruence is already available as Theorem 21;
- the earlier angle-order and free-angle calculus supports the neutral comparison stages;
- Theorem 89 supplies the structural model for a perpendicular to a plane through a prescribed point;
- Theorems 90–91 supply the local language for perpendicular planes and normals.

These results do not by themselves prove XI.23. The proposition is an *existence* theorem: from three prescribed plane angles it constructs a new noncoplanar trihedral configuration. The current Lean proof still needs the planar XI.22 chord triangle, its circumcenter, a strict radius comparison, and the spatial apex construction.

The exact kernel audit is

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

Thus XI.23 should not be grouped with XI.20–XI.21 as a continuity-free metric theorem. The neutral angle machinery survives, but the complete construction inherits circle–circle continuity.

This is also a useful control on source attribution. The Borsuk–Szmielw text supports the local synthetic ingredients, but the present audit has not identified a single theorem there whose statement should be advertised as “Borsuk–Szmielw XI.23.”

C.33 Checkpoint XI.23

By XI.23 the Borsuk–Szmielw comparison has three distinct roles.

For XI.4, XI.6, XI.11–XI.13, and parts of XI.18–XI.19 there are exact or near-exact structural theorem correspondences.

For XI.20–XI.21 the source validates the neutral metric ingredients without supplying a proposition-number match.

For XI.22–XI.23 the audit additionally exposes the construction boundary: the current formal route passes through circle continuity and then returns to spatial perpendicular/congruence geometry for the final apex.

The resulting lesson is not that Borsuk–Szmielw “contains XI.23” under a different number. It is that their layered synthetic organization helps separate the four kinds of mathematics that the proposition combines:

angle order + segment/chord geometry + circle existence + spatial perpendicularity.

Appendix D

Forder: Neutral Solid Geometry and Euclid Book XI

D.1 Purpose and scope

This appendix records the Book XI source audit against Henry George Forder, *Foundations of Euclidean Geometry* [4]. Forder is especially important for the present reconstruction because he does not merely mention solid geometry as an application of an already established Euclidean theory. He develops a synthetic theory of space in its own right, with explicit incidence, order, congruence, perpendicularity, dihedral angles, and trihedral angles.

The comparison is therefore different from a proposition-by-proposition commentary on Euclid. Forder reorganizes the material according to the logical structure of his axiomatic system. Several Euclidean Book XI propositions become general structural theorems, some are strengthened by uniqueness, and some are absorbed into definitions and reusable lemmas.

The present audit has two aims:

1. to identify Forder results that correspond directly or structurally to the Book XI propositions already reconstructed in CGJteamLab;
2. to record where Forder’s proof architecture supplies an independent synthetic route, especially when it avoids the Euclidean parallel axiom.

The most important checkpoint is Proposition XI.20. Forder’s Theorem 93 is a direct modern synthetic counterpart of that proposition and its proof is remarkably close to the proof graph isolated in the current Lean development.

D.2 Forder’s explicit relation to Euclid Book XI

The relevance of Forder is not inferred only from subject matter. At the transition from planar congruence theory to solid geometry he explicitly announces that he is turning to solid geometry and observes that, because no parallel axiom is being assumed, his proofs are usually different from those of Euclid Book XI.

This statement fixes the logical position of the chapter. The spatial theory developed there is intended to be neutral with respect to the parallel axiom. Consequently, when a Book XI result

is recovered in this part of Forder’s development, one should not automatically classify that result as Euclidean in the stronger parallel-postulate sense.

For the present project this is a particularly useful control source. The formal reconstruction repeatedly asks whether a Euclidean-looking spatial argument genuinely needs Group IV, or whether the theorem belongs already to incidence, order, and congruence. Forder gives an independent classical answer to that question for a substantial part of Book XI.

D.3 Order-theoretic preparation: half-spaces, dihedral angles, and trihedral angles

Before the spatial congruence theorems, Forder develops the order structure of space. In Chapter III he proves that a plane separates the remaining points into two convex regions and defines the corresponding half-spaces.

He then introduces dihedral and trihedral angles. If two half-planes have a common edge and do not lie in one plane, their union together with the edge is a dihedral angle. For three noncoplanar rays

$$OA, \quad OB, \quad OC$$

with no two lying on the same line, Forder defines the trihedral angle

$$O.ABC.$$

Its vertex is O , its edges are the rays OA, OB, OC , and its faces are the plane angular regions determined by the pairs of rays. Forder calls the dihedral angles between those faces the *angles* of the trihedral angle.

This terminology must be kept distinct from Euclid’s terminology in XI.20–XI.21. Euclid XI.20 compares the three *plane angles* at the vertex,

$$\angle AOB, \quad \angle BOC, \quad \angle COA.$$

Forder’s trihedral object contains these as its three face sectors, while his “angles of the trihedral angle” are the dihedral angles between the faces.

The distinction is important for formalization. The current Lean relation

```
HilbertTrihedralConfiguration
```

records the noncoplanar three-ray configuration needed for XI.20 without identifying plane face angles with dihedral angles.

D.4 The line–plane perpendicularity block

Forder’s Chapter IV contains a concentrated block of spatial perpendicularity results. This block is one of the closest modern synthetic counterparts of the first half of Euclid Book XI.

The first result of the block, Theorem 79, proves the following principle. Let OA, OB, OC be coplanar, with OA and OC distinct. If

$$OP \perp OA, \quad OP \perp OC,$$

then

$$OP \perp OB.$$

This is the universalization step at the heart of Euclid XI.4: perpendicularity to two distinct intersecting lines of a plane forces perpendicularity to every line of that plane through the common point. Forder then packages the corresponding line–plane relation explicitly.

After Theorem 80 he defines OP to be perpendicular to the plane OAB when

$$OP \perp OA, \quad OP \perp OB, \quad OA \neq OB.$$

Theorem 81 immediately recovers the universal form: if $OP \perp OAB$, then OP is perpendicular to every line OC of that plane through O .

Thus Forder separates the same two layers that the current Lean development separates:

$$\begin{array}{c} \text{two-line criterion} \\ \Downarrow \\ \text{line–plane perpendicularity} \\ \Downarrow \\ \text{universal perpendicularity in the plane.} \end{array}$$

The present formal predicate

```
HilbertLinePerpendicularPlaneAt
```

uses the universal formulation as its target. Euclid XI.4 proves that target from two line–line perpendicularities. Forder’s local theorem block confirms that this is a natural theorem of synthetic congruence geometry rather than a coordinate or scalar-product fact.

D.5 Proposition XI.5 and Forder Theorem 80

Forder Theorem 80 states, in modernized notation, that if

$$OP \perp OA, \quad OP \perp OB, \quad OP \perp OC,$$

then the four points

$$O, A, B, C$$

are coplanar.

This is directly recognizable as the structural content of Euclid XI.5. The Euclidean proposition says that if one straight line is perpendicular at a common point to three straight lines, then those three straight lines lie in one plane.

The difference is mainly one of packaging. Euclid expresses the conclusion in terms of the three lines. Forder expresses it through coplanarity of the four points that determine those lines.

The proof is also structurally informative. Forder forms two planes and uses their intersection line. The previously established two-line perpendicularity theorem then identifies the common line, forcing the third candidate ray into the plane of the first two.

For the Lean reconstruction this is a strong independent check on the role of XI.5. The proposition is not merely a convenient Euclidean lemma: it is a natural incidence–congruence theorem of neutral three-dimensional geometry.

D.6 Existence and uniqueness: Forder Theorems 82 and 83

Forder does not preserve Euclid's separation of the perpendicular construction problems into XI.11, XI.12, and XI.13. Instead he proves stronger structural theorems.

Theorem 82 states that through a given point there passes one and only one plane perpendicular to a given line.

Theorem 83 states that through a given point there passes one and only one line perpendicular to a given plane.

Theorem 83 treats both cases:

- the given point lies in the plane;
- the given point lies outside the plane.

Accordingly the relation to Euclid is

Euclid	Forder Theorem 83
XI.11	existence through a point outside the plane
XI.12	existence through a point in the plane
XI.13	uniqueness

This is closely parallel to the Borsuk–Szmielw Theorem 89 comparison, but Forder is independently valuable because his whole spatial block is explicitly developed without assuming a parallel axiom.

The comparison therefore sharpens an architectural lesson already visible in the current Lean project. A proof of XI.11 or XI.12 that passes through Euclidean parallel transport may be Euclid-faithful, but the abstract existence-and-uniqueness theorem is not thereby shown to require Group IV. Forder supplies a neutral synthetic route.

D.7 Perpendicular planes

Forder next introduces the metric relation between planes. His definition is made through perpendicular cross-sections, and the subsequent theorems establish the expected structural properties.

Theorem 85 proves symmetry:

$$\alpha \perp \beta \implies \beta \perp \alpha.$$

The surrounding results relate:

- a line perpendicular to a plane;
- a plane containing such a line;
- the common section of two planes;
- mutually perpendicular planes and their intersection lines.

Theorems 86–88 then give construction and characterization statements for perpendicular planes. These results occupy the same mathematical territory as the Euclidean development around XI.Def.4 and Propositions XI.18–XI.19, although the proposition boundaries are not the same.

For the present project the important source-level point is not a forced one-to-one numbering. It is that plane–plane perpendicularity is treated by Forder as derived synthetic metric structure built on the line–plane theory. This agrees with the current neutral module

```
Hilbert3DPlanePerpendicular.lean
```

where plane perpendicularity is defined from a common section and the appropriate line–plane perpendicularity behavior.

D.8 Two normals to a plane: the XI.6 neighborhood

Forder Theorem 90 states that two lines perpendicular to the same plane are coplanar.

This isolates the genuinely spatial half of Euclid XI.6. In three-space it is not enough to know that two lines do not meet: disjoint ambient lines could be skew. A proof of spatial parallelism must first produce a common plane.

The current formal relation

```
HilbertSpaceLinesParallel
```

makes exactly this distinction by packaging

$$\text{coplanarity} \quad + \quad \text{disjointness}.$$

Forder Theorem 90 supplies the coplanarity conclusion as a general theorem of his neutral perpendicularity theory. The remaining nonintersection component is then planar.

The comparison is therefore especially useful as an architectural control: the difficult spatial content of XI.6 is the production of the common carrier plane, not merely the familiar planar fact that two perpendiculars to the same line cannot meet.

D.9 Orthogonal projection

Theorems 89–91 develop orthogonal projection onto a plane. In particular, Forder shows that the family of perpendiculars to a plane through points of a line lies in a plane perpendicular to the reference plane, and he defines the projection of sets onto a line or plane.

This material is not tied to one Euclidean proposition number, but it is important for the later Book XI infrastructure. It provides a systematic language for replacing repeated proposition-local constructions by reusable spatial operations.

This is another respect in which Forder is useful for the formal project: Euclid often constructs individual perpendiculars as needed, whereas a modern axiomatic development benefits from a global projection operation together with existence, uniqueness, and carrier-plane theorems.

D.10 Proposition XI.20 and Forder Theorem 93

The strongest direct correspondence at the present checkpoint is Forder Theorem 93.

In Forder's notation, if

$$O, A, B, C$$

are not coplanar, then

$$\angle AOB + \angle AOC > \angle BOC.$$

Up to permutation of A, B, C , this is exactly Euclid XI.20: in a solid angle contained by three plane angles, the sum of any two of the plane angles is greater than the remaining one.

The noncoplanarity hypothesis is particularly significant. Forder does not leave the "solid angle" character of the configuration to the diagram. The three rays are required to determine a genuinely spatial configuration.

D.11 Forder's proof of Theorem 93

Forder's proof can be normalized into the following synthetic DAG.

Suppose the desired inequality is not already immediate. Choose a ray OX inside $\angle BOC$ such that

$$\angle BOA \cong \angle BOX,$$

and let this ray meet the segment BC at X .

Lay off

$$OA \cong OX.$$

The corresponding triangle congruence gives

$$AB \cong BX.$$

Apply the triangle inequality in triangle ABC :

$$AB + AC > BC.$$

Since X lies between B and C ,

$$BC = BX + XC.$$

Using $AB = BX$, cancellation yields

$$AC > XC.$$

The side-angle comparison theorem then gives

$$\angle AOC > \angle XOC.$$

But

$$\angle BOC = \angle BOX + \angle XOC$$

and

$$\angle BOX \cong \angle BOA.$$

Therefore

$$\angle BOA + \angle AOC > \angle BOX + \angle XOC = \angle BOC.$$

The proof is entirely synthetic:

$$\begin{array}{c}
 \text{interior ray and angle copy} \\
 \Downarrow \\
 OA = OX \\
 \Downarrow \\
 AB = BX \\
 \Downarrow \\
 \text{triangle inequality} \\
 \Downarrow \\
 AC > XC \\
 \Downarrow \\
 \text{side-angle comparison} \\
 \Downarrow \\
 \angle AOC > \angle XOC \\
 \Downarrow \\
 \text{angle decomposition} \\
 \Downarrow \\
 \angle AOB + \angle AOC > \angle BOC.
 \end{array}$$

No coordinate system, vector inner product, trigonometric representation, or numerical solid-angle measure is used.

D.12 Comparison with the current Lean XI.20 proof

The correspondence with the current formal reconstruction is unusually close.

The project uses the neutral configuration

```
HilbertTrihedralConfiguration
```

to record:

- the three face angles are proper;
- the four points determining the three rays are not coplanar.

The target relation

```
HilbertTwoAnglesGreaterThanAngle
```

does not introduce a numerical angle measure. Instead it represents the synthetic alternatives needed to express that the target angle is smaller than the sum of the other two.

The planar metric core has been isolated in

```
HilbertTrihedralAngleMetric.lean
```

and uses the same classical ingredients visible in Forder:

- copying an angle into another angle;
- laying off an equal segment;
- SAS;
- Euclid I.20 / triangle inequality;
- Euclid I.25 / side–angle comparison;
- angle decomposition and order transport.

The direct Hilbert spatial proof differs mainly in the bookkeeping forced by formalization. The auxiliary point laid off on the third ray must be shown not to collapse the relevant triangle. This is where the noncoplanarity of the trihedral configuration is used explicitly.

Thus the comparison is not merely thematic:

Forder Theorem 93	current Lean XI.20
noncoplanar O, A, B, C	<code>HilbertTrihedralConfiguration</code>
OX inside $\angle BOC$	<code>HilbertRayMeetsSegment</code>
$\angle BOA \cong \angle BOX$	<code>AngleCongruent</code>
$OA \cong OX$	segment construction
$AB \cong BX$	SAS helper
$AB + AC > BC$	Proposition I.20 infrastructure
$AC > XC$	segment-order cancellation
$\angle AOC > \angle XOC$	Proposition I.25 infrastructure
angle decomposition	Hilbert angle-decomposition API

Forder therefore provides an independent classical validation of the proof architecture chosen for the formal theorem.

D.13 Why Theorem 93 matters for the Hilbert/Wyler comparison

Theorem 93 also clarifies the dimension question behind XI.20.

The metric part of the proof is planar. The genuinely spatial information is the noncoplanarity of the three rays. This agrees with the current formal audit: almost the entire XI.20 proof lies in the common planar incidence/order/congruence layer, while the spatial step is used to rule out a degenerate auxiliary triangle.

This is precisely why XI.20 is a useful test for the parallel Hilbert and Wyler routes.

The direct Hilbert proof uses the established three-dimensional spatial incidence interface. The Wyler reconstruction can replace that one spatial step by the weaker dimension-free line–plane closure principles:

plane through three noncollinear points + line-in-plane closure.

Forder’s neutral proof gives strong historical support for treating the metric core independently of the ambient dimension machinery. It does not, by itself, prove the project’s dimension-free Wyler theorem; that is a separate formal result. But it confirms that no Euclidean parallel argument is mathematically intrinsic to XI.20.

D.14 Trihedral congruence beyond XI.20

Immediately after Theorem 93, Forder develops further spatial metric theory. Theorem 95 treats congruence of dihedral angles with equal measure, and Theorem 96 gives several congruence criteria for trihedral angles.

In modernized form, Theorem 96 says that suitable congruence data among faces and included dihedral angles determine the remaining corresponding faces and angles of a trihedral configuration.

This goes beyond the exact content of XI.20, but it is relevant to the subsequent theory of solid angles. It shows that Forder does not treat Theorem 93 as an isolated inequality: it belongs to a systematic synthetic theory of trihedral geometry.

Theorem 96 is not a direct counterpart of Euclid XI.21. It is a congruence theorem for trihedral angles, not a four-right-angle bound. The useful conclusion is more structural: Forder supplies a developed trihedral-angle language in which XI.20 sits naturally, but the theorem sequence immediately following Theorem 93 does not reproduce Euclid XI.21 as a separate proposition.

D.15 Proposition XI.21 and Forder’s angular-measure framework

The source audit for XI.21 produces a different result from XI.20.

Forder Theorem 93 is a direct counterpart of XI.20. Immediately afterwards, however, Theorem 94 concerns the right-bisector plane of an interval, Theorem 95 concerns congruence of dihedral angles with equal measure, and Theorem 96 gives congruence criteria for trihedral angles. No theorem in this local sequence is a direct restatement of Euclid XI.21.

Forder nevertheless has already built exactly the kind of arithmetic that would make Euclid’s printed XI.21 proof straightforward. In Chapter IV, Section 56 he defines the measure of an ordinary angle and observes the essential limitation:

the sum of two angle measures need not itself be an angle measure.

He explicitly notes that one common response is to introduce “angles” greater than two or four right angles, but rejects enlarging the notion of angle in that way.

Instead, Section 57 introduces an *angular measure*. In Forder’s notation an angular measure is an ordered pair representing an integral number of half-turns together with an ordinary angle measure. Addition is then defined on these objects; Forder records commutativity and associativity and states that angular measures form a condensed set of magnitudes. He also notes that signed angular measures form a fully ordered group under addition.

This is highly relevant to XI.21. Euclid’s printed proof performs exactly the kind of finite angle-sum bookkeeping for which Forder’s angular measures were designed:

$$6R - (\text{six lateral angles}) < 4R.$$

Thus Forder supplies a rigorous synthetic arithmetic in which the classical calculation can be expressed without identifying angles themselves with real numbers.

The present Lean reconstruction deliberately takes a weaker route. It does *not* formalize Forder’s full angular-measure calculus merely in order to prove XI.21. Instead it represents

$$\alpha + \beta + \gamma < 4R$$

by a complement certificate:

$$(2R - \alpha) + (2R - \beta) > \gamma.$$

The right-hand comparison is exactly of the XI.20 type and can be obtained by extending two rays through the vertex, applying XI.20 to the resulting opposite-ray trihedral configuration, and transporting the comparison by vertical-angle congruence.

The comparison between Forder and the current Lean architecture is therefore particularly informative:

Forder	current Lean XI.21
general angular measures	local complement certificate
global addition is available	no general angle addition introduced
signed angular measures form an ordered group	only XI.20 comparison and congruence transport are u
framework supports sums beyond one ordinary angle	the single $4R$ bound is encoded synthetically

This is not a rejection of Forder’s construction. On the contrary, it identifies a natural future extension of the library if later propositions genuinely require systematic angle arithmetic. XI.21 shows only that this stronger layer is not yet forced.

D.16 Book XI correspondence table

The source audit, including the limited XI.24–XI.25 comparison, can be summarized as follows.

Euclid	Forder	Source-audit interpretation
XI.4	Theorem 79, definition after Theorem 80, Theorem 81	Two-line criterion and universal line-plane perpendicularity.
XI.5	Theorem 80	Three lines perpendicular to one line at the same point are forced into one plane.
XI.6	Theorem 90 plus planar non-intersection	Forder isolates the genuinely spatial coplanarity of two normals to one plane.
XI.11	Theorem 83, off-plane case	Existence is included in a stronger existence-and-uniqueness theorem.
XI.12	Theorem 83, on-plane case	Again included in the same structural theorem.
XI.13	Theorem 83, uniqueness	The Euclidean three-proposition sequence is compressed into one theorem.
XI.18–XI.19	Theorems 84–88	Perpendicular-plane definitions, symmetry, and structural characterization; the proposition boundaries differ from Euclid.
XI.20	Theorem 93	Direct counterpart, with essentially the same synthetic proof graph.
XI.21	No direct counterpart located; Sections 56–57 plus Theorem 93	Forder’s angular-measure calculus supports the classical angle-sum bookkeeping, while the current Lean proof uses a weaker complement certificate based on XI.20.
XI.22	No direct counterpart located	The proposition is planar. The current Lean proof reduces the angle hypotheses to chord triangle inequalities and then uses I.22; its final existence step therefore lies on the circle-continuity side of the architecture rather than in Forder’s trihedral congruence block.
XI.23	No direct counterpart located; Theorem 96 is relevant only as later trihedral congruence infrastructure	Forder’s Theorem 96 gives congruence criteria for already existing trihedral angles. It is not an existence theorem from three prescribed face angles. The current Lean XI.23 instead constructs a chord triangle, circumcenter, normal apex, and then recovers the three faces by SSS.
XI.24	No direct theorem-number counterpart located; VI.36.7–36.8 gives the parallelepiped framework	Forder defines parallelepipeds by three pairs of parallel face planes and constructs them from three planes through one point. This supplies structural background for XI.24, but Forder does not reproduce Euclid’s opposite-face proof as a separately numbered theorem.
XI.25	No direct theorem-number counterpart located; Chapter VII and XII.24–28 are the relevant later theories	Forder develops geometric proportion for intervals in Chapter VII and only later develops numerical volume for polyhedra. His Chapter XII explicitly marks the limit of the earlier equivalence theory before introducing volume; this is a useful contrast with Euclid’s V.Def.5 proof and with the present non-numerical Lean reconstruction.
Later trihedral theory	Theorems 95–96 and later solid geometry	Dihedral and trihedral congruence infrastructure; not identified with XI.21 or XI.23 itself.

D.17 Comparison with the other Book XI source audits

Forder occupies a distinct position among the sources used in this volume.

Hilbert supplies the foundational spatial incidence architecture: the axioms governing lines, planes, dimensionality, and their intersections.

Borsuk–Szmielw give a systematic synthetic theory of three-dimensional perpendicularity and projection, often packaging several Euclidean propositions into stronger structural results.

Forder is especially valuable because he combines those two roles with an explicit comparison to Euclid Book XI. He develops solid geometry synthetically, states that no parallel axiom is assumed in this phase, and then proves theorems that can be matched directly to several Euclidean propositions.

For XI.20, Forder goes further: the theorem and proof are sufficiently close to the Euclidean configuration that they provide an independent classical control of the exact proof architecture.

D.18 Structural lessons for the formal project

The Forder audit suggests five durable lessons.

1. **Book XI should not be audited only by proposition number.** Modern axiomatic sources often reorganize several Euclidean propositions into one structural theorem.
2. **Line–plane perpendicularity belongs naturally to synthetic congruence geometry.** It need not be introduced analytically by vectors or scalar products.
3. **Existence and uniqueness should be audited separately from Euclid’s construction route.** A Euclid-faithful proof can use stronger intermediate infrastructure than the abstract target theorem intrinsically requires.
4. **Spatial parallelism has a coplanarity component.** Forder Theorem 90 isolates exactly the part that is invisible in a purely planar treatment.
5. **XI.20 has a largely planar metric core.** The genuinely spatial datum is noncoplanarity; the inequality itself is obtained from planar angle copying, SAS, triangle inequality, side–angle comparison, and angle decomposition.
6. **XI.21 separates available strength from necessary strength.** Forder already has a general arithmetic of angular measures, but the current formal theorem needs only a complement construction and XI.20. A stronger arithmetic layer should be added only when later geometry genuinely requires it.

These observations align closely with the current Hilbert/Wyler double-proof program. They also provide a source-based reason to keep neutral metric lemmas separate from dimension-specific spatial infrastructure.

D.19 Checkpoint XI.21

At Proposition XI.21 the Forder comparison has become more informative than a search for proposition-number matches.

Theorem 93 remains the exact source counterpart of XI.20. XI.21, by contrast, is not reproduced by the immediately following theorem sequence. Instead, the relevant source material lies earlier

in Forder’s general theory of angle measure: Sections 56–57 explain why ordinary angle measures are not closed under addition and introduce angular measures precisely to support larger finite sums.

This gives the project two mathematically legitimate design options:

build Forder-style angular arithmetic
or
use proposition-specific synthetic complement certificates.

The present XI.21 proof demonstrates that the second option is sufficient for the four-right-angle bound.

Consequently no general angular-measure object is added to the production library at this checkpoint. Forder’s construction is retained as the source-based blueprint for such a layer if subsequent propositions make it necessary.

D.20 Proposition XI.22: a planar theorem at the edge of the circle layer

No direct Forder theorem has been identified that should be numbered as the counterpart of Euclid XI.22.

This absence is structurally informative. XI.22 is placed in Book XI because it is needed for the following solid-angle construction, but the proposition itself is planar. In the current formalization its new mathematical core is the neutral implication

pairwise angle inequalities $\longrightarrow$ triangle inequalities for equal-radius chords.

The final triangle is then constructed by Euclid I.22.

Forder’s organization makes a related boundary visible. Theorem 96 is the last theorem in the congruence chapter’s local trihedral block; immediately after it begins Chapter V, *The Circle Axioms*. Forder explicitly describes that chapter as a continuation of the congruence theory devoted to circle and sphere properties and to the relation between circle axioms and congruence axioms.

The present project should not turn this structural proximity into a false theorem-number correspondence. What it does support is the architectural distinction already exposed by Lean: XI.22’s angle-to-chord comparison is neutral congruence geometry, while the chosen existence proof through I.22 uses circle-continuity machinery.

D.21 Proposition XI.23: existence is not trihedral congruence

No direct Forder theorem corresponding to Euclid XI.23 has been located in the audited source.

The nearest local theorem, Theorem 96, is a congruence theorem for *already existing* trihedral angles. In particular, suitable corresponding face/angle data determine the remaining corresponding parts. That result is valuable for uniqueness and transport questions, but it does not solve the XI.23 existence problem:

Given three plane angles satisfying the XI.20 and XI.21 inequalities, construct a trihedral angle having those three faces.

The production Lean theorem follows Euclid’s constructive architecture rather than deriving existence from Forder’s congruence criterion. It constructs the XI.22 chord triangle in one plane, obtains its circumcenter, proves that the common source radius is greater than the circumradius, erects a normal through the center, lays off the required height, and finally uses spatial SSS to recover the three face angles.

Thus the Forder comparison separates two logically different tasks:

Forder Theorem 96	Euclid / Lean XI.23
congruence of existing trihedra	existence of a trihedral realization
determine remaining corresponding parts	construct the apex and prove noncoplanarity
congruence-layer theorem	circle construction + spatial normal + SSS

This is a useful warning for the formal library. Strong congruence criteria should not be mistaken for existence principles.

The final Lean axiom audit adds another source-level distinction:

```
Geometry.hilbert_circle_circle_intersection
```

appears in the closure of XI.23. Hence the completed construction crosses a continuity boundary that was absent from the neutral XI.20–XI.21 proofs.

D.22 Checkpoint XI.23

The Forder audit through XI.23 now identifies three different roles for the source.

First, Theorem 93 is an exact and highly informative counterpart of XI.20.

Second, Sections 56–57 provide a stronger general angular arithmetic than is needed for XI.21, and therefore serve as a blueprint rather than a direct dependency.

Third, Theorems 95–96 show that Forder possesses a mature theory of dihedral and trihedral congruence, but they do not replace the XI.23 existence construction.

The boundary after Theorem 96 is especially suggestive: Forder passes from trihedral congruence into the circle axioms at exactly the point where the Euclidean XI.22–XI.23 construction becomes circle-dependent in the present formalization. This is a structural comparison, not a claim that Forder states XI.22 or XI.23 under those theorem numbers.

For the project the resulting lesson is precise: keep angle comparison, trihedral congruence, circle existence, and spatial realization as separate layers. XI.23 is where all four meet.

D.23 Propositions XI.24–XI.25: parallelepipeds before numerical volume

Forder contains material relevant to XI.24–XI.25, but it should not be turned into a fictitious proposition-number correspondence.

In Chapter VI, §36.7, Forder defines a parallelepiped as a figure formed by six quadrilaterals lying in three pairs of parallel planes, with the usual edge-sharing condition. Section 36.8 then

constructs such a figure from three distinct planes through a point by drawing the three opposite planes parallel to them. This is genuine structural background for XI.24: the parallel-plane shell of the Euclidean parallelepiped is present explicitly.

The more revealing comparison occurs much later, in Chapter XII on volumes of polyhedra. At XII.24 Forder first transfers the earlier notions of equivalence and equivalence by addition from polygons to polyhedra. He then remarks that two parallelepipeds on the same base and with congruent corresponding altitudes can be shown equivalent, but immediately adds that practically no further progress can be made on that basis alone.

This is important for reading XI.25. Forder does *not* simply identify Euclid's solid ratio with a previously available numerical volume. His numerical theory comes only after a separate dissection theory for polyhedra. After defining tetrahedral volume and proving its additivity, he defines the volume of an arbitrary polyhedron from a tetrahedral dissection and obtains, in XII.28.1,

$$V(\text{parallelepiped}) = \text{area of a face} \times \text{corresponding altitude}.$$

Once this later measure theory exists, a statement analogous to XI.25 can be expressed numerically. But that is not Euclid's route and it is not the route used by the present formal reconstruction. Euclid proves a Book V proportion before assigning numerical volumes; the Lean theorem likewise works with geometric base and solid magnitude classes, repeated positive multiples, and strict proper-part comparison.

The comparison therefore has a precise value. Forder confirms both sides of the architectural boundary:

parallelepiped incidence and equivalence
do not by themselves constitute a full volume theory,
whereas numerical volume requires an additional layer.

XI.25 sits deliberately on the first side of that boundary.

D.24 Forder checkpoint through XI.25

The new material does not alter the earlier XI.4–XI.23 correspondences. It adds one further source-level lesson: the passage from equality of parallelepipedal pieces to a theory of their ratios should not be confused with the later introduction of numerical polyhedral volume.

For XI.24 the Forder source supplies a clean structural parallelepiped framework but no exact proposition counterpart. For XI.25 it supplies a useful contrast between geometric proportion/equivalence and the later measure-theoretic theory of volume. That contrast is substantive; anything stronger would overstate the source.

Appendix E

Wyler: Flats, Closure, and the 3D Incidence Calculus

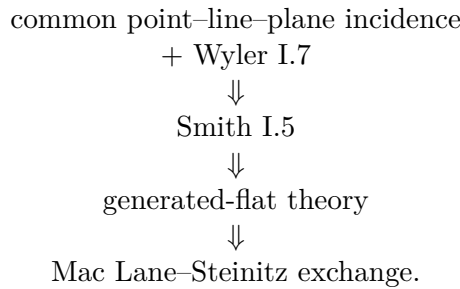
E.1 Purpose and current scope

This appendix records the Wyler-side architectural audit for Euclid Book XI through Proposition XI.25. Its purpose is not to claim that CGJteamLab is a line-by-line formalization of Oswald Wyler’s 1953 papers. The project uses Wyler as a guide to a different question:

Which parts of the spatial incidence machinery used in Book XI can be reorganized so that they remain meaningful beyond a globally three-dimensional Hilbert space?

The answer has changed substantially since the first Book XI flat experiment. The project now has a dimension-free Hilbert–Wyler incidence foundation, derives the Smith and exchange consequences above it, reconstructs local three-dimensional incidence inside derived E^4 hyperplanes, and uses the resulting architecture in independent Book XI proofs.

The current logical picture is



A central design decision is that the lower items in this diagram are *derived theory*. They are not silently promoted to axioms.

E.2 A source-critical note on Wyler

Wyler published two papers in 1953 relevant to the present program. *Incidence geometry* is the standard reference for the incidence-geometric program [5]. The companion paper *Order in*

projective and in descriptive geometry is valuable because it states an explicit point–line–plane incidence basis and defines flats and generated flats [6].

The historical terminology should be kept distinct from later matroid and geometric-lattice language. Modern closure and exchange terminology is an excellent interpretive framework, but the present appendix does not retroactively attribute every modern phrase to Wyler.

The later *Handbook of Incidence Geometry*, especially the chapter on dimensional linear spaces, gives the mature closure-space perspective: varieties form a closure system, and strong exchange leads to the usual finite-rank geometric-lattice structure [7, 8]. This is the natural modern context for the project, but it is a secondary structural comparison rather than a quotation of Wyler’s own terminology.

E.3 The original three-dimensional flat experiment

The first Wyler phase of CGJteamLab was deliberately conservative. Over the already established Hilbert 3D incidence API it introduced:

```
HilbertFlat3D
HilbertSpan3D
HilbertJoin3D
HilbertLineCarrier3D
HilbertPlaneCarrier3D
```

The definition of span used the standard closure principle:

$$\text{Span}(X) = \bigcap \{F : F \text{ is a flat and } X \subseteq F\}.$$

The resulting rank-three generator ladder was

$$\begin{aligned} \{A, B\}, A \neq B &\implies \text{Span}\{A, B\} = \text{line carrier}, \\ \{A, B, C\}, A, B, C \text{ noncollinear} &\implies \text{Span}\{A, B, C\} = \text{plane carrier}, \\ \{A, B, C, D\} \text{ noncoplanar} &\implies \text{Span}\{A, B, C, D\} = E^3. \end{aligned}$$

This layer remains useful as a record of the first structural decomposition of Book XI. It is no longer the final foundational boundary of the Wyler route.

E.4 The current foundation: HilbertWylerAxioms

The present dimension-free incidence foundation is

```
class HilbertWylerAxioms ...
```

in

```
CGJteamLab/HilbertWylerAxioms.lean
```

It retains the project’s ordinary point, line, and plane signatures:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
```

but replaces the dimension-specific Hilbert spatial package by a smaller point–line–plane theory.

The common incidence fields are:

- two distinct points on every line;
- three noncollinear points on every plane;
- existence of a plane through three noncollinear points;
- uniqueness of that plane;
- line-in-plane closure from two distinct points.

The final field is

```
wyler_i7_intersection_line
```

which gives the exact line of intersection in the characteristic higher-dimensional configuration built from two lines in one plane and an external point.

The file states the intended boundary explicitly. The class does *not* assume:

- Hilbert I.7 in its globally three-dimensional form;
- Hilbert I.8;
- Smith I.5;
- Mac Lane–Steinitz exchange;
- any ambient dimension bound.

This is the crucial improvement over the first 3D flat experiment.

E.5 Wyler I.7 is an axiom; Smith I.5 and exchange are theorems

The derived module

```
CGJteamLab/HilbertWylerTheory.lean
```

makes the dependency direction explicit.

First, the common part of `HilbertWylerAxioms` is exposed as the existing Smith incidence core:

```
hilbertWyler_smithIncidenceCore
```

Second, the Wyler I.7 field is exported in the already developed production form:

```
hilbertWyler_wylerI7IntersectionLine
```

Then the theory proves:

```
hilbertWyler_implies_smithI5
hilbertWyler_implies_smithSpanExchange
```

Thus the architecture is not

Wyler + Smith + exchange as unrelated axioms.

It is

Hilbert–Wyler common core + $I.7_{\text{Wyler}} \implies I.5_{\text{Smith}} \implies \text{exchange}$.

This correction is important for the documentation. Earlier project notes described exchange as a future possibility. That is no longer accurate: the exchange theorem has now been derived and belongs to the production theory.

E.6 Generated flats and dimension-free closure

The Smith/Wyler theory generalizes the original 3D closure experiment. The essential idea is unchanged: generated objects are defined by closure, not by coordinates.

The dimension-free development is valuable for two reasons.

First, it separates the logical role of incidence from metric geometry. A generated plane, a line carrier, or a span-exchange theorem does not need to know anything about right angles.

Second, the theory provides a language in which a higher-dimensional ambient space can contain ordinary local planes and local three-dimensional configurations without forcing the whole space to satisfy Hilbert’s three-dimensional axiom I.7.

This is exactly the distinction needed for the E^4 program.

E.7 Derived local three-space inside E^4

The module

```
CGJteamLab/HilbertWylerE4Local3D.lean
```

reconstructs the nontrivial local 3D incidence facts inside derived E^4 hyperplanes.

The file proves, in particular, the two facts needed by the historical local 3D API:

- two distinct ambient planes contained in one E^4 hyperplane and sharing a point have the appropriate further common incidence;

- every derived E^4 hyperplane contains four ambiently noncoplanar points.

The first argument uses a Wyler one-point generation formula derived through Smith I.5. No new axiom is introduced at this stage.

This establishes the intended architecture:

dimension-free ambient incidence $\longrightarrow$ derived E^4 hyperplane $\longrightarrow$ local 3D incidence inside the hyperplane.

The local 3D theory is therefore a theorem about a derived subgeometry, not a second global axiom system imposed on the ambient E^4 .

E.8 Why Book XI remains the right laboratory

Book XI repeatedly alternates among:

1. generated objects: the line or plane determined by given data;
2. intersections: exact meets of planes and carriers;
3. order: betweenness and Pasch;
4. metric structure: congruence and perpendicularity;
5. Euclidean structure: uniqueness of parallels.

A successful higher-dimensional architecture must keep these layers separate. The Wyler route is useful precisely because it prevents a theorem that needs only incidence closure from being packaged under a global three-dimensional metric class.

Propositions XI.16–XI.20 provide a particularly clear sequence of tests.

E.9 Proposition XI.16: exact meets and an incidence-only conclusion

XI.16 states that a cutting plane intersects two parallel planes in parallel section lines.

The current theorem constructs the two section lines by exact carrier identities:

$$\text{carrier}(\pi) \cap \text{carrier}(\sigma) = \text{carrier}(l),$$

$$\text{carrier}(\rho) \cap \text{carrier}(\sigma) = \text{carrier}(m).$$

The two lines lie in the cutting plane. If they had a common point, that point would belong to both parallel reference planes, a contradiction.

This proposition is an ideal carrier-calculus theorem:

$$\text{meet construction} + \text{plane disjointness} \implies \text{line parallelism}.$$

No order, congruence, perpendicularity, or Group IV uniqueness is required by the core proof.

E.10 Proposition XI.17: the affine endpoint

The classical statement of XI.17 is proportional: three parallel planes cut two transversals in the same ratio.

The canonical Hilbert theorem retains the Book V/VI style of the source. It uses XI.16 to obtain two section parallelisms, applies a plane-local VI.2 relation twice, and composes the resulting raw proportions.

The Wyler development now has a genuinely different endpoint:

```
euclid_proposition_11_17_wyler_affine
```

with conclusion

```
HilbertWylerSameDivision Geo A E B C F D
```

The relation `HilbertWylerSameDivision` records equality of affine division position of ordered triples. It is generated by the elementary parallel-section configuration and closed under the natural symmetry, reversal, and transitivity operations.

Hence the Wyler proof no longer needs to represent a ratio by a congruent acute angle. Its spatial/affine spine is:

$$\begin{array}{c}
 EO \parallel BD \\
 \Downarrow \\
 (A, E, B) \sim (A, O, D) \\
 AC \parallel OF \\
 \Downarrow \\
 (A, O, D) \sim (C, F, D) \\
 \Downarrow \\
 (A, E, B) \sim (C, F, D).
 \end{array}$$

The older raw-proportion theorem remains a compatibility bridge to the canonical Book V/VI interface; it is not the foundational endpoint of the Wyler route.

E.11 Proposition XI.18: two independent proofs of the same theorem

XI.18 is now formalized twice:

```
CGJteamLab/Proposition11_18.lean
CGJteamLab/Wyler/Proposition11_18.lean
```

with final theorems

```
euclid_proposition_11_18
euclid_proposition_11_18_wyler
```

The theorem types agree deliberately. The direct file contains a Hilbert 3D proof and does not import the Wyler Book XI route. The Wyler file rebuilds the argument using the Wyler-side normal and plane-perpendicular infrastructure.

Both proofs use the neutral definition in

```
Hilbert3DPlanePerpendicular.lean
```

so the comparison concerns proof architecture rather than theorem statement.

The mathematical decomposition is:

exact section + planar perpendicular comparison + transport of a normal across a parallel line.

XI.18 is therefore a useful test of how metric structure can sit above two different incidence organizations.

E.12 Proposition XI.19: global existence and neutral common-point core

XI.19 is also retained in two independent production forms.

Its architecture separates naturally.

First, one must prove that the two traces

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi$$

meet. If they were disjoint, they would be parallel in π . XI.Def.4 supplies normals to π inside ρ and σ ; XI.6 makes those normals parallel; XI.15 would then force

$$\rho \parallel \sigma,$$

contradicting the hypothesis that ρ, σ cut one another.

Second, once a point

$$D \in \rho \cap \sigma \cap \pi$$

is known, the remaining theorem is neutral. The Wyler-side module isolates this as

```
euclid_proposition_11_19_common_point_wyler
```

XI.Def.4 produces one normal in each perpendicular plane, XI.13 identifies the two normals at D , and exact meet data identify that common normal with the section $\rho \cap \sigma$.

Thus XI.19 distinguishes two problems that are easy to conflate:

global existence of the triple point vs. local uniqueness of the normal.

E.13 Proposition XI.20: a result below Wyler I.7

XI.20 is the sharpest dimensional-transfer result obtained so far.

The neutral metric core is isolated in

```
HilbertTrihedralAngleMetric.lean
```

and uses:

- synthetic angle order;
- SAS;
- Euclid I.20;
- Euclid I.25;
- same-ray transport.

Only one step is ambient: an auxiliary triangle must be shown nondegenerate. The direct Hilbert proof uses the ordinary 3D calls

```
HilbertSpaceIncidence.plane_through  
HilbertSpaceIncidence.line_in_plane
```

The independent Wyler theorem

```
euclid_proposition_11_20_wyler
```

replaces those calls by

```
A.plane_through  
A.line_in_plane
```

for

```
[A : HilbertWylerAxioms Geo]
```

Crucially, it does *not* use

```
A.wyler_i7_intersection_line
```

Nor does the proof use Smith I.5, exchange, or an ambient dimension bound.

Therefore XI.20 lies strictly below the genuinely new Wyler axiom. Its ambient requirement is only the common incidence core:

three noncollinear points determine a plane + two points of a line in a plane force the whole line into it.

This is stronger than saying that XI.20 can be reduced to a local 3-flat. No local 3-flat is required at all.

E.14 The E^n lesson of XI.20

XI.20 gives a clean test of what “dimension-free” should mean in the project.

The relevant contradiction takes place in a single ordinary plane. If an auxiliary point D were collinear with two points X, Y , then D would lie in the plane OXY . Since O, D lie in that plane and O, D, Z are collinear, line-in-plane closure would force Z into the same plane, contradicting noncoplanarity.

Nothing in this argument counts ambient dimensions.

Accordingly the same proof pattern is meaningful in E^4 and more generally E^n , provided the common point–line–plane incidence core and the planar metric theory are available.

This is exactly the kind of theorem the Hilbert/Wyler comparison was intended to reveal: one that looks “solid” in Euclid but whose formal ambient-space content is much weaker than global three-dimensionality.

E.15 Proposition XI.21: line-in-plane closure is enough

XI.21 strengthens the dimensional-transfer conclusion obtained from XI.20.

The independent Wyler theorem is

```
euclid_proposition_11_21_wyler
```

with hypotheses

```
[HilbertIncidence Geo]
[S : HilbertSpacePrimitive Geo]
[HilbertCongruence Geo]
[W : HilbertWylerAxioms Geo]
```

The proof does not call the direct Hilbert XI.21 theorem. It independently reconstructs the opposite-ray trihedral configuration and calls

```
euclid_proposition_11_20_wyler
```

Its new spatial obligation is particularly small. Extend AC through A to X . To prove that A, B, X, D remain noncoplanar, assume that they lie in a plane π . Since $A, X \in \pi$ and $A \neq X$, the whole line AX lies in π . Collinearity of C, A, X then forces $C \in \pi$, contradicting the original noncoplanarity of A, B, C, D .

Unlike XI.20, this argument does not even need to construct a plane through three noncollinear points. The plane π is supplied by the contradictory assumption. Consequently the only Wyler field used by the new XI.21 spatial step is

```
W.line_in_plane
```

The proof does not use

```
W.plane_through
W.wyler_i7_intersection_line
```

and it does not use Smith I.5, exchange, a generated local 3-flat, or an ambient dimension bound.

The rest of the theorem is neutral metric/order geometry:

$$\begin{array}{c}
 C - A - X, \quad D - A - Y \\
 \Downarrow \\
 A; D, X, B \text{ trihedral} \\
 \Downarrow \\
 \text{XI.20} \\
 \Downarrow \\
 \angle DAX + \angle XAB > \angle DAB \\
 \Downarrow \\
 \text{vertical-angle and congruence transport} \\
 \Downarrow \\
 \angle CAY + \angle BAX > \angle DAB \\
 \Downarrow \\
 \text{four-right-angle complement certificate.}
 \end{array}$$

E.16 The E^n lesson of XI.21

XI.21 is an even stronger transfer test than XI.20.

XI.20 needed two common incidence operations: a plane through three noncollinear points and line-in-plane closure. XI.21 itself needs only the second. Therefore no local 3-flat is required and no dimension-counting argument occurs.

The correct interpretation for later E^4 work is consequently not “reprove XI.21 in four dimensions.” Rather, XI.21 has become a reusable local theorem: whenever an E^4 Coxeter configuration contains three rays satisfying the neutral trihedral hypotheses, the same theorem can be applied directly.

This is precisely the transfer pattern the Hilbert/Wyler double route was designed to expose:

higher-dimensional configuration $\longrightarrow$ neutral local trihedral certificate $\longrightarrow$ existing XI.20/XI.21 library theorem

E.17 What is now formalized

The current Wyler-side production theory includes:

- a dimension-free common point–line–plane incidence core;
- Wyler I.7 as the additional dimension-free spatial incidence axiom;
- Smith I.5 as a theorem;
- Mac Lane–Steinitz exchange for the generated span as a theorem;

- generated-flat and span infrastructure;
- a derived E^4 hyperplane theory;
- local 3D incidence inside those derived E^4 hyperplanes;
- Book XI carrier/meet applications;
- an affine XI.17 endpoint;
- independent XI.18, XI.19, XI.20, and XI.21 proof routes;
- the planar XI.22 Wyler entry point;
- a production XI.23 Hilbert–Wyler route with an explicit 3D compatibility theorem.

This list supersedes the earlier Book XI checkpoint in which exchange and arbitrary-dimensional closure were still described as future work.

E.18 What remains open or deliberately incomplete

The present theory should still not be described as a completed higher-dimensional Euclidean metric geometry.

Among the remaining architectural questions are:

- a fully systematic dimension-free metric layer for perpendicularity and congruence of higher-dimensional flats;
- the most economical axiomatization of ambient dimension n beyond the concrete E^4 development;
- a general orthogonality theory for generated flats and hyperplanes;
- a systematic comparison between the Smith/Wyler closure theory and the local metric structures needed by later Euclidean books;
- identification, proposition by proposition, of which Book XI results genuinely need Wyler I.7 and which live already in the common core.

The project therefore has a real dimension-free incidence theory, but it does not claim that every metric construction of Euclidean E^n has already been reconstructed.

E.19 Relation to modern incidence geometry

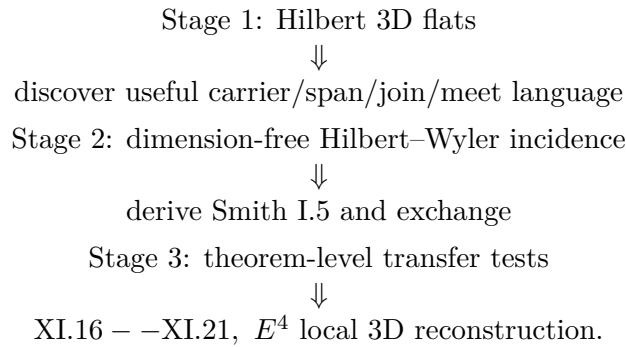
The modern theory of locally projective geometries provides an additional warning against overgeneralization. Dimension three has exceptional embedding behavior, while higher dimensions admit stronger general statements under suitable hypotheses [9].

For the present project the lesson is methodological. Success of a theorem inside E^3 does not by itself tell us which dimension-free axiom produced it. The proof term must be inspected.

The XI.20 and XI.21 results are exemplary because proof-term inspection shows that even Wyler I.7 is unnecessary; XI.21 itself needs only line-in-plane closure. Other propositions, especially those involving intersections of higher-dimensional generated objects, may lie higher in the Wyler–Smith–exchange hierarchy.

E.20 Checkpoint XI.21

The Wyler route has now moved through three stages.



The decisive refinement at XI.21 is that not every theorem useful in E^4 must pass through the local-3-flat machinery. XI.20 already lay below Wyler I.7; XI.21 lies lower still in its new spatial step, requiring only line-in-plane closure.

The current hierarchy is therefore genuinely useful as a proof-audit tool:

common incidence core $\subset$ Wyler I.7 layer $\subset$ derived exchange/local-flat theory,

with metric theorems attaching at the weakest level their proof terms actually require.

XI.20–XI.21 are the first clear examples of classical “solid geometry” theorems that should be carried forward as dimension-free reusable library results rather than rebuilt separately for E^4 .

E.21 Proposition XI.22: the Wyler route collapses to the common planar core

XI.22 is an important negative test for the Wyler hierarchy: no Wyler-specific spatial machinery is needed at all.

The independent Wyler-side entry point is

```
euclid_proposition_11_22_wyler
```

and its public assumptions are only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

In particular it does not use:

- HilbertSpacePrimitive;
- HilbertWylerAxioms;
- Wyler I.7;

- Smith I.5;
- exchange;
- a local 3-flat;
- an ambient dimension bound.

The direct Hilbert XI.22 theorem and the Wyler XI.22 theorem are independent entry points over the same neutral planar chord-comparison core; neither calls the other.

This is the cleanest possible transfer result. XI.22 does not need to be “generalized to E^4 ” because its theorem already lives below all spatial dimension machinery.

There is nevertheless an axiomatic caveat. The final I.22 construction used by both entry points depends on the project’s

```
hilbert_circle_circle_intersection
```

continuity axiom. Dimension independence and continuity independence are different questions, and XI.22 separates them sharply.

E.22 Proposition XI.23: the Hilbert–Wyler route is now production

XI.23 now has a production Hilbert–Wyler proof in

```
CGJteamLab/Wyler/Proposition11_23.lean
```

The final entry points are

```
euclid_proposition_11_23_wyler_core
euclid_proposition_11_23_wyler
```

The first theorem exposes the incidence architecture explicitly. It assumes

```
[W : HilbertWylerAxioms Geo]
```

and reconstructs the spatial incidence steps of XI.23 through that interface. The second theorem is the public Hilbert 3D result. It first derives the Wyler package using

```
hilbertWylerAxioms_of_hilbert3D
```

from

```
CGJteamLab/HilbertWyler3DCompatibility.lean
```

and then invokes the Wyler core.

This compatibility theorem is mathematically important. In three dimensions Wyler is not installed as a second independent axiom system. The dependency is

$$\text{Hilbert 3D} \implies \text{Hilbert--Wyler incidence} \implies \text{XI.23.}$$

No global typeclass instance is introduced; the conversion is explicit in the proof term.

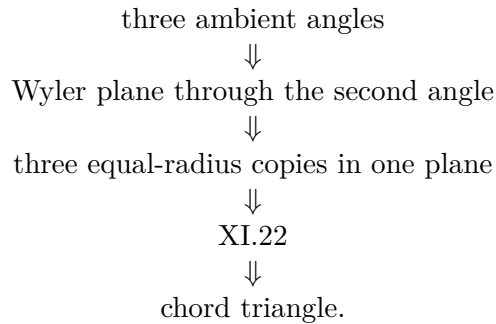
E.22.1 Where the Wyler route actually differs

The difference from the direct Hilbert proof is deliberately localized. The Wyler proof replaces the relevant ambient incidence calls by

```
W.plane_through
W.line_in_plane
W.plane_unique
```

The first operation chooses the common plane in which the three prescribed angles are normalized. The second keeps constructed rays and collinear points inside that plane without appealing directly to `HilbertSpaceIncidence.line_in_plane`. The third is used in the final noncoplanarity argument: if the apex and the three base points belonged to a second plane, uniqueness of the plane through the noncollinear base triangle would identify it with the base plane, contradicting that the apex lies outside it.

The normalization chain is therefore



After this incidence front end, the proof deliberately rejoins the common metric core. It reuses the already established planar and spatial machinery:

- the `PlaneGeo` transport layer;
- XI.22 and the chord-triangle noncollinearity argument;
- existence of the circumcenter;
- the circumradius comparison;
- the auxiliary right triangle determining the height;
- XI.12 to erect a normal to the base plane;
- spatial SAS to obtain equal lateral edges;

- SSS to recover the three prescribed face angles.

Thus the Wyler proof is not a duplicate metric development. It is an incidence refactor of XI.23 over a shared synthetic metric core.

E.22.2 What this result does and does not prove

The completed theorem is a genuine Wyler-routed XI.23 in three-dimensional Hilbert geometry. It is not merely an alias for the direct theorem: the normalization and final incidence arguments are independently routed through `HilbertWylerAxioms`.

It is equally important not to overstate the result. The production file does not use the earlier experimental E^4 local-hyperplane route. No `E4Dimension`, `E4Hyperplane`, or local hyperplane geometry is needed. The theorem remains in the Hilbert spatial metric environment because the apex construction still requires perpendicular-to-plane and spatial congruence theory.

The present result therefore resolves the 3D architectural question:

$$\text{Hilbert 3D metric geometry} + \text{Wyler-routed incidence} \implies \text{XI.23}.$$

The genuinely higher-dimensional question remains different:

$$\text{base plane} \subset \text{local metric 3-flat} \subset E^n.$$

To obtain an E^4 or general E^n XI.23 theorem one still needs the appropriate local perpendicularity and spatial congruence package on such a 3-flat. The incidence theory is no longer the missing part of the 3D proof; the missing higher-dimensional ingredient is metric localization.

E.23 Continuity and dimension are orthogonal axes at XI.23

The final kernel audits for the direct theorem, `euclid_proposition_11_23_wyler`, and `euclid_proposition_11_23` are identical:

```
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

This is the decisive audit result for the 3D Hilbert–Wyler refactor. The Wyler route changes the incidence organization of the proof, but it does not add a new kernel axiom. In particular, the explicit `HilbertWylerAxioms` dependency of the core theorem is discharged in the public theorem by the proved compatibility theorem `hilbertWylerAxioms_of_hilbert3D`; the resulting kernel dependency set is unchanged.

Thus XI.23 exposes two independent questions:

1. **Dimension transfer.** Can the spatial apex construction be confined to a suitable local 3-flat inside E^n ?

2. **Continuity.** Which circle-existence principles are required by the planar XI.22/circumcircle infrastructure?

The completed 3D Hilbert–Wyler theorem addresses the first axis at the level of incidence organization: the relevant plane and line operations are successfully routed through `HilbertWylerAxioms`. It does not remove the second axis. A future E^4 or general E^n XI.23 theorem may therefore be dimension-free in its incidence organization while still inheriting the same circle-continuity requirement in its planar slices.

This distinction is important for the project: “dimension-free” must never be used as a synonym for “neutral” or “continuity-free.”

E.24 Checkpoint XI.23

The Wyler-side transfer picture now has four distinct outcomes:

Proposition	transfer status
XI.20	dimension-free after a small common incidence argument
XI.21	even lower: line-in-plane closure suffices
XI.22	already purely planar; no Wyler spatial layer needed
XI.23	Wyler-routed 3D proof complete; metric localization in E^n open

XI.23 is therefore no longer an unfinished Wyler transfer in three dimensions. The production theorem demonstrates that its ambient incidence can be organized through the Hilbert–Wyler interface while the metric core is shared with the direct proof. In Hilbert 3D the Wyler package is derived, not added as an independent axiom.

At the same time XI.23 remains the first proposition in this sequence that genuinely tests the missing metric side of the E^n program. The next higher-dimensional target is now precise: equip a suitable local 3-flat with exactly the perpendicular-to-plane and spatial congruence infrastructure needed by the already completed XI.23 apex construction.

E.25 Proposition XI.24: carrier incidence and a shared metric core

Proposition XI.24 now has two production realizations:

```
euclid_proposition_11_24
euclid_proposition_11_24_wyler
```

Both the direct Hilbert theorem and the Hilbert–Wyler theorem have the same input configuration and the same conclusion:

```
HilbertParallelepipedConfiguration
HilbertXI24Conclusion
```

The configuration consists of three pairs of parallel planes,

$$\pi_0 \parallel \pi_1, \quad \rho_0 \parallel \rho_1, \quad \sigma_0 \parallel \sigma_1,$$

and the eight vertices determined by choosing one plane from each pair. The conclusion packages the six parallelogram faces and three explicit triangle-congruence witnesses for the three pairs of opposite faces.

The important architectural fact is that the direct and Wyler proofs differ at the incidence front end but deliberately rejoin at the metric core.

E.25.1 The direct route

The direct proof uses

```
euclid_proposition_11_16
```

twice on each face. For the face $ABCD$, for example, the pair ρ_0, ρ_1 cut by the face plane gives

$$AB \parallel CD,$$

while the pair σ_0, σ_1 gives

$$BC \parallel DA.$$

Hence $ABCD$ is a parallelogram.

This pattern is isolated in

```
hilbert_XI24_face_parallelogram
```

and then applied six times by

```
hilbert_XI24_all_faces_parallelograms
```

after permuting the three plane directions.

Thus the first stage of the direct proof is

$$\text{XI.16} \implies \text{six parallelogram faces.}$$

E.25.2 The Wyler route: exact carrier intersections

The Wyler realization replaces precisely this first stage.

Its one-face theorem is

```
hilbert_XI24_face_parallelogram_wyler
```

and it uses

```
euclid_proposition_11_16_wyler.
```

The Wyler XI.16 theorem returns exact carrier identities for the section lines. Schematically,

$$\text{carrier}(\rho_0) \cap \text{carrier}(\pi) = \text{carrier}(l_{AB}),$$

and

$$\text{carrier}(\rho_1) \cap \text{carrier}(\pi) = \text{carrier}(l_{CD}).$$

The endpoint incidences needed for the face proof are then recovered by rewriting through these carrier equalities. For example, from

$$B \in \rho_0, \quad B \in \pi$$

the proof obtains

$$B \in l_{AB}.$$

Only after the exact meet information has been used are the section lines converted to the project's point-defined relation

```
Geo.Parallel
```

and hence to a parallelogram.

This is a genuine proof difference. The Wyler XI.24 theorem is not an alias for the direct theorem and does not call the direct face-parallelogram lemma.

The six-face Wyler theorem is

```
hilbert_XI24_all_faces_parallelograms_wyler.
```

E.25.3 The common metric core

Once the six face parallelograms have been obtained, both routes use the same metric infrastructure.

For one pair of opposite faces, say

$$ABCD \quad \text{and} \quad EFGH,$$

the proof compares the diagonal triangles

$$BAC \quad \text{and} \quad FEG.$$

First, I.34 is applied inside the two relevant side planes. The helper

```
hilbert_XI24_face_opposite_sides_congruent
```

moves the ambient face to the induced geometry

```
PlaneGeo
```

and transports the resulting opposite-side congruences back to the ambient space. This gives

$$BA \cong FE, \quad BC \cong FG.$$

Second, the helper

```
hilbert_XI24_parallelogram_directed_opposite_sides
```

extracts the directed spatial parallel-segment data required by XI.10. For the first opposite pair the two side faces yield

$$AB \parallel_{\text{dir}} EF, \quad CB \parallel_{\text{dir}} GF.$$

The noncoplanarity assumption required by XI.10 is derived from uniqueness of the plane through a noncollinear triple together with

$$\pi_0 \parallel \pi_1.$$

If A, B, C, E, F, G lay in one plane, that plane would have to equal both π_0 and π_1 , contradicting their parallelism.

XI.10 therefore gives

$$\angle ABC \cong \angle EFG.$$

Finally, spatial SAS gives the complete congruence of the diagonal triangles. The proof uses

```
hilbert_space_sas_third_side
hilbert_space_sas_remaining_angles
```

and packages the result as

```
TriangleCongruenceResult Geo B A C F E G.
```

The direct theorem for this step is

```
hilbert_XI24_opposite_pi_faces_triangle_congruent
```

and the Wyler theorem is

```
hilbert_XI24_opposite_pi_faces_triangle_congruent_wyler.
```

E.25.4 The other two opposite-face pairs

Only one pair of opposite faces is proved metrically from scratch. The other two are obtained by permuting the three parallel-plane directions.

The direct theorem is

```
hilbert_XI24_all_opposite_faces_triangle_congruent
```

and the Wyler theorem is

```
hilbert_XI24_all_opposite_faces_triangle_congruent_wyler.
```

The three triangle witnesses are

opposite faces	triangle witness
π_0/π_1	$BAC \cong FEG,$
ρ_0/ρ_1	$BAF \cong CDG,$
σ_0/σ_1	$BCF \cong ADE.$

Thus neither the direct nor the Wyler file duplicates the full metric proof three times.

E.25.5 Why XI.24 stops at triangle congruence

Euclid concludes that opposite parallelograms are equal by using the fact that each parallelogram is twice one of its diagonal triangles.

The present library does not introduce a general theory of area or equality of planar figures merely for XI.24. Instead,

```
HilbertXI24Conclusion
```

stores the stronger synthetic witness that the corresponding diagonal triangles are congruent.

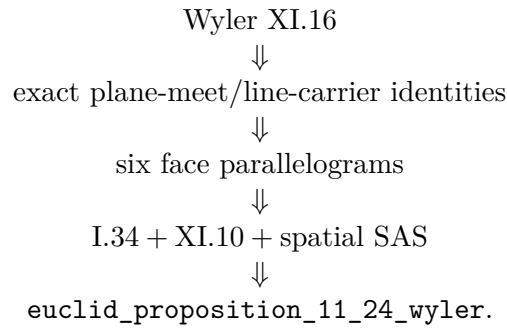
This preserves the geometric content needed by XI.24 while avoiding an unrelated measure layer. A future area theory can later recover Euclid’s final “double of equal triangles” sentence from these witnesses.

E.25.6 The two proof DAGs

The direct route is

$$\begin{array}{c}
 \text{XI.16} \\
 \Downarrow \\
 \text{six face parallelograms} \\
 \Downarrow \\
 \text{I.34 + XI.10 + spatial SAS} \\
 \Downarrow \\
 \text{three opposite-face triangle congruences} \\
 \Downarrow \\
 \text{euclid_proposition_11_24.}
 \end{array}$$

The Wyler route is



The lower metric part is intentionally shared.

E.25.7 Axiom audit

The kernel audit of the direct theorem is

```
[proptext,
  Classical.choice,
  Geometry.bookZero_nullSegment1,
  Quot.sound]
```

The audit of

```
euclid_proposition_11_24_wyler
```

is exactly the same:

```
[proptext,
  Classical.choice,
  Geometry.bookZero_nullSegment1,
  Quot.sound]
```

Thus the Wyler route introduces no additional kernel axiom.

The contrast with XI.23 is instructive. XI.24 does not depend on

```
Geometry.bookZero_nullSegment2
Geometry.hilbert_circle_circle_intersection
```

and therefore does not inherit the circle-circle continuity boundary of XI.23.

This equality of the two XI.24 audit profiles is strong evidence that the direct/Wyler difference lies in proof organization at the incidence layer, not in geometric strength.

E.25.8 Dimension status

The present Wyler XI.24 theorem is still proved in the existing Hilbert three-dimensional environment. It does not use E^4 , and it is not yet a general E^n theorem.

The result nevertheless advances the dimension-free program in a useful way: it shows that the carrier/meet formulation of XI.16 can be substituted at the top of a substantial solid-geometry theorem without changing the downstream metric proof or its kernel axiom profile.

The remaining higher-dimensional question is therefore not whether the XI.24 incidence front end can be expressed through carriers; that part is already explicit. The question is how much of the shared metric core can be localized inside an appropriate three-dimensional flat of a higher-dimensional ambient geometry.

E.26 Proposition XI.25: a Wyler front end with a shared magnitude core

XI.25 now has a production Hilbert–Wyler realization in

```
CGJteamLab/Wyler/Proposition11_25.lean
```

with terminal theorem

```
HilbertXI25SolidCutWitness.eudoxusProportion_from_seed_cut_wyler
```

Unlike XI.24, XI.25 does not introduce a new Wyler incidence theorem. Its architectural interest is more precise: the completed Wyler XI.24 theorem can be substituted everywhere XI.25 needs a parallelepiped face package, while the much larger magnitude and Eudoxus development remains shared.

The neutral XI.25 structures are imported from the direct production file. They include the two-slab and ordered-cut configurations, base and solid equivalence classes, geometric proper-part relations, repeated positive multiples, comparison-faithful families, canonical width points, section planes, and the generic Eudoxus machinery.

The Wyler route rebuilds only the layers whose proof genuinely depends on XI.24. In particular it provides

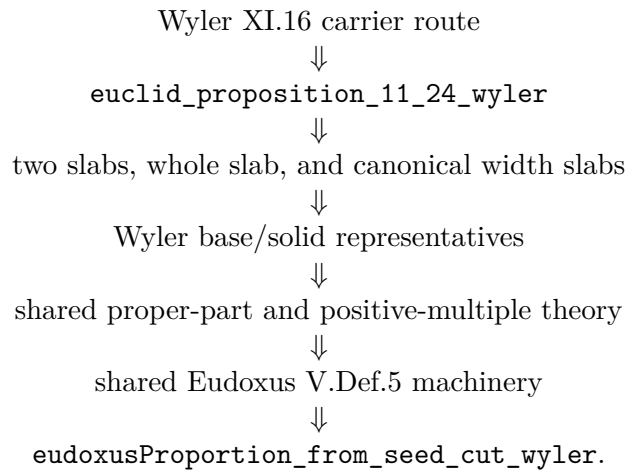
```
hilbert_XI25_twoSlab_XI24_wyler
hilbert_XI25_whole_XI24_wyler
hilbert_XI25_equal_width_all_edges_wyler
hilbert_XI25_adjacent_elementary_solids_equal_wyler
hilbert_XI25_adjacent_solids_equal_of_base_equal_wyler
HilbertXI25SolidCutWitness.widthXI24_wyler
HilbertXI25SolidCutWitness.widthSolid_wyler
HilbertXI25SolidCutWitness.widthBase_wyler
HilbertXI25SolidCutWitness.toWidthRealization_wyler
```

This list should be read structurally rather than as a second copy of XI.25. The point is that every place where the direct proof asks XI.24 for the geometry of a slab is rerouted through

`euclid_proposition_11_24_wyler`

and its exact carrier-intersection proof. Once the corresponding base and solid representatives have been obtained, the strict proper-part order, positive-multiple systems, comparison trichotomy, and V.Def.5 packaging are the common XI.25 theory.

The resulting proof diagram is therefore



This is a stronger test of the Hilbert–Wyler design than XI.24 alone. XI.24 showed that the incidence front end could be changed while retaining a shared metric core. XI.25 shows that the same replacement survives through a much larger theorem whose endpoint is not face congruence but a non-numerical proportion of solid magnitudes.

At the same time the result should not be overstated. The production theorem still lives in the existing three-dimensional Hilbert environment. It is not an E^n theorem, and XI.25 introduces no new Wyler axiom or dimension-free magnitude theory. The Wyler contribution remains localized where the project intends it to remain: in the organization of spatial incidence and carrier geometry.

E.27 Checkpoint through XI.25

The Wyler-side transfer picture is now

Proposition	transfer status
XI.20	dimension-free after a small common incidence argument
XI.21	line-in-plane closure suffices
XI.22	already purely planar
XI.23	Wyler-routed 3D proof; higher-dimensional metric localization open
XI.24	carrier-routed 3D proof; same metric core
XI.25	Wyler XI.24 front end; shared proper-part and Eudoxus core

The progression from XI.24 to XI.25 sharpens the project strategy. The aim is not to replace every mature Hilbert or Book V construction by a new Wyler-specific version. The incidence layer is changed where the carrier/flat formulation gives genuinely different information; the synthetic metric and magnitude layers are reused when their logical content is already dimension-neutral or proposition-independent.

For XI.25 this separation is now realized in production rather than merely proposed as an architectural goal.

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